

*Janne's Guide to*

# Russian Ground Equipment

Field reference for recognition and unit association

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54 systems: fires, air defence, radars, electronic warfare, command and signals, and engineers. What each looks like, which units hold it, how far it reaches.

**F**

FIRES

**D**

AIR DEFENCE

**R**

RADARS

**E**

ELECTRONIC  
WARFARE

**G**

ENGINEERS

**L**

SUPPORT  
VEHICLES

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# About this guide

**Status of this proof.** This proof holds 54 entries, all checked claim by claim against their sources. Figures may still be wrong. Do not use this proof operationally.

**How to read an entry.** The black box at top right is the mobility code. *Operated by* lists unit types that hold the equipment. Each row ends with a tag of three parts, such as `[D 2016 H]`: basis, date and confidence. Basis is D for doctrinal tables of organisation, O for observed with an identified unit, R for reported or claimed. Confidence is H, M or L. *Recognise* bullets follow a fixed order and start with the first letter of the feature. Unknown values are printed INA.

**Caution.** Since 2022, army and district assets are often parcelled out to groupings of forces, new formations are under strength, and legacy or civilian vehicles appear in any unit. Equipment indicates a unit type. It does not prove it.

- TR** Tracked, cross-country
- WH-AT** Wheeled all-terrain
- WH-LT** Wheeled, limited off-road
- TW** Towed
- AIR-RW** Helicopter

- TR-A** Tracked, swims without preparation
- WH-AT-A** Wheeled all-terrain, swims
- RD** Roadbound, hard surface only
- MP** Man-portable
- AIR-FW** Fixed-wing aircraft

## BM-21 Grad 122 mm MRL 5M-21 «Град»

NATO: M1964

WH-AT

Truck-mounted 40-tube 122 mm area-fire rocket launcher (divisional and brigade MRL) In service 1963

**OPERATED BY** motor rifle brigade, MRL battalion; 6 launchers per battery, 18 per battalion [D 2016 M]

artillery regiment, one Grad MRL battalion in division field artillery

regiment, being replaced by Tornado-G [D 2019 M]

motor rifle regiment, BTG, attached MRL element, 3 in one BTG

[O 2022-04 M] seen with 228th Motor Rifle Regiment, 90th Tank Division

motor rifle brigade, BTG, attached BM-21 battery, 1 battery

[O 2022-04 M] seen with 57th Motor Rifle Brigade, 5th Army

**INDICATES** A group of about 18 launchers points to a brigade MRL battalion; 6 launchers to one battery. A few launchers inside a battalion tactical group suggest an attached MRL element.

**RECOGNISE Cab** Unarmoured Ural 6x6 truck cab [Ural-375D; Ural-4320 on BM-21-1]. The 40-tube pack can be turned away from the cab.

**Body** Rear pack of 40 tubes in four rows of ten, each tube 3 m long, on a welded cradle; elevates 0 to +55 degrees. Traverse 102 left, 70 right of the vehicle.

**Axles** Three axles, 6x6 truck chassis, wheeled; not amphibious.

**Details** Stabilizing jacks at the rear for firing (stated for the BM-21V; the text says the BM-21 has the same). Reload rockets carried on separate resupply trucks.

**NOT TO BE CONFUSED WITH** BM-21V Grad-V (9P125) 12 tubes on a GAZ-66B 4x4 truck, airborne version

9P138 Grad-1 36 tubes on a ZIL-131 6x6 truck

RM-70 (Czechoslovak) 40 tubes on an 8x8 Tatra 813 with an armoured cab and room behind the cab for 40 spare rockets



MILRU (Ministry of Defence of the Russian Federation), CC BY 4.0

**MOBILITY** Unarmoured truck chassis on wheels with central tyre inflation. Fords 1.5 m without preparation. Not amphibious.

Weight, combat 13.7 t BM-21 on Ural-375D, loaded with crew

L × W × H 7.35 × 2.4 × 3.09 m travelling

Road speed 75 km/h maximum, hard road, loaded

Road range 750 km

Range, 9M22U HE-frag

20.4 km maximum, 9M22U (M-210F) HE-fragmentation rocket

Minimum range, 9M22U HE-frag

5 km 9M22U (M-210F) HE-fragmentation rocket

Range, 9M28F HE-frag

15 km maximum, 9M28F HE-fragmentation rocket

Minimum range, 9M28F HE-frag

1.5 km minimum effective, 9M28F HE-fragmentation rocket

**SOURCES** ATP CNA IIS25 MISS6 RUSIPL RWW WEG13 WEG2015 WP-EN13 WP-RU15

## Tornado-G 9K51M «Торнадо-Г»

WH-AT

122 mm 40-tube truck MRL; automated Grad upgrade for brigade and division artillery In service 2012

**OPERATED BY** motor rifle division, artillery regiment, MRL battalion (replacing Grad battalion) [D 2019 H]

tank division, artillery regiment, MRL battalion [D 2019 H]

motor rifle brigade, brigade artillery, MRL battalion, INA for Tornado-G; it replaces the BM-21, 18 per brigade MRL battalion (RWW p.100); 18

delivered to one brigade 2012 (RBASE) [D 2019 M]

motor rifle brigade, 18 [O 2012-04 M] seen with 20th Guards Separate

Motor Rifle Brigade

**INDICATES** Tornado-G replaces the Grad in the MRL battalion of motor rifle and tank brigades and of division artillery regiments. A group of them suggests a manoeuvre brigade or division artillery group nearby.

**RECOGNISE Cab** Ural-4320 6x6 truck; cab seats driver and two.

Launcher is laid and returned to travel position from inside the cab.

**Body** Single pack of 40 122 mm tubes, 10 × 4, on the rear body; traverses 102 deg left and 70 deg right, elevates 0 to 55 deg.

**Axles** Three axles, all driven (6x6); Ural-4320 has central tyre pressure regulation.

**Details** Rebuilt BM-21 Grad on Ural-4320 (about 20 rebuilt in 2019); additions are GLONASS navigation, a ballistic computer and a video monitor in the cab.

**NOT TO BE CONFUSED WITH** BM-21-1 Grad [Ural-4320] p. 4 Same truck and 40-tube pack. Tornado-G lays from the cab with no crew at the launcher; no reliable external mark found in text sources.

9A51 Prima 50 tubes instead of 40 on the same Ural-4320 chassis.

9K51S Tornado-S p. 4 12 tubes of 300 mm on an 8x8 MAZ-543M chassis, against 40 small tubes on a 6x6 Ural.



Vitaly V. Kuzmin, CC BY-SA 4.0

**MOBILITY** 6x6 truck; WEG claims small-arms protection, WEG 2015 BM-21 page gives none. Fords about 1.5 to 1.75 m. Can fire from an unsurveyed position and leave straight after firing.

Weight, combat 14 t

L × W × H 7.35 × 2.5 × 3.1 m travelling

Road speed 85 km/h maximum

Road range 650 km cruising range

Range, 9M521 HE-frag

40 km maximum, 9M521 extended-range HE-frag rocket; attribution to 9M521 from RUWIKI [Karpenko], WEG, ZR and WWW1 give 40 km as system maximum

Range, 9M522 HE-frag

37.5 km maximum, 9M522 rocket, separable HE-frag warhead

Range, 9M538 HE-frag

20 km maximum, 9M538 rocket, enhanced monobloc HE-frag warhead, claim

Range, 9M539 HE-frag

20 km maximum, 9M539 rocket, separable HE-frag warhead; Tecmash statement at IDEX 2019, claim

**SOURCES** CNA D24 EDR GS16 IIS25 JF18 MAILRU2 ORYX RBASE RDP5 RWW TASS12 WEG41 WEG2015 WMC16 WP-RU58 WWW13 ZR

## BM-27 Uragan 9П140 «Ураган»

NATO: M1977

WH-AT

220 mm 16-tube MRL on ZIL-135LM 8x8 for area fire and remote minelaying at 10–35 km In service 1975

**OPERATED BY** rocket artillery regiment, three rocket launcher battalions, 54 per regiment (3 battalions of 18) [D 1991 M] rocket artillery brigade, rocket launcher battalions, 54 per brigade (3 battalions of 18) [D 1991 M] self-propelled artillery regiment, Uragan rocket artillery battalion, 8 per battalion [R 2017-12 M]

**INDICATES** A 9P140 points to army-level rocket or artillery brigade fires, or to a Uragan battalion (8 launchers) in a division artillery regiment; the MoD also planned such battalions for combined arms brigades. In the Airborne Forces it indicates the artillery brigade.

**RECOGNISE** Cab ZIL-135LM cab with a blast shield that is raised during firing.

**Body** 16 tubes in three banks at the rear: two lower rows of six, upper row of four. In travel the pack faces rearward; muzzles covered by a sheet-metal cover when loaded.

**Axles** Four axles, 8x8; only the front and rear axles steer. Two petrol engines, each driving the wheels on one side.

**Details** Two manually emplaced hydraulic jacks at the rear of the chassis are lowered for firing. Newer launchers sit on a 6x6 BAZ-69092 chassis.

**NOT TO BE CONFUSED WITH** 9T452 transloader Same ZIL-135 chassis but a flatbed with 16 rockets in two stacks either side and a loading device between them; no tube pack.

**BM-21 Grad** p. 4 40 tubes on a Ural-375 truck, against 16 larger tubes on an 8x8 ZIL-135.

**9A52 Smerch** 12 tubes of 300 mm on a MAZ-543A chassis, against 16 tubes on a ZIL-135.



Unknown author, CC BY 4.0

**MOBILITY** Unarmoured 8x8 petrol truck; rear jacks must be lowered to fire; fords 1.2 m unprepared; 60% gradient.

|                |                                     |
|----------------|-------------------------------------|
| Weight, combat | 20 t                                |
| L × W × H      | 9.63 × 2.8 × 3.23 m travel position |
| Road speed     | 65 km/h maximum                     |
| Road range     | 500 km cruising                     |

|                              |                                     |
|------------------------------|-------------------------------------|
| Range, 9M27F HE-frag         | 35 km maximum, 9M27F HE-frag rocket |
| Minimum range, 9M27F HE-frag | 10 km 9M27F HE-frag rocket          |

|                              |                                                            |
|------------------------------|------------------------------------------------------------|
| Range, 9M27K cluster         | 35 km maximum, 9M27K rocket, 30 9N210 HE-frag submunitions |
| Minimum range, 9M27K cluster | 10 km 9M27K rocket, 30 9N210 HE-frag submunitions          |

|                                |                                                        |
|--------------------------------|--------------------------------------------------------|
| Range, 9M27K2 AT mines         | 35 km maximum, 9M27K2 rocket, 24 PTM-1 anti-tank mines |
| Minimum range, 9M27K2 AT mines | 10 km 9M27K2 rocket, 24 PTM-1 anti-tank mines          |

**SEEN WITH** 9T452 transloader; ZIL-135LM (1)

**SOURCES** CNA D242 FAS4 GS19 IISS25 IZ17 MIL31 MIL32 MIL110 MPER ORYX RUSIPL RWW W1R WEG46 WEG2015 WP-EN49 WP-EN50 WP-RU62

## Uragan-1M 9K512 «Ураган-1М»

WH-AT

Heavy 220/300 mm MRL with swappable rocket pods; successor to 9K57 Uragan In service 2007

**OPERATED BY** self-propelled artillery regiment, rocket artillery battalion (planned replacement for 9K57 Uragan, MoD statement; no holding confirmed) [R 2017-12 L]

artillery brigade, MRL battalion (Uragan 'being succeeded' by Uragan-M1; no holding confirmed) [R 2019-10 L]

**INDICATES** Planned for divisional artillery regiments and army artillery brigades as the Uragan successor. Few were built and front-line use is unconfirmed, so a sighting is notable and should be checked against the 6x6 9K57-1.

**RECOGNISE** Cab Forward-control cab on the MZKT-7930 8x8 chassis, seating the driver and two passengers.

**Body** Launcher on a turntable at the rear carrying two launcher pods instead of a fixed tube rack.

**Axles** Four axles (8x8), all driven; same chassis family as the 9T249 transporter-loader.

**Details** 300 mm pods hold 6 tubes each in two layers of three; 220 mm pods hold 15 tubes each. Pods can be mixed, one of each calibre.

**NOT TO BE CONFUSED WITH** 9K57-1 Uragan-1 p. 5 6x6 BAZ-69092 chassis with armoured cab, three axles not four; Russian press also calls it Uragan-1M.

**9P140 Uragan** p. 5 Fixed rack of 16 tubes on a ZIL-135 8x8; no detachable pods.

**9A52-2 Smerch** p. 6 Fixed 12-tube block in 3 rows of 4 on a MAZ-543M 8x8; Uragan-1M has two separate 6-tube pods.

**9A52-4 Tornado** p. 6 KamAZ-63501 8x8 truck with a replaceable 6-rocket module (WEG 2015); Uragan-1M carries two pods on the larger MZKT-7930.



Ministerstwo Obrony Rosyjskiej Federacji | Министерство обороны Российской Федерации | Ministry of Defence of the Russian Federation, CC BY 4.0

**MOBILITY** Heavy 8x8 truck; fords 1.4 m, climbs 45% gradient, crosses 2.0 m trench.

|                |                                |
|----------------|--------------------------------|
| Weight, combat | 40 t                           |
| L × W × H      | 13.0 × 3.0 × 3.29 m travelling |
| Road speed     | 70 km/h maximum, maximum, road |
| Road range     | 1,000 km maximum, road         |

|                                     |                                                        |
|-------------------------------------|--------------------------------------------------------|
| Range, 9M27F HE-frag 220 mm         | 35 km maximum effective, 9M27F unguided HE-frag rocket |
| Minimum range, 9M27F HE-frag 220 mm | 10 km minimum effective, 9M27F unguided HE-frag rocket |

|                             |                                                                            |
|-----------------------------|----------------------------------------------------------------------------|
| Range, 9M27K cluster 220 mm | 35 km maximum effective, 9M27K rocket, 30 9N210 fragmentation submunitions |
| Range, 9M55F HE-frag 300 mm | 70 km maximum, 9M55F HE-frag rocket with trajectory correction             |

|                                     |                            |
|-------------------------------------|----------------------------|
| Minimum range, 9M55F HE-frag 300 mm | 20 km 9M55F HE-frag rocket |
|-------------------------------------|----------------------------|

**SEEN WITH** 9T249

**SOURCES** AR22 CNA D24A D24B IISS25 IZ17 ORYX RAND RWW U244 W1A W1C W1E WEG45 WEG2015 WP-EN48 WP-RU61 ZR

## BM-30 Smerch 9A52-2 «Смерч»

NATO: 280mm M1983

WH-AT

Heavy 300 mm 12-tube rocket launcher for deep area fire; Tornado-S adds guided rockets In service 1987

**OPERATED BY** rocket artillery brigade, three groups of 4 x 9A52-2 in ATP fig. 5-16; whether each group is a battalion or a battery is not readable from the text, 12 x 9A52-2 shown in ATP fig. 5-16 (3 groups of 4) **D 2024 M**

rocket artillery brigade, 7 delivered April 2017, 5 more announced

**O 2017-04 M** seen with rocket artillery brigade, Tver Oblast, Western MD (named as the 79th Guards Rocket Artillery Brigade by Wikipedia only)

rocket artillery brigade **O 2026-09 M** seen with 79th Guards Rocket Artillery Brigade (Zapad group), Tornado-S, 2026-09, 232nd Guards Rocket Artillery Brigade (Tsentr group), Tornado-S, 2026-08 and 2026-09

**INDICATES** Smerch or Tornado-S points to an army-level rocket artillery brigade, not a manoeuvre brigade, whose MRL battalion holds BM-21. Expect deep fires out to 70-120 km and ammunition resupply by MAZ-543 transloaders.

**RECOGNISE** Cab MAZ-543M 8x8 truck; driver cab front left, commander behind him; launch and fire-control cabin to the rear of the driver cab, one door each side, two small front windows.

**Body** Single package of 12 long 300 mm tubes over the rear deck, 3 rows of 4 (two banks of four with four more on top); traverses 30 deg each side, elevates to 55 deg.

**Axles** Four axles, 8x8; hydraulic jacks between the wheels of the 3rd and 4th axles are lowered to steady the rear of the launcher when firing.

**Details** Tornado-S (9A54): satellite navigation antenna on the roof of the fire control cab; otherwise laid out like the 9A52-2.

**NOT TO BE CONFUSED WITH** BM-27 Uragan (9P140) p. 5 16 tubes of 220 mm on a ZIL-135 chassis; Smerch has 12 larger tubes on the MAZ-543.

**9A53 Uragan-1M** p. 5 Two quick-change pods of six tubes on an MZKT-7930 8x8; Smerch has one fixed 12-tube package.

**9A52-4 Tornado** KamAZ-63501 8x8 truck with a single six-tube pod; Smerch has one fixed 12-tube package on the MAZ-543M.

**9T234-2 and 9T255 transloaders** Same MAZ-543A family chassis; carry 12 rockets and a crane (850 kg on the 9T255) with no launch tube package.

**Vilkha (Ukraine)** Ukrainian development that uses the same Smerch launcher vehicle; not separable by shape alone.

**SEEN WITH** MAZ-543M 8x8 chassis (1 per launcher); 9T234-2 transloader (TZM) (INA; carries 12 rockets); 9T255 transloader (Tornado-S) (INA; carries 12 rockets); 1K123 Vivary automated fire control system (up to 3 battalion and 18 battery vehicles per brigade system); 9S936 command and staff vehicle (Tornado-S); 1V168 ASUNO-S fire control; 1T12-2M survey vehicle; 1B44 meteorological complex; Zoopark counter-battery radar (INA; reported cueing Smerch fire 2024-03)

**VARIANTS** 9A54 Tornado-S Satellite antenna on the fire control cab roof; same MAZ-543M chassis and 12-tube package. GLONASS, automated laying and fire control; fires 9M544 and 9M549 guided rockets to 120 km; crew 3. 9A52-4 Tornado KamAZ-63501 8x8 truck with one six-tube 300 mm pod. Half the load; pod replaced after firing; lighter and more mobile. ISS25 lists none in Russian service.



Vitaly V. Kuzmin, CC BY-SA 4.0

**MOBILITY** Heavy 43.7 t 8x8; fords 1.1 m unprepared; not amphibious; rear jacks must be lowered to fire.

Weight, combat, 9A52-2 **43.7 t** loaded  
Weight, empty **33.7 t** 9A52-2 (WEG) and 9A54 (MISSTS)  
L × W × H **12.1 × 3.05 × 3.05 m** travelling  
Road speed **60 km/h** maximum  
Road range **850 km** 9A52-2; also 850 km on the WEG  
Tornado-S page  
Fording **1.1 m** unprepared  
Time into action **3 min** from march to ready to fire  
Time out of action **3 min**  
Ground pressure **INA**

|                                       |                                                                                                                                                          |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range, 9M55K cluster                  | <b>70 km</b> maximum, unguided, range-corrected rocket with fragmentation submunitions                                                                   |
| Minimum range, 9M55K cluster          | <b>20 km</b> unguided, range-corrected rocket with fragmentation submunitions                                                                            |
| Range, 9M55F HE-frag                  | <b>70 km</b> maximum, unguided, separable HE-frag warhead                                                                                                |
| Minimum range, 9M55F HE-frag          | <b>20 km</b> unguided, separable HE-frag warhead                                                                                                         |
| Range, 9M528 HE-frag                  | <b>90 km</b> maximum, unguided extended-range rocket, separable HE-frag warhead, claim                                                                   |
| Range, 9M544 and 9M549 guided         | <b>120 km</b> maximum, Tornado-S guided cluster rockets, inertial and GLONASS, claim                                                                     |
| Minimum range, 9M544 and 9M549 guided | <b>20-40 km</b> Tornado-S guided rockets; crew statement, claim                                                                                          |
| Employment range, Smerch              | <b>35-70 km</b> practical, unguided rockets, Ukraine 2022, used at or near maximum range, observed                                                       |
| Employment range, Tornado-S           | <b>about 100 km</b> practical, guided rockets against a command post (2026-08) and a fortified strongpoint (2026-09); MoD figures via state media, claim |
| Crew, 9A52-2                          | <b>4</b>                                                                                                                                                 |
| Crew, 9A54 Tornado-S                  | <b>3</b>                                                                                                                                                 |
| Tubes                                 | <b>12 x 300 mm</b> 9A52-2 and 9A54                                                                                                                       |
| CEP, 9M55K cluster                    | <b>150 m</b> inertial range and attitude correction, no terminal guidance; also given for 9M528                                                          |
| CEP, 9M544 and 9M549 guided           | <b>5-10 m</b> inertial and GLONASS; official claim, claim                                                                                                |

## TOS-1A Solntsepek TOC-1A «Солнцезёк»

TR

Tracked 220 mm thermobaric and incendiary rocket launcher of the RKhBZ troops In service 1988

**OPERATED BY** RKhBZ brigade, flamethrower battalion of 3 batteries, each 6 BM-1 and 6 TZM-T; RWW places these battalions in NBC defence brigades and regiments and does not state the echelon, 18 per battalion (6 per battery) [\[D 2016 M\]](#)

RKhBZ brigade, TOS battalion of 3 companies of 6 launchers (ATP para 5-146), 18 per battalion [\[D 2024 M\]](#)

**INDICATES** TOS launchers belong to army or district RKhBZ units, not manoeuvre brigades. Seeing one suggests an attached flamethrower company or battalion supporting an attack or breach, since it must close to within about 6 km of the target.

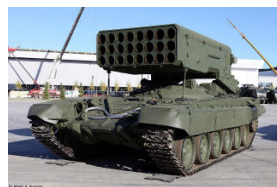
**RECOGNISE Hull** T-72 tank hull with no gun turret; a large launcher box fills the hull roof. Paired with look-alike T-72-based TZM-T reload vehicles.

**Armament** 24 tubes of 220 mm in a heavy armoured box. Launchers shipped from Nov 2025 add a factory armour shell over the tubes that lifts open when the tubes elevate.

**Turret** No turret: the launcher box sits on a rotating platform and elevates to fire. Improvised anti-drone cover was fitted in the field before factory kits.

**Suspension** Standard T-72 tracked running gear with large-diameter road wheels. Road wheel and return roller counts not confirmed in cited sources.

**NOT TO BE CONFUSED WITH** TOS-1 Buratino 30 tubes, without the heavy armoured housing added on TOS-1A, which has 24 tubes. TZM-T reload vehicle Same T-72 hull but carries two rocket racks with a crane between them instead of a trainable launcher.



Vitaliy V. Kuzmin, CC BY-SA 4.0

**MOBILITY** Tracked T-72 chassis, not amphibious; fords 1.2 m; must close to within 6 km of the target to fire the longest-range thermobaric rocket.

|                         |                      |
|-------------------------|----------------------|
| Weight, combat          | 44.3 t               |
| L × W × H               | 6.86 × 3.46 × 2.60 m |
| Road speed, maximum     | 60 km/h              |
| Off-road speed, maximum | 45 km/h              |

|                              |                                                                                  |
|------------------------------|----------------------------------------------------------------------------------|
| Range, thermobaric rocket    | 6,000 m maximum, MO.1.01.04M thermobaric rocket, 3.7 m, 217 kg                   |
| Range, incendiary rocket     | 6,000 m maximum effective, MO.1.01.04M.OP incendiary rocket, 3.7 m, 204 kg       |
| Range, original short rocket | 2,700 m maximum, MO.1.01.04 thermobaric rocket, 3.3 m, 173 kg, fired from TOS-1A |
| Range, minimum               | 600 m MO.1.01.04M rocket                                                         |
| Launch tubes                 | 24 × 220 mm                                                                      |

**SOURCES** AR20 AR21 ATP CNA EMP GS17 MIL28 MIL29 MIL30 MISS14 RWW TASS13 TASS14 TASS15 TASS16 WEG42 WMC17 WP-EN42 WP-EN44 WP-EN45

## 2S5 Giatsint-S 2C5 «Гиацинт-С»

NATO: M1981

TR

Army-level 152 mm self-propelled gun for long-range counter-battery and deep fire In service 1975

**OPERATED BY** artillery brigade, gun battalion (152 mm), 18 per battalion (24 in forward armies); 2S5 or 2A36 or M-46 [\[D 1991 M\]](#) unit not identified, Airborne Forces artillery; unit not named, 18 (IISS 2024 figures) [\[R 2024 M\]](#)

**INDICATES** In Soviet doctrine the 2S5 marked the gun battalion of an army artillery brigade. Russian MoD reporting (2026) also places it with motor rifle brigade artillery, so alone it no longer points to army-level artillery.

**RECOGNISE Hull** Low tracked hull, engine and driver at front; front dozer blade; large spade or baseplate at rear, lowered to the ground before firing.

**Armament** Very long 152 mm barrel with five-part multi-baffle muzzle brake, held by a travel lock; with gun forward, length 8.95 m against a hull of 8.33 m.

**Turret** No turret: gun on an open mount at the rear of the hull roof, crew exposed; 15 deg traverse each side. Commander's cupola carries a 7.62 mm PKT and a searchlight.

**Suspension** Six rubber-tyred double road wheels and four return rollers per side; drive sprocket front, idler rear; torsion bars, track shoes 484 mm wide.

**NOT TO BE CONFUSED WITH** 2S3 Akatsiya Large enclosed turret at hull rear, 360 deg traverse, cupola offset left; double-baffle muzzle brake and fume extractor. 2S5: open mount, no turret, multi-baffle brake.

**2S4 Tyulpan** p. 9 Same chassis family, but a 240 mm mortar travels folded with muzzle forward, not reaching past the first road wheel; no long barrel over the front.



One half 3544, Public domain

**MOBILITY** Tracked; fords about 1 m; rear spade must be lowered to fire; gun traverse only 15 deg each side, so the hull is laid on the target line.

|                |                       |
|----------------|-----------------------|
| Weight, combat | 28.2 t                |
| L × W × H      | 8.33 × 3.25 × 2.76 m  |
| Road speed     | 63 km/h maximum, road |
| Road range     | 500 km maximum, road  |

|                                      |                                                                         |
|--------------------------------------|-------------------------------------------------------------------------|
| Range, HE-frag 30F29                 | 28.4 km maximum, HE-frag 30F29, full charge                             |
| Range, HE-frag 30F29 reduced charge  | 21.5 km maximum, HE-frag 30F29 with reduced charge 4Zh48                |
| Range, HE-frag 30F59                 | 30.5 km maximum, HE-frag 30F59, improved aerodynamic shape, full charge |
| Range, rocket-assisted HE-frag 30F30 | 33 km maximum, rocket-assisted HE-frag                                  |
| Range, Krasnopol guided 30F39        | 20 km maximum, laser-guided HE-frag Krasnopol                           |

**SEEN WITH** 2A37 152 mm gun (1)

**SOURCES** BMPD2 FAS2 FM3 GS4 IISS25 IZ26 IZ233 JANESRG MF RIA22 RUSIPL RWW WEG3 WP-EN4 WP-RU7

## 2S19 Msta-S 2C19 «Mcma-C»

NATO: M1990 Farm

TR

152 mm tracked self-propelled howitzer of brigade, division and army artillery In service 1989

**OPERATED BY** motor rifle brigade, two self-propelled artillery battalions, each of three batteries of 6 howitzers; ATP notes 2S3 may be held instead, 18 per battalion (6 per battery), 36 per brigade; ATP: some battalions up to 24 systems [D 2024 M]

tank brigade, two howitzer (self-propelled artillery) battalions, three batteries of 6, 18 per battalion, 36 per brigade [D 2016 M]

**INDICATES** A battery of 6 or battalion of 18 Msta-S points to a self-propelled artillery battalion of a motor rifle or tank brigade, a division artillery regiment or an army artillery brigade. Wartime batteries may be smaller.

**RECOGNISE Hull** Tank-type tracked hull with T-72 hull geometry and T-80 running gear, built-in dozer blade, travel lock for the barrel. About 11.9 m long overall with the gun forward.

**Armament** Very long 152 mm L/47 barrel with double-baffle muzzle brake and a bore evacuator. 12.7 mm NSVT machine gun on the commander's cupola.

**Turret** Large turret over the hull centre; vehicle about 3 m high. Six 81 mm smoke launchers. 2S19M2: two stowage boxes on the right turret side.

**Suspension** T-80 type running gear: six road wheels and five return rollers per side, 580 mm tracks.

**NOT TO BE CONFUSED WITH** 2S3 Akatsiya (152 mm SP howitzer) Engine at front and turret toward the rear; much shorter L/27 barrel; four return rollers; about 28 t against 42 t. 2S5 Giatsint-S (152 mm SP gun) p. 7 152 mm 2A37 L/54 gun; most of the crew, all but the gunner, work outside the vehicle when firing; about 28 t.

**SOURCES** AG4 ARMYT ATP CNA GS9 IIS25 MIL17 ORYX RWW TASS4 TASS5 WEG20 WP-EN4 WP-EN7 WP-EN20 WP-EN21 WP-RU25



MILRU, CC BY 4.0

**MOBILITY** Heavy tracked vehicle, not amphibious. Fords 1.2 m unprepared; WEG gives 5 m with preparation. About 42 t, needs heavy bridging.

|                        |                                                           |
|------------------------|-----------------------------------------------------------|
| Weight, combat         | 42 t 2S19 and 2S19M1                                      |
| Weight, combat, 2S19M2 | 43.24 t 2S19M2                                            |
| L × W × H              | 11.917 × 3.584 m overall; 3.38 m without skirts × 2.985 m |
| Road speed, maximum    | 60 km/h                                                   |

|                           |                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------|
| Range, HE-frag            | 24.7 km maximum, 30F45 (OF-45) HE-frag, full charge                                         |
| Range, HE-frag base bleed | 29 km maximum, 30F61 HE-frag base bleed (round 3V0F91)                                      |
| Range, rocket-assisted    | 36 km HE rocket-assisted projectile (HERA/RAP), type not named                              |
| Range, guided             | 20 km maximum, Krasnopol laser-guided projectile; laser designation by an observer required |

## 2S35 Koalitsiya-SV 2C35 «Коалиция-СВ»

TR

152 mm self-propelled howitzers: tracked autoloader 2S35 and wheeled 8x8 2S43 In service 2020

**OPERATED BY** self-propelled artillery battalion, self-propelled howitzer battery, 6 per battery, 18 per battalion [D 2016 M] self-propelled artillery regiment, 8 2S35 on troop trials

[R 2020-05 M]

artillery brigade, self-propelled artillery battalions, INA; planned replacement of the brigade's 2S19 by 2S35, reported 2017, not carried out by late 2024 [R 2019 L]

**INDICATES** 2S35 or 2S43 suggests new-build guns above manoeuvre-brigade level: 2S35 was planned for army artillery brigades and trialled in a division SP artillery regiment; 2S43 is reported with the 273rd Artillery Brigade, 34th Artillery Division. Rare [IIS2: 8, 20].

**RECOGNISE Hull** 2S35: T-90-type tracked hull with reshaped front; driver, commander and gunner sit side by side in the front, three hatches across the hull front; engine at rear.

**Armament** 2S35: very long 152 mm 2A88 barrel (52 calibres) with muzzle brake, held by a travel lock on the glacis; Kord 12.7 mm remote weapon station on turret roof.

**Turret** 2S35: large unmanned turret mid-hull; 902 smoke dischargers on both sides and roof; laser warning sensors at front and rear of roof. From 2021 two boxes on turret front.

**Suspension** 2S35: six road wheels each side, idler front, drive sprocket rear, torsion bars; upper track covered by a skirt. Return roller count INA.

**NOT TO BE CONFUSED WITH** 2S19 Msta-S / 2S19M2 Msta-SM2 p. 8 2S19 has a crew of five with gun crew in the turret and a shorter 47-calibre barrel; 2S35 has three crew hatches across the hull front. 2S44 Giatsint-K Same BAZ 8x8 chassis as 2S43 but carries the gun of the 2A36 Giatsint-B towed gun.

**SOURCES** AG2 AR1 ATP BMPD1 CNA IIS25 I223 IZMAL K21 KZ2 ORYX REDSTAR RG24 RGMAL RGMAL2 RGPAP RWW SHEP WEG1 WP-EN7 WP-EN8 WP-RU4 WP-RU5 WSN2



Kirill Borisenko, CC BY-SA 4.0

**MOBILITY** 2S35 tracked, 48–55 t, not amphibious. 2S43 wheeled 32 t, for road and off-road use, fits IL-76; fords 1.4 m.

|                  |                     |
|------------------|---------------------|
| Weight, 2S35     | 48–55 t combat      |
| Weight, 2S43     | 32 t combat         |
| L × W × H        | 9.63 × 3.78 × 2.2 m |
| Road speed, 2S35 | 65 km/h maximum     |

|                                            |                                                                                                    |
|--------------------------------------------|----------------------------------------------------------------------------------------------------|
| Range, 2S35 standard HE-frag               | 40 km maximum, standard (unassisted) rounds                                                        |
| Range, 2S35 guided or extended-range shell | 80 km maximum, precision-guided or rocket-assisted projectile; munition not named, claim           |
| Range, 2S43 HE-frag                        | 24.5 km maximum, 2A64 gun; munition not stated by WEG or WPMALRU (Rostec: HE-frag more than 24 km) |
| Rate of fire, 2S35                         | 16 rds/min maximum, claim                                                                          |
| Rate of fire, 2S43                         | 7 rds/min maximum, claim                                                                           |

**SEEN WITH** 2F66-1 transport-loader vehicle (KamAZ-6560), 92 rounds, loads 2S35 in 15 min

## 2S7 Pion / 2S7M Malka 2C7 «Пион»

NATO: M-1975

TR

203 mm heavy self-propelled gun for long-range fire on high-value targets In service 1975

**OPERATED BY** high-power artillery brigade, 2S7 battalions (45th bde: two battalions of eight, 2017), ATP fig. 5-18: 24 x 2S7 per brigade, 8 x 2S7 battalion block (OCR uncertain); 16 reported in 45th bde, 2017

**D 2024-02 H**

high-power artillery brigade, 16 x 2S7 Pion (with 8 x 2S4 Tyulpan)

**O 2017-12 M**

artillery brigade, 2S7 battalion (IZ17, 2017); 2S7M Malka (mil.ru via

RDP18), 1 battalion; 12 x 2S7 (RUWIKI, 2019) **O 2018-04 M**

artillery brigade, 12 x 2S7 each (RUWIKI, 2019) **O 2019 L**

**INDICATES** A 203 mm tracked gun points to army-level artillery: a high-power artillery brigade or an army artillery brigade's heavy battalion, not a manoeuvre brigade's guns. Used on high-value targets, in urban fighting and, since 2022, for counter-battery fire.

**RECOGNISE Hull** Purpose-built tracked chassis using T-72/T-80 automotive parts. Four compartments: driver, power pack, crew, rear. Front dozer blade.

**Armament** Very long 203 mm 2A44 gun (about 12 m) on an open mount outside the hull; on the march it lies forward, overhanging the nose. No muzzle brake, no fume extractor.

**Turret** No turret. Gun on open external mount at hull rear. Large recoil spade at rear lowered to the ground when firing.

**Suspension** Seven twin rubber-tyred road wheels and six return rollers per side; drive sprocket front, idler rear, idlers lowered to the ground when firing. Torsion bar suspension.

**NOT TO BE CONFUSED WITH** 2S4 Tyulpan p.9 Also given the NATO name M-1975. A 240 mm self-propelled mortar, not a gun: no 12 m barrel overhanging the hull.



Boevaya mashina, CC BY-SA 3.0 de

**MOBILITY** Not amphibious. Needs firm ground to emplace; recoil spade and dozer blade used to dig in.

|                |                     |
|----------------|---------------------|
| Weight, combat | 46,500 kg           |
| L × W × H      | 10.5 × 3.38 × 3.0 m |
| Road speed     | 51 km/h maximum     |
| Road range     | 500 km cruising     |

|                           |                                                          |
|---------------------------|----------------------------------------------------------|
| Range, HE-frag            | 37.5 km maximum, Conventional HE-frag, 110 kg projectile |
| Range, HE rocket-assisted | 47.5 km maximum, RAP projectile, 103 kg                  |
| Rate of fire              | 1.5 rds/min (2S7); 2.5 rds/min (2S7M) maximum            |

|                    |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| Ammunition carried | 4 (2S7); 8 (2S7M) On the vehicle                                                        |
| Crew               | 7 (2S7); 6 (2S7M) on the vehicle; 2S7 crew is 14 with the auxiliary vehicle crew (WIKI) |

|                  |                                                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Elevation        | 0 to +60 deg; traverse 30 deg                                                                                                            |
| <b>SEEN WITH</b> | Ammunition transport vehicle with the rest of the crew; carries the main load (40 rounds, RUWIKI; WIKI: 4); 1V12M command vehicle (2S7M) |

**SOURCES** ATP CNA CNA21 DEF24 FAS3 IISS25 IZ17 IZ23B IZ23C LENTA16 LENTA23 MILN RDP1 TASS1 TWZ1 WEG4 WP-EN10 WP-RU8 WWW1

## 2S4 Tyulpan 2C4 «Тюльпан»

NATO: M-1975

TR

240 mm self-propelled breech-loading heavy mortar for destroying fortified targets In service 1971

**OPERATED BY** high-power artillery brigade, 2S4 battalion (ATP figure 5-18 shows two 2S4 groups; subunit name not given), ATP fig. 5-18: groups of 6 and 4 2S4 (OCR uncertain); 8 reported in 45th brigade (2017) **D 2024 H**

high-power artillery brigade, heavy mortar battalion (240-mm); brigade allocated to a front (Soviet RVGK) **D 1991 M**

**INDICATES** Held by high-power artillery brigades and by heavy battalions of army-level artillery brigades. A 2S4 sighting points to army-level heavy artillery tasked against fortified positions or built-up areas.

**RECOGNISE Hull** Turretless tracked hull; engine front right, driver front left, commander behind driver; ammunition drums in mid-hull; open, externally mounted mortar at rear.

**Armament** Very large 240 mm breech-loading mortar, barrel about 5 m, carried with its baseplate on the hull roof; in action it swings back onto a baseplate on the ground.

**Turret** No turret. Commander's rotating cupola behind the driver with an external 7.62 mm PKT and a searchlight.

**Suspension** Six rubber-tyred road wheels and four return rollers per side, drive sprocket front, idler rear, torsion bars. Road wheel size and spacing INA.

**NOT TO BE CONFUSED WITH** 2S3 Akatsiya Fully enclosed turret with 152 mm gun at the rear on a similar six-road-wheel chassis; 2S4 has no turret and a roof-carried mortar.

**2S5 Giatsint-S** p.7 Same chassis family, also turretless, but a long 152 mm gun overhangs the front; the 2S4 mortar muzzle is only level with the first road wheel, baseplate folded above it.



Vitaly V. Kuzmin, CC BY-SA 4.0

**MOBILITY** Tracked; heavy. Fords about 1 m unprepared. Mortar must be lowered onto a ground baseplate to fire.

|                |                                  |
|----------------|----------------------------------|
| Weight, combat | 27.5 t                           |
| L × W × H      | 7.94 × 3.25 × 3.225 m travelling |
| Road speed     | 60 km/h maximum                  |
| Road range     | 500 km                           |

|                                 |                                                |
|---------------------------------|------------------------------------------------|
| Range, 53-F-864 HE mine         | 9.65 km maximum, HE mine 130.7 kg, full charge |
| Minimum range, 53-F-864 HE mine | 0.8 km HE mine, smallest charge                |

|                                    |                                                                                                 |
|------------------------------------|-------------------------------------------------------------------------------------------------|
| Range, 3F2 rocket-assisted HE mine | 18 km maximum, 228 kg rocket-assisted HE mine; WEG and GS do not name the round, 3F2 per RUWIKI |
|------------------------------------|-------------------------------------------------------------------------------------------------|

|                                            |                                |
|--------------------------------------------|--------------------------------|
| Minimum range, 3F2 rocket-assisted HE mine | 0.8 km rocket-assisted HE mine |
|--------------------------------------------|--------------------------------|

|                                        |                                                                            |
|----------------------------------------|----------------------------------------------------------------------------|
| Range, 3F5 Smel'chak laser-guided mine | 9.2 km maximum, laser-guided mine, target designated by 1D15 from 0.2-5 km |
|----------------------------------------|----------------------------------------------------------------------------|

**SEEN WITH** 6x6 truck or MT-LB carrying the rest of the crew

**SOURCES** AG3 ATP CNA CUL DB17 FM32 GS3 IISS25 IZ17 IZ232 JANESRG MK26 ORYX PRIME26 RG17 RG22 WEG2 WP-EN9 WP-RU6 WZV2

## 2A36 Giatsint-B 152 mm towed gun 2A36 «Гуашын-Б»

NATO: M1976

TW

152 mm towed gun for long-range and counter-battery fire at army level In service 1976

**OPERATED BY** artillery brigade, gun battalion (152 mm) of the army artillery brigade; 2A36 or 2S5 or M-46, 36 per brigade (two 18-tube gun battalions) or 48 (four 24-tube battalions) [D 1991 M]

artillery brigade, gun brigade of the front artillery division; 2A36 or 2S7, 72 per brigade [D 1991 M]

unit not identified, IISS estimate: 150 in service, 350 in store [Ground Forces total] [R 2024-T2 M] seen with Russian Ground Forces, holdings by type unit not identified, IISS estimate: 50 across Naval Infantry; holding unit not stated [R 2024-T2 L] seen with Russian Naval Infantry, holdings by type

**INDICATES** ATP: towed artillery remains in some units. The 2A36 went to the gun battalions of artillery brigades and divisions at army and front level (FM 100-2-3, Victory Museum), so it indicates an artillery formation rather than a manoeuvre unit.

**RECOGNISE** Barrel Long 152 mm barrel, 8197 mm with muzzle brake; multi-slot slit-type muzzle brake; recoil brake and recuperator sit above the barrel.

**Shield** Shield on the upper carriage protects the crew against bullets and fragments.

**Carriage** Split box-section welded trails; gun rests on a base plate for firing, wheels raised by a hydraulic jack.

**Wheels** Two axles, twin pneumatic-tyred wheels each side (four wheels), torsion-bar suspension.

**NOT TO BE CONFUSED WITH** 2A65 Msta-B p. 10 2A65 has a two-wheeled carriage (2A36 has four wheels), a shield sloping to the rear over the wheels and a caster wheel on each trail.



Ministry of Defense of Ukraine, CC BY-SA 2.0

**MOBILITY** Towed by truck or tracked tractor; fired from a base plate with wheels raised, so needs a prepared firing position. Ground pressure and fording not found.

Weight, firing position 9,760 kg  
L × W × H 12.92 × 2.788 × 2.76 m travel position  
Road speed, towed 80 km/h maximum, on road  
Off-road speed, towed 35 to 45 km/h

Range, HE-frag 3VOF39 28.5 km maximum, HE-frag 30F29, full charge  
Range, rocket-assisted 33.5 km maximum, active-rocket projectile 30F30  
Minimum range, HE-frag 3VOF39 8.6 km ammunition not stated; listed with HE-frag maximum  
Range, direct fire 1,530 m ammunition and target not stated in source  
Rate of fire 6 rpm maximum  
Crew 8

**SEEN WITH** KrAZ-260 6x6 truck (or KrAZ-255B, Ural-4320)

**SOURCES** ATP FM3 GS1 IISS25 ORYX RUSARMY RUSPLN RWW VMUS WP-EN1 WP-EN2 WP-EN3 WP-EN4 WP-RU2 WSN1 not in WEG

## 2A65 Msta-B 2A65 «Мста-Б»

NATO: M1987

TW

Towed 152 mm howitzer for army-level artillery brigades and regiments In service 1986

**OPERATED BY** artillery brigade, artillery battalions (several per brigade; 2S19 or 2A65) [D 2019 M]

unit not identified, Naval Infantry artillery; brigade not named, 50 (IISS 2024 figures) [R 2024 M]

unit not identified, Russian deployment in Syria; unit not named, 12 (IISS 2024 figures) [R 2024 M]

**INDICATES** A towed 152 mm howitzer points to an army-level artillery brigade or a division artillery regiment. CNA says replaced 2S19s and 2A65s move to division regiments and brigade battalions.

**RECOGNISE** Gun Conventional towed howitzer, 152.4 mm, 47 calibres, monoblock barrel with a three-chamber muzzle brake and no bore evacuator.

**Carriage** Split trail with box-section trails, each with a caster wheel that swings up over the trail when firing. Hydraulic firing jack under the front.

**Shield** Gun shield on the upper carriage.

**Running gear** Sprung two-wheel running gear; springing cuts out automatically when the trails are spread.

**Towing** Towed by Ural-4320 6x6 or KamAZ-63501AT 8x8; in the travel position the base plate is raised and clamped to barrel and cradle.

**NOT TO BE CONFUSED WITH** 2S19 Msta-S p. 8 Self-propelled turret version of the same gun; the 2S19 barrel has a bore evacuator, the 2A65 does not.

2A36 Giatsint-B p. 10 Four-wheeled carriage, two wheels each side; the 2A65 has two-wheeled running gear.



One half 3544, public domain

**MOBILITY** Towed. Emplaced with hydraulic jack and spades. Towing speed 80 km/h on highway, 60 km/h on dirt roads, 20 km/h cross-country.

Weight, travel position 7,000 kg claim  
L × W × H 12.7 × 2.5 × 2.95 m claim  
Towing speed, road 80 km/h 20 km/h cross-country, 60 km/h dirt road, claim  
Time into action 2 to 2.5 min claim

Range, OF45 HE-frag 24.7 km maximum, OF45 (30F45) HE-frag projectile, claim  
Minimum range, OF45 HE-frag 6.5 km OF45 HE-frag projectile, claim  
Range, OF61 HE-frag base bleed 29 km maximum, OF61 (30F61) HE-frag with base-bleed gas generator, round 3VOF91, claim  
Rate of fire up to 7 rounds/min maximum, claim

**SOURCES** ATP CNA FM3 GS2 IISS25 JANESRG ORYX RWW WP-EN2 WP-EN5 WP-EN6 WP-RU3 not in WEG

## Iskander-M 9K720 «Искангеп-М»

NATO: SS-26 Stone

WH-AT

Army-level operational-tactical ballistic and cruise missile system for deep strike In service 2007

**OPERATED BY** missile brigade, 3 missile battalions of 2 batteries each; 2 launchers per battery, 12 launchers and 12 transloaders per brigade; 4 of each per battalion [D 2021 H](#)

**combined arms army**, army missile brigade, 1 missile brigade (12 launchers) per army [D 2024 H](#)

**tank army**, army missile brigade, 1 missile brigade (12 launchers) [O 2021 M](#)

**army corps**, corps missile brigade, 1 missile brigade (12 launchers) [O 2021 M](#)

**combined arms army**, army missile brigade, 12 launchers per brigade [O 2023 M](#)

**missile battalion**, 4 launchers [O 2008 M](#)

**INDICATES** Iskander vehicles indicate an army-level missile brigade of 12 launchers. The army commander holds it as a reserve for priority strikes; it is not part of brigade artillery groups, so it does not by itself mark a manoeuvre brigade.

**RECOGNISE Launcher** Iskander-M: two ballistic missiles lie in the rear body on separate lift arms, raised to 85 deg to fire. Iskander-K: 2 x 9M728 or 4 x 9M729 in canisters.

**Radar** No radar is described on the launcher; it has satellite navigation and its own fire-control post. Target data comes from the 9S920 and 9S552 vehicles.

**Chassis** MZKT-7930 8x8 heavy truck, about 13 m long and 3 m wide, first two axles steered. Four hydraulic jacks lower to level the vehicle before launch.

**Complex** Launchers move with 9T250 transloaders on the same 8x8 chassis, which carry a crane. Command, data preparation, maintenance and living vehicles are KamAZ trucks.

**NOT TO BE CONFUSED WITH** 9T250 transporter-loader (same complex) Same MZKT-7930 chassis and two missiles, but has a loading crane and no launch equipment.

**K-300P Bastion-P coastal launcher (MZKT-7930)** Coastal troops system: two Oniks anti-ship missiles fired vertically; battery of four launchers with KamAZ command vehicles.

**Bal coastal missile launcher (MZKT-7930)** Carries eight Kh-35 anti-ship missiles, not two large missiles.

**9A53 Uragan-1M MRL (MZKT-7930)** [p. 5](#) Rear carries two rocket pods of six tubes each instead of two missiles.

**SEEN WITH** 9P78-1 self-propelled launcher, MZKT-7930 8x8 (2 per battery, 4 per battalion, 12 per brigade); 9T250 transporter-loader, MZKT-7930 8x8 with crane, 2 missiles (2 per battery, 4 per battalion, 12 per brigade); 9S552 command and staff vehicle, KamAZ truck (1 per battery; 11 per brigade); 9S920 information preparation point (mission planning), KamAZ truck (1 per brigade); Maintenance vehicle (MRT0), KamAZ truck (1 per brigade in the 51-vehicle brigade set); Life support vehicle, KamAZ truck, living space for up to 8 (14 per brigade); 9M723 ballistic missile (2 per launcher, 2 per transloader; 48 per brigade); 9M728 (R-500) and 9M729 cruise missiles (2 x 9M728 or 4 x 9M729 per launcher); MZKT-7930 8x8 chassis

**VARIANTS** 9K720 Iskander-M with 9M723 Baseline: two ballistic missiles about 7.3 m long and 0.92 m wide on lift arms. Quasi-ballistic missile, up to about 500 km; reported to release decoys during flight. Iskander-K with 9M728 (R-500) Same 9P78-1 launcher, loaded with two cruise missile canisters instead of two ballistic missiles. Low-flying subsonic cruise missile; official range up to 500 km. Used against Chernihiv in Aug 2023. Iskander launcher with 9M729 Special launcher configuration with four cruise missile canisters; the missile is 530 mm longer than 9M728. Range disputed: US assessed over 500 km in breach of the INF Treaty; Russia said it was within limits.



Vitaly V. Kuzmin, CC BY-SA 4.0



launcher with 9M728 cruise missiles. Boevaya mashina, CC BY-SA 3.0

**MOBILITY** Heavy 8x8 truck: 70 km/h on roads, about 40 km/h on dirt roads. Must halt and level on jacks to fire. Fords 1.4 m. A battery has been airlifted by An-124.

Weight, launcher loaded 42.85 t

Weight, 9T250 transloader 40 t

loaded

L × W × H

Road speed

Road range

Fording

Time into action

13.07 × 3.07 × 3.55 m

70 km/h

1,000 km

1.4 m

16 min to first launch from the march; 4 min from highest readiness claim

41 Derived: 42.85 t wheeled x 0.94, rounded up, estimated

30 to 80 s to level on jacks; about 20 s to raise missiles

Military load class

Levelling and erecting

Range, 9M723 ballistic missile

Minimum range, 9M723

Range, 9M728 (R-500) cruise missile

Accuracy (CEP), 9M723

Warhead mass, 9M723

Warhead types, 9M723

Launch weight, 9M723

Missile length and diameter, 9M723

Missiles per launcher

Interval between launches

Crew

Brigade vehicle set

Range, 9M729 cruise missile

up to 500 km maximum, 9M723, domestic Iskander-M

50 km From the MIS Iskander-E (export) data table, max 280 km; domestic 9M723 minimum not confirmed

500 km official maximum, 9M728 cruise missile, Iskander-K, claim

10 to 30 m reported, With electro-optical terminal seeker

480 to 700 kg Depends on warhead type

Cluster, unitary HE, penetrator, fuel-air and others; nuclear capable 3,800 to 4,020 kg Depends on payload

7.3 m; 0.92 m

2 ballistic (9M723) or 2 cruise (9M728); 4 x 9M729 in special configuration claim

about 1 min Two missiles from one launcher

Launcher 3; transloader 2

51 vehicles 12 launchers, 12 transloaders, 11 command, 14 life support, 1 maintenance, 1 information preparation

INA maximum, 9M729 cruise missile

# S-400 Triumph 40P6 «Триумф»

NATO: SA-21 Growler

WH-AT

Long-range SAM system against aircraft, cruise missiles and ballistic missiles In service 2007

**OPERATED BY** military district, SAM regiments of the district's Air Force and Air Defence Army, IISS late 2024, SAM regiments listed with S-400 in each district's Air Force and Air Defence Army: 45th (Leningrad MD) 5, plus 4 listed with S-300PS, S-300PM1 and S-400; 14th (Central MD) 4; 4th (Southern MD) 3; 11th (Eastern MD) 4 listed with S-300PS and S-400, plus 1 with S-300V and S-400; 6th (Moscow MD) none (Moscow area: 5 regiments under Aerospace Defence Command). Launchers per district INA [D 2024 H](#)

air defence missile regiment, about 12 launchers reported at the unit site; 4 launchers and 96L6 and 92N6 radars reported destroyed

[O 2025-11 M](#)

air defence missile regiment, air defence missile battalion (division), usually 2 battalions of 8 launchers each (16 per regiment); 2012 press report relaying MoD press service [R 2012 M](#)

aerospace forces air defence regiment, IISS: 13 Aerospace Forces regts with S-400 and Pantsir-S1, plus 2 mixed with S-300 [R 2024 M](#)

air defence missile regiment, Navy: regiments under fleet air defence divisions, IISS: 64 S-400 in Navy; 3 SAM regts: 2 Baltic Fleet, 1 Pacific Fleet (naval aviation) [R 2024 M](#)

air defence missile battalion, IISS: 2 SAM bn in Belarus; 1 SAM bty in Syria [R 2024 M](#)

**INDICATES** S-400 belongs to Aerospace Forces or Navy SAM regiments, not ground-force brigades. A regiment usually has two battalions of eight launchers; launchers imply a 92N6 engagement radar nearby, and often Pantsir-S1 in the same regiment.

**RECOGNISE Launcher** Four large missile canisters on an erector, raised vertical to fire. One canister can hold four small 9M96 missiles instead of one 48N6.

**Radar** 92N6 Grave Stone engagement radar: phased array on MZKT-7930 8x8. 91N6 acquisition radar: back-to-back phased array, MZKT-7930 tractor. 96L6 may sit on a 40V6M mast.

**Chassis** Launchers self-propelled on MAZ-543M 8x8 (5P85SM2-01), or on a semitrailer towed by a BAZ-6402-series 6x6 tractor (5P85T2; export 5P85TE2 on BAZ-64022).

**Complex** 92N6 and 96L6 radars on MZKT-7930 8x8; 55K6 command post on Ural 8x8 truck. A battalion site holds about eight launchers around one 92N6.

**NOT TO BE CONFUSED WITH S-300PM1/PM2 (SA-20 Gargoyl)** Same 5P85 launcher family, 96L6 radar and a look-alike command post; S-400 has the redesigned 92N6 Grave Stone, not the 30N6 Flap Lid/Tomb Stone.

**64N6E Big Bird (S-300PMU family)** Same two-faced array layout; the S-400 91N6E is readily told apart by its new-build MZKT-7930 tractor (APA).

**SEEN WITH** 55K6E command post (30K6E control system) (1 per 30K6E system of up to 6 fire units (export composition)); 91N6E Big Bird acquisition radar (1 per 30K6E system (export composition)); 92N6E Grave Stone engagement radar (domestic 92N6A) (1 per 98Zh6E fire unit); 96L6E all-altitude radar (domestic 96L6-TsP); 5P85SM2-01 / 5P85T2 launcher (APA export list: 5P85SE2 / 5P85TE2) (8 per battalion (Russian regiments, 2012); up to 12 per fire unit (export composition)); BAZ-64022 6x6 tractor (1 per 5P85TE2 semitrailer launcher); 40V6M/MD mast (INA (optional)); 5T58-2 missile transporter and 22T6-2 loading crane; 96K6 Pantsir-S1 (INA; listed in the same regiments)

**VARIANTS** 5P85SM2-01 Self-propelled launcher on MAZ-543M 8x8; no separate tractor. INA. 5P85T2 Launcher on a semitrailer towed by a BAZ-6402-series 6x6 tractor. INA. 5P90S Self-propelled TEL on BAZ-6909-022; new gantry, folding mast with directional antenna, satnav antennas on a longer baseline. Intended to carry a heavier missile payload than the MAZ-79100-series TELs (APA). 51P6 Launcher on MZKT-7930 chassis (RU Wikipedia only); Oryx lists 51P6 TELs among S-400 losses. INA.

**SOURCES** APA1 ATP CSIS6 ISS25 JREW02 LENT2 MIL25 MIL26 ORYX RUSIAD STRASAM TASS11 WEG34 WP-EN37 WP-RU51 WP-RU52



Vitaly V. Kuzmin, CC BY-SA 4.0



radar.vitaly kuzmin, CC BY-SA 4.0

**MOBILITY** Wheeled 8x8 vehicles and tractor-semitrailer launchers; WEG estimates 30 km/h on dirt roads for the semitrailer launcher.

Road speed, 5P85TE2 launcher  
Cruising range, 5P85TE2 launcher  
Time into action

60 km/h road; dirt road 30 km/h, estimated  
800 km estimated

5-10 min from travelling position; 3 min to ready from deployed position

Weight, launcher  
Fording  
Time out of action  
Ground pressure  
Length, launcher  
Width, launcher

INA  
INA  
INA  
INA  
INA  
INA

Range, 40N6

400 km maximum, aerodynamic targets (STRASAM); APA brochure figure is for export 40N6E, 215 nmi, claim

Altitude, 40N6

30-35 km maximum, 40N6; the 27 km often quoted is reported for the 48N6DM missile specifically, claim

Range, 48N6DM (48N6E3)

250 km maximum, aerodynamic targets; RUSIAD names export 48N6E3 (Army Technology lineage); TASS 2018 state claim: 48N6DM up to 250 km, claim

Altitude, S-400 (missile not stated)  
Range, 48N6 (ballistic targets)  
Range, 9M96E2

0.1-27 km reported, reported engagement altitude band  
60 km ballistic missile targets, 48N6 series

Altitude, 9M96E2

120 km export 9M96E2; aerodynamic targets; APA gives 64.8 nmi, claim

Range, 9M96E

0.005-30 km range, export 9M96E2; APA gives 15 ft to 100 kft, claim

Altitude, 9M96E

40 km export 9M96E; aerodynamic targets; APA gives 21.6 nmi, claim

Detection range, 91N6E

0.005-20 km range, export 9M96E; APA gives 15 ft to 66 kft, claim

Detection range, 96L6E

580-600 km detection/tracking, target RCS >10 m<sup>2</sup> (strategic bomber)  
300 km max, export 96L6E in S-300PMU family, target not stated; JREW-p1877, PDF p. 1877, claim

Fire unit distance from command post  
Warhead, 48N6

up to 100 km maximum, 98Zh6E fire unit from 55K6E, export composition, claim  
143 kg HE-frag

Missiles per launcher, 48N6 family

4 or up to 16 9M96E/E2, or mixes such as 3 x 48N6 + 4 x 9M96

## S-300V4 C-300BA

NATO: SA-23 **TR**

Tracked long-range SAM and theatre anti-ballistic-missile system of Army air defence In service 1988

**OPERATED BY** air defence missile brigade, air defence missile battalion (up to 4 per brigade, under Polyana-D4M1 brigade CP); battalion CP 9S457M controls up to 4 batteries; WEG: an upper-level 9S52 Polyana-D4 command post may integrate S-300V with Buk into one brigade, S-300V battery: 1 9S32, 4 9A83, 2 9A82, 1 9A84, 2 9A85; S-300V4 battery counts INA [\[D 2017 M\]](#)

air defence missile brigade, INA. IIS lists S-300V and S-300V4 in Army holdings without a count and counts 16 SAM brigades, but does not say which hold them; Izvestia reports S-300V4 brigades under the OSK [\[R 2024 M\]](#)

**INDICATES** S-300V4 vehicles point to an Army air defence missile brigade: a front-level asset directly under a district (OSK) or front commander, covering main troop groupings. Older S-300V also serves in Aerospace Forces SAM regiments and at the Armenia base (IIS).

**RECOGNISE Launcher** 9A83 TELAR: four large tubes carried flat, raised vertical to fire; illuminator radar at the front on a folding mast. 9A82: two tubes, radar fixed over the cab.

**Launcher** Loader-launchers (9A84, 9A85) look like TELARs but carry a hydraulic crane in place of the radar. In S-300V4 heavy missiles ride only on 9A84-2 loader-launchers.

**Radar** 9S32 guidance station: large phased array raised hydraulically on its own tracked vehicle. S-300VM/V4 9S32M uses a larger High Screen-type space-fed array.

**NOT TO BE CONFUSED WITH** S-300PS/PM, S-400 [SA-10, SA-20, SA-21] p. 12 Launchers are wheeled: S-300P on 8x8 trucks, S-400 on trailers behind 6x6 tractors. The S-300V family is on tracked vehicles.



TELAR, Alexey Ivanov, CC BY 4.0



radar, Vitaly V. Kuzmin, CC BY-SA 4.0

**MOBILITY** Heavy tracked vehicles, not amphibious; ford 1.2 m. Deploy from march and fire from unprepared sites in about 5 min; ready to move about 5 min after shutdown.

|                    |                      |                   |
|--------------------|----------------------|-------------------|
| Weight, 9A83 TELAR | 50 t                 | combat weight     |
| L × W × H          | 9.30 × 4.50 × 3.40 m | WEG 'hull length' |
| Road speed         | 53 km/h              | maximum           |
| Road range         | 250 km               | cruising range    |

|                     |             |                                               |
|---------------------|-------------|-----------------------------------------------|
| Range, 9M83         | 75 km       | maximum, aircraft targets; S-300V             |
| Minimum range, 9M83 | 6 km        | aircraft targets; S-300V                      |
| Altitude, 9M83      | 250 m–25 km | maximum and minimum, aircraft targets; S-300V |
| Range, 9M82         | 100 km      | maximum, aircraft and jammer targets; S-300V  |
| Minimum range, 9M82 | 13 km       | aircraft targets; S-300V, claim               |
| Altitude, 9M82      | 1–30 km     | maximum and minimum, aircraft targets; S-300V |

**SEEN WITH** 9S457M command post (1 per battalion); 9S15M/9S15MV Obzor-3 all-round surveillance radar (Bill Board) (1 per battalion)

**SOURCES** APA10 AR16 ATP BMP07 CSIS5 IIS25 IZV3 JREW02 MISS9 RUSIAD RWW TASS10 VKS WEG4 WEG33 WEG34 WP-EN34 WP-RU48 WP-RU49 WSP0T11 WSP0T12

## S-350 Vityaz C-350 «Бумраг»

NATO: SA-28 **WH-AT**

Wheeled medium-range SAM against aircraft, cruise missiles, UAVs and ballistic missiles In service 2019

**OPERATED BY** air defence missile battalion, Battalion set: 1 command post 50K6A, 2–3 multifunction radars 50N6A, 12 launchers 50P6A, 12 launchers (12 missiles each) per battalion, per Russian MoD statement [\[R 2020 M\]](#)

unit not identified, training centre of the Anti-aircraft Missile Forces, Zhukov Military Academy of Aerospace Defence, first S-350 set (crews and instructors training) [\[R 2020-02 M\]](#)

**INDICATES** An S-350 battalion covers important sites or groups of forces against cruise missiles and aircraft. Russian statements tie deliveries to VKS air defence regiments and brigades and to Navy units; units seen near the front in 2022–25 are unidentified.

**RECOGNISE Launcher** 50P6 launcher carries 12 missiles in vertical-launch containers; the container pack is raised upright to fire ('cold' vertical launch).

**Launcher** From photograph: forward-control boxy cab; container pack in two horizontal tiers behind a machinery box. Four axes (8x8 in text sources): two front, gap, two rear.

**Radar** 50N6A multifunction radar: rotating phased-array antenna, 360° cover, circular and sector modes; unmanned, remote-controlled from the command post.

**Chassis** Launcher on 8x8 BAZ-6909 (WEG, Almaz-Antey 2023). WEG puts the radar on 6x6 BAZ-69092; Aif puts all modules, launcher included, on BAZ-69092-012.

**NOT TO BE CONFUSED WITH** S-300PS/PM and S-400 launcher p. 12 Four missile canisters per launcher, against twelve containers on the 50P6.

**50P6T launcher-loader** Described as very similar to the 50P6 launcher; no distinguishing mark found in text sources.

**SOURCES** AIF5 AINF2 AR17 ATO GS13 GS14 IIS25 IZ21 IZ202 RG2 RUSIAD TW27 WEG35 WMC14 WMC15 WP-EN35 WP-EN36 WP-RU50



TELAR, Vitaly V. Kuzmin, CC BY-SA 4.0



radar, Vitaly V. Kuzmin, CC BY-SA 4.0

**MOBILITY** Not amphibious; fords about 1.2 m (WEG). Wheeled 8x8 launcher; into and out of action about 5 minutes. Cannot fire on the move.

|                                |                      |                           |
|--------------------------------|----------------------|---------------------------|
| Weight, launcher vehicle empty | 22 t                 | BAZ-6909 launcher vehicle |
| L × W × H                      | 11.3 × 2.75 × 2.85 m | BAZ-6909 launcher vehicle |
| Road speed                     | 70 km/h              | maximum                   |
| Road range                     | 1,000 km             | maximum                   |

|                       |              |                                                    |
|-----------------------|--------------|----------------------------------------------------|
| Range, 9M96E2         | 120 km       | maximum, aerodynamic target; export missile, claim |
| Minimum range, 9M96E2 | 1 km         | export missile, claim                              |
| Altitude, 9M96E2      | 5 m to 30 km | maximum, aerodynamic target; export missile, claim |
| Range, 9M96E          | 40 km        | maximum, aerodynamic target; export missile, claim |
| Minimum range, 9M96E  | 1 km         | export missile, claim                              |
| Altitude, 9M96E       | 5 m to 20 km | maximum, aerodynamic target; export missile, claim |

**SEEN WITH** 50K6A command post (1 per battalion)

## Buk-M1 / Buk-M2 / Buk-M3 9K37M1; 9K317; 9K317M «Byk-

M1»; «Byk-M2»; «Byk-M3»

NATO: SA-11 Gadfly, SA-17 Grizzly, SA-27

TR

Tracked medium-range SAM complex; army-level air defence of ground forces In service 1983

**OPERATED BY** air defence missile brigade, 4 air defence missile battalions, each 3 batteries; brigade command post Polyana-D4, Buk-M1: 6 TELARs, 3 loader-launchers, 1 CP, 1 search radar per battalion; 24 TELARs per 4-battalion brigade (JIT: 53rd Bde had 3 operational battalions in 2014) [D 2016 H]

air defence missile brigade, battalion TELAR, INA per battalion; brigade had 3 operational battalions in 2014 (JIT) [O 2014-07 M]

air defence missile brigade, air defence missile battalion (brigade set of Buk-M3), Buk-M3: up to 6 TELARs or 9S36M radars and up to 12 launchers per battalion [R 2019 M]

aerospace forces air defence regiment, 4 Aerospace Forces regiments with Buk-M1-2, Buk-M2 and S-300V; about 80 Buk-M1-2/Buk-M2 in Aerospace Forces holdings [R 2024 M]

air defence missile brigade, air defence missile battalion [R 2018 L]

**INDICATES** Buk vehicles usually indicate an army-level air defence missile brigade, nominally of four battalions (the 53rd had three in 2014). The army may detach Buk subunits to its divisions, so a battery alone does not fix the brigade's position.

**RECOGNISE** **Launcher** M1 and M2 TELAR: four bare missiles on a trainable launcher above the hull, radar at its front. M3 TELAR 9A317M: six missiles in cylindrical containers.

**Launcher** Loader-launcher has no radar. M1/M2: 8 missiles (4 on rails, 4 on cradles) and a crane. M3 9A316M: 12 containers on two arms. TELAR: 4 missiles (M3: 6).

**Radar** M1 TELAR carries the 9S35 radar, not a phased array. M2 and M3 TELARs carry a flat phased-array panel. Tell M2 from M3 by missiles: 4 bare vs 6 in tubes.

**Radar** 9S18M1 search radar: flat array on a tracked hull, 8 m high when raised, turns 360 deg. M2/M3 9S36: array on a telescopic mast up to 21–22 m, no missiles.

**Chassis** All combat vehicles on similar heavy tracked chassis, about 8–9.6 m long, 3.3 m wide, 30–38 t. M1/M2: GM-567M, 569, 577, 579. M3: GM-5969.

**NOT TO BE CONFUSED WITH** **Tor-M1/M2 (SA-15)** p. 15 One self-contained vehicle with its own search radar; missiles in 4-round container modules, not on an arm. No separate radar vehicle; battery has two transloader trucks.

**2K12 Kub (SA-6)** 2P25 launcher carries three missiles on a lighter 19.5 t chassis and has no radar; radar is on a separate 1S91 vehicle.

**S-300V 9A83/9A82 (SA-12)** p. 13 Launcher carries four (9A83) or two (9A82) large missiles fired vertically, on an MT-T-based chassis. Buk fires from an inclined arm.

**SEEN WITH** 9S52 Polyana-D4 brigade command post (1 per brigade); 9S470M1 command post (Buk-M1) (1 per battalion); 9S18M1 Kupol-M1 target acquisition radar (Buk-M1) (1 per battalion); 9A310M1 TELAR (Buk-M1) (2 per battery, 6 per battalion); 9A39M1 loader-launcher (Buk-M1) (1 per battery, 3 per battalion); 9S510 command post (Buk-M2) (1 per battalion; controls up to 6 fire sections); 9S18M1-3 target acquisition radar (Buk-M2) (1 per battalion); 9A317 TELAR (Buk-M2) (Up to 4 per battalion, each with 1 loader-launcher); 9S36 mast-mounted illumination and guidance radar (Buk-M2) (Up to 2 per battalion, each with 2 loader-launchers)

**VARIANTS** 9K37M1 Buk-M1 TELAR 9A310M1 with 9S35 radar and four bare 9M38M1 missiles; loader-launcher 9A39M1 with 8 missiles. 9K37M1-2 Buk-M1-2 Looks like Buk-M1: TELAR 9A310M1-2 with 9S35M1-2 radar and bare missiles. 9K317 Buk-M2 TELAR 9A317 with flat phased-array radar, four bare missiles; adds 9S36 radar on a 21–22 m mast. 9K317M Buk-M3 Missiles in containers: 6 on TELAR 9A317M, 12 on launcher 9A316M. All vehicles on GM-5969 chassis.



Buk-M2 TELAR. Boevaya mashina, CC BY-SA 3.0



Buk-M3 TELAR. Ministry of Defence of the Russian Federation, CC BY 4.0



9S36M radar on mast. Boevaya mashina, CC BY-SA 4.0

**MOBILITY** Heavy tracked, not amphibious; fords 1 m. TELAR section ready 5 min from march; 9S36 mast-radar section needs 10–15 min.

|                    |                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------|
| Weight, combat     | <b>34 t</b> Buk-M1; WEG does not say which vehicle (MISM1 gives 35 t for the 9S18M1 radar)                 |
| L × W × H          | <b>9.3 × 3.25 × 3.8 m</b> Buk-M1; WEG does not say which vehicle (MISM1 gives 9.59 m for the 9S18M1 radar) |
| Road speed         | <b>65 km/h</b> maximum                                                                                     |
| Road range         | <b>500 km</b> maximum                                                                                      |
| Fording            | <b>1 m</b>                                                                                                 |
| Time into action   | <b>5 min</b> TELAR/battery from march                                                                      |
| Time out of action | <b>5 min</b>                                                                                               |
| Ground pressure    | <b>0.8 kg/cm2</b>                                                                                          |
| Survivability move | <b>20 s</b> reposition 100–200 m with equipment running                                                    |

|                                                         |                                                                                                                                                           |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range, 9M38M1 (Buk-M1)                                  | <b>3.25–34 km</b> maximum, air target                                                                                                                     |
| Altitude, 9M38M1 (Buk-M1)                               | <b>30 m–22 km</b> intercept altitude, air target                                                                                                          |
| Range, 9M317 (Buk-M2)                                   | <b>3–50 km</b> engagement zone, aircraft                                                                                                                  |
| Range, 9M317 vs ballistic and cruise missiles           | <b>20 km</b> maximum, ALCM-type cruise missile at 30 m (MISM2); ROE gives 100 m, unchecked; Lance-type tactical ballistic missile 15–20 km (MISM2), claim |
| Altitude, 9M317 (Buk-M2)                                | <b>15 m–25 km</b> intercept altitude, air target                                                                                                          |
| Range, 9M317M (Buk-M3)                                  | <b>2.5–70 km</b> maximum effective, air target                                                                                                            |
| Altitude, 9M317M (Buk-M3)                               | <b>35 km</b> maximum, air target                                                                                                                          |
| Detection range, 9S18M1 search radar (M1)               | <b>100–150 km</b> target size and altitude not stated                                                                                                     |
| Detection range, 9S18M1-3 / 9S18M3 search radar (M2/M3) | <b>160 km</b> RCS 1–2 m2; Buk-M2 9S18M1-3                                                                                                                 |
| Detection range, 9S35 TELAR radar (M1)                  | <b>80 km</b> RCS 2 m2; 100 km vs RCS 3 m2                                                                                                                 |
| Detection range, 9A317 / 9A317M TELAR radar (M2 / M3)   | <b>120 km</b> RCS 1–2 m2 at 3 km altitude and 18–20 km vs same target at 10–15 m (M2); M3: RCS 1–2 m2, altitude not stated                                |
| Detection range, 9S36 mast radar (M2)                   | <b>120 km</b> RCS 1–2 m2 at 3 km altitude; 30–35 km vs same target at 10–15 m                                                                             |
| Detection range, 9S36M mast radar (M3)                  | <b>70 km</b> cruise missile flying at 5 m                                                                                                                 |

**SOURCES** ATP BMPD3 BMPD4 CNA IISS25 JIT MIL8 MISS1 MISS2 MISS3 MISS4 MISS5 ROE RWW TWZ3 WEG10 WEG11 WEG12 WP-EN12 WP-RU14

## Tor 9K330 «Top»

NATO: SA-15 Gauntlet

TR

Short-range tracked SAM with own radars; defends troops against aircraft, PGMs and UAVs In service 1986

**OPERATED BY** motor rifle division, air defence missile regiment, 16 per regiment (4 batteries of 4, bmpd); ATP: two battalions; ru.wikipedia: Tor-M2 'divisional' set of 3 batteries, 12 vehicles [\[D 2024 H\]](#)

motor rifle brigade, air defence missile battalion, three batteries, 12 launcher vehicles per battalion (4 per battery) [\[D 2024 M\]](#)

tank brigade, air defence missile battalion, three batteries, 12 launcher vehicles per battalion (4 per battery) [\[D 2024 M\]](#)

motor rifle division, air defence missile regiment, one regimental set of Tor-M2 (16 combat vehicles) [\[O 2017-11 M\]](#) seen with 933rd Air Defence Missile Regiment, 150th Motor Rifle Division (Millerovo)

tank division, air defence missile regiment, one regimental set of Tor-M2 (16 combat vehicles) [\[O 2017-03 M\]](#) seen with 538th Guards Air Defence Missile Regiment, 4th Guards Tank Division

motor rifle brigade, air defence missile battalion (Tor-M2DT); brigade link by location and retraining report only [\[R 2019-11 L\]](#) seen with Tor-M2DT battalion of Northern Fleet ground and coastal troops, on duty in Pechenga district where a motor rifle brigade is based (number not given)

**INDICATES** Four Tor vehicles together suggest a battery of a brigade air defence missile battalion (12 vehicles) or of a division's air defence missile regiment (16). Doctrine sites Tor-M1 5-10 km behind the forward edge.

**RECOGNISE Launcher** Missiles stand vertically inside the central unmanned turret. Tor/Tor-M1: two boxes of four (8). Tor-M2: 16 missiles.

**Radar** Rotating search antenna at turret rear, lowered behind turret when stowed. Phased-array tracking radar on turret front folds down; dome antenna and TV sight nearby.

**Chassis** Tracked GM-355 chassis (Tor-M2: GM-5955): three crew in front, turret amidships, engine at rear. Not amphibious. Road wheel count INA.

**Complex** Battery: command post, four combat vehicles, two transloaders on Ural-4320 trucks. Tor-M2DT rides a two-section articulated tracked DT-30 Vityaz carrier.

**NOT TO BE CONFUSED WITH** 2K22 Tunguska (SA-19 Grison) p. 17 Tunguska carries two 30 mm cannon on its turret; Tor has no guns and its missiles sit inside the turret.

96K6 Pantsir-S1 (SA-22 Greyhound) p. 16 Pantsir-S1 combines guns with 12 missiles and is usually on a wheeled chassis; Tor (tracked) has no guns and its missiles sit inside the turret.

9K33 Osa (SA-8 Gecko) p. 17 Osa guidance group has two small dishes flanking a central tracker; Tor has one flat phased-array tracking antenna.

**SEEN WITH** 9S737M Ranzhir-M battery command post (9S737MK with Tor-M2E) (1 per battery); 9T244 transloader (2 per battery); 9T245 transport vehicle; 9V887M2U / 9V887-1M2U maintenance vehicles (9V887M2K / -1M2K in the Tor-M2E export set); GM-355 / GM-5955 tracked chassis (1 per combat vehicle); DT-30 Vityaz articulated tracked carrier (Tor-M2DT; DT-30PM per Army Recognition) (1 per combat vehicle)

**VARIANTS** 9K331 Tor-M1 Missiles in two four-round containers (9Ya281) in the turret; some vehicles on GM-5955 chassis (9A331-1). Second target channel: 2 targets at once; crew 3. 9K331MU Tor-M1-2U INA. Interim upgrade from 2012: 4 targets at once, ceiling 10 km. 9K332 / 9K331M Tor-M2 New surveillance antenna; GM-5955 chassis; 2025 build adds armour and EW fit. 16 9M338K missiles, 4 targets at once, fires on the move; claimed 32 km detection and 16 km range. 9K331MDT Tor-M2DT Two-section articulated tracked DT-30 carrier; weapon module on rear section. Arctic mobility; seen in Ukraine from December 2022.



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**MOBILITY** Tracked, not amphibious; fords 1.0 m. Tor-M1 halts to fire; Tor-M2 claimed to fire on the move. Tor-M2DT for snow and bog.

|                    |                                                                |
|--------------------|----------------------------------------------------------------|
| Weight             | 32 t combat; WEG figure for the 9K330 page, variant not stated |
| L × W × H          | 8.0 × 3.2 × 4.0 m                                              |
| Road speed         | 65 km/h maximum                                                |
| Road range         | 580 km cruising range                                          |
| Fording            | 1.0 m                                                          |
| Swimming           | not amphibious                                                 |
| Time into action   | less than 3 min                                                |
| Time out of action | less than 5 min                                                |
| Bridge load class  | 36 derived from 32 t tracked x 1.10, estimated                 |

|                                                              |                                                                                           |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Range, 9M330/9M331                                           | 12 km maximum, aircraft and PGM targets, Tor/Tor-M1                                       |
| Minimum range, 9M330/9M331                                   | 1.5 km Tor/Tor-M1                                                                         |
| Altitude, 9M330/9M331                                        | 10 m to 6 km maximum, Tor/Tor-M1                                                          |
| Range, 9M338K                                                | 16 km maximum, Tor-M2 domestic, air targets, claim                                        |
| Altitude, 9M338K                                             | 10 m to 10 km maximum, Tor-M2 domestic, claim                                             |
| Range, 9M331 (Tor-M2E)                                       | 1 to 12 km maximum, Tor-M2E export; target 300 m/s, claim                                 |
| Altitude, 9M331 (Tor-M2E)                                    | 10 m to 10 km maximum, Tor-M2E export; target 300 m/s (10 m to 6 km at 600 m/s), claim    |
| Detection range, surveillance radar (Tor/Tor-M1)             | 25-27 km maximum, aircraft; WEG does not state the variant                                |
| Detection range, surveillance radar (Tor/Tor-M1), helicopter | 12 km maximum, helicopters                                                                |
| Detection range, surveillance radar (Tor/Tor-M1), UAV        | 9-15 km maximum, UAVs                                                                     |
| Detection range, surveillance radar (Tor-M2)                 | 32 km air targets, size not stated, claim                                                 |
| Detection range, tracking radar (Tor/Tor-M1)                 | 15 km maximum, 60-degree sector; target and variant not stated                            |
| Detection range, TV sight                                    | 20 km maximum, target not stated                                                          |
| Deployment distance, Tor-M1 in defence                       | 5-10 km behind forward edge practical, first defensive position; 5-10 km between vehicles |

## Pantsir-S1 96K6 «Панцирь-С1»

NATO: SA-22 Greyhound

WH-AT

Short-range gun and missile air defence of S-400/S-300 battalions and key sites In service 2012

**OPERATED BY** aerospace forces air defence regiment, Pantsir vehicles deployed with each S-400 battalion (Armada); Lenta: batteries attached to S-400 regiments, 3 per S-400 battalion (Armada, 2024)

**D 2024 H**

aerospace forces air defence regiment, Pantsir-S battery, 6 Pantsir-S delivered to Eastern MD air defence units on Kamchatka, Dec 2015; the regiment was reported due one battery **O 2015-12 M**

aerospace forces air defence regiment, Pantsir-S1 gun-missile battalion; each battery covers one S-400 battalion **O 2020-02 M** air defence missile regiment, 30 held by the Navy (IIS p. 187)

**R 2024 M**

aerospace forces air defence regiment, INA per regiment; IIS lists 9 SAM regts with Pantsir-S1 in the 45th, 4 in the 14th, 3 in the 4th and 4 in the 11th AF&AD Army **R 2024 M**

air defence missile regiment, INA (1 SAM regt with S-400 and Pantsir-S1) **R 2024 M**

**INDICATES** Usually marks a nearby S-400 (or S-300) battalion or a defended strategic site: VKS regiments deploy Pantsir with their long-range SAM battalions (three per S-400 battalion, Armada). IIS lists none in Ground Forces holdings.

**RECOGNISE Launcher** Turret with two blocks of six missile tubes, one each side, and a twin-barrel 30 mm 2A38M gun beside each block; 12 missiles ready.

**Radar** Flat rectangular search array, about 1.8 x 0.9 m, rotating on the turret roof; folds forward flat on the roof when stowed.

**Radar** Tracking radar on the turret front face; small electro-optical post on a pedestal on the turret top, ahead of the search radar.

**Chassis** Russian systems on an 8x8 KAMAZ-6560 truck; four hydraulic jacks level the vehicle to fire. Some later ones on MZKT-7930 with armoured two-piece-windshield cab.

**Complex** Battery of up to six combat vehicles linked by data link, with one transloader truck per two combat vehicles.

**NOT TO BE CONFUSED WITH** 2K22M1 Tunguska-M1 (SA-19 Grison) p. 17 Tracked chassis and 8 missiles; Pantsir-S1 in Russian service is on an 8x8 truck with 12 missiles.

**Pantsir-SM-SV** Same Pantsir turret family but on a tracked GM-5970-series chassis.

**SEEN WITH** S-400 (SA-21 Growler) battalion (one Pantsir battery per S-400 battalion (Red Star); 3 Pantsir per battalion (Armada)); Transloader vehicle (designation not confirmed) (1 per 2 combat vehicles); Maintenance, repair, alignment and spares vehicles; KAMAZ-6560 8x8 truck (1 per combat vehicle); 1L122 Garmon radar (INA; reported ringing Pantsir positions in Ukraine)

**VARIANTS** Pantsir-S2 INA. Modernised Pantsir for Russian forces, in service 2015; IIS lists VKS holdings as Pantsir-S1/S2. Details INA. **Pantsir-SM** New multifunction radar with active phased array; reported on a new 8x8 KAMAZ with armoured cab. KBP claim: engagement range 40 km and detection 75 km, against 20 km and 40 km for Pantsir-S1. **Pantsir-SM-SV** Tracked GM-5970-series chassis instead of a truck. For Ground and Airborne Forces; 1RS2-3 radar; 57E6M and TKB-1055 missiles; imagery with army units reported March 2025 (likely taken 2024). **Pantsir-S1 (MZKT-7930)** MZKT-7930 8x8 chassis with armoured cab and two-piece windshield; one destroyed near Mariupol Feb 2026. Same 72V6 combat module as the KAMAZ version; chassis ordered from Belarus in 2024. **Pantsir-S1M** INA. Export version of Pantsir-SM; new missile with 30 km range (claim). Oryx records one destroyed in Ukraine.



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**MOBILITY** 8x8 truck, fast on roads. KAMAZ chassis has limited off-road mobility and tends to roll over (high centre of gravity, narrow track). Jacked level to fire guns accurately.

|                    |                                                                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Weight             | 14.85 t WEG 'curb weight'; basis unclear also up to 30 t depending on chassis WP-RU29                                               |
| L × W × H          | 10.3 × 2.5 × 3.27 m                                                                                                                 |
| Road speed         | 90 km/h maximum                                                                                                                     |
| Road range         | 1,000 km maximum also 500 km WP-RU29                                                                                                |
| Fording            | 1.8 m                                                                                                                               |
| Time into action   | 3 min WEG 'time in action'; WEG also gives deployment time 5 min; which is time into action is not stated also about 15 min WP-RU29 |
| Time out of action | INA                                                                                                                                 |
| Ground pressure    | INA                                                                                                                                 |

|                                        |                                                                                                                      |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Range, 57E6 missile                    | 20 km maximum, aircraft-type targets; KBP figures for export 57E6-E, claim also 12 km CSIS4                          |
| Minimum range, 57E6 missile            | 1.2 km KBP figures for export 57E6-E, claim also 1.5 km CSIS4                                                        |
| Altitude, 57E6 missile                 | 5 m to 15 km maximum, KBP figures for export 57E6-E; 5 m only in optical mode over water, claim also 10 km CSIS4     |
| Range, 30 mm 2A38M gun                 | 4 km maximum, air and ground targets, claim                                                                          |
| Minimum range, 30 mm 2A38M gun         | 200 m air targets, claim                                                                                             |
| Altitude, 30 mm 2A38M gun              | 0 to 3 km maximum, air targets, claim                                                                                |
| Detection range, 1RS1 search radar     | 36 km maximum, 2 m2 target; S-band; up to 20 targets tracked, claim also 47 km APA8                                  |
| Detection range, 1RS2-1 tracking radar | 28 km maximum, target size not stated; WEG and Armada, claim also 24 km MISS8 APA8                                   |
| Detection range, optical post          | 17-26 km auto-tracking, F-16 autotracking, 10 km visibility; 13-15 km for HARM missile, claim                        |
| Missile speed, 57E6                    | 1,300 m/s maximum, claim                                                                                             |
| Target speed                           | 1,000 m/s maximum, claim                                                                                             |
| Range, Pantsir-SM missile              | 40 km claimed, Pantsir-SM with new missile; KBP chief designer, 2016, claim                                          |
| Detection range, Pantsir-SM radar      | 75 km claimed, target size not stated, claim                                                                         |
| Range, Pantsir-S1M missile             | 30 km claimed, export Pantsir-S1M; new missile, designation not given by source; High Precision Systems, 2019, claim |

**SOURCES** APA8 AR8 AR9 ARMADA7 CSIS4 IIS25 LENTA MIL21 MISS8 ORYX RSTAR RUSIAD RWW TASS7 TASS8 WEG23 WP-EN22 WP-RU29

## Tunguska 2K22 «Тунгуска»

NATO: SA-19 Grison

TR

Tracked short-range gun and missile air defence vehicle, 30 mm guns and SAMs In service 1982

**OPERATED BY** motor rifle brigade, air defence missile-artillery battalion, Tunguska battery, 6 launcher vehicles per battery

[D 2016 M](#)

**tank brigade**, anti-aircraft missile-artillery battalion, SA-19 battery; ATP 4-161 gives this battalion to tank brigades, but the battery figure is for brigades generally, 6 per battery (generic brigade figure: RWW 'maneuver brigade', ATP fig. 4-19) [D 2024 L](#)

**INDICATES** A Tunguska battery in a forward position points to a manoeuvre brigade's air defence missile-artillery battalion covering its first-echelon battalions; doctrine places it in brigade air defence, not in area air defence units.

**RECOGNISE Launcher** 2S6M and 2S6M1 carry eight missile tubes, four each side of the turret; original 2S6 had four, two each side (APA and FM 100-2-3: eight).

**Radar** Search radar at turret rear, folded over the rear when travelling; tracking radar at turret front; search antenna a cylindrical parabolic section, tracker a dish.

**Chassis** Tracked, six road wheels per side, three return rollers, rear sprocket; driver front left, engine at rear. GM-352 chassis; GM-5975 on 2S6M1.

**NOT TO BE CONFUSED WITH** **Pantsir-S1** p. 16 Usually an 8x8 truck with 12 missile tubes in two blocks of six, one each side of the turret; the tracked GM-352M1E Pantsir is a rarer offer with 8 missiles. **ZSU-23-4 Shilka** Gun-only predecessor, four 23 mm barrels, no missile tubes; Tunguska has two twin 30 mm guns with missile tubes outboard and a second radar at the turret front.



Vitaliy V. Kuzmin, CC BY-SA 4.0

**MOBILITY** Fires guns on the move but must be stationary to fire missiles. Not amphibious (WEG); fords 1.0 m (WEG 2S6M asset, RUWIKI).

|                     |                                                           |
|---------------------|-----------------------------------------------------------|
| Weight, combat      | <b>34 t</b>                                               |
| L × W × H           | <b>7.93 × 3.24 × 4.02 m</b> radar up; 3.36 m radar down m |
| Speed, maximum road | <b>65 km/h</b>                                            |
| Range, road         | <b>500 km</b> cruising range                              |

|                                 |                                                                                |
|---------------------------------|--------------------------------------------------------------------------------|
| Range, 9M311-M1 missile         | <b>10 km</b> maximum, Tunguska-M1, air target                                  |
| Minimum range, 9M311-M1 missile | <b>2.5 km</b> Tunguska-M1, air target; APA gives 2.5 km for 2K22 and 2K22M too |
| Altitude, 9M311-M1 missile      | <b>3.5 km</b> maximum, air target; ceiling for 9M311 family                    |
| Range, 30 mm 2A38M guns         | <b>4 km</b> maximum, air target                                                |
| Minimum range, 30 mm 2A38M guns | <b>200 m</b> air target                                                        |

**SEEN WITH** 2F77 / 2F77M transporter-loader (6 per battery (WEG; RUWIKI: one per 2S6); APA 2 to 3, WIKI up to 3)

**SOURCES** APAB ATP FM32 IISS25 JREW02 MISS8 ROE2 RWW WEG43 WP-EN46 WP-RU59

## Strela-10 and Osa short-range SAM 9K35, 9K33 «Смерел-10», «Оса»

NATO: SA-13 Gopher

TR

Short-range SAM: tracked Strela-10 (optical/IR) and wheeled Osa (radar command guidance) In service 1971

**OPERATED BY** motor rifle brigade, air defence missile-artillery battalion, missile battery (SA-13), 6 launchers per battery [D 2024 M](#)

**tank brigade**, air defence missile-artillery battalion, missile battery (SA-13), 6 launchers per battery [D 2024 L](#)

**air defence missile regiment**, division air defence regiment (Osa-AKM, being deactivated in 2019) [D 2019 L](#)

**INDICATES** SA-13 in a brigade fits the missile battery of its air defence missile-artillery battalion, six launchers in doctrine; two platoons cover the artillery group. Osa-AKM fits a division air defence regiment, being withdrawn in 2019; it covers troops on the march.

**RECOGNISE Launcher** Strela-10: turret with four missile canisters in two pairs, range radar between them. Osa-AK/AKM: six ribbed missile containers; first Osa had four exposed missiles.

**Radar** Strela-10 9A35 adds Flat Box-B passive antennas on hull and turret; 9A34 has none. Osa: surveillance array folds back 90 degrees at rear, tracking group at front.

**Chassis** Strela-10: modified MT-LB, tracked, six equally spaced road wheels, no return rollers. Osa: BAZ-5937, 6x6 wheeled, amphibious; 4.20 m high with mast stowed.

**Complex** Strela-10 battery: one 9A35, three 9A34, PPRU-1M command post with Dog Ear radar. Osa battery: four launch vehicles and two reload vehicles with crane.

**NOT TO BE CONFUSED WITH** **9K31 Strela-1 (SA-9 Gaskin)** Four-wheeled BRDM-2 chassis; Strela-10 is on a tracked MT-LB. **2K22 Tunguska (SA-19 Grison)** p. 17 Radar-directed gun and missile hybrid; Strela-10 is visually aimed and has no cannon, only a 7.62 mm MG.



Vitaliy V. Kuzmin, CC BY-SA 4.0

**MOBILITY** Strela-10 is tracked and floats only with fitted buoys (M2). Osa is a wheeled amphibious 6x6 (WH-AAT-A class); it fires from short stops, not on the move.

|                |                                                |
|----------------|------------------------------------------------|
| Weight, combat | <b>12.1 t</b> Strela-10                        |
| Weight, combat | <b>18.68 t</b> Osa-AKM, APA8 comparative table |
| L × W × H      | <b>6.45 × 2.85 × 2.22 m</b> Strela-10          |
| Road speed     | <b>60 km/h</b> maximum, Strela-10              |

|                                               |                                                                                                                                         |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Range, 9M37M/9M333                            | <b>0.8 to 5 km</b> engagement envelope, Strela-10 engagement zone, all variants; target closing at up to 415 m/s or receding at 310 m/s |
| Altitude, 9M37M/9M333                         | <b>25 m to 3.5 km</b> engagement envelope, Strela-10; 9M333 minimum 10 m                                                                |
| Range, 9M33M3                                 | <b>15 km</b> operational range, Osa-AKM missile                                                                                         |
| Altitude, 9M33M3                              | <b>4 km</b> maximum flight altitude, Osa-AKM missile; WEG figure                                                                        |
| Detection range, Land Roll surveillance radar | <b>30 km</b> acquisition range, Osa; target type not stated in source                                                                   |

**SOURCES** APA11 APA12 ATP CNA IISS25 JANESRG JREW02 MISS12 RWW WEG38 WP-EN40 WP-EN41

## Nebo-M 55Ж6М «He60-M»

NATO: Tall Rack **WH-AT**

Mobile multiband air surveillance radar complex; paired low-altitude radar **Podlet-K1** In service 2012

**OPERATED BY** unit not identified, 4 Nebo-M and Nebo-UM combined **R 2015 L**

unit not identified, air defence and radio-technical crews with a division-level command post (MoD release; unit not named)

**R 2017-09 L**

**INDICATES** Nebo-M or Podlet-K1 indicates an Aerospace Forces radio-technical (radar) unit building the air picture, often near an S-300 or S-400 position that the radars cue. One reported Podlet-K1 site (Kherson, Jul 2022) lay over 100 km behind the front.

**RECOGNISE** **Antenna** RLM-M: very large VHF array of short Yagi elements, 24 x 7. RLM-D: flat L-band panel, 38 x 48 elements; outer segments fold and stow flat under clamshell doors.

**Antenna** Podlet-K1: one large upper phased array, two smaller back-to-back IFF arrays below it, a narrow anti-jam antenna beside it. Antenna lies lowered on the truck in travel.

**Mast** Nebo-M antennas raised and folded hydraulically on a mast with a compact equipment cabin at its base; small navigation antenna package at the top of each array.

**Chassis** Nebo-M: each module on a BAZ-6909-015 8x8 truck, the chassis of the S-400 TEL. Podlet-K1: KamAZ-6350 8x8 truck.

**NOT TO BE CONFUSED WITH** 55Zh6U Nebo-U Transportable, not self-propelled: six transport units and hours to deploy. RLM-M is one 8x8 truck that deploys in about 15 min.

**11L19** Nebo-SVU Box-shaped T/R module enclosures on the mast and rear backlobe elements; both absent on RLM-M.

**59N6E** Protivnik-GE Semi-trailer mounted L-band radar; RLM-D uses a smaller antenna on a self-propelled 8x8.



Nebo-M RLM-M. Ministry of Defence of the Russian Federation, CC BY 4.0



RLM-D and KU-RLK control cabin. Ministry of Defence of the Russian Federation, CC BY 4.0

**MOBILITY** Self-propelled 8x8 trucks; Nebo-M can also move by rail. Each Nebo-M module has its own 100 kW generator. Hydraulic levelling and antenna deployment.

|                                    |                                                               |
|------------------------------------|---------------------------------------------------------------|
| Road speed, KamAZ-6350             | 95 km/h maximum, base truck, Podlet-K1 carrier                |
| Height, Podlet-K1 antenna deployed | 12 m Podlet-K1, radar fully deployed                          |
| Time into action, Nebo-M           | 15 min Nebo-M, manufacturer cited stow and deploy time, claim |
| Time into action, Podlet-K1        | 20 min Podlet-K1, deployment or stowage, claim                |

Detection range, Nebo-M 600 km maximum, complex; target type, size and altitude not stated, claim

Altitude coverage, Nebo-M 600 km maximum, complex, ballistic targets implied; not stated, claim

Detection range, Podlet-K1 200 km maximum, main mode, 10-200 km; target size not stated, claim

Minimum range, Podlet-K1 10 km air targets, lower limit of stated 10-200 km detection zone, claim

**SOURCES** AIF ANNA APA6 APA7 ARMADA6 ESP GS10 JREW02 RUSIAD TRUCK TWZ5 TWZ6 U242 VPK4 WEG22 WP-RU26 WP-RU27 ZVEZ

## 96L6 all-altitude acquisition radar 96Л6 «Всевысотный

обнаружуватель»

NATO: Cheese Board **WH-AT**

3D C-band acquisition and target designation radar for S-300PM/PMU2 and S-400 SAM units In service 1997

**OPERATED BY** aerospace forces air defence regiment, 3rd SAM battalion, INA (photographed deployed and stowed) **O 2010-02 M**

aerospace forces air defence regiment, S-400 'battery' in Armada's usage; APA export diagram lists it as a battery component, 1 per S-400 battery (Armada, term undefined; APA lists 96L6E as a battery component without a count); S-300PMU2 count INA **R 2014 M**

**INDICATES** A 96L6 usually marks a nearby long-range SAM fire unit (S-300PM/PMU2 or S-400) of the Aerospace Forces, which are district-level assets. The maker also offers it to radio-technical units, so on its own it does not confirm a SAM site.

**RECOGNISE** **Antenna** Deployed: one flat rectangular array, slightly taller than wide, near-vertical, tilted back a little, on a rotating pedestal just behind the cab; lattice IFF array below.

**Antenna, stowed** Stowed: array folds forward and lies flat and horizontal over the pedestal and cab roof, giving a long, low, box-like vehicle outline.

**Mast** Normally no mast. Antenna post can be lifted onto a 40V6M (23.8 m) or 40V6MD lattice mast for low-level cover; the roughly 2 h fitting time is an unverified maker figure.

**NOT TO BE CONFUSED WITH** 76N6 Clam Shell Split antenna with a large beak-like shroud, usually raised on a 40V6M mast. A mast-mounted 96L6 shows one flat rectangular array.

**64N6E / 91N6E** Big Bird Janus-faced (back-to-back) space-fed array on a semi-trailer towed by an MZKT-7930 or BAZ-6403 tractor. 96L6 is a single self-propelled vehicle with one array.



Vitaly V. Kuzmin, CC BY-SA 4.0

**MOBILITY** Self-propelled 8x8. Single-vehicle version ready 5 min from march. Two-trailer and mast versions are semi-mobile; maker figures of 30 min to about 2 h not verified. Not amphibious. L x W x H 11.7 x 3.07 x 2.92 m WEG figure for MAZ-543 chassis; radar vehicle is MZKT-7930

|            |                                                 |
|------------|-------------------------------------------------|
| Road speed | 65 km/h maximum, WEG figure for MAZ-543 chassis |
| Road range | 650 km cruising, WEG figure for MAZ-543 chassis |
| Fording    | 1.3 m WEG figure for MAZ-543 chassis            |

Detection range 300 km instrumented, Range band 5 to 300 km for detected targets; target size and altitude not stated, claim

Maximum target altitude 100 km All-altitude mode, aerodynamic targets, claim

Band C-band claim

Frequency 5.25 to 5.925 GHz

Elevation cover, all-altitude mode -1.5° to +20° 360° azimuth

**SOURCES** APA1 APA2 APA3 ARMADA1 ATP GS5 JREW02 MILRU ORYX PVO RS RT RUSI1 RUSIAD RWW U241 WEG5 WMC1 WP-RU9

## Zoopark-1 / Zoopark-1M 1A219, 1A219M; 1A260 «3oonapk-1», «3oonapk-1M»

TR

Tracked radar that locates enemy guns, mortars, rockets and missile launchers In service 1970

**OPERATED BY** motor rifle brigade, fire control and artillery reconnaissance battery, radar platoon (with SNAR-10M1) [D 2016 M] tank brigade, fire control and artillery reconnaissance battery, radar platoon [D 2016 M] motor rifle division [D 2024 L]

**INDICATES** Doctrinally held in the radar platoon of a manoeuvre brigade's fire control battery and by artillery formations. Seeing one suggests brigade-level or higher artillery is nearby and set up for counter-battery fire.

**RECOGNISE** **Antenna** One flat rectangular phased-array panel. 1L219/1L219M: raised upright at rear of roof. 1L260: a much taller panel.

**Mast** No separate mast. Panel is lifted and turned on the roof by a lift and rotation mechanism. For travel it folds down flat along the roof.

**Chassis** 1L219/1L219M: low MT-LBu tracked carrier, flat sides, raised rear hull. 1L260: larger GM-5955 tracked chassis, 7 road wheels a side, boxy hull with side lockers.

**Complex** Radar vehicle can work alone. Support: 1I30 (1L219M) or 1I38 (1L260) maintenance vehicle and a generator; same brigade fire control battery as SNAR-10M1 and PRP-4A.

**NOT TO BE CONFUSED WITH** SNAR-10 / SNAR-10M1 (1RL232) battlefield surveillance radar p. 20 Same MT-LBu hull but has a turret with a PKT machine gun and a rotating parabolic radar, not a large flat panel.

**ARK-1 / ARK-1M Rys (1RL239) counter-battery radar** p. 19

Predecessor on the same MT-LBu hull and looks alike. No reliable distance cue found; confirm by markings or close imagery.



Boevaya mashina, CC BY-SA 3.0 de

**MOBILITY** Tracked, cross-country capable. Emplaces or displaces in about 5 min. Swimming ability not documented for either chassis.

|                       |                            |
|-----------------------|----------------------------|
| Weight, 1L261 (1L260) | <b>38.1 t</b>              |
| Road speed, 1L219M    | <b>62 km/h</b> maximum     |
| Road speed, 1L260     | <b>70 km/h</b> maximum     |
| Road range            | <b>500 km</b> Both chassis |

Detection range, mortar, 1L219M

**20 km** detection of launch position, Mortar firing position; calibre not stated

Detection range, tube artillery, 1L219M

**15 km** detection of launch position, Field artillery firing position

Detection range, rocket artillery, 1L219M

**30 km** detection of launch position, MLRS firing position

Detection range, tactical missile, 1L219M

**40 km** detection of launch position, Tactical missile launch position

**SOURCES** AR24 AR25 AR26 ARMADA9 ATP BMPD10 DOGS FISH GWARA IZ3 MIL34 MIL35 MIL36 ORYX RUMOD RWW SAIDPVO WEG2015 WMC21 WMC22 WP-RU63 not in WEG

## ARK-1M Rys 1PA239 «PblCb»

NATO: Small Fred

TR

Artillery weapon-locating and fire-control radar In service 1977

**OPERATED BY** unit not identified, division's reconnaissance battery (WEG wording, identical to Wikipedia; Soviet-era) [R 2024 L]

**INDICATES** A single sighting points to an artillery reconnaissance element supporting a formation's guns. Sources give no current Russian subunit or count, so it says little about formation type.

**RECOGNISE** **Antenna** Parabolic antenna (JREW02); radiator in a compact radio-transparent housing plus a large receiving antenna (TOPWAR). Sizes INA.

**Mast** No mast described in any read source. Two emitters and a receiving antenna sit outside the hull (MILN); TOPWAR names one emitter housing. Height INA.

**Chassis** MT-LBu unarmed tracked carrier; electronics mostly inside the armoured hull.

**Complex** Single self-contained vehicle with own navigation, topographic survey, communications and power supply (ARK-1M generator). Crew 4.

**NOT TO BE CONFUSED WITH** 1L219 Zoopark-1 p. 19 Also MT-LB based; Zoopark-1 uses a phased array antenna, ARK-1M a parabolic antenna.

**1L260 Zoopark-1M** p. 19 Built on the GM-5955 tracked chassis with a phased array antenna.

**1RL232 SNAR-10** p. 20 Same MT-LBu hull, parabolic antenna and near-identical size; SNAR-10 has a turret with a 762 mm PKT MG. No source arms the ARK-1M.

**1K148 Yastreb-AV** Four-axle BAZ-6910-025 wheeled chassis, not tracked; antenna post at the rear.



ShinePhantom, CC BY-SA 3.0

**MOBILITY** Tracked MT-LBu chassis; WEG says it may not be amphibious given radar weight. Forging depth INA.

|            |                                             |
|------------|---------------------------------------------|
| Weight     | <b>15.7 t</b>                               |
| L × W × H  | <b>7.61 × 2.85 × 2.72 m</b> Chassis overall |
| Road speed | <b>61 km/h</b> maximum                      |
| Road range | <b>500 km</b> cruising                      |

Detection range, ARK-1M vs artillery (howitzers) **8 km** unspecified, Artillery (WEG), howitzers (JREW02); target and mode not stated by WEG

Detection range, ARK-1M vs mortars **13 km** unspecified, Mortar (WEG); mode not stated

Detection range, ARK-1M vs MRL rockets **25 km** unspecified, Rocket (WEG); mode not stated

Detection range, ARK-1M vs battlefield missiles **30 km** unspecified, Missile (WEG)

Frequency band **NATO H band (6–8 GHz)**  
Track capacity **3 tracks/min**

**SEEN WITH** MT-LBu tracked chassis [1]

**SOURCES** ATP JREW02 MIL2 MIL3 ORYX PMUL TOPWAR1 TOPWAR2 WEG6 WEG2015 WP-EN11

## 1B75 Penicillin 1B75 «Пенициллин»

WH-AT

Passive acoustic-thermal gun locator (1B75); man-portable counter-mortar radar (1L271) In service 2018

**OPERATED BY** self-propelled artillery battalion, battalion fire control network, conceptual ATP fig. 5-3 (1L271) [D 2024 M]  
 tank division, 1L271; holding subunit not stated [O 2018-01 M]  
 combined arms army, 1L271 in rocket, missile and tube artillery units [R 2017 L]

**INDICATES** A 1B75 points to an artillery regiment or brigade counter-battery element, as deliveries were planned for those units. A 1L271 was issued down to company and battalion level and marks forward mortar or artillery fire control.

**RECOGNISE** **Antenna** 1L271: transceiver on hand-turned tripod pedestal; reflector described as single-curvature parabolic or as phased array with mirror. 1B75 has no radar antenna.

**Mast** 1B75: stabilised head with six TV and six thermal cameras on a raising mast. Per GlobalSecurity only, the mast folds rearward onto the van roof in transit.

**Chassis** 1B75: KamAZ cross-country truck, 8x8 per GlobalSecurity and Wikipedia; van body and four jacks per GlobalSecurity only. 1L271: man-carried or on a UAZ-2966.

**Complex** 1B75: two KamAZ vehicles plus remote ground acoustic sensors, up to six vehicles. 1L271: 135 kg set of radar, processor, power unit, console and radio.

**NOT TO BE CONFUSED WITH** **Zoopark-1 / Zoopark-1M** p.19 Radar complex on one armoured amphibious vehicle; 1B75 is a KamAZ truck with a camera mast and no radar.

**1B33M AZI-7M** Older sound-ranging system, acoustic only; lacks the 1B75 thermal and TV camera mast.



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**MOBILITY** 1B75 on KamAZ 8x8 truck; mast raised and ground sensors laid out at site, crew relocates daily. 1L271 man-portable (135 kg in loads) or vehicle-carried.

|                         |                                                                 |
|-------------------------|-----------------------------------------------------------------|
| Weight, 1L271           | <b>135 kg</b> complete set: radar, processor, power unit, claim |
| Road speed, 1B75        | <b>90 km/h</b> maximum, KamAZ chassis, claim                    |
| Time into action, 1L271 | <b>5 min</b> maximum, preparation for work, claim               |
| Weight, 1B75            | <b>INA</b>                                                      |

|                            |                                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------------------|
| Detection range, 1B75      | <b>up to 25 km</b> maximum, radius; artillery, mortar and MLRS firing positions, sound and thermal, claim |
| Target location time, 1B75 | <b>5 s</b> maximum, single firing target, claim                                                           |
| Radio range, 1B75          | <b>40 km</b> maximum, to external subscribers, claim                                                      |
| Detection range, 1L271     | <b>20 km</b> maximum, tank-type moving ground target, claim                                               |

**SOURCES** AIF2 AR10 ATP BMPD5 BMPD6 DEFEX GS11 IZV20 IZV22 ORYX RADTUT RG21 RG152 RG242 RWW SHEP2 TASS9 WP-EN23 WP-EN24 WP-RU30 WP-RU31 not in WEG

## SNAR-10 / SNAR-10M1 Leopard CHAP-10M1 «Aeonapp»

NATO: Big Fred

TR

Tracked battlefield surveillance and artillery-adjustment radar In service 1975

**OPERATED BY** motor rifle brigade, fire control and artillery reconnaissance battery, radar platoon [D 2016 M]  
 tank brigade, fire control and artillery reconnaissance battery, radar platoon [D 2016 M]

military district, artillery units of the tank formation stationed in the South Urals (MoD wording; formation not named) [R 2019-01 M] seen with Central Military District

military district, artillery reconnaissance subunits (SNAR-10M Panthera, deliveries by end March 2016) [R 2016-03 L] seen with Western Military District

**INDICATES** A SNAR-10M1 points to a maneuver brigade's fire control battery radar platoon, so an artillery-adjustment and ground-surveillance element is working for that formation. Ukraine also holds the SNAR-10, so a sighting alone is not proof of a Russian unit.

**RECOGNISE** **Antenna** Rectangular reflector nearly vehicle width, raised electromechanically for use; folds forward when travelling.

**Mast** No mast described: radar is built into the vehicle's turret, antenna on the roof (SNAR-10); on the M1 the antenna module sits on the rear roof.

**Chassis** MT-LB family tracked hull; the Big Fred version is a roof-mounted radar on an MT-LB.

**Complex** Single vehicle with its own diesel generator (M1); sources describe no separate power or command vehicle.

**NOT TO BE CONFUSED WITH** **MT-LBu command vehicle** MT-LBu has canvas cover extending to rear and extra radios, no folding radar reflector on roof.

**PRP-4A Argus** p. 36 Built on the BMP-1 infantry fighting vehicle chassis with a mast-mounted radar; SNAR-10 is on the MT-LB tractor hull.

**SOURCES** AR18 ARMYINF3 ATP BMPD8 FOCUS2 JANESRG JREW02 MILUA PERISC RWW SHEPH UKRIN VPK8 VPK9 WEG27 WEG37 WEG2015 WP-RU54



Milru / Yevgeny Kel, CC BY 4.0

**MOBILITY** Tracked MT-LB derivative; radar deploys from halt, about 5 min into action from prepared position.

|                  |                                                                                  |
|------------------|----------------------------------------------------------------------------------|
| Weight, combat   | <b>12.7 t</b> SNAR-10M1, developer figure, claim                                 |
| Road speed       | <b>60 km/h</b> maximum, SNAR-10 (WEG); SNAR-10M1 'up to 60 km/h' (developer)     |
| Road range       | <b>500 km</b> cruising, SNAR-10                                                  |
| Time into action | <b>5 min</b> claimed, SNAR-10, prepared position; up to 20 min unprepared, claim |

|                                  |                                                                                                         |
|----------------------------------|---------------------------------------------------------------------------------------------------------|
| Detection range, SNAR-10 radar   | <b>16 km</b> maximum, moving vehicles, without MTI; 10 km with MTI, claim                               |
| Detection range, SNAR-10 radar   | <b>30 km</b> maximum, moving ships, without MTI, claim                                                  |
| Detection range, SNAR-10 radar   | <b>10 km</b> maximum, shell impact (100 mm), without MTI, claim                                         |
| Detection range, SNAR-10M1 radar | <b>up to 40 km</b> maximum, tanks and vehicles; Russian MoD figure; instrumented range 0.2-40 km, claim |

## Krasukha-2 / Krasukha-4 1A269; 1PA257 «Kpacuxa-2»; «Kpacuxa-4»

WH-AT

Truck radar jammers: Krasukha-2 vs S-band AEW radar; Krasukha-4 vs X/Ku-band radars In service 2011

**OPERATED BY** EW brigade, REB-S [counter-air] EW battalion, INA; usually one Krasukha-2 and one Krasukha-4 per brigade (reported 2022) [D 2018 H]

**EW brigade**, part of 1 Krasukha-4 complex (command-post container found near Makariv); Armada estimates 4 complete Krasukha systems in theatre [R 2022-03 M]

**EW battalion**, independent REB-S battalion of an Air and Air Defence Army (VKS); FOI infers, does not name Krasukha there [R 2018 L]

**unit not identified**, 1 Krasukha-4 reported hit [R 2023-11 L]

**unit not identified**, 1 Krasukha-2 reported [R 2018-08 L]

**INDICATES** *Krasukha jammers belong to EW brigades (REB-S battalions): four under the military districts, the 15th (Tambov) under the General Staff. Manoeuvre brigades have EW companies. A sighting points to an EW brigade element or a protected site such as an air base.*

**RECOGNISE** **Antenna** Krasukha-2: one very large solid circular dish, about twice a wheel diameter across, on a turntable at the rear; rotates 360° and tilts. Stowed look not documented.

**Antenna** Krasukha-4: box-shaped rotating turret on a van-body roof with three dishes, two large on the sides, one small on the front; all far smaller than the Krasukha-2 dish.

**Mast** Krasukha-4: telescopic mast with a small antenna head behind the turret; the second vehicle also carries a mast antenna. Krasukha-2: no tall mast in photos.

**Stowed** Krasukha-4: in travel position the turret is stored inside the shelter (AR); in action two hydraulic stabilisers are lowered on each side of the shelter.

**Chassis** Krasukha-2: BAZ-6910-022 8x8, angular flat-faced cab, axles 2+2 with a wide gap. Krasukha-4: KamAZ-6350 8x8, standard KamAZ cab-over, spare wheel behind cab.

**Complex** Krasukha-4: two KamAZ-6350 trucks, jammer plus command-post container vehicle. Krasukha-2: three vehicles (one source); probably procured with Moskva-1 (FOI).

**NOT TO BE CONFUSED WITH** 1L269 Krasukha-2 vs 1RL257 Krasukha-4 Krasukha-2: one big dish on a BAZ-6910 with angular cab. Krasukha-4: turret with three smaller dishes on a KamAZ-6350 van body.

**Pelena-1** (older S-band counter-AEW jammer) Antenna is a tall curved rectangular reflector on a towed wheeled trailer (APA photos), not a solid circular dish on a self-propelled 8x8 truck.

1L267 Moskva-1 ELINT and command complex p. 22 Three KamAZ trucks (FOI: one ELINT, two command; 3G: KamAZ-6350-1335, third with a jamming control station); no Krasukha-4 style dish turret.

GT-01 Murmansk-BN HF jamming complex p. 22 Has KamAZ-6350 support trucks, but its signature is very tall extendable HF antenna masts on KamAZ-53501 trucks, not dishes.

55Zh6M Nebo-M radar modules (BAZ-6909-015) p. 18 Same heavy BAZ 8x8 family, but antennas are a flat rectangular planar array or a Yagi array, not a solid circular dish.

**VARIANTS** 1L269 Krasukha-2 (Krasukha-20) BAZ-6910-022 8x8 with one large solid circular dish at the rear. S-band jammer aimed mainly at AEW radars such as E-3 and E-8. Analogue equipment per ru.wikipedia. 1RL257 Krasukha-4 (Krasukha-S4) KamAZ-6350 8x8 van with a three-dish turret, plus a second KamAZ-6350 carrying the command post. X/Ku-band jammer vs fighter fire-control and ISR radars, UAVs and LEO satellites. Digital equipment per ru.wikipedia.



Krasukha-4. The Ministry of Defence of the Russian Federation, CC BY 4.0



Krasukha-2. The Ministry of Defence of the Russian Federation, CC BY 4.0

**MOBILITY** Heavy 8x8 trucks. Must halt and emplace to jam: Krasukha-4 needs under 20 min to emplace or displace; Krasukha-2 lowers jacks before operating.

|                              |                                                                 |
|------------------------------|-----------------------------------------------------------------|
| Krasukha-4: weight           | <b>39,000 kg</b><br>also 12,450 kg AR6                          |
| Krasukha-2: weight           | <b>40,000 kg</b> BAZ-6910-022 chassis full mass; kerb mass 18 t |
| Krasukha-4: road speed       | <b>95 km/h</b> maximum, KamAZ-6350 truck figure (AR)            |
| Krasukha-2: road speed       | <b>80 km/h</b> maximum                                          |
| Krasukha-4: road range       | <b>1,000 km</b> KamAZ-6350 truck figure (AR)                    |
| Krasukha-2: road range       | <b>1,000 km</b>                                                 |
| Krasukha-4: fording          | <b>1.75 m</b> KamAZ-6350 truck figure (AR), claim               |
| Krasukha-2: fording          | <b>1.4 m</b> BAZ-6910-022 chassis figure (REACH)                |
| Krasukha-4: time into action | <b>less than 20 min</b>                                         |

|                                  |                                                                                                            |
|----------------------------------|------------------------------------------------------------------------------------------------------------|
| Krasukha-2: jamming range        | <b>up to 250 km</b> up to, air target: AEW aircraft radar; target altitude not stated                      |
| Krasukha-4: jamming range        | <b>150–300 km</b> maximum, air and ground-based radar targets; target size and altitude not stated, claim  |
| Krasukha-2: target band          | <b>S-band</b><br>also 2.3–2.7 GHz CRFS                                                                     |
| Krasukha-2: main targets         | <b>AEW aircraft radars (E-3 Sentry, E-8 JSTARS) air targets</b>                                            |
| Krasukha-4: operating band       | <b>8–18 GHz (X and Ku band)</b><br>also 8.5–10.7 and 13.4–17.7 GHz 3G                                      |
| Krasukha-4: main targets         | <b>Fighter and interceptor radars, UAVs, reconnaissance aircraft, LEO satellites</b> air and space targets |
| Krasukha-4: coverage             | <b>360° azimuth; 1° to 60° elevation</b>                                                                   |
| Krasukha-4: output power         | <b>24 dBW (8 GHz); 23 dBW (13 GHz); 21 dBW (18 GHz)</b>                                                    |
| Krasukha-4: power consumption    | <b>30 kW</b>                                                                                               |
| Krasukha-4: continuous operation | <b>24 h</b>                                                                                                |
| Krasukha-2: crew                 | <b>INA</b><br>also 7 or 3 REACH                                                                            |
| Krasukha-4: crew                 | <b>INA</b><br>also 3 AR6                                                                                   |

**SEEN WITH** Krasukha-4 command-post vehicle (KamAZ-6350 with container) (1 per Krasukha-4 complex); 1L267 Moskva-1 ELINT and command complex (INA; one Moskva-1 can link up to nine Krasukha-type jammers); KamAZ-6350 8x8 truck (2 per Krasukha-4 complex); BAZ-6910-022 8x8 chassis (1 per Krasukha-2 jammer vehicle); RB-109A Bylina EW brigade command system (brigade-level EW C2, not Krasukha-specific)

## Murmansk-BN «Мурманск-БН»

WH-AT

Strategic HF (3-30 MHz) communications intercept and jamming complex, multi-truck In service 2014

**OPERATED BY** EW brigade, Probably a company of the REB-N (anti-ground) battalion [FOI assessment] [D 2018 H](#)  
 unit not identified, Radio-jamming node (y3ea paguonome); its 1st section named at the 471st (KZ), 2 complexes at the Kamchatka centre (KZ, reported Jan 2016; FOI: delivered 2014) [O 2019 H](#)

**EW brigade**, At least 1 complex per brigade (Armada); 2 antenna groups observed at 16th EW bde, vehicle count at 19th suggests 2 (PAS imagery, 2019) [O 2019 M](#)

**INDICATES** Seeing Murmansk-BN indicates an Army EW brigade (IISS lists 5; 15th under General Staff) or a fleet EW centre, with long-range HF intercept and jamming in place; it takes about 72 hours to deploy, so a raised complex marks a deliberate, fixed position.

**RECOGNISE** **Antenna** Wire HF antennas and mast guy lines run from the masts to the ground; from afar like a giant spider's web. Antenna field about 800 m a side. Stowed: carried on trucks.

**Mast** Four hydraulic telescopic masts up to 32 m, guyed. Armada: antenna trucks typically about 35 m apart across a 90-degree arc. Mast lowered for travel.

**Chassis** Heavy multi-axle KamAZ (or Ural) trucks with unified box bodies (kungs); some masts reported on towed trailers.

**Complex** Seven trucks: four mast vehicles plus generator, stowed-antenna and command post vehicles. Kola site seen with two groups of four antenna vehicles about 350 m apart.

**NOT TO BE CONFUSED WITH** Krasukha-2 / Krasukha-4 p. 21 Each is housed on a single truck; Murmansk-BN is spread over several trucks with tall guyed masts.

**Samarkand** Stationary coastal jamming complex, probably a fixed version of Murmansk-BN; not truck-mounted.

**SOURCES** 3G ARMADA5 B02 CRFS FOI ICDS ISS25 IZ KZ3 MIL18 PAS RG16 RG173 RWW UKRINF2 WEG21



Министерство обороны России, CC BY 4.0

**MOBILITY** Road-mobile truck column; needs a prepared antenna field about 800 m a side, marking and engineer work; 72 hours to deploy. Trucks up to 35 t need category E drivers.

|                    |            |                                           |
|--------------------|------------|-------------------------------------------|
| Weight, vehicle    | up to 35 t | Loaded KamAZ trucks of the complex, claim |
| Time into action   | 72 h       | Normative deployment time, claim          |
| Number of vehicles | 7          | Complex in travel order, claim            |
| Time out of action | INA        |                                           |

|                                  |                        |                                                                                                           |
|----------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------|
| Jamming range, HF communications | 5,000 km               | effective (Armada's term), HF (shortwave) radio links, skywave; ship, aircraft and ground emitters, claim |
| Frequency band                   | 3-30 MHz (HF)          |                                                                                                           |
| Power                            | up to 400 kW           | In some operating modes (KZ); generator output (Armada), claim                                            |
| Mast height                      | 32 m                   | 4 telescopic masts, fully raised, claim                                                                   |
| Deployed position area           | 640,000 m <sup>2</sup> | Antenna field about 800 x 800 m, claim                                                                    |

EW M

## Moskva-1 1A267 «Mockba-1»

WH-AT

Passive ELINT and automated control of counter-air jammers and air defence cueing In service 2014

**OPERATED BY** EW brigade, REB-S (counter-air EW) battalion, reported [D 2018 H](#)

**EW brigade**, INA [ARM22 lists one Moskva-1 among five brigade systems] [R 2023-09 L](#)

unit not identified, 1 claimed destroyed [R 2022-09 L](#)

**INDICATES** Moskva-1 is held by Ground Forces EW brigades, four of which answer to the military districts. Seeing it suggests a brigade counter-air EW element and Krasukha-type jammers at other sites, up to nine of which one complex can task.

**RECOGNISE** **Antenna** No automatically raised antenna array, unlike Krasukha. The ELINT module (1L265) was withheld from press; its antenna is not described in text (INA).

**Chassis** KamAZ-6350 8x8 trucks with plain box bodies (kung); a Russian reporter wrote the vehicles are hard to tell from other trucks.

**Complex** Three KamAZ vehicles per complex: an ELINT module and control vehicles; one account names a power-station truck as the third.

**Complex** Deployed: the control module stands on hydraulic legs on the ground while its truck drives 30-100 m away.

**NOT TO BE CONFUSED WITH** 1RL257 Krasukha-4 Krasukha deploys an automatically raised antenna block; Moskva-1 vehicles show plain box bodies.

1L269 Krasukha-20 p. 21 Krasukha-20 is on the larger BAZ-6910 chassis, not KamAZ.

**Other KamAZ-6350 box-body vehicles** No reliable external mark in text sources; FOI says Krasukha-20 is probably procured together with Moskva-1, so Krasukha jammers nearby are the main clue.



Vitaly V. Kuzmin, CC BY-SA 4.0

**MOBILITY** Road and track mobile truck complex; manufacturer gives 45 min to deploy. Control module can be set down on legs apart from its truck.

|            |                    |              |
|------------|--------------------|--------------|
| Weight     | 12,450 kg          | chassis only |
| L × W × H  | 9.3 × 2.5 × 3.25 m | chassis      |
| Road speed | 95 km/h            | maximum      |
| Road range | 1,000 km           | cruising     |

|                                      |                              |                                                                                                                        |
|--------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Detection range, airborne emitters   | 400 km                       | up to, passive; emitting aircraft and cruise missiles; 360 deg sector, claim                                           |
| Jamming range, airborne S-band radar | 250 km                       | up to, air targets; ARM23 says the 1L267 'can engage targets' per unnamed sources; Moskva-1 itself has no jammer (FOI) |
| Sector of view                       | 360 deg                      | claim                                                                                                                  |
| Frequency coverage                   | UHF, L, S, C, X and Ku bands | ELINT module 1L265                                                                                                     |
| Controlled systems                   | 9                            | EW or air defence complexes tasked simultaneously, claim                                                               |

**SOURCES** 3G AEX17 ARMADA3 ARMADA4 ATP CH97 DLG22 FOI ICDS IEEE22 IZ234 RIA15 RWW SDU13 TOPWAR5 VPK3 WEG19 WP-RU22 WP-RU23 WP-RU24

## Borisoglebsk-2 РБ-3015 «Борисоглебск-2»

TR-A

Tactical HF/VHF/UHF radio intercept, direction-finding and jamming complex In service 2013

**OPERATED BY** motor rifle brigade, EW company [D 2024-02 H]

tank brigade, EW company [D 2024-02 H]

motor rifle division, separate EW battalion [O 2022-01 M] seen with 3rd

Motor Rifle Division (Vislenskaya), separate EW battalion, Belgorod oblast

motor rifle brigade, EW subunit [O 2018 M] seen with 74th Separate Guards

Motor Rifle Brigade

unit not identified, EW units [O 2018 M] seen with 201st Military Base, Tajikistan (R-934BMV stations)

unit not identified [O 2022-03 L] seen with 69th Separate Covering Brigade (R-330BMV captured, Kyiv oblast, 2022-03)

**INDICATES** Usually indicates the EW company of a motor rifle or tank brigade or division within reach; since 2022 also division EW battalions. Vehicles of the complex may deploy singly rather than as a full set.

**RECOGNISE** Antenna Antennas on a mast, reported for one R-934BMV or R-378BMV station hit by drone. No text source gives antenna shape or size: INA.

**Mast** Vehicle-mounted antenna mast; height and stowed position INA in text sources.

**Chassis** MT-LBu tracked amphibious armoured carrier, modified for the EW equipment. One source says one R-325UMV rides on a Ural truck.

**Complex** Up to 9 vehicles: R-330KMV command post plus pairs of R-378BMV, R-330BMV, R-934BMV and R-325UMV jammers, with diesel generators. May deploy in part.

**NOT TO BE CONFUSED WITH** R-330Zh Zhitel p. 26 Ural-43203 wheeled truck (Diabazol complex), not tracked MT-LBu.

RB-341V Leer-3 p. 27 Single vehicle working with up to three Orlan-10 UAVs, not a tracked multi-vehicle set.

1L262 Rtut-BM Also on MT-LBu tracked chassis (3G table, tertiary); no text source found giving a visible difference.

SNAR-10 (Big Fred) p. 20 MT-LBu with turret-mounted radar and parabolic antenna; antenna folds forward on the roof when travelling.



Russian Ministry of Defense, CC BY 4.0

**MOBILITY** Tracked, amphibious MT-LBu chassis. About 15 min to deploy or pack up the complex on arrival.

**Weight** 11,000 kg WEG figure for base MT-LB chassis, not MT-LBu

also 12.6 t WEG2015

**L × W × H** 6.45 × 2.86 × 1.86 m WEG figure for base MT-LB chassis

**Road speed** 60 km/h maximum, base MT-LB chassis also 61.5 km/h WEG2015

**Road range** 500 km cruising, base MT-LB chassis also 500 km WEG2015

**Swimming speed** 5 km/h maximum

**Fording** Amphibious

**Time into action** 15 min or less whole complex, on arrival at site

**Time out of action** 15 min or less whole complex

**Ground pressure** 0.46 kg/cm2 standard track; 0.28 kg/cm2 wide track

**Employment distance from line of contact** 15 to 30 km practical, Donbas 2014-15, with Leer-3, Zhitel, R-934UM, observed

**Crew, R-330KMV command post** 4 2 of them at automated workstations

**Vehicles per complex** up to 9

**Jamming stations controlled** up to 4 pairs per R-330KMV command post

**Frequency coverage** HF, VHF, UHF intercept and jamming of ground and air radio; also satcom, GPS and cellular terminals

also 3 MHz to 3 GHz 3G

**Frequency-hopping signals** up to 300 hops/s VHF, detection and jamming, 'per open sources', claim

**Performance vs R-330 Mandat** 2x bandwidth, up to 100x faster HF and UHF bands; reported, same lineage

**Power supply** own diesel generators or 220/380 V 50 Hz mains

**Jamming range, ground radio** INA not published

**Detection range, radio emitters** INA

also 30 km 3G

**Armament** INA no source other than WEG names a

weapon on the EW vehicles

**SEEN WITH** R-330KMV command post (1 per complex); R-378BMV HF jamming station (2 per complex); R-330BMV VHF jamming station (2 per complex); R-934BMV airborne VHF/UHF radio jamming station (2 per complex); R-325UMV jamming station (2 per complex); ATO-40 service and repair vehicle; Diesel generators; MT-LBu (up to 9 per complex)

## R-330B Mandat-B and R-934BMV P-3305; P-9345MB «Mangam-B»

WH-AT

Ground VHF/UHF radio and air-communication jammers, tactical level In service 1996

**OPERATED BY** EW company, EW company of a motor rifle or tank brigade: R-330B squad in the VHF-band jamming platoon; R-934B squad [VHF aircraft] in the radio-communications jamming platoon

D 2016 H

unit not identified, R-934B (2024), Borisoglebsk-2 R-934 station (2023-11) and R-934BMV (2023-10, 2024-11, 2025-04) reported hit by Ukrainian units; no Russian holding unit named [R 2025 M]

**INDICATES** Seen with a brigade or division, it fits the organic EW company. RWW and FOI place these jammers in the EW company of motor rifle and tank brigades. It shows radio and air-link jamming is planned, not the size of the force.

**RECOGNISE** **Antenna** R-934UM2: one receiving and direction-finding antenna-feeder system plus seven transmitting antenna-feeder systems. Sizes INA.

**Mast** R-330T/B carry a telescoping mast system that is raised when deployed; R-330U stows its antenna in a rectangular housing. Mast height INA.

**Chassis** R-934B and R-330B: MT-LBu tracked, or Ural-4320-31 6x6 with K1-4320 body (R-934B); R-330T on KamAZ-43101 or Ural-43203-31 [K2-4320].

**Complex** Wheeled versions tow a generator trailer [16 kW for R-934B, ED-30 for R-330]; R-934UM2 has a truck with two generators and a shelter.

**NOT TO BE CONFUSED WITH** R-378BMV HF jammer [Borisoglebsk-2] p. 23 Same MT-LB carrier in the same nine-vehicle set; no text source gives an external difference. INA.

R-330Zh Zhitel (Diabazol) p. 26 Ural-43203 wheeled carrier, not tracked MT-LB; no external antenna difference given in sources. INA.

**SOURCES** 3G APAS ATP CRFS DEFBLOG FOI ICDS MIL6 MIL7 MIL9 MIL10 RWW TOPWAR4 UKRINF VRF3 WEG14 WP-RU13



Денис Абрамов [Denis Abramov], CC BY 4.0

**MOBILITY** Ural-4320 6x6 for R-934B/R-330T/R-934UM2 (truck-class off-road); tracked MT-LBu R-934B/R-330B. Figures below are chassis-class, not station-verified.

Weight, combat 11,000 kg R-330B on MT-LBu; WEG figure, unverified

Road speed 85 km/h R-934B on Ural-43203; WEG chassis figure

Road speed 60 km/h R-330B on MT-LBu; WEG figure, unverified

Road range 1,000 km R-934B on Ural-43203; WEG chassis figure

Frequency range, R-934B 100-400 MHz Export sheet; aircraft links 100-150 and 220-400 MHz, claim

Transmitter power, R-934B 500 W Export sheet, claim

Jammed links, R-934B 4 at fixed frequencies; 1 with frequency hopping Export sheet, claim

RMS direction-finding error, R-934B 2 deg Export sheet figure copied by WEG;

RWW gives not more than 3 deg, claim

Crew, R-934B 3 Export sheet, claim

**SEEN WITH** R-330KMV command post [1 per Borisoglebsk-2 set]

EW M

## Leer-2 85Я6 «Aeep-2»

WH-AT

Mobile technical control, radio reconnaissance and jamming vehicle on Tigr-M chassis In service 2012

**OPERATED BY** airborne division, EW subunit (FOI: VDV EW companies formed from 2016); INA; MoD announced 10 complexes for VDV EW units, Dec 2012 (title only, state claim) [R 2018 M] seen with 106th Guards Airborne Division (Tula), shown Nov 2012, Tula and Ivanovo airborne divisions, trials Sep 2012, Ivanovo and Ulyanovsk VDV formations, in use by 2014 unit not identified, Ground Forces EW units ('possibly' in FOI; state media: Southern MD column Mar 2018) [R 2018 M] seen with Russian military base in Abkhazia, EW subunit (MoD 2017, article titles only)

unit not identified, Position 17 km behind line of contact,

Novoluhanske, Donetsk region, 1 vehicle [R 2023-T1 M] seen with No Russian unit named

unit not identified, Navy EW centres [R 2018 L] seen with Navy EW centres, no unit named

**INDICATES** A Leer-2 most often indicates an Airborne Troops EW subunit; Navy EW centres, the Strategic Rocket Forces and Ground Forces EW units also hold it. It says nothing about the parent unit type.

**RECOGNISE** **Antenna** All EW antennas sit on a platform on the vehicle roof. Antenna shapes and sizes: INA, no text source.

**Mast** No separate mast reported. Two telescopic antennas are raised on the roof when static (3G calls them mast antennas). Height INA.

**Chassis** GAZ Tigr-M 4x4 armoured car: armoured bonnet, bulletproof hull and door windows, electric winch (4,000 kg, 20 m rope). No armament.

**Complex** Single vehicle. A forum report [2014] says an earlier army version of the same equipment is on a Kamaz-4350 truck with trailer.

**NOT TO BE CONFUSED WITH** Leer-3 [RB-341V] p. 27 Leer-3 works through Orlan-10 UAVs from a command vehicle; Leer-2 is a single armoured Tigr-M car with roof antennas.

**SOURCES** 3G AIRBASE AR7 CNA FOI GAGADG ICDS MIL15 ORYX RG2018 VPK2 WEG18 WEG2015 WP-RU20



Ministry of Defence of the Russian Federation, CC BY 4.0

**MOBILITY** Fords 1.3 m; not amphibious. Chassis figures are Tigr-M, from WEG.

Weight, combat 7,800 kg

L × W × H 5.7 × 2.3 × 2.3 m

Speed, maximum road 140 km/h

Road range 1,000 km cruising

Jamming frequency band 0.02-2.7 GHz Radio emitters incl. mobile communications; 3G table copied from an Airbase.ru forum image, 2014

Signal analysis band 0.1 MHz-18 GHz Radio reconnaissance; direction finding 0.02-18 GHz

Simultaneous suppression 12 emission sources Quasi-simultaneous; VDV EW chief, Nov 2012, via forum, claim

Direction-finding accuracy 3 deg roof antennas, 2 deg mast antennas

Crew 2 [driver, operator] claim

**SEEN WITH** GAZ Tigr-M [1 per vehicle]

## Palantin «ПАЛАТИН»

WH-AT

Operational-tactical HF/VHF comms jamming and SIGINT complex; networks other EW systems In service 2019

**OPERATED BY** combined arms army, EW units of a Western MD combined arms army in Voronezh Oblast (army not named)

[R 2019-04 M]

combined arms army, EW troops of a Guards combined arms army, Western MD, Luga range, Leningrad Oblast (Palantin-K, Oct 2019); Feb 2022 exercise in Leningrad Oblast names only the Western MD

[R 2022-02 M]

unit not identified, EW subunits of the Eastern MD in Zabaykalsky Krai (planned deliveries by end 2020) [R 2020-07 L]

**INDICATES** First delivered to EW units of a combined arms army; no source read names the holding subunit or echelon. A Palantin suggests an EW node that may network Moskva, Zhitel or Dzyudoist; in April 2026 one was reported covering air defence assets in Zaporizhzhia.

**RECOGNISE Antenna** Antenna devices carried on masts raised to 15 m on deployment. From photograph: compact multi-element antenna head at the top of a slender mast.

**Mast** Antenna masts of 15 m raised at the site. From photograph: masts stand beside the box body, braced by guy lines while being raised.

**Chassis** Four-axle KamAZ truck. From photograph: large box-body shelter with a side ladder. Generators are started on deployment.

**Complex** Reported as 22 mobile stations spread along the front; each starts generators, raises masts and joins a data network.

**NOT TO BE CONFUSED WITH** Murmansk-BN p. 22 Murmansk-BN uses 32 m telescopic masts holding a large HF antenna net between trucks; Palantin masts are about 15 m.

**Borisoglebsk-2** p. 23 Borisoglebsk-2 is on MT-LB tracked carriers; Palantin is on four-axle KamAZ trucks.

**R-330Zh Zhitel** p. 26 Zhitel (Diabazol) is on the Ural-43203 truck; Palantin is on a four-axle KamAZ.



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**MOBILITY** Truck-mounted on four-axle KamAZ; about 30 min from march to operation including raising 15 m masts.

Time into action about 30 min from the march, including raising 15 m antennas, claim

Chassis four-axle KamAZ truck

Weight INA

Time out of action INA

Jamming range, UAV control links up to 20 km maximum, air target: UAV radio control links, claim also 20 km MIL20

Jamming frontage, radio communications about 1,000 km wide strip claimed, ground target: radio communications and radio reconnaissance; battalion tactical group exercise near Voronezh, Sep 2019, claim

Jamming frequency range 3–30 MHz (HF) and 30–300 MHz (VHF) ground target: radio communications

Jamming targets HF/VHF radio incl. SDR-based, cellular and trunked radio, navigation, UAV control links claim

Reconnaissance function radio-electronic reconnaissance; detects and locates transmitters links Moskva, Zhitel and Dzyudoist EW systems into one network claim

Composition 22 mobile stations Palantin-K, claim

Antenna mast height 15 m claim

Crew INA

**SEEN WITH** Palantin mobile station (four-axle KamAZ) (22 per complex (reported, Palantin-K)); Moskva EW complex; R-330Zh Zhitel; MKTK-1A Dzyudoist

E EW

## R-330Zh Zhitel P-330Ж «ЖукеАб»

WH-AT

Truck-and-trailer jammer and direction finder against satcom, GPS and GSM user equipment In service 2008

**OPERATED BY** EW company, EW company, jamming platoon (radio, GPS, cellular, radio-controlled fuses), jamming squad, 1 (single station on RWW chart; no count stated) [D 2016 M]

**EW brigade**, REB-S (anti-aircraft communications and navigation) battalion [D 2018 M]

**EW company**, EW company of motor rifle or tank division (sources confirm the company exists; Zhitel holding not stated) [D 2018 L]  
**unit not identified**, 25 lost in Ukraine Feb 2022 to Sep 2026 (visually confirmed) [R 2026-09 H]

**EW company**, EW company deployed to Ukraine, 2 in the deployed company [R 2014 L]

**motor rifle brigade** [R 2014 L] seen with 18th motor rifle brigade (Army Recognition claim, undated; not in WEG; AR article dated 2014 per RUSIAD footnote)

**INDICATES** Doctrinally one Zhitel sits in the EW company of a manoeuvre brigade, so a sighting usually points to such a formation nearby. It is also held by the REB-S battalions of district EW brigades, so it can indicate an attached higher-level EW element.

**RECOGNISE** **Antenna** Four active phased-array transmit antennas, one atop each trailer mast. Truck has a separate DF array: ring of about 8 vertical elements.

**Mast** Stowed: four telescopic masts upright at trailer corners, just above the box. Deployed: four guyed masts, raised by hand winch.

**Chassis, truck** Ural-4320-family or KamAZ-43114 6x6 with full-length box body; air-conditioner box at front of body above cab; DF mast at body rear. Houses all operator equipment.

**Chassis, trailer** Two-axle trailer carrying a tall rectangular box with the antennas and a diesel generator; outrigger jacks at the corners.

**Complex** In action the trailer is uncoupled and parked well away from the truck, cable-linked. May work in pairs (master and slave) or under an automated EW command post.

**NOT TO BE CONFUSED WITH** R-934BMV jamming station (Borisoglebsk-2) p. 23 Listed on tracked MT-LB chassis, no trailer; Zhitel is a wheeled 6x6 truck towing a four-mast antenna trailer.  
 RB-301B Borisoglebsk-2 complex (R-330KMM, R-378BMV, R-330BMV, R-934BMV, R-325BMV) p. 23 Group of up to nine tracked MT-LB vehicles; Zhitel is a single wheeled truck plus trailer.  
 R-934UM, R-378UM and other R-330M1P Diabazol stations p. 24 Share the Ural-43203 box-body base. Confirm Zhitel by the separate two-axle trailer with four corner masts; appearance of the others INA.  
 RB-341V Leer-3 (KamAZ box-body truck) p. 27 KamAZ box-body truck working with Orlan-10 UAVs, no antenna trailer; Zhitel has the four-mast trailer and a DF ring mast on the body.

**SEEN WITH** Antenna trailer with four phased-array antennas and diesel generator (1 per station); Second R-330Zh (paired master/slave for synchronous DF) (1 when paired); R-330KMA EW command post (controls up to 20 stations); R-330M1P Diabazol automated jamming complex (with R-934UM, R-378UM, Altaets-AM); Ural-43203 6x6 (1); KamAZ-43114 6x6 (1)

**VARIANTS** R-330Zh on Ural-4320-family chassis Bonneted Ural cab with long nose; spare wheel on rear of box body. None reported. FOI and WEG give Ural-43203 as the standard carrier. R-330Zh on KamAZ-43114 chassis Flat-fronted KamAZ cab-over cab. Seen at Army-2015 and in Ukraine 2022. None reported.



MIL.ru (D. Abramov), CC BY 4.0

**MOBILITY** 6x6 truck, but the towed antenna trailer is fragile; WEG expects it near roads, not cross-country. Masts raised by hand, up to 40 min to deploy.

|                    |                      |                                                 |
|--------------------|----------------------|-------------------------------------------------|
| L × W × H          | 7.36 × 2.55 × 2.71 m | Truck; trailer not included (unclear in source) |
| Road speed         | 85 km/h              | maximum, Truck, maximum road                    |
| Time into action   | 40 min or less       | Full deployment including masts, claim          |
| Time out of action | 40 min or less       | Full march order, claim                         |
| Payload            | 4,500 kg             | maximum, Truck chassis                          |
| Weight, combat     | INA                  |                                                 |
| Road range         | INA                  |                                                 |
| Fording            | INA                  |                                                 |
| Ground pressure    | INA                  |                                                 |

|                                  |                                                                                                          |                                                                                                                |
|----------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Jamming range                    | 25 km                                                                                                    | maximum effective, Ground user equipment (WEG gives no target type; matches manufacturer's ground-user figure) |
| Jamming range, airborne users    | 50 km or more                                                                                            | claimed minimum, User equipment installed on aircraft, claim                                                   |
| Area covered by a pair           | About 40 × 20 km                                                                                         | claimed, Two paired stations, jamming ground communications and navigation                                     |
| Frequency band, intercept and DF | 100-2,000 MHz                                                                                            | claim                                                                                                          |
| Jamming bands                    | 1227.6 MHz; 1575.42 MHz; 1500-1900 MHz                                                                   | GPS L2 and L1, Inmarsat/Iridium and GSM-1800 bands                                                             |
| Targets                          | Inmarsat and Iridium user terminals; GPS receivers; GSM base stations (WEG: GSM 1900; RWW: GSM-900/1800) | claim                                                                                                          |
| Transmitter output               | 6,000 W or more                                                                                          | Total energy potential of the four phased arrays, claim                                                        |
| Simultaneous jamming signals     | 12 or fewer                                                                                              | Quasi-simultaneous, claim                                                                                      |
| DF accuracy                      | 2 degrees or better                                                                                      | RMS bearing error, claim                                                                                       |
| Scan speed                       | 800 MHz/s or more (detection); 400 MHz/s or more (DF)                                                    | claim                                                                                                          |
| Beam width, azimuth              | -90 to 120 deg                                                                                           | WEG labels this azimuth search coverage                                                                        |
| Beam width, elevation            | -15 to +20 deg                                                                                           | WEG labels this elevation search coverage                                                                      |
| Jamming signal types             | Spot, spot-barrage and barrage                                                                           | claim                                                                                                          |
| Power supply                     | 220/380 V 50 Hz from towed diesel generator or mains; 24 V batteries for emergency                       | claim                                                                                                          |

**SOURCES** 3G AR23 ATP CRFS FOI ICDS MIL33 ORYX PRO RUSIAD RWW TOPWAR7 TOPWAR8 VRF8 WEG48 WMC8 WMC18 WMC19 WMC20

## Leer-3 P5-341B «Aeep-3»

WH-AT

UAV-borne cellular (GSM) jamming, phone location and fake-SMS system using Orlan-10 drones In service 2015

**OPERATED BY** EW brigade, Leer-3 platoon, platoon operates the complex; number per brigade INA [D 2024 H]

EW battalion, Leer-3 drone/EW company [R 2019 M]

**unit not identified** [R 2015 M] seen with 4th Military Base, South Ossetia  
**unit not identified**, Leer-3 crews [R 2022-11 M] seen with Western Military District

**motor rifle division**, separate EW battalion [R 2022-01 M] seen with 3rd Motor Rifle Division (Vislenskaya), separate EW battalion, Belgorod oblast

**unit not identified** [R 2020-11 M] seen with Russian peacekeeping contingent, Nagorno-Karabakh

**INDICATES** A Leer-3 control truck points to an EW brigade platoon, an army EW battalion company or a division EW battalion. Expect Orlan-10 flights that jam or imitate cell networks, locate phones for artillery and send SMS within about 6 km of the drone.

**RECOGNISE** Antenna Control truck carries a command-telemetry radio antenna system for the UAVs. From photograph: small drum-shaped radome on a thin vertical pole. Size INA.

**Mast** From photograph: the antenna pole rises from the rear top corner of the box body; no large telescopic mast seen on the exhibited vehicle. Height INA.

**Chassis** KamAZ-5350 6x6 truck, closed box body with operator workstations. From photograph: bonneted cab, chamfered roof edges, side door, small windows, spare wheel on rear wall.

**Complex** Up to three Orlan-10 fixed-wing UAVs, catapult-launched and recovered by parachute; wingspan over 3 m.

**Complex** From photograph: Orlan-10 has a slim fuselage, straight wing and tail fin.

**NOT TO BE CONFUSED WITH** Leer-2 p. 24 Leer-2 is built on a Tigr-M 4x4 armoured car; the Leer-3 control vehicle is a KamAZ-5350 6x6 truck with a box body.

**Orlan-10** Same airframe also flies artillery reconnaissance with UAV companies; the airframe alone does not show the EW role. Look for the Leer-3 control truck or cellular jamming.



Vitaly V. Kuzmin, CC BY-SA 4.0



Orlan-10 UAV, Vitaly V. Kuzmin, CC BY-SA 4.0

**MOBILITY** Control vehicle is a KamAZ-5350 6x6 truck; complex deploys in about 30 minutes. Leer-3 vehicle weight, dimensions and fording not found.

**Road speed** 100 km/h maximum, KamAZ-5350 base truck at maximum permitted mass; not a Leer-3 figure, claim also 120 km/h WP-RU21

**Road range** 1,550 km minimum at reference consumption, KamAZ-5350 base truck, 420 l tanks, at reference fuel consumption; not a Leer-3 figure, claim also 1,000 km WP-RU21

**Time into action** 30 min whole complex, claim

**Weight** INA

**Time out of action** INA

**Jamming range, cellular terminals (ground)** up to 6 km up to, radius of the zone of blocked GSM terminals around the UAV, claim

**Detection range, GSM base stations** 6 km within, cellular base stations, GSM and possibly 3G/4G, measured from the UAV

**Practical range, Orlan-10 from control vehicle** 120 km up to, radius of UAV employment from the control point, claim also over 100 km IZ222 MASH

**Endurance, Orlan-10** 10 h maximum, maximum flight time of the Orlan-10 in the Leer set, claim also about 12 h (half a day) IZ222

**Altitude, Orlan-10** up to 5,000 m up to, operating altitude, claim

**Frequency band, jamming and imitation** 890-915, 935-960, 1710-1785, 1805-1880 MHz GSM 900 and GSM 1800 networks, claim

**Frequency band, UAV control channel** 902-922 MHz claim

**Networks attacked** GSM 900, GSM 1800 3G and 4G capability claimed after a 2017 upgrade, claim also GSM, 3G, 4G IZ20

**Cellular subscribers blocked simultaneously** 2,000 claim

**Cellular operators blocked simultaneously** 3 claim

**UAVs controlled simultaneously** 2 claim

**Crew** 5 whole complex, claim also 4 RG15

**Speed, Orlan-10** 70-130 km/h depends on modification, claim also 90-150 km/h MIL16

**Take-off weight, Orlan-10** 18 kg claim also 14 kg MIL16

**SEEN WITH** Orlan-10 (3 per complex); KamAZ-5350 (1 per complex); UAV catapult launcher (1 per complex); Expendable ground jammers

E EW

Pole-21 P-340PП «Поле-21»

NA

Networked GNSS jamming posts protecting fixed sites from GNSS-guided weapons and UAVs In service 2016

**OPERATED BY** combined arms army, Army EW specialists; first used in an exercise with battalion tactical groups, with R-934BMV and Silok-01 (MoD via AiF) [\(O 2021-04 M\)](#)

**INDICATES** A post on a cell tower or fixed mount suggests a protected fixed site such as a strategic facility or command post. A single R-340RP in the field indicates a GNSS-jamming layer near that position.

**RECOGNISE Antenna** Transmit and receive antennas on cellular communications towers. Small antennas resembling civilian cell antennas, 1 to 3 per post (Ukrainian expert, 2023). Size: INA.

**Mast** Posts attach to existing cell towers or buildings, or to a stationary ground mount. Height: INA.

**Chassis** Also shown as a wheeled mobile unit and as a unit deployed in the field. Vehicle type: INA.

**NOT TO BE CONFUSED WITH** R-330Zh Zhitel p. 26 Zhitel is a vehicle-borne station; Pole-21 posts sit on towers or fixed mounts. FOI notes similarities in function. Focus.ua wrongly calls Pole-21 'Zhitel'.

**MOBILITY** Posts are fixed on masts or mounts. A wheeled mobile unit is also reported. Weights and dimensions: INA.

Power supply 220 V, 50 Hz; 600 W consumption WEG (Pole-21E)

Operating temperature -40 to 50 C Pole-21E per WEG

Weight INA

Jamming range, GNSS receivers (ground or air not stated) up to 25 km effective, WEG (Pole-21E); Russian press gives 25 km per jamming module for the domestic Pole-21

Jamming frequency band 1.176 to 1.602 GHz (GPS, GLONASS, Galileo, BeiDou)

Coverage area of one complex 150 km2 WEG; number of posts not stated

Output power per module 300 to 1,000 W variable max radiation pattern, single frequency; Pole-21E

**SEEN WITH** Jamming posts (up to 100 per complex); jamming modules (1 to 3 per post)

**SOURCES** 3G AIF ATP DPSU FERG FERG2 FOCUS FOI IZRU IZV KPOST TOPWAR6 UKRINF1 WEG25 WP-UK2

Repellent-1 Репеллент-1

WH-AT

Wheeled counter-UAV EW complex: detects and jams small drone control and GNSS links In service 2016

**INDICATES** Sightings so far (Donbas 2018, Kherson 2022) name no holding unit type. Its presence shows a counter-UAV EW asset is deployed in that area, not which formation owns it.

**RECOGNISE Antenna** Sensor and jamming head on a collapsible telescopic mast; upper section holds ELINT and jamming equipment and a panoramic camera. Size: INA.

**Mast** Long collapsible telescopic mast mounted beside the control cabin on the rear of the truck. Height: INA.

**Chassis** Three-axle truck, reported as MAZ-6317; WEG says Kamaz-6317 8x8 and Armada says Kamaz-6350. Chassis maker varies by customer (Wikipedia).

**Complex** Single vehicle with rear control cabin (two operator monitors, small-arms and NBC protection). Mobile and stationary versions are offered. Crew 3.

**MOBILITY** Not amphibious (WEG). Fords 1.0 m; 60% gradient, 0.6 m step (WEG). Chassis type disputed, so figures may differ by chassis.

Weight, combat 23,650 kg WEG figure, Kamaz-6317 chassis

L × W × H 9.4 × 2.7 × 3.35 m

Road speed 85 km/h maximum

Road range 1,000 km maximum

Detection range, UAV control-link signals up to 30 km up to, ground-station command signals; UAV telemetry up to 10 km; target size and altitude not given, claim

Jamming range, small UAV control and telemetry links up to 30 km up to, air target: small UAVs; telemetry and satellite-navigation links; control link up to 10 km, claim

Simultaneous UAV targets up to 12; detect and jam in up to 25 s claimed, brochure claim, claim

**SOURCES** 3G ARMAD8 ARMYINF2 BCAT IZV2 LIGA OSCE RUSIPL UASV UKRINF4 UKRINF22 WEG32 WP-EN33

EW m

## Infauna РБ-531Б «Инфауна»

WH-AT-A

Vehicle-mounted radio jammer and IED jammer, with smoke self-screening In service 2012

**OPERATED BY** unit not identified, dedicated EW units and manoeuvre reconnaissance units [D 2024 L]

EW battalion, EW battalions of the Airborne Forces (VDV) (WEG 2015); RIA 2012 and FOI say VDV EW units [R 2015 M]

unit not identified, Russian military base in Armenia: MoD release title, 27 May 2015 (via FOI); filmed at Alag'yz range, 2019 (state media)

[R 2019-04 M]

**INDICATES** Presence suggests a VDV or other EW subunit escorting a column or protecting a battalion-level position against radio-controlled IEDs and guided munitions. WEG 2015 places it in VDV EW battalions; no source gives counts.

**RECOGNISE** **Antenna** Main antenna in a domed housing on the roof, raised on a telescopic mast when deployed. Size INA. Auxiliary antenna on starboard side.

**Mast** Telescopic mast raises the roof antenna. Mast height INA.

**Chassis** Modified BTR-80 (K1Sh1), 8x8, turret removed. Middle hull larger and higher than BTR-80; four hatches at front, one door at each side centre.

**Complex** Starboard side carries an optoelectronic sensor and smoke grenade launchers (WEG: 6 x 81 mm in 2 x 3). No crew-served weapon.

**NOT TO BE CONFUSED WITH** Kushetka-B command vehicle (BTR-80 chassis) Small turret, and a long rectangular box where Infauna has a camera; a 2018 Horlivka sighting was misidentified as Infauna.

**Standard BTR-80** Infauna has no turret; roof carries an antenna dome and a raised middle hull.



Иллюстрация: Pygwenko, CC BY 4.0

**MOBILITY** Amphibious 8x8 on BTR-80 derivative; operates on the move and escorts columns.

**Weight** **13,614 kg** WEG: combat weight, listed under BTR-80 chassis

**L × W × H** **7.65 × 2.90 × 2.46 m**

**Road speed** **80 km/h** maximum

**Road range** **600 km** maximum

**Jamming range, target not stated** **3 km** disruption range, WEG 'Disruption Range'; target and conditions not given

**Smoke screen distance** **130 m** up to, aerosol against laser and TV-guided weapons

**Jamming band, communications** **20-2020 MHz** frequency range as given by WEG, WEG 2015 and RWW; sub-bands not given

**Output power** **60 W**

**Crew** **3**

**SEEN WITH** K1Sh1 chassis (BTR-80 derivative) (1 per vehicle); Lesochek portable EW system

**SOURCES** 3G AR4 ATP DFRLAB FOI GS8 ICDS RDP2 RG RWW VRF4 WEG16 WEG2015 WP-RU18

## PMP heavy folding pontoon bridge park ПМП

WH-AT

Truck-carried folding pontoon park for 60-t floating bridges and 40-170 t rafts In service 1961

**OPERATED BY** motor rifle division, engineer battalion, pontoon bridge company (also in tank divisions), half set: 16 river and 2 shore links and 1 service vehicle on KrAZ-214; 6 BMK-90/150 boats per company **D 1991 M**  
**engineer-sapper brigade**, pontoon bridge battalion of the army or front engineer brigade (FM 100-2-3 'Engineer Brigade, Army or Front'), one full set: 32 river and 4 shore links on KrAZ-255, 15 BMK-150 boats **D 1991 M**

**pontoon-bridge regiment**, two pontoon bridge battalions (army or front/district level), three full sets: 96 river and 12 shore links on KrAZ-255 **D 1991 M**

**motor rifle brigade**, engineer battalion, pontoon bridge company, one pontoon park; RWW calls it 'PP-61' but its figures (268 m at 60 t; 90- and 120-t bridges) match PP-91, not PMP 6 BMK boats, 7 PTS-2 **D 2016 L**

**unit not identified**, INA; Oryx lists 47 PMP(-M) river links and 3 shore links lost **R 2026-09 M**

**INDICATES** A half set (16 river links, 2 shore links, about 6 boats) matches a division engineer battalion's pontoon company. Over 100 link carriers, or three full sets, matches an army or front pontoon-bridge regiment. Either points to a planned bridged river crossing.

**RECOGNISE Chassis** 6x6 truck carriers: KrAZ-214, later KrAZ-255B. Ural-4320 and KamAZ-53501 carriers also seen. Separate trucks carry river links, shore links and boats.

**Working equipment** Roller bed on the truck; a folding jib and a winch behind the cab recover the link. Truck backs to the water, brakes hard and the link slides off the rollers.

**Stowed** River link: four hinged steel pontoons folded into one tall block filling the truck bed, 6.75 m long. Shore link is shorter, 5.6 m, with a ramp end.

**Stowed, boats** BMK-90/150 tug-motor boats ride on their own truck carriers, 6 per half set and 12 per full set; PMP-M sets have 16.

**Deployed** Link unfolds on hitting the water to a flat deck 6.75 m long, about 8 m wide. Links pin into a continuous steel ribbon, 6.5 m roadway, held in line by boats.

**Deployed, 20-t** For 20-t loads each link is split into half-links, giving a narrower 3.29 m roadway and a longer bridge.

**NOT TO BE CONFUSED WITH** PP-91 / PP-2005 (PP-91M) pontoon parks Larger links (7.36 x 8.28 m) on KrAZ-260G/Ural-53236 or, for PP-2005, KamAZ-63501 8x8. Truck alone does not settle it: VPK lists Ural-4320-30 for PP-91 too.

**TMM-3 truck-launched scissors bridge** p. 31 Same KrAZ-255B chassis, but carries a folded 10.5 m scissors span with a launching girder and folding trestle legs, not a pontoon block.

**PMM-2 self-propelled ferry** p. 31 Tracked, boat-bowed hull with seven road wheels and two box pontoons stacked on top; one vehicle forms a ferry.

**VARIANTS** PMP-M Same folding river links; shore link has a streamlined deck; hydrodynamic shield kit. Carried on KrAZ-255B; KamAZ-53501 carriers also recorded. Bridge usable in currents up to 3 m/s against 2 m/s for PMP; set lists 16 boats instead of 12. PMP on Ural-4320 carriers PMP links on Ural-4320 6x6 trucks in place of KrAZ-255B. None known; same links. PP-91 Bigger river link, 7.36 m long and 8.28 m wide, on KrAZ-260G (Ural-53236) carriers; BMK-225 boats. 60-, 90- and 120-t bridges and double-width roadway; lengths per set differ between sources [see open questions]. PP-2005 (PP-91M, PP-2005M) PP-91-type links on KamAZ-63501 8x8 carriers. Boats: BMK-MT with PP-2005M (VPK); BMK-3 seen with PP-91M (AIF, 2025). Bridges to 120 t; rafts 90-360 t (PP-91) or 90/190/380 t (PP-2005M per VPK). In a Group Centre pontoon-bridge company, 2025-02.



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**MOBILITY** Carriers must back to the water's edge to launch and recover links. PMP bridge limited to currents of 2 m/s, PMP-M to 3 m/s.

River link weight **6,790 kg** one river link  
 Shore link weight **7,252 kg** one shore (ramp) link  
 Time into action, bridge **7 m of bridge per minute** ideal conditions

Time out of action, per link **a few minutes** retrieval of one link by winch and jib

River link, unfolded **6.75 m long, 8 m wide**

Weight, carrier with link **INA**

Road speed **INA** maximum

Road range **INA** cruising

Fording **INA**

Bridge, full set, 60-t **227 m** maximum, 32 river + 4 shore links, 6.5 m roadway

Bridge, full set, 20-t **382 m** maximum, split half-links, 3.29 m roadway

Maximum current **2 m/s** PMP

Bridge, half set, 60-t **119 m** 16 river + 2 shore links

Bridge, half set, 20-t **191 m** 16 river + 2 shore links, split

Load capacity, bridge **60 t** full-width bridge; 20 t on split-link bridge

Roadway width **6.5 m** 60-t bridge

Rafts **40, 60, 80, 110, 170 t** 2, 3, 4, 5+1 shore, 8+1 shore links

Raft assembly, 40-t **8 min** 2 river links, working party 6

Tank speed on bridge **up to 30 km/h** tanks on 60-t bridge

Crew per carrier **3** WEG figure, in the block copied from the PMM-2 page; WEG 2015 KrAZ-255B page gives 3 in cab

Working party, full bridge **approx. 65** full bridge set

**SEEN WITH** River link on KrAZ-214 or KrAZ-255B (or Ural-4320, KamAZ-53501) carrier (32 per full set, 16 per half set); Shore (ramp) link on carrier truck (4 per full set, 2 per half set); Roadway mats (vystilki) on trucks (2 per full set); BMK-90/150 tug-motor boats on truck carriers (12 per full set (16 for PMP-M), 6 per half set); PTS-2 (or K-61) tracked amphibious transporter (7 in a brigade engineer battalion; 28 in an army engineer brigade (1991)); GSP tracked self-propelled ferry (12 half-ferries per division engineer battalion; 24 in an army engineer brigade (1991)); PMM-2 self-propelled amphibious ferry (replacing GSP); TMM-3 truck-launched bridge (4 per engineer battalion)

## TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge

WH-AT

Brigding: TMM-3 truck scissors bridge; MTU-72 tank-chassis bridge layer In service 1970

**OPERATED BY** engineer-sapper battalion, brigade engineer battalion; sets almost always attached to the movement support detachment (OOD), 4 TMM-3 truck-launched bridge sets per brigade engineer battalion, each laying a 10 m span; together a 42 m bridge

**D** 2024-02 **M**

engineer-sapper regiment, engineer and bridge subunit (press-release wording) **O** 2020-11 **M**

**INDICATES** TMM-3 sets are held by the brigade engineer battalion and almost always attached to its movement support detachment, which bridges minor gaps ahead of the main body. Four trucks together suggest a route or a crossing of up to 42 m being prepared.

**RECOGNISE** **Chassis** TMM-3: KrAZ-255B 6x6 soft-skin truck; spare wheel on a steel frame above the cab. MTU-72: turretless T-72 hull: six road wheels, three return rollers, exhaust left rear.

**Working equipment** TMM-3: hydraulic slewing mast with forked end and winch lifts the folded span. MTU-72: single bridge launched from the hull, hydraulic front blade. Cantilever type.

**Stowed** TMM-3: each 10.5 m span folds to 5.2 m; spans ride on the truck bed. MTU-72: bridge folded 9.42 m long on a hull 11.64 m long, 3.38 m high.

**NOT TO BE CONFUSED WITH** MTU-90 T-90 chassis; RWW gives a 25 m bridge, 50 t class, versus 20 m on MTU-72; held in tank brigades per RWW.

MTU-20 and MT-55A T-54/55-family chassis (MT-55A is on T-55) rather than T-72 chassis.



TMM-3. Mil.ru (Ministry of Defence of the Russian Federation), CC BY 4.0



MTU-72. Vitaly V. Kuzmin, CC BY-SA 4.0

**MOBILITY** TMM-3 is an unarmoured 6x6 truck (WH-AT). MTU-72 is tracked (TR), armoured. TMM-3 span: gap to 40 m, 3 m deep.

Weight, TMM-3 truck with bridge

19,500 kg

Weight, MTU-72

40,000 kg Combat weight

L × W × H

9.3 × 3.2 × 3.15 m

Road speed, TMM-3

71 km/h

Obstacle, TMM-3 bridge set

up to 40 m wide, 3 m deep Set of four spans; ATP gives 42 m total length

Load class, TMM-3

60 t Wheeled and tracked loads; tracked vehicles at 15 km/h on bridge

Bridge geometry, TMM-3

4 spans of 10.5 m; roadway 3.8 m

Spans can be connected to 10.5, 21, 31.5 or 42 m (RWW); ATP gives 10 m per set and 42 m together

Crew, TMM-3

1 + 1 in cab (driver, passenger) per truck

**SOURCES** ARMYINF4 ATP IIS25 JANESRG JFRC KOLESA MIL27 ORYX RWW TRENCH WEG39 WEG2015 WP-EN42 WP-EN43 WP-RU55 WP-RU56 WSN3

## PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry

TR-A

Amphibious carrier for cargo and vehicles (PTS-4); self-propelled river ferry (PMM-2) In service 2007

**OPERATED BY** motor rifle brigade, engineer battalion, 7 PTS-2 per engineer battalion (RWW names PTS-2, not PTS-4) **D** 2016 **M**

tank brigade, engineer battalion, 7 PTS-2 per engineer battalion (RWW names PTS-2, not PTS-4) **D** 2016 **M**

unit not identified, INA (PMM-2: held by Russian Army; no count given)

**R** 2024 **L**

**INDICATES** PTS-2 amphibians are a brigade engineer battalion asset in RWW, so PTS-type carriers suggest engineer river-crossing support. IIS25 lists PMM-2 without unit type; no unit is tied to PTS-4 here.

**RECOGNISE** **Chassis** PTS-4: tracked amphibious hull with forward armoured cab; T-80 tracks and torsion bars, T-72 engine and gearbox. Road wheel count INA.

**Working equipment** PTS-4: open rear cargo area, platform 8.28 × 3.3 m, one remote-controlled 12.7 mm machine gun. PMM-2: boat-like bow, PTS-2-style cab.

**Stowed** PMM-2: MT-T-type chassis, seven road wheels and four return rollers per WEG; two large box pontoons carried one above the other on the hull.

**Deployed** PMM-2: pontoons hinge out to each side, with wave deflectors and ramps; deck 10 m long, 4.2 m wide; two 5 m loading ramps; two propellers lowered aft.

**NOT TO BE CONFUSED WITH** PTS-2 / PTS-3 Older, on MT-T chassis with T-64 components; PTS-3 has higher sides. PTS-4 uses T-80 and T-72 units and has a more heavily armoured cab.

**PMM-2 vs PTS-2** PMM-2 has similar cab and bow but carries two stacked box pontoons on top of the hull.



Aleksei Kumaev (Aleksy Kitayev), CC BY-SA 4.0

**MOBILITY** Amphibious tracked. PMM-2 fording 1.2 m, vertical step 2.8 m (WEG15; implausible, see open questions). PTS-4 vertical step 0.65 m, trench 2.5 m (WEG).

Weight, PTS-4

33.1 t empty, per WEG

Weight, PMM-2

36 t combat weight, per WEG15; ru Wikipedia: ferry mass 36 t

L × W × H

8.3 × 3.3 × 3.2 m WEG; ru Wikipedia gives 8.28 m for the cargo platform only

Road speed, PTS-4

60 km/h maximum

Payload on land, PTS-4

12 t

Payload on water, PTS-4

18 t

Payload, PMM-2 single ferry

42.5 t WEG; RUWM gives 42.5 t for the PMM-2M

Payload, PMM-2 two joined

85 t WEG gives 85 t for two PMM-2; RUWM gives 85 t for two PMM-2M, assembly 8 min

**SOURCES** AR12 AR13 IIS25 RWW WEG28 WEG2015 WP-EN29 WP-RU36 WP-RU37 WP-RU38 WWW2 WWW12

## IMR-2 / IMR-2M / IMR-3M combat engineer vehicle IMP-2

TR

Tank-based heavy engineer vehicle for obstacle and minefield clearing In service 1980

**OPERATED BY** engineer-sapper battalion, mine clearing platoon (brigade engineer battalion), 2 IMR-3M per platoon **D 2016 M** engineer-sapper brigade, 2 delivered Nov 2018, 1 more due first half 2019 (Central MD press service, state claim) **R 2018-11 L** engineer-sapper brigade, INA; listed with UR-77, BMR-3 and GMZ-3 (English Wikipedia, citing a 2023 Russian regional news item) **R 2023-06 L**

**INDICATES** An IMR-3M in a column suggests the mine clearing platoon of a brigade engineer battalion; its elements may join the movement support detachment (OOD) ahead of the main body to clear routes and minefield lanes. Separate engineer brigades also hold it.

**RECOGNISE** **Chassis** Tank chassis: T-72 for IMR-2 (six road wheels a side, torsion bar); T-90 for IMR-3M. No gun; superstructure carries a crane.

**Working equipment** Wide dozer blade at the front; telescopic boom with grab or bucket on a rotating superstructure where a turret would be.

**Stowed** Blade raised onto the hull roof; boom drawn back and turned. A KMT track-width mine plough may be fitted.

**Deployed** Blade works as a dozer, V-blade or grader; boom traverses 360 deg. IMR-3M has a remote 12.7 mm gun mount; IMR-2 a machine gun on the operator's cab.

**NOT TO BE CONFUSED WITH** BAT-2 Also a tracked bulldozer with adjustable blade and boom crane; RWW gives 7.3 m boom reach and 700 hp against about 8.4 m and 840 hp for IMR.

**MTU-72** p. 31 Also a T-72 hull with turret removed; carries a bridge-launching system, not a dozer blade and crane.



Kirill Borisenko, CC BY-SA 4.0

**MOBILITY** Fords 1.2 m unprepared; 5 m with snorkel equipment (Russian Wikipedia). Not amphibious. Heavy: 44 to 50 t.

|                   |                       |                                  |
|-------------------|-----------------------|----------------------------------|
| Weight, IMR-2     | 44.3 t                | combat weight                    |
| Weight, IMR-3M    | 49.5 t                | combat weight with KMT-R3 plough |
| L × W × H         | 9.55 × 3.735 × 3.68 m |                                  |
| Road speed, IMR-2 | 50 km/h               | maximum                          |

|                       |                                                |
|-----------------------|------------------------------------------------|
| Crew                  | 2 (commander-operator, driver-mechanic)        |
| Boom lifting capacity | 2 t                                            |
| Boom reach            | 8.4 m IMR-2M                                   |
| Dozer capacity        | 200 to 250 m <sup>3</sup> /h IMR-2 earthmoving |
| Armament, IMR-3M      | 12.7 mm NSVT or Kord remote mount              |
| Armament, IMR-2       | 7.62 mm PKT original IMR-2                     |

**SEEN WITH** UR-77 mine clearing vehicle (2 per mine clearing platoon); BAT-2 combat engineer vehicle (4 per brigade engineer battalion)

**VARIANTS** IMR T-54/55 tank chassis; articulating dozer blade and telescoping crane.

**SOURCES** ATP IIS25 RWW TASS2 TSAMTO WEG15 WEG2015 WP-EN14 WP-EN15 WP-RU16 WP-RU17

## UR-77 Meteorit YP-77 «Memeopum»

NATO: M1979

TRA

Tracked mine-clearing vehicle firing rocket-towed explosive line charges In service 1978

**OPERATED BY** engineer-sapper battalion, mine clearing platoon (organic engineer battalion of a maneuver brigade), 2 per platoon **D 2016 M**

**engineer-sapper brigade** **R 2023 L** seen with 1st Guards Engineer Brigade (Murm, MUN 11105)

**INDICATES** Doctrine places two per engineer mine clearing platoon, so one suggests engineers preparing a breach or route clearing. In Ukraine it has also been fired as improvised short-range artillery.

**RECOGNISE** **Chassis** Tracked hull derived from the 2S1 Gvozdika, itself built on MT-LB parts; driver and engine compartment front left on the 2S1.

**Working equipment** Turret-like superstructure with launch ramps at top rear of hull. RWW gives two launch ramps, TWZ two charges; JANESRG says three rockets.

**Stowed** Charges carried on board: hull holds two launchers and compartments for the line charges (UZP-77 or UZ-67).

**Deployed** Rocket tows a hose about 7 cm thick and 93 m long (UZP-77) across the minefield, then it is detonated; lane about 6 m wide.

**NOT TO BE CONFUSED WITH** 2S1 Gvozdika Same chassis; the 2S1 has a fully enclosed turret with a 122 mm howitzer, the UR-77 a launcher superstructure.

**UR-67** Older predecessor on a BTR-50PK chassis.

**UR-07M** BMP-3 chassis, not 2S1; ordered in small numbers, not in series production as of 2024 (GSEC). Named Razgon (GSEC) or Peresortirovka (RUWIKI).

**UR-83P** Same line-charge role but not mounted on a vehicle chassis: towed or trailer-based.



Mil.ru / Dmitry Golovko, CC BY 4.0

**MOBILITY** Tracked, 2S1-derived, amphibious: WEG gives 5 km/h swimming, GlobalSecurity 4.5 km/h. Fording depth and ground pressure INA. Reload after firing takes 30 to 40 minutes.

|                |                   |
|----------------|-------------------|
| Weight, combat | 15.5 t            |
| L × W × H      | 8.4 × 2.8 × 3.1 m |
| Road speed     | 60 km/h           |
| Road range     | 500 km            |

|                           |              |                                                                            |
|---------------------------|--------------|----------------------------------------------------------------------------|
| Range, UZP-77 line charge | 200 to 500 m | not stated, charge launch distance, UZ-77; UZ-67 charge 200 to 350 m (WEG) |
|---------------------------|--------------|----------------------------------------------------------------------------|

|                             |            |                                                |
|-----------------------------|------------|------------------------------------------------|
| Crew                        | 2          |                                                |
| Cleared lane length, UZP-77 | 80 to 90 m | anti-tank minefield; UZ-67 charge 75 to 80 m   |
| Cleared lane width          | up to 6 m  | one line charge                                |
| Line charge length          | 93 m       | UZP-77; UZ-67 is 83 m                          |
| Cleared lane length, ZRSchh | 100 m      | lane 6 m wide; charge in 6 m sections of 41 kg |

**SEEN WITH** UZP-77 / UZ-67 line charge (2 per vehicle)

**SOURCES** ESPR GS18 IIS25 JANESRG MILUA2 RBTH RWW TWZ8 TWZ9 WEG44 WEG2015 WP-EN15 WP-EN47 WP-RU60 WWW14

## ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer ИСДМ «Земледелец»

WH-AT

Remote minelaying by 122 mm rockets (ISDM) and tracked mechanical minelaying (GMZ-3) In service 2020

**OPERATED BY** engineer-sapper battalion, obstacle platoon (brigade engineer battalion per RWW; division engineer battalion per WEG), 3

GMZ-3 per obstacle platoon **D 2016 M**

engineer-sapper brigade **D 2024 L**

engineer-sapper regiment **D 2024 L**

**INDICATES** An ISDM points to an engineer brigade or army engineer regiment remote-mining subunit preparing minefields 5–15 km out. A group of three GMZ-3 suggests a brigade engineer obstacle platoon, likely a mobile obstacle detachment.

**RECOGNISE Chassis** ISDM: four-axle KamAZ-6560 8x8 truck; cab armoured per Za Rulem (WEG: armour none). The transloader uses the same chassis.

**Chassis** GMZ-3: tracked, on the SA-4 Ganef chassis; seven road wheels, front drive sprocket, rear idler, four return rollers. Driver front left, engine to his right.

**Working equipment** ISDM: launcher on rear deck carrying two box containers, each of 25 tubes in five rows of five. Transloader carries a small crane (blog source).

**Working equipment** GMZ-3: operator compartment at rear; mines leave through ports in the rear hull plate onto a conveyor and plough. PKT MG; three 81 mm smoke launchers each side.

**NOT TO BE CONFUSED WITH** BM-21 Grad p. 4 Ural 6x6 truck with 40 open tubes; ISDM is an 8x8 KamAZ with two 25-tube box containers. UMZ ZiL-131 6x6 truck with six firing modules of 30 tubes (180 tubes) scattering mines 30–60 m from the truck.



Mil.ru, CC BY 4.0

**MOBILITY** ISDM: 8x8 truck, not amphibious, fords 1.8 m per WEG. GMZ-3: tracked, not amphibious, 28.5 t; lays surface mines at up to 16 km/h, buries at 6 km/h.

Weight, GMZ-3 **28.5 t** combat  
L × W × H **10.3 × 2.5 × 3.27 m**

Road speed, ISDM **90 km/h** maximum, KamAZ-6560 chassis

Road speed, GMZ-3 **60 km/h** maximum

Range, ISDM mine rocket **15 km** maximum, cassette rocket with AP or AT mines, claim

Minimum range, ISDM mine rocket **5 km** cassette rocket with AP or AT mines, claim

Laying speed, GMZ-3 **16 km/h surface; 6 km/h buried** maximum

Range, 762 mm PKT (GMZ-3) **2,000 m** max effective, day

Rocket calibre, ISDM **122 mm** claim

**SOURCES** ARMINF ATP IISS25 IZV222 JANES2 M360 MPER2 ORYX OVC RWW TI2 WEG47 WEG2015 WP-EN51 WP-EN52 WWW15 ZR16 ZR24

## R-149 command and staff vehicles P-149

WH-AT-A

Tactical command and staff vehicles with HF/VHF radios and data terminals (ESU TZ) In service 1990

**OPERATED BY** motor rifle regiment, R-149MA3  2022-09 M

command brigade, R-149AKSh-1  2022-03 L

motor rifle division, R-149AKSh-1  2022-03 L

**INDICATES** An R-149MA1 or R-149AKSh-1 suggests a brigade, regiment, division or army control-brigade command post nearby. An R-149MA3 is meant for company to squad command and was lost with motor rifle and tank regiments.

**RECOGNISE Cab** R-149AKSh-1: KamAZ-5350 6x6 truck cab. BTR-based R-149MA1, MA3 and BMR: no cab; commander and driver sit at front of an 8x8 armoured hull.

**Body** AKSh-1: van body split into electronics compartment (3 workstations) and officer compartment (5). MA1 and BMR: K1Sh1 hull, larger than BTR-80, unarmed turret.

**Axles** AKSh-1: three axles. BTR-based: four large tyred wheels a side; hatch/door between 2nd and 3rd wheels, upper part opens to right, lower folds down (BTR-80 base).

**Details** R-149MA3: WEG lists a 14.5 mm KPVT turret (BTR-80 sheet text); one lost MA3 was turretless. R-149BMR carries an ASH-4 telescopic mast, AZI NVIS HF and discone antennas.

**NOT TO BE CONFUSED WITH** BTR-80 APC R-149MA1 and BMR use the K1Sh1 hull, larger than BTR-80 with an unarmed turret; WEG lists the MA3 with the standard armed turret.

**BMM-80 ambulance and 1V152 on K1Sh1** p. 35 WEG 2015: the Kuchetka-B/R-149BMR 'shares the same configuration as BMM medical vehicles and the 1V152/Kapustnik-B ACRV'.



Ministry of Defence of the Russian Federation, CC BY 4.0

**MOBILITY** BTR-80-based versions swim (WEG). R-149AKSh-1 on KamAZ-5350 is not amphibious; fords 1.75 m (WEG).

|                                |                     |                |
|--------------------------------|---------------------|----------------|
| Weight, R-149MA1               | 15,000–15,050 kg    | maximum, claim |
| Weight, R-149BMR               | 13.6 t              | combat         |
| L × W × H                      | 7.65 × 2.9 × 2.41 m |                |
| Road speed, BTR-based versions | 80 km/h             | maximum        |

|                                            |        |                                                                          |
|--------------------------------------------|--------|--------------------------------------------------------------------------|
| Range, VHF R-168-100UE-2, R-149MA1         | 25 km  | maximum, on the move, vehicle antenna, claim                             |
| Range, VHF R-168-100UE-2 on mast, R-149MA1 | 55 km  | maximum, halted, mast antenna, claim                                     |
| Range, HF R-168-100KBE, R-149MA1           | 350 km | maximum, halted, mast antenna; 50 km (20 km moving) on ASH-4 whip, claim |
| Range, HF air-ground Prima, R-149MA1       | 450 km | maximum, HF to aircraft, claim                                           |
| Range, VHF air-ground Prima, R-149MA1      | 120 km | maximum, VHF to aircraft, claim                                          |

**SOURCES** ACONF2 AR14 AR15 ARMYINF ATP ELSIG GAGA JANESRG KI2 MIL24 ORYX RUSARMY3 UNIAN3 WEG31 WEG2015 WP-EN32 WP-RU46 WP-RU47 WSPOT3 WSPOT4 WSPOT5 WSPOT6 WSPOT7 WSPOT8 WSPOT9 WSPOT10

## R-145BM Chayka command and staff vehicle P-1455M «Чаўка»

WH-AT-A

Turretless radio command and staff vehicle for mobile command posts In service 1960

**OPERATED BY** anti-tank battalion, ACV listed as 'BTR-60PU/BRDM-2'; battery placement unclear in OCR, 5 per battalion (BTR-60PU or BRDM-2)  1991 L

artillery brigade, 1 per brigade  1994 L

unit not identified, RKhBZ (NBC protection) troops; Wikipedia names the vehicle BTR-60PU/PUM and equates BTR-60PU with R-145BM

 2022 L

**INDICATES** Built for radio links at mobile command posts, so it points to a command post or communications node. A Soviet-era source places it at tank, motor rifle and artillery regiment and formation HQs; the current echelon is not fixed.

**RECOGNISE Cab** BTR-60 hull (BTR-60PA base) with no turret; two compartments, command section forward and radio section aft.

**Body** Collapsible frame antenna on left side of roof; cylindrical box (TZK) on the roof; characteristic 'horns' on left and right upper edges of front armour.

**Axles** 8x8 wheeled, amphibious, individual torsion-bar suspension.

**Details** Hinged antenna leg supports on side armour upper edge, mid-length; 10 m telescopic mast and AB-1-P/30 roof generator (Jane's: BTR-60PU has roof generator, 10 m antenna).

**NOT TO BE CONFUSED WITH** BTR-60PU-12 (9S482) air defence command vehicle p. 37 Telescopic mast on right of front roof, generator on the turret-ring blanking plate, large stowage box on right of hull; turretless like R-145BM.

BTR-60 1V18/1V19 Unarmed turret with observation devices; R-145BM has no turret.



VargaA, CC BY-SA 4.0

**MOBILITY** Amphibious 8x8, light armour; swims (no fording depth given). Petrol engines.

|                |                      |
|----------------|----------------------|
| Weight, combat | 10.3 t               |
| L × W × H      | 7.56 × 2.83 × 2.31 m |
| Road speed     | 80 km/h              |
| Road range     | 500 km               |

|                              |                                                |                                                            |
|------------------------------|------------------------------------------------|------------------------------------------------------------|
| Range, R-130M zenith antenna | 75 km                                          | HF, halted or moving, symmetrical zenith-radiation antenna |
| Range, R-130M dipole         | 350 km                                         | HF, halted only, symmetrical dipole                        |
| Range, R-123MT 4 m whip      | 20 km                                          | VHF, halted or moving                                      |
| Range, R-111 3.4 m whip      | 35 km                                          | VHF, halted or moving                                      |
| Crew                         | 2                                              |                                                            |
| Radio fit                    | 2 × R-111, R-123MT, R-130M, R-012M call device |                                                            |

**VARIANTS** BTR-60PUM Not stated in source. Russian late-2000s refit of BTR-60PU with new engines adapted from the BTR-80 series.

**SOURCES** ACONF ARINF FM32 IIS25 JANESRG ORYX RUSARMY2 UNIAN2 WEG30 WEG2015 WP-EN30 WP-EN31 WP-EN32 WP-RU43 WP-RU44 WP-RU45

## BMP-1KSh command and staff vehicle БМП-1КШ «Помок-2»

NATO: M1978

TR

Command and staff vehicle for regimental and battalion commanders In service 1972

**OPERATED BY** motor rifle regiment, regimental command post, 1 in regimental HQ of a BMP-equipped motor rifle regiment (FM 100-2-3 table, OCR scrambled) [D 1991 M]

**tank regiment**, regimental command post, INA (FM 100-2-3 table reads 0 for the tank regiment HQ) [D 1990 M]

**motor rifle regiment**, 501st signal company [O 1986 L]

**INDICATES** A BMP-1KSh marks a regimental command post; one source also gives one per battalion. Sources describe it for motor rifle and tank regiments. Oryx records 15 Russian losses 2022-26 but names no unit.

**RECOGNISE Hull** BMP-1 hull: pointed nose, almost horizontal ribbed glacis, driver front left, engine louvres roof right, two bulged rear doors. KSh keeps only two roof hatches.

**Armament** No gun or missile rail; turret fixed; one 7.62 mm PK fired from the rear-door port. Most firing ports blocked. Tube case for mast parts on right rear.

**Turret** Fixed turret cap carries the telescopic antenna mast (10 m per WEG and VALKA) in place of the gun; only two roof hatches behind the turret.

**Roof and rear** Box-shaped generator set centre rear of hull roof; four elevatable whip antennas at rear, two each side.

**Suspension** BMP-1 running gear: six road wheels each side, front sprocket, rear idler, three return rollers, skirts over upper track.

**NOT TO BE CONFUSED WITH** BMP-1K Keeps the 73 mm gun and rotating turret; K1/K2 have a semi-telescopic antenna, K3 a 10 m mast (WEG). KSh has a fixed turret with the mast in place of the gun.

**PRP-3 artillery reconnaissance vehicle** Enlarged two-man turret set to the rear with a folding rectangular radar antenna.

**SOURCES** FM23 GS7 IIS25 JANESRG ORYX VALKA2 VALKACZ WEG8 WEG2015 WP-RU11 WP-RU12



Vitaly V. Kuzmin, CC BY-SA 4.0

**MOBILITY** Amphibious, swims. Radios run on the move from an engine-driven generator; a portable generator set is used at halts (RUWK2).

|                |                      |
|----------------|----------------------|
| Weight, combat | 13 t                 |
| L × W × H      | 6.74 × 2.94 × 2.15 m |
| Road speed     | 65 km/h              |
| Road range     | 550 km               |
| Swimming speed | 7 km/h               |
| Fording        | Amphibious, swims    |

|          |                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------------------------|
| Crew     | 3 crew plus 4 staff officers                                                                                    |
| Armament | 1 x 7.62 mm PK, 2,000 rounds carried machine gun, fired through the rear-door port (RUWK2); WEG lists it as PKT |

|                     |                                          |
|---------------------|------------------------------------------|
| Antenna mast height | 10 m AMU Hawkeye telescopic mast, raised |
| Radios              | 2 x R-111, R-123MT, R-130M               |

|            |                                    |
|------------|------------------------------------|
| Navigation | TNA-3 gyroscopic navigation device |
|------------|------------------------------------|

**VARIANTS** BMP-1KShM Extra power sources on top of the troop compartment (VALKA). MP-31 INA.

## 1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) 1В12 «Машина-С»

NATO: M1974

TR-A

Tracked command, observation and fire direction vehicles, SP howitzer battalion In service 1972

**OPERATED BY** self-propelled artillery battalion, each battery: one 1V13 (battery FDC) and one 1V14 (battery COP); battalion HQ: one 1V15 (battalion COP) and one 1V16 (battalion FDC), 8 per battalion: 3 1V13, 3 1V14, 1 1V15, 1 1V16 [D 2016 H]

self-propelled artillery regiment, howitzer battalion with 2S19M1, at least 3 1V13 and 3 1V14 seen with one battalion [R 2021-09 L]

**INDICATES** A 1V13 marks a battery firing position, where it stands behind the guns. A 1V14 or 1V15 marks a battery or battalion command observation post, usually with the supported manoeuvre unit. A full set of eight serves one SP howitzer battalion.

**RECOGNISE Hull** High box-like MT-LBu hull with steep glacis; straight vertical rear with a single door. Three or four antennas on the hull roof.

**Armament** 1V13 and 1V16: 12.7 mm DShKMT machine gun on the turret. 1V14 and 1V15: 7.62 mm PKMB machine gun on a mount on the turret roof.

**Turret** Flat, round rotating turret shaped as a truncated cone on the rear half of the hull. On 1V14 and 1V15 it carries a laser rangefinder and optical sights.

**Suspension** Seven road wheels each side, no return rollers. The MT-LB it derives from has one road wheel fewer.

**NOT TO BE CONFUSED WITH** MT-LB Six road wheels, lower hull and a small machine-gun turret at the front over the driver's compartment instead of a turret at the rear.

**ARK-1M Rys p. 19** Same MT-LBu chassis but carries a parabolic radar antenna.



Simon Q from United Kingdom, CC BY 2.0

**MOBILITY** Amphibious, propelled by its tracks in water.

|            |                                     |
|------------|-------------------------------------|
| Weight     | 14.8 t 1V13 and 1V14, combat weight |
| L × W × H  | 7.475 × 2.85 × 2.425 m hull         |
| Road speed | 62 km/h maximum, claim              |
| Road range | 500 km maximum, claim               |

|                                              |                             |
|----------------------------------------------|-----------------------------|
| Detection range, 1V14 optical reconnaissance | up to 10,000 m maximum, day |
|----------------------------------------------|-----------------------------|

|                                                      |                              |
|------------------------------------------------------|------------------------------|
| Detection range, 1V14 optical reconnaissance (night) | up to 2,500 m maximum, night |
|------------------------------------------------------|------------------------------|

|                                         |                                                               |
|-----------------------------------------|---------------------------------------------------------------|
| Range, laser target illumination (1V14) | up to 7,000 m maximum, for laser-guided (corrected) munitions |
| Radio range                             | 20-25 km stated, 1V13 and 1V14, vehicle radios                |
| Radio range, R-111                      | 220-250 km stated, 1V15 and 1V16, vehicle radio               |

**SOURCES** AG1 ARMF13 ARMF14 ARMF15 ARMF16 FAS1 JANESRG JREW02 MILINF ORYX RWW VRF1 WEG2015 WP-RU1 not in WEG

## PRP-4A Argus / PRP-4M Deytery ПРП-4А «Арзус»

TR-A

Mobile artillery reconnaissance post on BMP-1 chassis (target location, fire adjustment) In service 1988

**OPERATED BY** motor rifle brigade, fire control and artillery reconnaissance battery, artillery spotting platoon **D 2016 M**  
 tank brigade, fire control and artillery reconnaissance battery, artillery spotting platoon **D 2016 L**  
 self-propelled artillery battalion, battalion fire control platoon, artillery spotter squad **D 2016 L**

**INDICATES** Doctrinally a brigade- or regiment-level artillery reconnaissance asset feeding a fire direction centre. In defence it sits behind first-echelon companies; in offence it moves with forward reconnaissance detachments up to 20 km ahead of the main body.

**RECOGNISE Hull** BMP-1 infantry fighting vehicle chassis, amphibious; 4 crew.

**Armament** One 7.62 mm PKTM machine gun and four smoke dischargers only. PRP-4M is reported with a fake gun mantlet and offset barrel, resembling a BMP-2.

**Turret** PRP-4/4M/4MU: retractable radar; 4M antenna mount on roof behind radar hatch; 4MU left optical flap hinged at top. Argus: WEG says must-mounted radar.

**Turret sides** PRP-4 family: extra fairing on right side of turret (GlobalSecurity); protected mountings either side of turret house optical devices (Wikipedia).

**NOT TO BE CONFUSED WITH** BMP-2 PRP-4M with a fake gun mantlet has no real 30 mm gun, only a 7.62 mm machine gun.

**PRP-3 Val** Earlier post; carries a 90 mm 2P130-1 launcher, which PRP-4 removed.

**PRP-4 Nard** Earlier base model (1V121, in service 1984); radar 1RL133-1 'Tall Mike' (Wikipedia, WEG 2015) vs WEG's '1L120-1' on Argus. No visible mark in a text source.

**SOURCES** AR11 ATP GS7 JANSERG JREW02 MIL23 ORYX RWW WEG27 WEG2015 WP-EN28 WP-RU35



Vitaly V. Kuzmin, CC BY-SA 4.0

**MOBILITY** Amphibious on BMP-1 chassis; swim speed 7 km/h (WEG). Armour reported to stop 12.7 mm rounds all round (WEG).

Weight, combat **13.8 t**  
 L × W × H **6.74 × 2.94 × 2.15 m**  
 Road speed **65 km/h** maximum, paved road  
 Road range **550 km** paved road

Detection range, radar (moving tank-type target) **12 km** up to, moving tank-type target

Detection range, radar (person) **7 km** up to, single soldier

Detection range, stationary tank, engine off (sensor not stated) **2 km** no less than, stationary tank-type target, engine not running; ARMYR and RWW name no sensor, WEG lists it under radar, claim

Detection range, optical-electronic (tank-type target, day) **8 km** up to, tank-type target, daytime

## Polyana-D4 9C52 «Поляна-Д4»

WH-AT

Automated command post of an Army air defence missile brigade (S-300V, Buk, Tor) In service 1986

**OPERATED BY** air defence missile brigade, brigade command post (combat control post MP06 and command-staff vehicle MP02)

**D 2021-12 H**

air defence missile brigade, brigade command post; vehicle captioned as 9S52M Polyana-D4M **O 2010-12 M**

combined arms army, army air defence and aviation command post (SKP PVO i aviatsii) **R 2021-12 M**

**INDICATES** Indicates the command post of an Army air defence missile brigade (S-300V/V4 or Buk), or an army or district AD command post. Up to four SAM battalions, a radar post and a point-defence CP report to a brigade Polyana.

**RECOGNISE Cab** MP06 combat control post on BAZ-6950 8x8: forward fibreglass two-door cab, three reverse-raked windscreens, oval vent hatch in centre of flat front wall.

**Body** Engine compartment behind cab. SKN-6950 'Rodinka' metal-frame van body with wide chamfered sides was built for this chassis for C2 complexes; MP06 body not described.

**Axles** Four axles in two widely spaced pairs (base 1.5 × 3.8 × 1.5 m); front pair steers. Large single tyres 1300x530-533.

**Details** Complex adds MP02 command-staff van on Ural-375 or Ural-43203 6x6 with KP4 trailer, MP45 maintenance van and two generator trucks (KamAZ-4310 on D4M).

**NOT TO BE CONFUSED WITH** 9S457 S-300V battalion command post p. 13 Tracked MT-T chassis shared with other S-300V vehicles, with telescoping datalink mast; Polyana vehicles are wheeled trucks.

**Baikal-1** air defence command post Carried on MAZ-7910 8x8 with integrated FL-94 telescoping mast and 5S15 turbine generator; Polyana MP06 is on BAZ-6950 with separate generator trucks.

**SOURCES** APA4 APA9 ATP IIS25 KZ2019 KZ2021 NASHAVT RDP4 RUSAR1 RUSARMM VPK6 VRF7 WEG26 WP-EN26 WP-EN27 WP-RU34



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**MOBILITY** Wheeled truck complex with generator trucks; needs a road or firm track. Deploys or packs up in about 20 to 35 min. Fording 1.2 m (Ural) and 1.3 m (BAZ) chassis figures.

Weight, gross, Ural-375D chassis (MP02) **13.2 t** chassis figure; MP02 as fitted INA

Weight, gross, BAZ-6950 chassis (MP06) **29.4 t** chassis figure; MP06 as fitted INA

L × W × H **7.36 × 2.67 × 2.68 m** cargo truck  
 Road speed, Ural-375D **75 km/h** maximum

SAM battalions controlled **up to 4** 9S52 and 9S52M; S-300V (9S457) or Buk (9S470) battalion CPs  
**up to 250** 9S52M1

Air targets tracked simultaneously  
 Targets received **up to 1,000 per 10 s** 9S52M1, claim  
 Linked objects, total **up to 14** 9S52M1; AEI aircraft, subordinate SAM unit CPs, higher and cooperating CPs, claim  
 Superior CP interfaces **1** variant not stated

## PU-12M7 9C482M7 PY-12M7

WH-AT-A

Air defence battery command post: fuses radar data and cues SHORAD fire units In service 1977

**OPERATED BY** motor rifle regiment, air defence missile-artillery battery; also regimental air defence chief's post (ARMF: PU-12 family 1977-95; ORUZH: undated PU-12M7 sheet), 1 per battery (PU-12 family)

**D 1995 M**

tank regiment, air defence missile-artillery battery; also regimental air defence chief's post (ARMF: PU-12 family 1977-95; ORUZH: undated PU-12M7 sheet), 1 per battery (PU-12 family) **D 1995 M**

motor rifle regiment, regiment on BTR, Soviet 1980s establishment, 3 PU-12 per regiment **D 1989 L**

**INDICATES** A PU-12-series or Ranzhir vehicle marks the command post of a short-range air defence battery (Strela-10, Tunguska, Tor, Osa, Shilka or Iglu teams, up to 4-f fire units) or a regimental air defence chief's post nearby.

**RECOGNISE** Body BTR-80 based K1Sh1 hull: raised centre section and extra roof hatches; no turret (from photograph).

**Mast** Telescopic mast at hull rear, raised for work; dome-shaped head on top and a box halfway up (from photograph). WEG: passive antennas mid-mast, day/night camera on top.

**Axles** Four axles, 8x8, front two axles steer; single waterjet at hull rear; amphibious (BTR-80 base).

**Details** Roof-mounted box sensor with two lenses on a pan-tilt head (from photograph); BTR-80 side door between second and third axles.

**NOT TO BE CONFUSED WITH** PPRU-1M / PPRU-M1 Also on MT-LBu but carries its own 9S80 Dog Ear radar; 9S737 Ranzhir takes data from external radars such as P-19 or 9S18M1.

**BTR-60PU-12** p. 34 BTR-60 hull with large stowage box on right side, generator and telescopic mast; PU-12M6/M7 use the BTR-80 hull.



Vitaly V. Kuzmin, CC BY-SA 4.0

**MOBILITY** PU-12M7 is amphibious on the BTR-80 base. 9S737 and 9S737M are tracked; 9S737M fords 1.0 m.

Weight, PU-12M7

13.6 t

Weight, 9S737 (WEG figure)

25000 kg WEG lists a Tata LPT 2518 truck chassis; see open\_questions

Weight, 9S737M

31 t GM-5965.05 tracked chassis; RUSA2 gives 'not more than 31,000 kg'  
7.7 × 2.9 × 2.41 m

L × W × H

Coordinates measurement envelope, PU-12M7 (connected radars)

up to 100 km maximum, coordinates measurement envelope from connected radars; PU-12M7 has no radar of its own, claim

Data link range, PU-12M7 by radio

up to 40 km maximum, radio link to higher and subordinate posts, claim

Data link range, PU-12M7 by field cable

up to 15 km maximum, field telephone cable, claim

Data link range, 9S737 to higher post

up to 30 km maximum, radio, claim

**SOURCES** APA4 ARMF JANESRG ORUZH ORYX ROE2 RUSA1 RUSA2 VPK7 WEG29 WMC12 WMC13 WP-RU39 WP-RU40 WP-RU41 WP-RU42 WP-RU43

## Barnaul-T «Барнаул-Т»

TR-A

Automated command and control set for tactical short-range air defence, battery to brigade In service 2015

**OPERATED BY** motor rifle brigade, brigade AD subunit; BO quotes a 'missile division' (battalion) commander **O 2021-12 M**

airborne division, air defence missile regiment, one full divisional set **O 2018-06 M**

airborne division, air defence missile regiment **O 2018 L**

**INDICATES** A 9S932-1 marks a short-range air defence battery command post. Strela-10, Tor, Osa, Tunguska or MANPADS fire units it controls are likely nearby; in VDV units the kit belongs to the division's AD regiment.

**RECOGNISE** Antenna 1L122-1E rectangular L-band antenna array, 1.2 × 0.8 m (maker's figure), on the top rear of the hull; folded down in transit.

**Mast** INA: no text source describes a mast on the 9S932-1.

**Chassis** Modified MT-LBu tracked amphibious carrier (9S932-1). Airborne version on BTR-MD/BTR-MDM tracked carrier.

**Complex** Tracked 9S932-1 at battery, 9S931 on MT-LBu, 9S931-1 on KamAZ 6x6 with trailer at brigade; 9S933 and 9S935 are portable MANPADS kits.

**NOT TO BE CONFUSED WITH** PPRU-1M / PPRU-M1 Also an MT-LBu AD battery command post, but carries a 9S80-series X-band radar, not the 1.2 × 0.8 m 1L122 L-band array.

1L122-2E Self-propelled Garmon radar on MT-LBu with its antenna on a lifting boom; antenna 1.9 × 1.8 m per TW, but GS gives 1.2 × 0.8 m for both versions.

9S737 Ranzhir p. 37 Older AD battery command post on MT-LBu-derived chassis; check for the 1L122 array on the rear hull.



Doomych, Public domain

**MOBILITY** Amphibious tracked MT-LBu; swims 5 km/h maximum. The BTR-MDM (VDV) version is parachute-droppable.

Weight, combat

11,000 kg MT-LBu chassis figure; 9S932-1 weight INA

L × W × H

6.45 × 2.86 × 1.86 m MT-LBu chassis

Road speed

60 km/h maximum, maximum

Road range

500 km cruising, cruising

Detection range, 1L122-1E

40 km maximum (1-40 km), aircraft, helicopters, cruise missiles, UAVs at up to 700 m/s; export 1L122-1E brochure figure, claim

Minimum detection range, 1L122-1E

1 km export 1L122-1E, claim

Detection altitude, 1L122-1E

10 km maximum, export 1L122-1E; 'at least 10 km', claim

Data link range, 9S932-1

20 km communication range, stationary; 10 km on the move (9S933: 3 km / 1 km)

Radar band, 1L122-1E

L-band claim

**SEEN WITH** 1L122-1E Garmon (1 per 9S932-1)

**SOURCES** APA4 BO DB GS6 KZ MIL4 ORYX SDU TOPWAR3 TW22 VALKA WEG7 WMC2 WP-RU10

Table 1. Master list of designations

Every system, vehicle, component and munition in this guide's lexicon, 809 in all, sorted by designation (digits sort as numbers). NATO reporting names and codes and US designations are set in **BLUE CAPITALS**, for example **STONE**, **SS-26** or **M1974**. NATO designations and reporting names can also be looked up in Table 1b. **(UA)** marks a type reported in Russian use in Ukraine since 2014. A **bold** page number leads to a full entry; a *grey italic* one to the entry of the system it belongs to. Look up other names and Cyrillic forms in Tables 1b and 1c.

| DESIGNATION                        | NAME                                   | CYRILLIC                     | WHAT IT IS                                                                  | PAGE |
|------------------------------------|----------------------------------------|------------------------------|-----------------------------------------------------------------------------|------|
| 1B44                               |                                        | 1B44                         | meteorological station - of 9K58 Smerch                                     | 6    |
| 1B75 1B75E                         | Penicillin                             | «Пенициллин»                 | acoustic-thermal locating system <a href="#">(UA)</a>                       | 20   |
| 1K113 3F5, 3VF4                    | Smelchak                               | 1K113 «Смелычак», 3Ф5        | guided mortar bomb - of 2S4 Tyulpan <a href="#">(UA)</a>                    | 9    |
| 1K148                              | Yastreb-AV                             | 1K148 «Ястреб-АВ»            | counter-battery radar <a href="#">(UA)</a>                                  |      |
| 1L13-3 1L13                        | Nebo-SV<br><b>BOX SPRING</b>           | 1Л13, «Небо-СВ»              | air surveillance radar                                                      |      |
| 1L119                              | Nebo-SVU                               | «Небо-СВУ» 1Л119             | air surveillance radar <a href="#">(UA)</a>                                 |      |
| 1L122                              | Garmon, Garmony                        | 1Л122 «Гармонь»              | portable air surveillance radar <a href="#">(UA)</a>                        |      |
| 1L125                              | Niobium-SV                             |                              | multifunction radar <a href="#">(UA)</a>                                    |      |
| 1L219 Zoopark-1                    | Zoopark-1                              | 1Л219 «Зоопарк-1»,<br>1Л219М | counter-battery radar <a href="#">(UA)</a>                                  | 19   |
| 1L222                              | Avtobaza-M                             | «Автомобаз»                  | ELINT system                                                                |      |
| 1L259                              |                                        | 1Л259                        | counter-battery radar <a href="#">(UA)</a>                                  | 19   |
| 1L260 Zoopark-1M                   | Zoopark-1M                             | 1Л260, «Зоопарк-1М»          | counter-battery radar <a href="#">(UA)</a>                                  | 19   |
| 1L261                              |                                        |                              | counter-battery radar <a href="#">(UA)</a>                                  | 19   |
| 1L262E SPR-2M                      | Rtut-BM, Mercury-BM                    | 1Л262З «Ртуть-БМ»            | proximity-fuze jammer <a href="#">(UA)</a>                                  |      |
| 1L265                              |                                        | 1Л265                        | ELINT system                                                                | 22   |
| 1L266                              |                                        | 1Л266                        | EW command post                                                             | 22   |
| 1L267                              | Moskva-1                               | «Москва-1»                   | EW command post <a href="#">(UA)</a>                                        | 22   |
| 1L269 Krasukha-20                  | Krasukha-2                             | 1Л269 «Красуха-2»            | radar jammer <a href="#">(UA)</a>                                           | 21   |
| 1L271                              | Aistenok, Aistyonok                    | 1Л271 «Аистенок»             | portable counter-battery radar <a href="#">(UA)</a>                         | 20   |
| 1L277                              | Sobolyatnik                            |                              | portable ground surveillance radar <a href="#">(UA)</a>                     |      |
| 1RL144M 1RL144M1, 1RL144, 1RL138   | <b>HOT SHOT</b>                        | 1РЛ144М1                     | fire-control radar <a href="#">(UA)</a>                                     | 17   |
| 1RL239 ARK-1, ARK-1M               | Rys<br><b>SMALL FRED</b> (disputed)    | 1РЛ239 / АРК-1 / «Рысь»      | counter-battery radar <a href="#">(UA)</a>                                  | 19   |
| 1RL257                             | Krasukha-4, Krasukha-4S                | 1РЛ257 «Красуха-4»           | radar jammer <a href="#">(UA)</a>                                           | 21   |
| 1T12-2M                            |                                        | 1Т12-2М                      | topographic survey vehicle - of 9K58 Smerch                                 | 6    |
| 1V13 1V13M, M1974/1                |                                        | 1В13                         | artillery battery fire control centre <a href="#">(UA)</a>                  | 35   |
| 1V14 1V15, 1V15M, 1V16             |                                        |                              | artillery command vehicle <a href="#">(UA)</a>                              | 35   |
| 1V18 1V19, <b>M1979-2B</b>         | Klyon-1, Klen-1, Klen-2                |                              | artillery command vehicle <a href="#">(UA)</a>                              |      |
| 1V110                              |                                        |                              | artillery command vehicle - of 9K51 Grad <i>legacy</i> <a href="#">(UA)</a> | 4    |
| 1V119                              | Spektr                                 | 1В119                        | artillery command vehicle <a href="#">(UA)</a>                              |      |
| 1V126 1V126M, 1V152, 1V153, 1V152M | Kapustnik-B, Kapustnik-BM              | 1В126 «Капустник-Б»          | artillery fire control system - of 9K59 Prima                               |      |
| 1V152                              |                                        | 1В152                        | artillery command vehicle                                                   |      |
| 1V1003                             |                                        |                              | artillery command vehicle <a href="#">(UA)</a>                              |      |
| 2A36 <b>M1976</b>                  | Giatsint-B                             | 2А36 «Гиацинт-Б»             | 152 mm towed gun <a href="#">(UA)</a>                                       | 10   |
| 2A45M 2A45MR                       | Sprut-B                                |                              | 125 mm towed anti-tank gun <i>legacy</i>                                    |      |
| 2A65 <b>M1987</b>                  | Msta-B                                 | 2А65 «Мста-Б»                | 152 mm towed howitzer <a href="#">(UA)</a>                                  | 10   |
| 2B9 2B9M                           | Vasilek                                | 2В9 «Василёк»                | 82 mm automatic gun-mortar <a href="#">(UA)</a>                             |      |
| 2B11                               |                                        | 2В11                         | 120 mm mortar - of 2S12 Sani <a href="#">(UA)</a>                           |      |
| 2B14                               | Podnos                                 | 2В14 «Поднос»                | 82 mm mortar                                                                |      |
| 2B16                               | Nona-K                                 | 2В16 «Нона-К»                | 120 mm towed gun-mortar <a href="#">(UA)</a>                                |      |
| 2B17M                              | Tornado-G                              | 2В17М                        | 122 mm multiple rocket launcher <a href="#">(UA)</a>                        | 4    |
| 2B26                               | Grad-K                                 | 2В26                         | 122 mm multiple rocket launcher - of 9K51 Grad <a href="#">(UA)</a>         | 4    |
| 2F77 2F77M                         |                                        | 2Ф77, 2Ф77М                  | transloader - of 2K22 Tunguska <a href="#">(UA)</a>                         | 17   |
| 2K22 2K22M, 2K22M1, <b>SA-19</b>   | Tunguska, Tunguska-M1<br><b>GRISON</b> | 2К22, 2К22М «Тунгуска-М»     | gun-missile air defence system <a href="#">(UA)</a>                         | 17   |
| 2S1 <b>M1974</b>                   | Gvozdika                               | 2С1 «Гвоздика»               | 122 mm self-propelled howitzer <a href="#">(UA)</a>                         | 35   |
| 2S3 Object 303, <b>M1973</b>       | Akatsiya                               | 2С3 «Акация»                 | 152 mm self-propelled howitzer <a href="#">(UA)</a>                         |      |
| 2S3M 2S3M1, 2S3M2                  | Akatsiya                               | 2С3М                         | 152 mm self-propelled howitzer - of 2S3 Akatsiya <a href="#">(UA)</a>       |      |
| 2S4 Object 305, <b>M1975</b>       | Tyulpan                                | 2С4 «Тюльпан»                | 240 mm self-propelled mortar <a href="#">(UA)</a>                           | 9    |
| 2S5 Object 307, <b>M1981</b>       | Giatsint-S                             | 2С5 «Гиацинт-С»              | 152 mm self-propelled gun <a href="#">(UA)</a>                              | 7    |

| DESIGNATION                                 | NAME                              | CYRILLIC               | WHAT IT IS                                                                   | PAGE |
|---------------------------------------------|-----------------------------------|------------------------|------------------------------------------------------------------------------|------|
| 2S5M                                        |                                   | 2C5M                   | 152 mm self-propelled gun · of 2S5 Giatint-S <i>legacy</i>                   | 7    |
| 2S6 2S6M, 2S6M1                             |                                   | 2C6, 2C6M              | gun-missile air defence vehicle <a href="#">UA</a>                           | 17   |
| 2S7 Object 216, <b>M1975</b>                | Pion                              | 2C7 «Пион»             | 203 mm self-propelled gun <a href="#">UA</a>                                 | 9    |
| 2S7M                                        | Malka                             | 2C7M «Малка»           | 203 mm self-propelled gun <a href="#">UA</a>                                 |      |
| 2S9 2S9-1                                   | Nona-S                            | 2C9 «Нона-С»           | 120 mm self-propelled gun-mortar <a href="#">UA</a>                          |      |
| 2S12 2S12A                                  | Sani                              | 2C12 «Сани», 2C12A     | 120 mm mortar <a href="#">UA</a>                                             | 8    |
| 2S19 Object 316, <b>M1990</b>               | Msta-S <b>FARM</b>                | 2C19 «Мсма-С»          | 152 mm self-propelled howitzer <a href="#">UA</a>                            |      |
| 2S19M1                                      | Msta-S                            | 2C19M1                 | 152 mm self-propelled howitzer <a href="#">UA</a>                            |      |
| 2S19M2 2S33                                 | Msta-SM                           | 2C19M2, 2C33 «Мсма-СМ» | 152 mm self-propelled howitzer <a href="#">UA</a>                            | 8    |
| 2S23                                        | Nona-SVK                          | 2C23 «Нона-СВК»        | 120 mm self-propelled gun-mortar <a href="#">UA</a>                          | 8    |
| 2S25                                        | Sprut-SD                          | 2C25                   | 125 mm self-propelled anti-tank gun                                          |      |
| 2S31                                        | Vena                              | 2C31 «Вена»            | 120 mm self-propelled gun-mortar <a href="#">UA</a>                          |      |
| 2S34 2S1M                                   | Khosta, Chosta                    | 2C34 «Хоста»           | 120 mm self-propelled gun-mortar <a href="#">UA</a>                          | 8    |
| 2S35                                        | Koalitsiya-SV                     | 2C35 «Коалиция-СВ»     | 152 mm self-propelled howitzer <a href="#">UA</a>                            |      |
| 2S38 ZAK-57                                 | Derivatsiya-PVO                   | 2C38, 3AK-57           | 57 mm self-propelled anti-aircraft gun <i>trial</i> <a href="#">UA</a>       |      |
| 2S40                                        | Floks                             | 2C40, «Флокс»          | 120 mm self-propelled gun-mortar <a href="#">UA</a>                          | 8    |
| 2S43                                        | Malva                             | 2C43 «Мальва»          | 152 mm wheeled self-propelled howitzer <a href="#">UA</a>                    |      |
| 3BK13                                       |                                   | 3BK13                  | artillery round · of 2S1 Gvozdika <i>legacy</i>                              |      |
| 3BK19                                       |                                   | 3BK19                  | artillery round                                                              | 9    |
| 3F2 3VF2                                    |                                   | 3Ф2                    | mortar bomb · of 2S4 Tyulpan                                                 |      |
| 3K55 K-300P, K300P, <b>SSC-5</b>            | Bastion, Bastion-P <b>STOOGEE</b> | 3K55, Бастуион         | coastal missile system <a href="#">UA</a>                                    |      |
| 3K60 <b>SSC-6</b>                           | Bal, Bal-E <b>SENIGHT</b>         | 3K60, Бал              | coastal missile system <a href="#">UA</a>                                    | 4    |
| 3M14 3M14T, 3M14K, <b>SS-N-30A, SS-N-27</b> | Kalibr <b>SAGARIS</b>             | Калибр, 3М-14          | cruise missile <a href="#">UA</a>                                            |      |
| 3M16                                        |                                   | 3M16                   | 122 mm cluster rocket · of 9K51 Grad <i>legacy</i>                           |      |
| 3-O-8 3VO11                                 |                                   | 3-O-8                  | mortar bomb · of 2S4 Tyulpan <a href="#">UA</a>                              | 9    |
| 3-O-13 3VO13, 3VO14                         |                                   | 3-O-13                 | artillery round <a href="#">UA</a>                                           | 35   |
| 3-O-23 3VO28, 3VO29, 3VO30                  |                                   | 3-O-23                 | artillery round                                                              |      |
| 3OF24                                       |                                   | 3OF24                  | artillery round · of 2S1 Gvozdika                                            |      |
| 3OF25 3VOF32, 3VOF33                        |                                   | 3OF25                  | artillery round <i>legacy</i>                                                | 7    |
| 3OF29 3VOF39, 3VOF40                        |                                   | 3OF29                  | artillery round · of 2S5 Giatint-S                                           |      |
| 3OF38 3VOF63, 3VOF66                        | Sanimetr                          | 3OF38                  | guided artillery round                                                       |      |
| 3OF39 3VOF64, 3VOF93, 2K25                  | Krasnopol                         | 3OF39, «Краснополь»    | guided artillery round <a href="#">UA</a>                                    | 7    |
| 3OF39M 3VOF64M, 2K25M                       | Krasnopol-M1                      | 3OF39M                 | guided artillery round · of 3OF39 Krasnopol <a href="#">UA</a>               |      |
| 3OF43 3VOF34, 3VOF42                        |                                   | 3OF43                  | artillery round · of 2S7 Pion                                                |      |
| 3OF44 3VOF35                                |                                   | 3OF44                  | artillery round · of 2S7 Pion                                                | 35   |
| 3OF45 3OF45-1, 3VOF58, 3VOF72, 3VOF73       |                                   | 3OF45                  | artillery round · of 2S19 Msta-S                                             |      |
| 3OF49 3VOF49                                |                                   | 3OF49                  | artillery round                                                              |      |
| 3OF50                                       |                                   | 3OF50                  | artillery round                                                              | 7    |
| 3OF56                                       |                                   | 3OF56                  | artillery round · of 2S1 Gvozdika                                            |      |
| 3OF59 3VOF86, 3VOF87                        |                                   | 3OF59                  | artillery round · of 2S5 Giatint-S                                           |      |
| 3OF61 3VOF91                                |                                   | 3OF61                  | artillery round                                                              | 7    |
| 3OF64 3VOF96, 3VOF97, 3VOF98                |                                   | 3OF64                  | artillery round                                                              |      |
| 3OF69 Kitolov-2                             | Kitolov-2                         | «Кумолов-2»            | guided mortar bomb <a href="#">UA</a>                                        |      |
| 3OF69M Kitolov-2M                           | Kitolov-2M                        | «Кумолов-2М»           | guided mortar bomb · of 3OF69 Kitolov-2 <a href="#">UA</a>                   | 12   |
| 3OF75                                       | Sanimetr-M                        | 3OF75                  | guided artillery round                                                       |      |
| 3RB30 3VRB36, 3VRB37, 3VRB38                |                                   | 3PE30                  | communications jammer shell                                                  |      |
| 3S55 K-340P                                 |                                   | K-340П                 | missile transporter erector launcher · of 3K55 Bastion <a href="#">UA</a>    | 12   |
| 5N63S 3ON6                                  | <b>FLAP LID B</b>                 | 5H63C                  | fire-control radar · of S-300PS S-300PS <i>legacy</i> <a href="#">UA</a>     |      |
| 5N84A Oborona-14                            | Oborona-14 <b>TALL KING C</b>     | 5H84A «Оборона»        | early warning radar <a href="#">UA</a>                                       |      |
| 5P85SD 5P85SM, 5P85S, 5P85D                 |                                   | 5P85C, 5P85A           | SAM launcher · of S-300PS S-300PS <i>legacy</i> <a href="#">UA</a>           | 12   |
| 5P85SM2-01                                  |                                   | 5P85CM2-01             | SAM launcher <a href="#">UA</a>                                              |      |
| 5P85T2                                      |                                   | 5P85T2                 | SAM launcher <a href="#">UA</a>                                              |      |
| 5V55R 5V55K, 5V55RM, V-500R                 |                                   | 5B55P, 5B55K           | surface-to-air missile · of S-300PS S-300PS <i>legacy</i> <a href="#">UA</a> | 12   |
| 6P50 6P49, 6P57, 6P58, 6P59                 | Kord                              | Kopg                   | 12.7 mm heavy machine gun <a href="#">UA</a>                                 |      |

| DESIGNATION                   | NAME                             | CYRILLIC                                  | WHAT IT IS                                                       | PAGE |
|-------------------------------|----------------------------------|-------------------------------------------|------------------------------------------------------------------|------|
| 9A33B 9A33BM3                 |                                  | 9A33B                                     | SAM launcher - of 9K33 Osa <a href="#">UA</a>                    | 17   |
| 9A35                          |                                  | 9A35                                      | SAM launcher - of 9K35 Strela-10 <a href="#">UA</a>              | 17   |
| 9A39 9A39M1                   |                                  | 9A39, 9A39M1                              | SAM loader-launcher - of 9K37 Buk stored <a href="#">UA</a>      | 14   |
| 9A39M1-2                      |                                  |                                           | SAM loader-launcher - of 9K37M1-2 Buk-M1-2 <a href="#">UA</a>    | 14   |
| 9A51                          | Prima                            | 9A51                                      | 122 mm multiple rocket launcher - of 9K59 Prima legacy           |      |
| 9A52 BM-30                    | Smerch                           | 9A52                                      | 300 mm multiple rocket launcher legacy <a href="#">UA</a>        | 6    |
| 9A52-2 BM-30                  | Smerch-M                         | 9A52-2                                    | 300 mm multiple rocket launcher <a href="#">UA</a>               | 6    |
| 9A52-2T                       |                                  | 9A52-2T                                   | 300 mm multiple rocket launcher - of 9K58 Smerch export          | 6    |
| 9A52-4                        | Kama                             | 9A52-4                                    | 300 mm multiple rocket launcher trial                            | 6    |
| 9A53                          | Uragan-1M                        | 9A53                                      | 220/300 mm multiple rocket launcher <a href="#">UA</a>           | 5    |
| 9A54 9A53-S                   | Tornado-S                        | 9A54                                      | 300 mm multiple rocket launcher <a href="#">UA</a>               | 6    |
| 9A82                          |                                  | 9A82                                      | SAM launcher - of S-300V S-300V                                  | 13   |
| 9A83 9A83-1, 9A83M2           |                                  | 9A83                                      | SAM launcher - of S-300V S-300V <a href="#">UA</a>               | 13   |
| 9A83M2                        |                                  |                                           | SAM launcher - of S-300V4 S-300V4 <a href="#">UA</a>             | 13   |
| 9A84                          |                                  | 9A84                                      | SAM loader-launcher - of S-300V S-300V <a href="#">UA</a>        | 13   |
| 9A84-2                        |                                  |                                           | SAM loader-launcher - of S-300V4 S-300V4 <a href="#">UA</a>      | 13   |
| 9A85 9A85-2                   |                                  | 9A85                                      | SAM loader-launcher - of S-300V S-300V <a href="#">UA</a>        | 13   |
| 9A85-2                        |                                  |                                           | transloader - of S-300V4 S-300V4 <a href="#">UA</a>              | 13   |
| 9A310 9A310M1, 9A38           |                                  | 9A310, 9A38                               | SAM launcher - of 9K37 Buk stored <a href="#">UA</a>             | 14   |
| 9A310M1-2                     |                                  | 9A310M1-2                                 | SAM launcher - of 9K37M1-2 Buk-M1-2 <a href="#">UA</a>           | 14   |
| 9A316                         |                                  | 9A316                                     | SAM loader-launcher - of 9K317 Buk-M2 <a href="#">UA</a>         | 14   |
| 9A317                         |                                  | 9A317                                     | SAM launcher - of 9K317 Buk-M2 <a href="#">UA</a>                | 14   |
| 9A317M                        |                                  | 9A317M                                    | SAM launcher - of 9K317M Buk-M3 <a href="#">UA</a>               | 14   |
| 9A330                         |                                  | 9A330                                     | SAM launcher legacy <a href="#">UA</a>                           | 15   |
| 9A331 9A331-1                 |                                  | 9A331, 9A331-1                            | SAM launcher <a href="#">UA</a>                                  | 15   |
| 9A331M 9A331M2                |                                  | 9A331M2                                   | SAM launcher <a href="#">UA</a>                                  | 15   |
| 9A4172 9K121, AT-16           | Vikhr<br>SCALLION                |                                           | air-launched anti-tank guided missile <a href="#">UA</a>         |      |
| 9K11 9M14, AT-3               | Malyutka, Malyutka-2<br>SAGGER   | 9K11, 9M14                                | anti-tank guided missile system legacy                           |      |
| 9K32 9K32M, SA-7              | Strela-2<br>GRAIL                | 9K32, «Смерел-2»                          | MANPADS legacy                                                   |      |
| 9K33 9K33M2, 9K33M3, SA-8     | Osa, Osa-AK, Osa-AKM<br>GECKO    | 9K33, 9K33M2                              | short-range SAM system <a href="#">UA</a>                        | 17   |
| 9K34 SA-14                    | Strela-3<br>GREMLIN              | 9K34, «Смерел-3»                          | MANPADS                                                          |      |
| 9K35 9K35M3, SA-13            | Strela-10, Strela-10M3<br>GOPHER | 9K35 «Смерел-10»,<br>9K35M3 «Смерел-10M3» | short-range SAM system <a href="#">UA</a>                        | 17   |
| 9K37 9K37M1, SA-11            | Buk, Buk-M1<br>GADFLY            | 9K37, 9K37M1 «Бук-M1»                     | medium-range SAM system stored <a href="#">UA</a>                | 14   |
| 9K37M1-2 SA-11                | Buk-M1-2<br>GADFLY               | 9K37M1-2 «Бук-M1-2»                       | medium-range SAM system <a href="#">UA</a>                       | 14   |
| 9K38 SA-18                    | Igla<br>GROUSE                   | 9K38, «Игла»                              | MANPADS <a href="#">UA</a>                                       |      |
| 9K51 M-21, M1964              | Grad                             | 9K51 «Град»                               | 122 mm multiple rocket launcher system <a href="#">UA</a>        | 4    |
| 9K51M                         | Tornado-G                        | 9K51M «Торнадо-Г»                         | 122 mm multiple rocket launcher system <a href="#">UA</a>        | 4    |
| 9K51M 2B17M, 9A53-G           | Tornado-G                        | 9K51M, Торнадо-Г                          | 122 mm 40-tube truck mrl; automated grad upgrade for brigade     | 4    |
| 9K52 FROG-7                   | Luna-M                           | 9K52, Луна-M                              | unguided rocket system legacy                                    |      |
| 9K54 BM-21V, M-21V, M1975     | Grad-V                           | 9K54 «Град-В»                             | 122 mm multiple rocket launcher system legacy                    |      |
| 9K55 BM-21b, M1976            | Grad-1                           | 9K55 «Град-1»                             | 122 mm multiple rocket launcher system stored <a href="#">UA</a> |      |
| 9K55-1                        | Grad-1                           | 9K55-1 «Град-1»                           | 122 mm multiple rocket launcher system legacy <a href="#">UA</a> | 4    |
| 9K57 BM-27, M1977             | Uragan                           | 9K57 «Ураган»                             | 220 mm multiple rocket launcher system <a href="#">UA</a>        | 5    |
| 9K57-1                        | Uragan-1                         |                                           | 220 mm multiple rocket launcher legacy <a href="#">UA</a>        | 5    |
| 9K58 BM-30                    | Smerch                           | 9K58 «Смерч»                              | 300 mm multiple rocket launcher system <a href="#">UA</a>        | 6    |
| 9K59                          | Prima                            | 9K59 «Прима»                              | 122 mm multiple rocket launcher system legacy                    |      |
| 9K72 8K14, R-17, SS-1c, R-300 | Elbrus<br>SCUD B                 | 9K72, P-17                                | ballistic missile system stored                                  |      |
| 9K79 9K79-1, OTR-21, SS-21    | Tochka, Tochka-U<br>SCARAB       | 9K79, Точка-У                             | ballistic missile system <a href="#">UA</a>                      |      |

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| 9K111 9K111M, 9P135, <b>AT-4</b>                                | Fagot<br><b>SPIGOT</b>                                        | 9K111, Фагот       | anti-tank guided missile system <a href="#">(UA)</a>                                |      |
| <b>9K111-1</b> 9K111-1M, 9K113, 9P135M, <b>AT-5</b>             | Konkurs, Konkurs-M<br><b>SPANDREL</b>                         | 9K111-1, Конкурс   | anti-tank guided missile system <a href="#">(UA)</a>                                |      |
| 9K112 9M112, <b>AT-8</b>                                        | Kobra<br><b>SONGSTER</b>                                      |                    | gun-launched anti-tank missile system <i>legacy</i>                                 |      |
| 9K114 <b>AT-6</b>                                               | Shturm<br><b>SPIRAL</b>                                       | 9K114, Штурм       | anti-tank guided missile system <a href="#">(UA)</a>                                |      |
| 9K115 9P151, <b>AT-7</b>                                        | Metis<br><b>SAXHORN</b>                                       | 9K115, Метис       | anti-tank guided missile system <a href="#">(UA)</a>                                |      |
| <b>9K115-1 AT-13</b>                                            | Metis-M<br><b>SAXHORN-2</b>                                   | 9K115-1, Метис-М   | anti-tank guided missile system                                                     |      |
| <b>9K115-2 AT-13</b>                                            | Metis-M1<br><b>SAXHORN-2</b>                                  | 9K115-2, Метис-М1  | anti-tank guided missile system · of 9K115-1<br>Metis-M                             |      |
| <b>9K116</b> 9K116-1, 9K116-2, 9K116-3,<br>3UBK10, <b>AT-12</b> | Kastet, Bastion, Sheksna<br><b>SWINGER</b>                    | 9K116, Кастет      | gun-launched anti-tank missile system                                               |      |
| 9K119 9K119M, 9K120, <b>AT-11</b>                               | Svir, Refleks, Refleks-M<br><b>SNIPER</b>                     | 9K119M, Рефлекс-М  | gun-launched anti-tank missile system <a href="#">(UA)</a>                          |      |
| <b>9K123</b> 9K123-1, <b>AT-15</b>                              | Khrizantema, Khrizantema-S<br><b>SPRINGER</b>                 | 9K123, Хризантема  | anti-tank guided missile system <a href="#">(UA)</a>                                |      |
| <b>9K132</b> 9P149M, <b>AT-9</b>                                | Shturm-SM<br><b>SPIRAL-2</b>                                  | 9K132, 9П149М      | anti-tank guided missile system · of 9K114<br>Shturm <a href="#">(UA)</a>           |      |
| <b>9K135</b> 9K129, <b>AT-14</b>                                | Kornet, Kornet-E, Kornet-EM<br><b>SPRIGGAN</b>                | 9K135, Корнет      | anti-tank guided missile system <a href="#">(UA)</a>                                |      |
| 9K310 <b>SA-16</b>                                              | Igla-1<br><b>GIMLET</b>                                       | 9K310, «Игла-1»    | MANPADS                                                                             |      |
| <b>9K317 SA-17</b>                                              | Buk-M2<br><b>GRIZZLY</b>                                      | 9K317 «Бук-М2»     | medium-range SAM system <a href="#">(UA)</a>                                        | 14   |
| <b>9K317M SA-27</b>                                             | Buk-M3<br><b>GOLLUM</b>                                       | 9K317M «Бук-М3»    | medium-range SAM system <a href="#">(UA)</a>                                        | 14   |
| <b>9K330 SA-15</b>                                              | Tor<br><b>GAUNTLET</b>                                        | 9K330              | short-range SAM system <i>legacy</i> <a href="#">(UA)</a>                           | 15   |
| 9K331 9K331M, <b>SA-15</b>                                      | Tor-M1<br><b>GAUNTLET</b>                                     | 9K331 «Тор-М1»     | short-range SAM system <a href="#">(UA)</a>                                         | 15   |
| <b>9K331MDT SA-15</b>                                           | Tor-M2DT<br><b>GAUNTLET</b>                                   | «Тор-М2ДТ»         | short-range SAM system <a href="#">(UA)</a>                                         | 15   |
| <b>9K331MU SA-15</b>                                            | Tor-M1-2U, Tor-M2U<br><b>GAUNTLET</b>                         | «Тор-М1-2У»        | short-range SAM system <a href="#">(UA)</a>                                         | 15   |
| <b>9K332 SA-15</b>                                              | Tor-M2<br><b>GAUNTLET</b>                                     | 9K332 3PK «Тор-М2» | short-range SAM system <a href="#">(UA)</a>                                         | 15   |
| <b>9K333 SA-29</b>                                              | Verba<br><b>GIZMO</b>                                         | 9K333, «Верб»      | MANPADS <a href="#">(UA)</a>                                                        |      |
| <b>9K334</b>                                                    | Gibka-S                                                       | «Гибка-С»          | MANPADS launcher vehicle <a href="#">(UA)</a>                                       |      |
| <b>9K338 SA-24</b>                                              | Igla-S<br><b>GRINCH</b>                                       | 9K338, «Игла-С»    | MANPADS                                                                             |      |
| <b>9K512</b>                                                    | Uragan-1M                                                     | 9K512 «Ураган-1М»  | 220/300 mm multiple rocket launcher system<br><a href="#">(UA)</a>                  | 5    |
| <b>9K515</b>                                                    | Tornado-S                                                     | 9K515              | 300 mm multiple rocket launcher system <a href="#">(UA)</a>                         | 6    |
| <b>9K720 SS-26, SSC-7, SSC-8</b>                                | Iskander-M, Iskander,<br>Iskander-K<br><b>STONE, SOUTHPAW</b> | 9K720, Искангеп    | ballistic missile system <a href="#">(UA)</a>                                       | 11   |
| <b>9M22</b>                                                     |                                                               | 9M22               | 122 mm HE-fragmentation rocket · of 9K51 Grad<br><i>legacy</i>                      | 4    |
| <b>9M22M</b>                                                    | Malysh                                                        | 9M22M              | 122 mm rocket · of 9P132 Grad-P <i>legacy</i>                                       |      |
| <b>9M22S</b>                                                    |                                                               | 9M22C              | 122 mm incendiary rocket · of 9K51 Grad <i>legacy</i><br><a href="#">(UA)</a>       | 4    |
| <b>9M22U</b> 9M22U-1, M-210F                                    |                                                               | 9M22Y              | 122 mm HE-fragmentation rocket · of 9K51 Grad<br><a href="#">(UA)</a>               | 4    |
| <b>9M27D</b>                                                    |                                                               | 9M27A              | 220 mm leaflet rocket · of 9K57 Uragan <i>legacy</i>                                | 5    |
| <b>9M27F</b>                                                    |                                                               | 9M27D              | 220 mm HE rocket · of 9K57 Uragan                                                   | 5    |
| <b>9M27K</b>                                                    |                                                               | 9M27K              | 220 mm cluster rocket · of 9K57 Uragan <a href="#">(UA)</a>                         | 5    |
| <b>9M27K1</b>                                                   |                                                               | 9M27K1             | 220 mm cluster rocket · of 9K57 Uragan <a href="#">(UA)</a>                         | 5    |
| <b>9M27K2</b>                                                   |                                                               | 9M27K2             | 220 mm scatterable-mine rocket · of 9K57<br>Uragan                                  | 5    |
| <b>9M27K3</b>                                                   |                                                               | 9M27K3             | 220 mm scatterable-mine rocket · of 9K57<br>Uragan                                  | 5    |
| <b>9M27S</b>                                                    |                                                               | 9M27C              | 220 mm incendiary rocket · of 9K57 Uragan <i>legacy</i>                             | 5    |
| <b>9M28D</b>                                                    |                                                               | 9M28A              | 122 mm leaflet rocket · of 9K51 Grad <i>legacy</i>                                  | 4    |
| <b>9M28F</b>                                                    |                                                               | 9M28D              | 122 mm HE-fragmentation rocket · of 9K51 Grad<br><i>legacy</i> <a href="#">(UA)</a> | 4    |
| <b>9M28K</b>                                                    |                                                               | 9M28K              | 122 mm cluster rocket · of 9K51 Grad <i>legacy</i>                                  | 4    |

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| 9M33 9M33M2, 9M33M3, 9M33M                   |                             | 9M33, 9M33M3    | surface-to-air missile                                                      | 17   |
| 9M37 9M37M, 9M37MD                           |                             | 9M37            | surface-to-air missile <a href="#">UA</a>                                   | 17   |
| 9M38                                         |                             | 9M38            | surface-to-air missile · of 9K37 Buk <a href="#">UA</a>                     | 14   |
| 9M39                                         |                             | 9M39            | surface-to-air missile · of 9K38 Igla <a href="#">UA</a>                    |      |
| 9M42                                         |                             | 9M42            | 122 mm illumination rocket · of 9K51 Grad <i>legacy</i>                     | 4    |
| 9M43                                         |                             | 9M43            | 122 mm cluster rocket · of 9K51 Grad <i>legacy</i>                          | 4    |
| 9M51                                         |                             | 9M51            | 220 mm thermobaric rocket · of 9K57 Uragan <i>legacy</i>                    | 5    |
| 9M53F                                        |                             | 9M53D           | 122 mm HE rocket · of 9K59 Prima <i>legacy</i>                              |      |
| 9M53K                                        |                             | 9M53K           | 122 mm cluster rocket · of 9K59 Prima <i>legacy</i>                         |      |
| 9M53S                                        |                             | 9M53C           | 122 mm incendiary rocket · of 9K59 Prima <i>legacy</i>                      |      |
| 9M55F                                        |                             | 9M55D           | 300 mm HE-fragmentation rocket · of 9K58 Smerch                             | 6    |
| 9M55K                                        |                             | 9M55K           | 300 mm cluster rocket · of 9K58 Smerch <a href="#">UA</a>                   | 6    |
| 9M55K1                                       |                             | 9M55K1          | 300 mm cluster rocket · of 9K58 Smerch <a href="#">UA</a>                   | 6    |
| 9M55K4                                       |                             | 9M55K4          | 300 mm scatterable-mine rocket · of 9K58 Smerch                             | 6    |
| 9M55K5                                       |                             | 9M55K5          | 300 mm cluster rocket · of 9K58 Smerch                                      | 6    |
| 9M55S                                        |                             | 9M55C           | 300 mm thermobaric rocket · of 9K58 Smerch                                  | 6    |
| 9M59                                         |                             | 9M59            | 220 mm scatterable-mine rocket · of 9K57 Uragan                             | 5    |
| 9M79 9M79-1, SS-21                           | SCARAB                      | 9M79            | ballistic missile · of 9K79 Tochka <a href="#">UA</a>                       |      |
| 9M82 9M82M, 9M82MD, SA-12B                   | GIANT                       | 9M82, 9M82M     | surface-to-air missile · of S-300V S-300V <a href="#">UA</a>                | 13   |
| 9M83 9M83M, SA-12A                           | GLADIATOR                   | 9M83            | surface-to-air missile · of S-300V S-300V <a href="#">UA</a>                | 13   |
| 9M96 9M96E, 9M96E2, 9M96M                    |                             | 9M96E2, 9M96E   | surface-to-air missile · of S-400 S-400 <a href="#">UA</a>                  | 12   |
| 9M100                                        |                             | 9M100           | surface-to-air missile · of S-350 S-350                                     | 13   |
| 9M111 9M111-2, 9M111M, AT-4                  | Fagot, Faktoriya SPIGOT     | 9M111           | anti-tank guided missile · of 9K111 Fagot <a href="#">UA</a>                |      |
| 9M113 AT-5                                   | Konkurs SPANDREL            | 9M113           | anti-tank guided missile · of 9K111-1 Konkurs <a href="#">UA</a>            |      |
| 9M113M AT-5B                                 | Konkurs-M SPANDREL-B        | 9M113M          | anti-tank guided missile · of 9K111-1 Konkurs                               |      |
| 9M114 AT-6                                   | Kokon SPIRAL                | 9M114           | anti-tank guided missile · of 9K114 Shturm <a href="#">UA</a>               |      |
| 9M115 AT-7                                   | Metis SAXHORN               | 9M115           | anti-tank guided missile · of 9K115 Metis <a href="#">UA</a>                |      |
| 9M117 9M117M, 9M117M1, AT-10                 | Bastion, Kan, Arkan STABBER | 9M117           | gun-launched anti-tank guided missile · of 9K116 Kastet <a href="#">UA</a>  |      |
| 9M119 9M119M, 9M119M1, AT-11B, AT-11C, AT-11 | Svir, Refleks, Invar SNIPER | 9M119M, 9M119M1 | gun-launched anti-tank guided missile · of 9K119 Svir <a href="#">UA</a>    |      |
| 9M120 AT-9                                   | Ataka SPIRAL-2              | 9M120           | anti-tank guided missile · of 9K132 Shturm-SM                               |      |
| 9M120-1                                      | Ataka-1                     | 9M120-1         | anti-tank guided missile · of 9K132 Shturm-SM                               |      |
| 9M123 9M123F, AT-15                          | Khrizantema SPRINGER        | 9M123, 9M123D   | anti-tank guided missile · of 9K123 Khrizantema <a href="#">UA</a>          |      |
| 9M131 9M131M, AT-13                          | Metis-M SAXHORN-2           | 9M131           | anti-tank guided missile · of 9K115-1 Metis-M                               |      |
| 9M133 9M133F-1, AT-14                        | Kornet SPRIGGAN             | 9M133           | anti-tank guided missile · of 9K135 Kornet <a href="#">UA</a>               |      |
| 9M217                                        |                             | 9M217           | 122 mm cluster rocket · of 9K51M Tornado-G                                  | 4    |
| 9M218 9M541                                  |                             | 9M218           | 122 mm cluster rocket · of 9K51M Tornado-G                                  | 4    |
| 9M311 9M311-1M                               |                             | 9M311           | surface-to-air missile <a href="#">UA</a>                                   | 17   |
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| 9M317M                                       |                             |                 | surface-to-air missile · of 9K317M Buk-M3 <a href="#">UA</a>                | 14   |
| 9M330 9M330-2                                |                             | 9M330           | surface-to-air missile <i>legacy</i> <a href="#">UA</a>                     | 15   |
| 9M331                                        |                             | 9M331           | surface-to-air missile <a href="#">UA</a>                                   | 15   |
| 9M333                                        |                             | 9M333           | surface-to-air missile <a href="#">UA</a>                                   | 17   |
| 9M335                                        |                             | 9M335           | surface-to-air missile · of 9K333 Verba <a href="#">UA</a>                  | 16   |
| 9M336                                        |                             | 9M336           | surface-to-air missile · of 9K333 Verba <a href="#">UA</a>                  |      |
| 9M338 9M338K                                 |                             | 9M338           | surface-to-air missile <a href="#">UA</a>                                   | 15   |
| 9M340                                        |                             | 9M340           | surface-to-air missile · of 9M340 Sosna <i>trial</i>                        |      |
| 9M340                                        | Sosna, Sosna-R              | «Cочна»         | short-range SAM system <i>trial</i> <a href="#">UA</a>                      |      |
| 9M342                                        |                             | 9M342           | surface-to-air missile · of 9K338 Igla-S                                    |      |

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| 9M521                                     |                          | 9M521              | 122 mm extended-range HE rocket - of 9K51M Tornado-G                                  | 4    |
| 9M522                                     |                          | 9M522              | 122 mm extended-range HE rocket                                                       |      |
| 9M525                                     |                          | 9M525              | 300 mm cluster rocket - of 9K58 Smerch <i>trial</i>                                   | 6    |
| 9M528                                     |                          | 9M528              | 300 mm HE-fragmentation rocket - of 9K58 Smerch                                       | 6    |
| 9M529                                     |                          | 9M529              | 300 mm thermobaric rocket - of 9K58 Smerch <i>trial</i>                               | 6    |
| 9M531                                     |                          | 9M531              | 300 mm cluster rocket - of 9K58 Smerch <i>trial</i>                                   | 6    |
| 9M533                                     |                          | 9M533              | 300 mm cluster rocket - of 9K58 Smerch <i>trial</i>                                   | 6    |
| 9M534                                     |                          | 9M534              | 300 mm rocket with reconnaissance UAV - of 9K58 Smerch <i>trial</i>                   | 6    |
| 9M542                                     |                          | 9M542              | 300 mm guided rocket - of 9K515 Tornado-S <i>trial</i>                                | 6    |
| 9M544                                     |                          | 9M544              | 300 mm guided cluster rocket - of 9K515 Tornado-S <a href="#">(UA)</a>                | 6    |
| 9M549                                     |                          | 9M549              | 300 mm guided cluster rocket - of 9K515 Tornado-S <a href="#">(UA)</a>                | 6    |
| 9M723 9M723K1, SS-26                      | STONE                    | 9M723              | ballistic missile <a href="#">(UA)</a>                                                | 11   |
| 9M728 R-500, SSC-7                        | SOUTHPAW                 | 9M728, P-500       | cruise missile <a href="#">(UA)</a>                                                   | 11   |
| 9M729 SSC-8                               | SCREWDRIIVER             | 9M729              | cruise missile <a href="#">(UA)</a>                                                   | 11   |
| 9P78-1 9P78                               |                          | 9П78-1             | missile transporter erector launcher <a href="#">(UA)</a>                             | 11   |
| 9P122                                     | Malyutka-M               | 9П122              | ATGM launcher vehicle - of 9K11 Malyutka <i>legacy</i>                                |      |
| 9P125 BM-21V, M1975                       |                          | 9П125, 5M-21B      | 122 mm multiple rocket launcher - of 9K54 Grad-V <i>legacy</i>                        |      |
| 9P129 9P129-1M                            |                          | 9П129              | missile transporter erector launcher - of 9K79 Tochka <a href="#">(UA)</a>            |      |
| 9P132 DKZ-66, BM-21P, Grad-1P, 9K510      | Grad-P, Partizan         | 9П132              | man-portable 122 mm rocket launcher - of 9K51 Grad <i>legacy</i> <a href="#">(UA)</a> | 4    |
| 9P133                                     | Malyutka-P               | 9П133              | ATGM launcher vehicle - of 9K11 Malyutka <i>legacy</i>                                |      |
| 9P137                                     |                          |                    | ATGM launcher vehicle - of 9K111-1 Konkurs <i>legacy</i>                              |      |
| 9P138 BM-21b, M1976                       | Grad-1                   | 9П138              | 122 mm multiple rocket launcher - of 9K55 Grad-1 stored <a href="#">(UA)</a>          |      |
| 9P139                                     |                          | 9П139              | 122 mm multiple rocket launcher - of 9K55-1 Grad-1 <i>legacy</i> <a href="#">(UA)</a> | 4    |
| 9P140 BM-27, BM-22, M1977                 | Uragan                   | 9П140              | 220 mm multiple rocket launcher <a href="#">(UA)</a>                                  | 5    |
| 9P148                                     | Konkurs                  | 9П148              | ATGM launcher vehicle - of 9K111-1 Konkurs <i>legacy</i> <a href="#">(UA)</a>         |      |
| 9P149                                     | Shturm-S                 | 9П149              | ATGM launcher vehicle - of 9K114 Shturm <a href="#">(UA)</a>                          |      |
| 9P157-2 9P157                             | Khrizantema-S            | 9П157, 9П157-2     | ATGM launcher vehicle - of 9K123 Khrizantema <a href="#">(UA)</a>                     |      |
| 9P162 9K128-1, 9P163M-1                   | Kornet-T, Kornet-LR      | 9П162, 9K128-1     | ATGM launcher vehicle - of 9K135 Kornet <a href="#">(UA)</a>                          |      |
| 9P163-3 Kornet-D1                         | Kornet-D                 | 9П163-3            | ATGM launcher vehicle - of 9K135 Kornet                                               |      |
| 9P163M-1                                  | Kornet-P                 | 9П163M-1           | man-portable ATGM launcher - of 9K135 Kornet <a href="#">(UA)</a>                     |      |
| 9P163M-2 9P163-2                          | Kvartet                  | 9П163M-2           | vehicle-mounted ATGM launcher module - of 9K135 Kornet                                |      |
| 9S15M 9S15MT2E, 9S15                      | Obzor-3 BILL BOARD       | 9C15M              | SAM surveillance radar - of S-300V S-300V                                             | 13   |
| 9S18 9S18M1, 9S18M1-1, 9S18M1-2, 9S18M1-3 | Kupol SNOW DRIFT         | 9C18, 9C18M1       | SAM surveillance radar - of 9K37M1-2 Buk-M1-2 <a href="#">(UA)</a>                    | 14   |
| 9S19M2 9S19                               | Imbir HIGH SCREEN        | 9C19M2             | SAM surveillance radar - of S-300V S-300V <a href="#">(UA)</a>                        | 13   |
| 9S32 9S32M1                               | GRILL PAN                | 9C32               | fire-control radar - of S-300V S-300V <a href="#">(UA)</a>                            | 13   |
| 9S36 9S36M                                |                          | 9C36M              | fire-control radar - of 9K317 Buk-M2 <a href="#">(UA)</a>                             | 14   |
| 9S52 9S52M, 9S52M1                        | Polyana-D4, Polyana-D4M1 | Поляна-Д4, 9C52    | air defence command post                                                              | 36   |
| 9S80 9S80-1, PPRU-1, PPRU-1M              | Sborka, Ovod             |                    | air defence command post - of 9K35 Strela-10 <a href="#">(UA)</a>                     | 17   |
| 9S457                                     |                          | 9C457              | SAM command post - of S-300V S-300V                                                   | 13   |
| 9S470 9S470M1, 9S470M1-2                  |                          | 9C470              | SAM command post - of 9K37M1-2 Buk-M1-2 <a href="#">(UA)</a>                          | 14   |
| 9S510 9S510M                              |                          | 9C510M             | SAM command post - of 9K317 Buk-M2                                                    | 14   |
| 9S552                                     |                          | 9C552              | command and staff vehicle - of 9K720 Iskander-M                                       | 11   |
| 9S729M1                                   | Slepok-1                 | 9C729M1 «Сленок-1» | artillery fire-control system - of 9K58 Smerch <a href="#">(UA)</a>                   | 6    |
| 9S737                                     | Ranzhir                  |                    | air defence command post                                                              | 37   |

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| 9S920                              |                           | 9C920                                  | missile data preparation vehicle - of 9K720 Iskander-M            | 11   |
| 9S931 MP                           |                           | 9C931                                  | planning module vehicle                                           | 37   |
| 9S931-1 MP-K                       |                           | МП-K                                   | planning module vehicle                                           | 37   |
| 9S932-1 9S932-2, MRU-B             | Barnaul-T                 | 9C932-1                                | air defence command post <a href="#">UA</a>                       | 37   |
| 9S933                              |                           | ПМЮ                                    | portable fire control module - of Barnaul-T                       | 37   |
| 9S935                              |                           | KCAC                                   | squad automation complex - of Barnaul-T <a href="#">UA</a>        | 37   |
| 9S936                              |                           |                                        | command and staff vehicle - of 9K515 Tornado-S <a href="#">UA</a> | 6    |
| 9T217 9T217B                       |                           | 9T217Б                                 | transloader - of 9K33 Osa <a href="#">UA</a>                      | 17   |
| 9T218 9T228-1                      |                           | 9T218                                  | transloader - of 9K79 Tochka                                      |      |
| 9T232M                             |                           | 9T232M                                 | transloader - of 9K59 Prima legacy                                |      |
| 9T234                              |                           | 9T234                                  | transloader - of 9K58 Smerch legacy <a href="#">UA</a>            | 6    |
| 9T234-2                            |                           | 9T234-2                                | transloader - of 9K58 Smerch <a href="#">UA</a>                   | 6    |
| 9T234-2T                           |                           | 9T234-2T                               | transloader on Tatra chassis - of 9K58 Smerch export              | 6    |
| 9T234-4                            |                           | 9T234-4                                | transloader on KamAZ chassis - of 9A52-4 Kama trial               | 6    |
| 9T243M                             |                           |                                        | transloader - of 9K317M Buk-M3 <a href="#">UA</a>                 | 14   |
| 9T244                              |                           | 9T244                                  | transloader - of 9K331 Tor-M1 <a href="#">UA</a>                  | 15   |
| 9T248                              |                           |                                        | life support vehicle - of 9K720 Iskander-M <a href="#">UA</a>     | 11   |
| 9T249                              |                           | 9T249                                  | transloader - of 9K512 Uragan-1M                                  | 5    |
| 9T250 9T250E                       |                           | 9T250                                  | transloader - of 9K720 Iskander-M <a href="#">UA</a>              | 11   |
| 9T254                              |                           |                                        | ammunition carrier - of 9K51 Grad legacy                          | 4    |
| 9T255                              |                           | 9T255                                  | transloader - of 9K515 Tornado-S                                  | 6    |
| 9T450                              |                           | 9T450                                  | transloader - of 9K55 Grad-1 stored                               |      |
| 9T451                              |                           | 9T451                                  | transloader on MT-LBu chassis - of 9K55-1 Grad-1 legacy           | 4    |
| 9T452                              |                           | 9T452                                  | transloader - of 9K57 Uragan <a href="#">UA</a>                   | 5    |
| 35N6                               | Kasta                     | 35Н6 «Каста»                           | air surveillance radar <a href="#">UA</a>                         |      |
| 39N6 39N6E, Kasta-2E2              | Kasta-2-2<br>SQUAT EYE    | 39Н6 «Каста-2-2»,<br>39Н6Е «Каста-2Е2» | low-altitude detection radar <a href="#">UA</a>                   |      |
| 40N6 40N6E                         |                           | 40Н6Е                                  | surface-to-air missile - of S-400 S-400 <a href="#">UA</a>        | 12   |
| 48N6 48N6E, 48N6E2, 48N6E3, 48N6DM |                           | 48Н6Е, 48Н6Е2                          | surface-to-air missile - of S-400 S-400 <a href="#">UA</a>        | 12   |
| 48Ya6-K1 Podlet-K1                 | Podlet-K1                 | 48Я6-K1 «Погнём»                       | low-altitude detection radar <a href="#">UA</a>                   | 18   |
| 50K6A 50K6E                        |                           | 50К6А                                  | SAM command post                                                  | 13   |
| 50N6A 50N6E                        |                           | 50Н6А                                  | fire-control radar <a href="#">UA</a>                             | 13   |
| 50P6A 50P6E                        |                           | 50П6А                                  | SAM launcher <a href="#">UA</a>                                   | 13   |
| 51P6                               |                           | 51П6                                   | SAM launcher - of S-400 S-400 <a href="#">UA</a>                  | 12   |
| 55K6 55K6A                         |                           | 55К6, 55К6Е                            | SAM command post - of S-400 S-400 <a href="#">UA</a>              | 12   |
| 55Zh6M 55Zh6ME                     | Nebo-M                    | 55Ж6М «Небо-М»,<br>55Ж6МЕ              | multi-band air surveillance radar <a href="#">UA</a>              | 18   |
| 55Zh6U                             | Nebo-U<br>TALL RACK       |                                        | air surveillance radar <a href="#">UA</a>                         |      |
| 55Zh6UM                            | Nebo-UM                   |                                        | air surveillance radar                                            |      |
| 57E6 57E6-E, 57E6M                 |                           | 5736Е                                  | surface-to-air missile - of 96K6 Pantsir-S1 <a href="#">UA</a>    | 16   |
| 59N6 59N6-E                        | Protivnik-G, Protivnik-GE | Противник-ГЕ                           | 3-D air surveillance radar                                        |      |
| 64N6 64N6E                         | BIG BIRD                  |                                        | target acquisition radar - of S-300PM S-300PM                     |      |
| 67N6 67N6E                         | Gamma-DE                  | Гамма-ДЕ                               | UHF 3-D air surveillance radar export                             |      |
| 77Ya6-DM                           | Voronezh-DM               | «Воронеж-ДМ»                           | early warning radar <a href="#">UA</a>                            |      |
| 85Ya6                              | Leer-2                    | 85Я6 «Леер-2»                          | communications jammer <a href="#">UA</a>                          | 24   |
| 91N6A 91N6E                        |                           | 91Н6Е                                  | air surveillance radar - of S-400 S-400 <a href="#">UA</a>        | 12   |
| 92N6A 92N6E                        |                           | 92Н6Е                                  | fire-control radar - of S-400 S-400 <a href="#">UA</a>            | 12   |
| 96K6 SA-22                         | Pantsir-S1<br>GREYHOUND   | 96К6, Панцирь-С1                       | gun-missile air defence system <a href="#">UA</a>                 | 16   |
| 96L6 96L6E, 96L6-TsP               | CHEESE BOARD              | 96Л6Е                                  | air surveillance radar <a href="#">UA</a>                         | 18   |
| AGS-17                             | Plamya                    | АГС-17, Пламя                          | 30 mm automatic grenade launcher <a href="#">UA</a>               |      |
| AGS-30                             |                           | АГС-30                                 | 30 mm automatic grenade launcher                                  |      |
| AGS-40                             | Balkan                    | АГС-40, Балкан                         | 40 mm automatic grenade launcher                                  |      |
| Akatsiya-E                         | Akatsiya-E                |                                        | air defence command and control complex                           |      |
| Akatsiya-M                         | Akatsiya-M                |                                        | command and control automation system                             |      |
| Altaets                            | Altaets, Altaets-AM       | «Алмаец»                               | communications jammer - of R-330M1P Diabazol                      |      |
| Alurgit                            | Alurgit                   |                                        | GNSS jammer                                                       |      |

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| AMN-233114 GAZ-233114            | Tigr-M                   | Тигр-М, АМН-233114    | armoured car <a href="#">UA</a>                                                | 24   |
| AMN-590951                       | VPK Ural                 | АМН-590951, ВПК-Урал  | armoured car <a href="#">UA</a>                                                |      |
| Andromeda-D                      | Andromeda-D              |                       | command and control system                                                     |      |
| APA-80                           |                          |                       | engine starting vehicle <a href="#">UA</a>                                     |      |
| Argus-Antidron                   | Argus-Antidron           |                       | UAV jammer                                                                     |      |
| Armata                           | Armata                   | Армата                | tracked armoured platform <a href="#">trial</a> <a href="#">UA</a>             |      |
| ARS-14 ARS-14KM                  |                          | АРС-14KM              | decontamination vehicle <a href="#">UA</a>                                     |      |
| AS-4350.1                        |                          |                       | ambulance <a href="#">UA</a>                                                   |      |
| AShT-4310                        |                          |                       | topographic command and staff vehicle <a href="#">UA</a>                       |      |
| ASN-233115                       | Tigr-M SpN               | АШН-233115            | armoured car · of AMN-233114 Tigr-M <a href="#">UA</a>                         | 24   |
| Asteis-7020                      | Patrul                   | Патруль               | armoured car <a href="#">UA</a>                                                |      |
| ASUNO-S 1V168                    |                          |                       | artillery fire control system                                                  |      |
| ATO-40                           |                          |                       | repair vehicle · of RB-301B Borisoglebsk-2                                     | 23   |
| ATS-59G                          |                          | АТС-59Г               | tracked artillery tractor <a href="#">legacy</a> <a href="#">UA</a>            |      |
| Auriga-1.2V                      |                          |                       | satellite communications terminal <a href="#">UA</a>                           |      |
| AZK-7                            | AZK-7                    |                       | acoustic artillery reconnaissance system                                       |      |
| B-4 <a href="#">M1931</a>        |                          | Б-4                   | 203 mm towed howitzer <a href="#">legacy</a>                                   |      |
| Barnaul-T                        | Barnaul-T                | Барнаул-Т             | air defence command and control system <a href="#">UA</a>                      | 37   |
| BAT-2 Object 454                 |                          | БАТ-2                 | route-clearance vehicle <a href="#">UA</a>                                     |      |
| BAT-M                            |                          |                       | route-clearance vehicle <a href="#">legacy</a>                                 |      |
| BAZ-6909 BAZ-69092               |                          |                       | heavy truck chassis <a href="#">UA</a>                                         |      |
| BAZ-69092                        |                          |                       | transloader · of S-400 S-400 <a href="#">UA</a>                                | 12   |
| BLA-07                           |                          |                       | reconnaissance UAV                                                             |      |
| BM-1                             |                          | БМ-1                  | 220 mm heavy flamethrower launcher <a href="#">UA</a>                          | 7    |
| BM-13                            | Katyusha                 | БМ-13                 | 132 mm multiple rocket launcher <a href="#">stored</a>                         |      |
| BM-21 2B5, <a href="#">M1964</a> |                          | БМ-21, 2Б5            | 122 mm multiple rocket launcher <a href="#">UA</a>                             | 4    |
| BM-21-1 2B17, 2B17-1             |                          | БМ-21-1, 2Б17         | 122 mm multiple rocket launcher <a href="#">UA</a>                             |      |
| BM-35                            | Italmas                  | Италмас               | one-way attack drone <a href="#">UA</a>                                        |      |
| BMD-1                            |                          | БМД-1                 | airborne fighting vehicle <a href="#">stored</a> <a href="#">UA</a>            |      |
| BMD-1KSh-A                       |                          |                       | armoured command vehicle <a href="#">UA</a>                                    |      |
| BMD-2                            |                          | БМД-2                 | airborne fighting vehicle <a href="#">UA</a>                                   |      |
| BMD-3 Object 950                 |                          | БМД-3, Объект 950     | airborne fighting vehicle <a href="#">legacy</a>                               |      |
| BMD-4                            |                          | БМД-4                 | airborne fighting vehicle <a href="#">legacy</a> <a href="#">UA</a>            |      |
| BMD-4M                           |                          | БМД-4М                | airborne fighting vehicle · of BMD-4 <a href="#">UA</a>                        |      |
| BMK-130M BMK-150                 |                          |                       | towing and motor boat <a href="#">UA</a>                                       |      |
| BMK-460                          |                          | БМК-460               | towing and motor boat <a href="#">UA</a>                                       |      |
| BMK-MO                           |                          |                       | towing and motor boat <a href="#">UA</a>                                       |      |
| BMK-MT                           |                          |                       | towing and motor boat <a href="#">UA</a>                                       |      |
| BMO-T Object 564                 |                          | БМО-Т, Объект 564     | armoured personnel carrier <a href="#">UA</a>                                  |      |
| BMP-1 Object 765, BMP Model 1970 |                          | БМП-1, Объект 765     | infantry fighting vehicle <a href="#">UA</a>                                   |      |
| BMP-1AM                          | Basurmanin               | БМП-1АМ, Басурманин   | infantry fighting vehicle · of BMP-1 <a href="#">UA</a>                        |      |
| BMP-1KSh BMP-1KShM               |                          |                       | command and staff vehicle <a href="#">UA</a>                                   | 35   |
| BMP-1P                           |                          | БМП-1П                | infantry fighting vehicle · of BMP-1 <a href="#">UA</a>                        |      |
| BMP-1U                           | Shkval                   | БМП-1У, Шквал         | infantry fighting vehicle · of BMP-1 <a href="#">legacy</a> <a href="#">UA</a> |      |
| BMP-2 Object 675                 |                          | БМП-2, Объект 675     | infantry fighting vehicle <a href="#">UA</a>                                   |      |
| BMP-2D                           |                          | БМП-2Д                | infantry fighting vehicle · of BMP-2 <a href="#">UA</a>                        |      |
| BMP-2M                           | Berezhok                 | БМП-2М, Бережок       | infantry fighting vehicle · of BMP-2 <a href="#">UA</a>                        |      |
| BMP-3 Object 688                 |                          | БМП-3, Объект 688     | infantry fighting vehicle <a href="#">UA</a>                                   |      |
| BMP-3F                           |                          | БМП-3Ф                | infantry fighting vehicle · of BMP-3                                           |      |
| BMP-3M                           |                          | БМП-3М                | infantry fighting vehicle · of BMP-3                                           |      |
| BMPT                             | Terminator               | БМПТ, Терминатор      | tank support fighting vehicle <a href="#">UA</a>                               |      |
| BMPT-72 Object 183               | Terminator-2, Terminator | БМПТ-72, Терминатор-2 | tank support fighting vehicle · of BMPT Terminator <a href="#">UA</a>          |      |
| BMR-3M BMR-3MA, Object 197       | Vepr                     | БМР-3М, Вепрь         | mine-clearing vehicle <a href="#">UA</a>                                       |      |
| BRDM-2 GAZ-41-08                 |                          | БРДМ-2, ГАЗ-41-08     | armoured reconnaissance vehicle <a href="#">UA</a>                             |      |
| BRDM-2A                          |                          | БРДМ-2А               | armoured reconnaissance vehicle · of BRDM-2 <a href="#">UA</a>                 |      |
| BRDM-2M                          |                          | БРДМ-2М               | armoured reconnaissance vehicle · of BRDM-2 <a href="#">UA</a>                 |      |

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| BRDM-2MS                                    |                              | БРДМ-2МС             | armoured reconnaissance vehicle - of BRDM-2 <a href="#">UA</a>           |      |
| BRDM-2RKhb BRDM-2RKh                        |                              |                      | CBRN reconnaissance vehicle <a href="#">UA</a>                           |      |
| BREM-1 Object 608                           |                              | БРЭМ-1               | armoured recovery vehicle <a href="#">UA</a>                             |      |
| BREM-2                                      |                              | БРЭМ-2               | armoured recovery vehicle <a href="#">UA</a>                             |      |
| BREM-4                                      | BREM-Ch                      |                      | armoured recovery vehicle <a href="#">UA</a>                             |      |
| BREM-64                                     |                              |                      | armoured recovery vehicle                                                |      |
| BREM-D                                      | BREhM-D                      | БРЭМ-Д               | armoured recovery vehicle <a href="#">UA</a>                             |      |
| BREM-K                                      |                              | БРЭМ-К               | armoured recovery vehicle <a href="#">UA</a>                             |      |
| BREM-L                                      | Beglyanka, Beglianka         | БРЭМ-Л               | armoured recovery vehicle <a href="#">UA</a>                             |      |
| BRM-1K BRM-1KM, Object 676, BMP M1976/2     |                              | БРМ-1К, БРМ-1        | armoured reconnaissance vehicle <a href="#">UA</a>                       |      |
| BRM-3K Object 501                           | Rys                          | БРМ-3К, Рысь         | armoured reconnaissance vehicle                                          | 19   |
| BS-3 M1944                                  |                              | БС-3                 | 100 mm field gun stored <a href="#">UA</a>                               |      |
| BTM-3                                       |                              | БТМ-3                | trench digger <a href="#">UA</a>                                         |      |
| BTR-50                                      |                              | БТР-50               | tracked armoured personnel carrier legacy <a href="#">UA</a>             |      |
| BTR-50PK                                    |                              | БТР-50ПК             | tracked armoured personnel carrier - of BTR-50 legacy <a href="#">UA</a> |      |
| BTR-60PB BTR-60                             |                              | БТР-60ПБ, БТР-60     | armoured personnel carrier legacy <a href="#">UA</a>                     | 34   |
| BTR-70                                      |                              | БТР-70               | armoured personnel carrier <a href="#">UA</a>                            |      |
| BTR-70M                                     |                              | БТР-70М              | armoured personnel carrier - of BTR-70 <a href="#">UA</a>                |      |
| BTR-80 GAZ-5903                             |                              | БТР-80, ГАЗ-5903     | armoured personnel carrier <a href="#">UA</a>                            |      |
| BTR-80A                                     |                              | БТР-80А              | armoured personnel carrier - of BTR-80                                   |      |
| BTR-82                                      |                              | БТР-82               | armoured personnel carrier - of BTR-80 <a href="#">UA</a>                |      |
| BTR-82A BTR-82AM                            |                              | БТР-82А, БТР-82АМ    | armoured personnel carrier - of BTR-80 <a href="#">UA</a>                |      |
| BTR-82AT                                    |                              | БТР-82АТ             | armoured personnel carrier - of BTR-80 <a href="#">UA</a>                |      |
| BTR-90 GAZ-5923                             | Rostok                       | БТР-90, ГАЗ-5923     | armoured personnel carrier <a href="#">UA</a>                            |      |
| BTR-D                                       |                              | БТР-Д                | armoured personnel carrier <a href="#">UA</a>                            |      |
| BTR-MDM                                     | Rakushka                     | БТР-МДМ, Пакушка     | armoured personnel carrier <a href="#">UA</a>                            |      |
| BTR-T                                       |                              |                      | heavy armoured personnel carrier legacy                                  |      |
| BTS-4A                                      |                              | БТС-4А               | armoured recovery vehicle <a href="#">UA</a>                             |      |
| BTS-4V                                      |                              | БТС-4В               | armoured recovery vehicle <a href="#">UA</a>                             |      |
| Buran                                       | Buran, RIDA Buran, Buran ASV |                      | armoured utility vehicle <a href="#">UA</a>                              |      |
| Chistyulya                                  | Chistyulya                   | Чистюля              | UAV jammer <a href="#">UA</a>                                            |      |
| D-1 M1943                                   |                              | Д-1                  | 152 mm towed howitzer stored <a href="#">UA</a>                          |      |
| D-20                                        |                              | Д-20                 | 152 mm towed gun-howitzer <a href="#">UA</a>                             |      |
| D-30 2A18, D-30A, 2A18M                     |                              | Д-30, Д-30А          | 122 mm towed howitzer <a href="#">UA</a>                                 |      |
| D-44                                        |                              | Д-44                 | 85 mm towed gun legacy <a href="#">UA</a>                                |      |
| DIM                                         |                              |                      | vehicle-mounted mine detector legacy                                     |      |
| DShK 38/46                                  | DShK, DShKM                  | ДШК                  | 12.7 mm heavy machine gun legacy <a href="#">UA</a>                      |      |
| DT-10PM                                     | Vityaz                       |                      | articulated tracked carrier <a href="#">UA</a>                           | 13   |
| DT-30 DT-30PM, DT-10PM                      | Vityaz                       | ДТ-30, ДТ-30ПМ       | articulated tracked carrier <a href="#">UA</a>                           | 13   |
| Eleron-3 Eleron-3SV                         |                              | Элерон-3, Элерон-3СВ | reconnaissance UAV <a href="#">UA</a>                                    |      |
| Eleron T-16                                 |                              |                      | reconnaissance UAV <a href="#">UA</a>                                    |      |
| EOV-3521                                    |                              |                      | military excavator <a href="#">UA</a>                                    |      |
| EOV-3523                                    |                              |                      | military excavator <a href="#">UA</a>                                    |      |
| EOV-4421                                    |                              |                      | military excavator <a href="#">UA</a>                                    |      |
| ESU TZ                                      | YeSU TZ                      | ЕСУ ТЗ               | tactical-level command and control system                                |      |
| Fateh-360                                   | Fath-360                     |                      | short-range ballistic missile                                            |      |
| Forpost                                     | Forpost, Searcher II         | Форпост              | reconnaissance UAV <a href="#">UA</a>                                    |      |
| Forpost-R                                   | Forpost-R                    |                      | reconnaissance and strike UAV <a href="#">UA</a>                         |      |
| Garpun-3                                    | Garpun-3, Harpoon-3          | Гарпун-3             | UAV jammer <a href="#">UA</a>                                            |      |
| GAZ-51                                      |                              | ГАЗ-51               | cargo truck 4x2 legacy <a href="#">UA</a>                                |      |
| GAZ-66                                      |                              | ГАЗ-66               | cargo truck 4x4 legacy <a href="#">UA</a>                                |      |
| GAZ-3307                                    |                              | ГАЗ-3307             | cargo truck 4x2 <a href="#">UA</a>                                       |      |
| GAZ-3308 GAZ-3308I                          | Sadko, Eger-II               | ГАЗ-3308 «Cagko»     | cargo truck 4x4 <a href="#">UA</a>                                       |      |
| GAZ-3344-20                                 | Aleut                        |                      | articulated tracked carrier <a href="#">UA</a>                           |      |
| GAZ-39371                                   | Vodnik                       |                      | armoured utility vehicle <a href="#">UA</a>                              |      |
| GAZ-233014 GAZ-2330, VPK-233136, GAZ-233036 | Tigr                         | ГАЗ-233014, ГАЗ-2330 | armoured car <a href="#">UA</a>                                          |      |

| DESIGNATION                                 | NAME                                 | CYRILLIC              | WHAT IT IS                                                     | PAGE |
|---------------------------------------------|--------------------------------------|-----------------------|----------------------------------------------------------------|------|
| GAZ-233036 SPM-2                            | Tigr SPM-2                           | ГАЗ-233036, СПМ-2     | armoured car · of GAZ-233014 Tigr <a href="#">UA</a>           |      |
| Geran-1 Shahed-131, Shahed 131              | Geran-1                              | Герань-1              | one-way attack drone <a href="#">UA</a>                        |      |
| Geran-2 Shahed-136, Shahed 136              | Geran-2                              | Герань-2              | one-way attack drone <a href="#">UA</a>                        |      |
| Geran-3                                     | Geran-3                              |                       | one-way attack drone <i>trial</i> <a href="#">UA</a>           |      |
| Gerbera                                     | Gerbera                              | Гербера               | one-way attack drone <a href="#">UA</a>                        |      |
| GMZ-3 Object 318                            |                                      | ГМЗ-3                 | minelayer <a href="#">UA</a>                                   | 33   |
| Granat-1                                    |                                      | Гранат-1              | reconnaissance UAV <a href="#">UA</a>                          |      |
| Granat-4                                    |                                      | Гранат-4              | reconnaissance UAV <a href="#">UA</a>                          |      |
| Groza.04.K                                  | Groza-04K                            | Гроза.04.K            | UAV jammer <a href="#">UA</a>                                  |      |
| GSP Object 55                               |                                      | ГСП                   | ferry <i>legacy</i>                                            |      |
| GT-01                                       | Murmansk-BN                          | Мурманск-БН           | HF communications jammer <a href="#">UA</a>                    | 22   |
| Harpya                                      | Harpya                               |                       | UAV jammer                                                     |      |
| IMR IMR-1                                   |                                      |                       | combat engineer vehicle <i>legacy</i> <a href="#">UA</a>       |      |
| IMR-2 IMR-2M, IMR-2M1, Object 637           |                                      | ИМР-2, ИМР-2М         | combat engineer vehicle <a href="#">UA</a>                     | 32   |
| IMR-3 IMR-3M                                |                                      | ИМР-3М                | combat engineer vehicle <a href="#">UA</a>                     | 32   |
| Irbis                                       | Irbis                                |                       | counter-battery radar                                          |      |
| IRM                                         | Zhuk                                 |                       | engineer reconnaissance vehicle <a href="#">UA</a>             |      |
| ISDM                                        | Zemledeliye                          | ИСДМ, Земледелие      | truck-mounted remote mining system <a href="#">UA</a>          | 33   |
| Izdeliye 51 Izdeliye 52                     | Lancet, Lancet-3                     | Ланцет, Изделие 51/52 | loitering munition <a href="#">UA</a>                          |      |
| Izdeliye 53                                 |                                      |                       | loitering munition · of Izdeliye 51 Lancet <a href="#">UA</a>  |      |
| K1Sh1 GAZ-59032                             | UNSh                                 | К1Ш1                  | command vehicle chassis [BTR-80]                               |      |
| K155M                                       | Krasnopol-M2                         |                       | guided artillery round · of 30F39 Krasnopol <a href="#">UA</a> |      |
| KamAZ-4310                                  |                                      | КАМАЗ-4310            | cargo truck 6x6 <i>legacy</i>                                  |      |
| KamAZ-4386                                  | Typhoon-VDV                          |                       | mine-resistant ambush-protected vehicle <a href="#">UA</a>     |      |
| KamAZ-5350                                  | Mustang                              | КАМАЗ-5350            | cargo truck 6x6 <a href="#">UA</a>                             |      |
| KamAZ-6350                                  |                                      | КАМАЗ-6350            | cargo truck 8x8 <a href="#">UA</a>                             |      |
| KamAZ-6560                                  |                                      | КАМАЗ-6560            | heavy truck chassis 8x8                                        |      |
| KamAZ-43269 BPM-97                          | Vystrel                              | КАМАЗ-43269, БПМ-97   | armoured car <a href="#">UA</a>                                |      |
| KamAZ-53501                                 |                                      | КАМАЗ-53501           | cargo truck 6x6 <a href="#">UA</a>                             |      |
| KamAZ-53949 Typhoon-K 4x4                   | Typhoon-K                            | КАМАЗ-53949, Тайфун-К | mine-resistant ambush-protected vehicle <a href="#">UA</a>     |      |
| KamAZ-53949                                 | Linza                                | КАМАЗ-53949, Линза    | armoured ambulance <a href="#">UA</a>                          |      |
| KamAZ-63501                                 |                                      | КАМАЗ-63501           | cargo truck 8x8 <a href="#">UA</a>                             |      |
| KamAZ-63968 Typhoon-K 6x6                   | Typhoon-K                            | КАМАЗ-63968, Тайфун-К | mine-resistant ambush-protected vehicle <a href="#">UA</a>     |      |
| KamAZ-65225                                 |                                      | КАМАЗ-65225           | tractor truck 6x6                                              |      |
| Kapustnik-BM                                | Kapustnik-BM                         | Капустник-БМ          | artillery fire-control system · of 9K51M Tornado-G             | 4    |
| Kh-22 AS-4                                  | Burya <a href="#">KITCHEN</a>        | Х-22 «Буря»           | air-launched anti-ship missile <a href="#">UA</a>              |      |
| Kh-32                                       |                                      | Х-32                  | air-launched cruise missile <a href="#">UA</a>                 |      |
| Kh-47M2 AS-24                               | Kinzhal <a href="#">KILLJOY</a>      | Х-47М2, Кинжал        | air-launched ballistic missile <a href="#">UA</a>              |      |
| Kh-55 Kh-55SM, AS-15                        | <a href="#">KENT</a>                 | Х-55                  | air-launched cruise missile <a href="#">UA</a>                 |      |
| Kh-59 Kh-59M, AS-13, AS-18                  | Ovod <a href="#">KINGBOLT, KAZOO</a> | Х-59 «Овод»           | air-launched cruise missile <a href="#">UA</a>                 |      |
| Kh-69                                       |                                      | Х-69                  | air-launched cruise missile <a href="#">UA</a>                 |      |
| Kh-101 AS-23A                               | <a href="#">KODIAK</a>               | Х-101                 | air-launched cruise missile <a href="#">UA</a>                 |      |
| KM-8                                        | Gran                                 | КМ-8, «Грань»         | guided mortar bomb <a href="#">UA</a>                          |      |
| KMM                                         |                                      |                       | bridgelayer <i>legacy</i>                                      |      |
| KMT-5                                       |                                      |                       | mine plough and roller set                                     |      |
| KMT-7                                       |                                      |                       | mine roller                                                    |      |
| KMT-8                                       |                                      |                       | mine plough                                                    |      |
| KMT-10                                      |                                      |                       | mine roller                                                    |      |
| KN-23 Hwasong-11A                           |                                      |                       | short-range ballistic missile <a href="#">UA</a>               |      |
| Korsar                                      | Korsar                               | Корсар                | reconnaissance UAV <a href="#">UA</a>                          |      |
| KPV KPVT                                    |                                      | КПВ                   | 14.5 mm heavy machine gun <a href="#">UA</a>                   |      |
| KrAZ-255 KrAZ-255B                          |                                      | КраЗ-2555             | cargo truck 6x6 <i>legacy</i> <a href="#">UA</a>               |      |
| KrAZ-260                                    |                                      | КраЗ-260              | cargo truck 6x6 <i>legacy</i> <a href="#">UA</a>               |      |
| Kredo-1E Credo-1E, RP-200, 1RL133-3, PSNR-8 | Kredo-1, Credo-1                     | 1РА133-3 «Крего-1»    | ground surveillance radar                                      |      |
| Kredo-M1 PSNR-8M, PSNR-9                    | Kredo-M1                             |                       | portable ground surveillance radar <a href="#">UA</a>          |      |
| KRUS-VR                                     | Strelets, Strelets-M                 | Стрелец, Стрелец-М    | soldier command and control terminal <a href="#">UA</a>        |      |

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|----------------------------------------|--------------------|---------------------|-------------------------------------------------------------------------------------|------|
| KRVD                                   |                    |                     | temporary road deployment kit <a href="#">UA</a>                                    |      |
| KT-L                                   |                    |                     | recovery vehicle <a href="#">UA</a>                                                 |      |
| KUB-BLA                                | Kub-BLA            | Куб-БЛА             | loitering munition <a href="#">UA</a>                                               |      |
| Kurganets-25                           | Kurganets-25       | Курганец-25         | infantry fighting vehicle <i>trial</i>                                              |      |
| KZKT-7428                              | Rusich             |                     | tank transporter <i>legacy</i>                                                      |      |
| Ladoga                                 | Ladoga, VTS Ladoga |                     | heavy command and reconnaissance vehicle <a href="#">UA</a>                         |      |
| Lancet launcher                        | Lancet launcher    |                     | UAV control station - of Izdeliye 51 Lancet <a href="#">UA</a>                      |      |
| LPD-801                                |                    |                     | UAV jammer <a href="#">UA</a>                                                       |      |
| M-30 M-1938                            |                    | M-30                | 122 mm towed howitzer <i>stored</i> <a href="#">UA</a>                              |      |
| M-46                                   |                    | M-46                | 130 mm towed gun <a href="#">UA</a>                                                 |      |
| M-160 M1949, 52-M-853                  |                    | M-160               | 160 mm mortar <i>stored</i>                                                         |      |
| M-1938 PM-38                           |                    | ПМ-38               | 120 mm mortar <i>stored</i>                                                         |      |
| M1977                                  |                    |                     | armoured recovery vehicle                                                           | 5    |
| MAZ-537                                |                    | MA3-537             | tank transporter <i>legacy</i> <a href="#">UA</a>                                   |      |
| MAZ-543 MAZ-7310                       |                    | MA3-543/MA3-7310    | heavy truck chassis 8x8 <i>legacy</i>                                               |      |
| MAZ-63172                              |                    |                     | tractor truck <a href="#">UA</a>                                                    |      |
| MAZ-74101                              | Oplot-T            |                     | tractor truck <a href="#">UA</a>                                                    |      |
| MAZ-531605                             |                    |                     | cargo truck <a href="#">UA</a>                                                      |      |
| MDK-3                                  |                    | МДК-3               | trench digger <a href="#">UA</a>                                                    |      |
| Merlin-VR                              | Merlin-VR          |                     | reconnaissance UAV <a href="#">UA</a>                                               |      |
| MKTK-1                                 | Plavsk             | «Плавск»            | radio monitoring station                                                            |      |
| MKTK-1A                                | Dzyudoist, Judoist | «Дзюдоист»          | SIGINT system <a href="#">UA</a>                                                    |      |
| ML-20 M-1937                           |                    | MA-20               | 152 mm towed gun-howitzer <i>stored</i> <a href="#">UA</a>                          |      |
| MLZh-VF-VT                             | MLZh-VF-VT         |                     | railway pontoon bridge <a href="#">UA</a>                                           |      |
| MO.1.01.04                             |                    |                     | 220 mm thermobaric rocket - of TOS-1A Sointsepek                                    | 7    |
| MO.1.01.04M                            |                    |                     | 220 mm thermobaric rocket, extended range - of TOS-1A Sointsepek <a href="#">UA</a> | 7    |
| MO.1.01.04M2                           |                    |                     | 220 mm thermobaric rocket, heavier warhead - of TOS-2 Tosochka                      | 7    |
| Mohajer-6                              |                    |                     | reconnaissance and strike UAV <a href="#">UA</a>                                    |      |
| Molniya-1 Molniya-2                    | Molniya            |                     | FPV drone <a href="#">UA</a>                                                        |      |
| MON-50                                 |                    | МОН-50              | directional anti-personnel mine <a href="#">UA</a>                                  |      |
| MON-90                                 |                    | МОН-90              | directional anti-personnel mine <a href="#">UA</a>                                  |      |
| MON-100                                |                    |                     | directional anti-personnel mine <a href="#">UA</a>                                  |      |
| MON-200                                |                    |                     | directional anti-personnel mine <a href="#">UA</a>                                  |      |
| MP-32M1 MP32M1                         |                    |                     | command post vehicle - of 9S729M1 Slepok-1 <a href="#">UA</a>                       |      |
| MRO-A                                  | Borodach           |                     | 72.5 mm disposable thermobaric launcher <a href="#">UA</a>                          |      |
| MRT0                                   | MRT0               | MPT0                | maintenance vehicle - of 9K720 Iskander-M                                           | 11   |
| MT-12 2A29, T-12A                      | Rapira             | MT-12, 2A29         | 100 mm anti-tank gun <a href="#">UA</a>                                             |      |
| MT-12R 2A29R                           | Ruta               |                     | 100 mm anti-tank gun - of MT-12 Rapira <i>legacy</i>                                |      |
| MT-55A                                 |                    |                     | bridgelayers                                                                        |      |
| MT-LB                                  |                    | MT-ЛБ               | tracked armoured transporter <a href="#">UA</a>                                     |      |
| MT-LBM 6MA MT-LBM-6MA, Object 6MA      |                    | Объект 6МА          | tracked armoured transporter - of MT-LB <a href="#">UA</a>                          |      |
| MT-LBM 6MB MT-LBM-6MB, Object 6MB      |                    | Объект 6МБ          | tracked armoured transporter - of MT-LB <a href="#">UA</a>                          |      |
| MT-LBV                                 |                    | MT-ЛБВ              | tracked armoured transporter - of MT-LB <i>legacy</i> <a href="#">UA</a>            |      |
| MT-LBVM MT-LBVMK, MT-LB VM1K, MT-LBVMK |                    | MT-ЛБВМ, MT-ЛБВМК   | tracked armoured transporter - of MT-LB <a href="#">UA</a>                          |      |
| MTK-2 MTK                              |                    |                     | mine-clearing vehicle <a href="#">UA</a>                                            |      |
| MT0-AT                                 |                    |                     | repair vehicle <a href="#">UA</a>                                                   |      |
| MT0-UB2 MT0-UB                         |                    |                     | repair vehicle <a href="#">UA</a>                                                   |      |
| MTP-A2.1 MTP-A2                        |                    |                     | recovery vehicle <a href="#">UA</a>                                                 |      |
| MTP-LB                                 |                    | МТП-ЛБ              | repair vehicle                                                                      |      |
| MTU-20 Object 602                      |                    | MTY-20              | bridgelayers <a href="#">UA</a>                                                     |      |
| MTU-72 Object 632                      |                    | MTY-72              | bridgelayers <a href="#">UA</a>                                                     | 31   |
| MTU-90                                 |                    | MTY-90              | bridgelayers                                                                        |      |
| MZKT-7930                              | Astrolog           | МЗКТ-7930, Астролог | heavy truck chassis 8x8                                                             | 16   |
| Nebo-T                                 | Nebo-T             |                     | air surveillance radar                                                              |      |
| NSV NSVT, NSVS                         | Utyos              | НСВ, Умёс           | 12.7 mm heavy machine gun <a href="#">UA</a>                                        |      |

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| Object 10                                   | MT-LBu                       | MT-Абу                   | tracked armoured transporter · of MT-LB <a href="#">UA</a>    |      |
| OF-843B 53-OF-843B, 30F843B                 |                              | 53-0Ф-843Б               | mortar bomb                                                   |      |
| Oreshnik RS-SS-28 MOD                       | Oreshnik                     | Орешник                  | intermediate-range ballistic missile <a href="#">UA</a>       |      |
| Orion                                       | Orion, Inokhodets            | Орион, Иноходец          | reconnaissance and strike UAV <a href="#">UA</a>              |      |
| Orlan-10                                    |                              | Орлан-10                 | reconnaissance UAV <a href="#">UA</a>                         |      |
| Orlan-30                                    |                              | Орлан-30                 | reconnaissance UAV <a href="#">UA</a>                         |      |
| OZM-72                                      | Lyagushka, Vedma             | ОЗМ-72                   | bounding anti-personnel mine <a href="#">UA</a>               |      |
| P-18 1RL131, P-18M                          | Terek<br>SPOON REST D        | П-18 «Терек», 1РЛ131     | early warning radar <a href="#">UA</a>                        |      |
| P-19 1RL134                                 | Dunai<br>FLAT FACE B         | П-19 «Дунай», 1РЛ134     | air surveillance radar                                        |      |
| P-35 P-37, 1RL110, 1RL139                   | Saturn, Mech<br>BAR LOCK     | П-35 «Сатурн»            | early warning radar <a href="#">UA</a>                        |      |
| P-240 P-240I-4                              | Pereselenets, Pereselenetz   |                          | digital communications vehicle <a href="#">UA</a>             |      |
| P-260-U P-260                               |                              |                          | signals vehicle · of Redut-2US <a href="#">UA</a>             |      |
| P-800 3M55, SS-N-26                         | Oniks<br>STROBILE            | П-800 «Онукс»            | anti-ship cruise missile · of 3K55 Bastion <a href="#">UA</a> |      |
| PAK-200 PAK-200M                            |                              |                          | field kitchen <a href="#">UA</a>                              |      |
| Palantin                                    | Palantin, Palantin-K         | «Палантин», «Палантин-К» | communications jammer <a href="#">UA</a>                      | 25   |
| Pantsir-S1M                                 | Pantsir-SM, Pantsir-S1M      | Панцирь-СМ               | gun-missile air defence system <a href="#">UA</a>             | 16   |
| Pantsir-S2 SA-22                            | Pantsir-S2<br>GREYHOUND      | Панцирь-С2               | gun-missile air defence system <a href="#">UA</a>             | 16   |
| Pantsir-SM-SV                               | Pantsir-SM-SV                | Панцирь-СМ-СВ            | gun-missile air defence system                                | 16   |
| Pantsir-SMD-E                               | Pantsir-SMD-E                | Панцирь-СМД-Е            | short-range SAM system                                        |      |
| Parodist                                    | Parodist                     |                          | radar and communications jammer <a href="#">trial</a>         |      |
| PAZ-32053                                   |                              |                          | bus <a href="#">UA</a>                                        |      |
| PDM-1                                       |                              |                          | anti-landing mine <a href="#">legacy</a>                      |      |
| PFM-1 PFM-1S                                | Lepestok                     | ПФМ-1, ПФМ-1С            | scatterable anti-personnel mine <a href="#">UA</a>            |      |
| PKM                                         |                              |                          | man-portable minelaying kit                                   |      |
| Plastun-SN                                  | Plastun-SN                   |                          | light tracked all-terrain vehicle <a href="#">UA</a>          |      |
| PMM-2 PMM-2M                                |                              | ПММ-2, PMM-2M            | amphibious ferry                                              | 31   |
| PMN PMN-1, PMN-2, PMN-4                     |                              | ПМН                      | anti-personnel blast mine <a href="#">UA</a>                  |      |
| PMP PMP-M                                   |                              | ПМП, ПМП-М               | heavy folding pontoon bridge <a href="#">UA</a>               | 30   |
| PMR-3                                       |                              |                          | towed minelayer <a href="#">legacy</a>                        |      |
| PMZ-4                                       |                              |                          | towed minelayer                                               |      |
| POM-2 POM-2S                                |                              | ПОМ-2                    | scatterable anti-personnel mine <a href="#">UA</a>            |      |
| POM-2R                                      |                              | ПОМ-2Р                   | scatterable anti-personnel mine · of POM-2 <a href="#">UA</a> |      |
| POM-3                                       | Medalyon                     |                          | scatterable anti-personnel mine <a href="#">UA</a>            |      |
| PP-91                                       |                              | ПП-91                    | pontoon park <a href="#">legacy</a> <a href="#">UA</a>        | 30   |
| PP-91M PP-2005                              |                              | ПП-91М, ПП-2005          | pontoon park <a href="#">UA</a>                               | 30   |
| PPS-84                                      |                              | ППС-84                   | pontoon park <a href="#">legacy</a>                           |      |
| Predel-E                                    | Predel-E                     |                          | coastal defence radar <a href="#">UA</a>                      |      |
| Prokhod-1                                   | Prokhod-1                    |                          | remote-controlled mine-clearing system <a href="#">UA</a>     |      |
| PRP-3 1Zh3, Object 767                      | Val<br>SMALL FRED (disputed) | ПРП-3 «Вал»              | artillery reconnaissance vehicle <a href="#">UA</a>           |      |
| PRP-4 1V121, 1V125, Object 779              | Nard                         | ПРП-4 Нард               | artillery reconnaissance vehicle <a href="#">UA</a>           | 35   |
| PRP-4A                                      | Argus                        | ПРП-4А «Аргус»           | artillery reconnaissance vehicle <a href="#">UA</a>           | 36   |
| PRP-4M PRP-4MU, 1V145                       | Deyteriy                     | ПРП-4М «Дейтерий»        | artillery reconnaissance vehicle <a href="#">UA</a>           | 36   |
| PSNR-5 PSNR-5M, 1RL133                      | Kredo-M<br>TALL MIKE         |                          | ground surveillance radar · of PRP-4M Deyteriy                | 36   |
| PT-76                                       |                              | ПТ-76                    | light amphibious tank <a href="#">legacy</a>                  |      |
| PTM-1 PTM-1S, PTM-1G                        |                              | ПТМ-1                    | scatterable anti-tank mine <a href="#">UA</a>                 |      |
| PTM-3 KPTM-3                                |                              | ПТМ-3                    | scatterable anti-tank mine <a href="#">UA</a>                 |      |
| PTM-4                                       | Nokaut-2                     | ПТМ-4, Нокгайт-2         | scatterable anti-tank mine <a href="#">UA</a>                 |      |
| PTS-2 PTS, PTS-M                            |                              | ПТС-2, ПТС-М             | amphibious transporter <a href="#">UA</a>                     | 31   |
| PTS-3                                       |                              | ПТС-3                    | amphibious transporter <a href="#">legacy</a>                 | 31   |
| PTS-4                                       |                              | ПТС-4                    | amphibious transporter                                        | 31   |
| PU-12 BTR-60PU-12, PU-12M, PU-12M6, PU-12M7 |                              | ПУ-12                    | air defence command post <a href="#">UA</a>                   | 37   |
| PZM-2                                       |                              | ПЗМ-2                    | trench digger <a href="#">UA</a>                              |      |
| R-142N R-142NM, R-142NMR, R-142NSA          | Deimos-N                     | Р-142Н, Деѳѳмос-Н        | command and staff vehicle <a href="#">UA</a>                  |      |
| R-145BM R-145BM1, R-145BMA                  |                              |                          | command and staff vehicle <a href="#">UA</a>                  | 34   |

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| R-149AKSh-1                        |                                     |                          | command and staff vehicle <a href="#">UA</a>                             | 34   |
| R-149BMR R-149BMRg                 | Kushetka-B                          | P-149BMP, Кушетка-Б      | command and staff vehicle <a href="#">UA</a>                             | 34   |
| R-149MA3 R-149MA1                  |                                     |                          | command and staff vehicle <a href="#">UA</a>                             | 34   |
| R-161 R-161A2, R-161A2M            | Poisk                               | P-161, Пойск             | radio station <a href="#">UA</a>                                         |      |
| R-166 R-166-0.5, R-166S            | Artek                               | P-166, Артек             | radio station <a href="#">UA</a>                                         |      |
| R-168 R-168E                       | Akveduk                             | P-168, Акведук           | radio set family <a href="#">UA</a>                                      |      |
| R-187 R-187-P1                     | Azart                               |                          | radio set family <a href="#">UA</a>                                      |      |
| R-325UMV R-325BMV                  |                                     | P-325YMB                 | communications jammer · of RB-301B Borisoglebsk-2 <a href="#">UA</a>     | 23   |
| R-330                              | Mandat                              | P-330 «Мандат»           | communications jammer complex <i>legacy</i>                              |      |
| R-330B                             | Mandat-B                            | P-330Б «Мандат-Б»        | communications jammer <i>legacy</i>                                      | 24   |
| R-330BMV                           |                                     | P-330БMB                 | communications jammer <a href="#">UA</a>                                 | 24   |
| R-330KMA                           |                                     | P-330KMA                 | EW command post · of R-330M1P Diabazol                                   |      |
| R-330KMV                           |                                     | P-330KMB                 | EW command post · of RB-301B Borisoglebsk-2                              | 23   |
| R-330M1P                           | Diabazol                            | P-330M1П «Диабазол»      | communications jammer complex <a href="#">UA</a>                         |      |
| R-330T                             |                                     | P-330T                   | communications jammer <i>export</i>                                      | 24   |
| R-330Zh                            | Zhitel                              | P-330Ж «Житель»          | GNSS and cellular jammer <a href="#">UA</a>                              | 26   |
| R-340RP Pole-21M, Pole-21E         | Pole-21                             | P-340RP «Поле-21»        | GNSS jammer <a href="#">UA</a>                                           | 28   |
| R-378A R-378B                      |                                     | P-378A(Б)                | communications jammer · of R-330 Mandat <i>legacy</i> <a href="#">UA</a> |      |
| R-378BMV                           |                                     | P-378BMB                 | communications jammer · of RB-301B Borisoglebsk-2                        | 23   |
| R-378UM                            |                                     |                          | communications jammer · of R-330M1P Diabazol                             |      |
| R-381T R-381TM, R-381T2M           | Taran, Taran-M                      | P-381T, «Таран»          | SIGINT system <a href="#">UA</a>                                         |      |
| R-410-7.5                          |                                     |                          | tropospheric scatter station <a href="#">UA</a>                          |      |
| R-412A R-412AM                     | Torf, Rezoner                       | P-412AM, Резонер         | tropospheric scatter station <a href="#">UA</a>                          |      |
| R-416GM                            |                                     |                          | radio relay station <a href="#">UA</a>                                   |      |
| R-417 R-417S                       | Baget                               |                          | tropospheric scatter station <a href="#">UA</a>                          |      |
| R-431AM                            |                                     |                          | radio relay antenna module · of Redut-2US <a href="#">UA</a>             |      |
| R-438                              |                                     |                          | satellite communications terminal                                        |      |
| R-439 R-439-P, R-439-BK, R-439-KUL | Legenda-2                           | P-439, Легенда-2         | satellite communications terminal <a href="#">UA</a>                     |      |
| R-934B                             | Sinitsa                             | P-934Б «Синица»          | communications jammer <a href="#">UA</a>                                 | 24   |
| R-934BMV                           |                                     | P-934БMB                 | communications jammer <a href="#">UA</a>                                 | 24   |
| R-934UM                            | Sinitsa                             |                          | communications jammer <a href="#">UA</a>                                 | 24   |
| RB-105B                            | Less                                |                          | SIGINT system                                                            |      |
| RB-108S                            |                                     |                          | EW command post                                                          |      |
| RB-109A                            | Bylina                              | PБ-109A «Былина»         | EW command post <a href="#">UA</a>                                       |      |
| RB-301B                            | Borisoglebsk-2                      | PБ-301Б «Борисоглебск-2» | communications jammer complex <a href="#">UA</a>                         | 23   |
| RB-341V                            | Leer-3                              | PБ-341В «Лер-3»          | cellular jammer <a href="#">UA</a>                                       | 27   |
| RB-531B                            | Infaua                              | PБ-531Б «Инфауна»        | counter-IED jammer <a href="#">UA</a>                                    | 29   |
| RB-636AM2 RB-636                   | Svet-KU                             | PБ636 «Свет-КУ»          | SIGINT system <a href="#">UA</a>                                         |      |
| RB-636 Svet-VSG                    | Svet-VSG                            |                          | SIGINT system · of RB-636AM2 Svet-KU                                     |      |
| REBovets Valley                    | REBovets Valley                     |                          | UAV jammer <i>trial</i>                                                  |      |
| Redut-2US                          |                                     |                          | signal and communications system <a href="#">UA</a>                      |      |
| REM-KL                             |                                     |                          | recovery vehicle <a href="#">UA</a>                                      |      |
| Repellent-1                        | Repellent-1                         | Репеллент-1              | UAV jammer <a href="#">UA</a>                                            | 28   |
| Repeynik                           | Repeynik                            |                          | portable counter-UAS radar <a href="#">UA</a>                            |      |
| Rezonans-N                         | Rezonans-N, Resonans-N, Rezonans-NE |                          | early warning radar                                                      |      |
| RKhM-4                             |                                     | PXM-4                    | CBRN reconnaissance vehicle <a href="#">UA</a>                           |      |
| RKhM-6                             | Povozka                             | PXM-6                    | CBRN reconnaissance vehicle <a href="#">UA</a>                           |      |
| RKhM-8                             |                                     | PXM-8                    | CBRN reconnaissance vehicle                                              |      |
| RKhM-9                             |                                     | PXM-9                    | CBRN reconnaissance vehicle                                              |      |
| RLM-D                              |                                     | PAM-Д                    | UHF radar module · of 55Zh6M Nebo-M <a href="#">UA</a>                   | 18   |
| RLM-M                              |                                     | PAM-M                    | VHF radar module · of 55Zh6M Nebo-M <a href="#">UA</a>                   | 18   |
| RLM-S                              |                                     | PAM-C                    | S-band radar module · of 55Zh6M Nebo-M <a href="#">UA</a>                | 18   |
| RM-G                               |                                     |                          | armoured recovery vehicle                                                |      |
| RP-377L RP-377LA                   | Lorandit, Lorandit-M                |                          | communications jammer <a href="#">UA</a>                                 |      |
| RP-377UVM1 RP-377UVM2, RP-377UM2   | Lesochek                            |                          | counter-IED jammer <a href="#">UA</a>                                    |      |
| RPG-7 RPG-7V, RPG-7V2, 6G3         |                                     | РПГ-7                    | anti-tank rocket launcher <a href="#">UA</a>                             |      |

| DESIGNATION                                             | NAME                               | CYRILLIC                            | WHAT IT IS                                                                | PAGE |
|---------------------------------------------------------|------------------------------------|-------------------------------------|---------------------------------------------------------------------------|------|
| RPG-22                                                  | Netto                              | РПГ-22, Нетто                       | 72.5 mm disposable anti-tank launcher <i>legacy</i> <a href="#">UA</a>    |      |
| RPG-26                                                  | Aglen                              | РПГ-26, Азлень                      | disposable anti-tank rocket launcher <a href="#">UA</a>                   |      |
| RPG-27                                                  | Tavolga                            | Таволга                             | 105 mm disposable anti-tank launcher <a href="#">UA</a>                   |      |
| RPG-28                                                  | Klyukva                            | РПГ-28, Клюкба                      | 125 mm disposable anti-tank launcher <a href="#">UA</a>                   |      |
| RPG-29 TKB-0175                                         | Vampir                             | РПГ-29 «Вампир», ТKB-0175           | 105 mm anti-tank rocket launcher                                          |      |
| RPG-30                                                  | Kryuk                              | РПГ-30, Крюк                        | 105 mm disposable anti-tank launcher                                      |      |
| RPG-32                                                  | Barkas, Hashim, Nashshab           | РПГ-32                              | 105 mm reusable anti-tank launcher                                        |      |
| RPMK-1 1B44, 1B44-1                                     | Ulybka, Ulibka-M                   |                                     | meteorological station <a href="#">UA</a>                                 |      |
| RPO-A RPO-M, RPO                                        | Shmel                              | РПО-А «Шмель»                       | thermobaric rocket launcher <a href="#">UA</a>                            |      |
| Rys                                                     | Rys, Iveco LMV                     | Рысь                                | armoured utility vehicle <a href="#">UA</a>                               | 19   |
| S-60 AZP-57                                             |                                    | C-60, АЗП-57                        | 57 mm anti-aircraft gun <a href="#">UA</a>                                |      |
| S-300PM S-300PM1, S-300PM2, S-300PMU1, S-300PMU2, SA-20 | S-300PM, Favorit GARGOYLE          | C-300ПМ, C-300ПМУ1                  | long-range SAM system <a href="#">UA</a>                                  |      |
| S-300PS 75R6, S-300PMU, SA-10                           | S-300PS GRUMBLE                    | C-300ПС, 75Р6                       | long-range SAM system <i>legacy</i> <a href="#">UA</a>                    |      |
| S-300V 9K81, SA-12A, SA-12B                             | S-300V, Antey-300 GLADIATOR, GIANT | C-300B, 9K81                        | long-range SAM system <a href="#">UA</a>                                  | 13   |
| S-300V4 S-300VMD, SA-23                                 | S-300V4                            | C-300B4                             | long-range SAM system <a href="#">UA</a>                                  | 13   |
| S-350 50R6A                                             | S-350, Vityaz                      | C-350 «Витязь», 50Р6А               | medium-range SAM system <a href="#">UA</a>                                | 13   |
| S-400 40R6, SA-21                                       | S-400, Triumph GROWLER             | C-400 «Триумф», 40Р6                | long-range SAM system <a href="#">UA</a>                                  | 12   |
| S-500 55R6M                                             | S-500, Prometey, Triumphator-M     | C-500 «Прометей», 55Р6М             | long-range SAM and missile defence system <i>trial</i> <a href="#">UA</a> |      |
| SA-14RM/SM                                              |                                    |                                     | encryption command post <a href="#">UA</a>                                |      |
| Saniya                                                  | Saniya, Sania                      | «Саня»                              | UAV jammer <a href="#">UA</a>                                             |      |
| Sapfir                                                  | Sapfir                             |                                     | UAV jammer <a href="#">UA</a>                                             |      |
| Sarmat-2 Sarmat-3, LSTC-1944                            | Sarmat, STS Sarmat                 |                                     | armoured utility vehicle <a href="#">UA</a>                               |      |
| SBA-60K2                                                | Bulat                              | «Булат»                             | armoured personnel carrier                                                |      |
| SBR-5M                                                  | Fara-VR                            |                                     | portable battlefield surveillance radar <a href="#">UA</a>                |      |
| SERP-VS6D                                               | Serp                               | СЕРП-ВС6Д                           | UAV jammer <a href="#">UA</a>                                             |      |
| Shipovnik-Aero                                          | Shipovnik-Aero                     | «Шиповник-Аэро»                     | UAV jammer <a href="#">UA</a>                                             |      |
| Silok-01                                                | Silok                              | «Силок-01»                          | UAV jammer <a href="#">UA</a>                                             |      |
| SNAR-10 1RL232, 1RL-127-1                               | Leopard BIG FRED                   | СНАР-10 «Леопард», 1РА232 «Леопард» | battlefield surveillance radar <a href="#">UA</a>                         | 20   |
| SNAR-10M SNAR-10M1, 1RL232-1M                           | Pantera                            | СНАР-10М, 1РА232-1М «Пантера»       | battlefield surveillance radar <a href="#">UA</a>                         | 20   |
| Sopka-2 12A6                                            | Sopka-2                            |                                     | air surveillance radar                                                    |      |
| Sozvezdie                                               | Sozvezdie                          |                                     | tactical command and control system <i>legacy</i>                         |      |
| SPG-9 SPG-9M, SPG-9D                                    | Kopye                              | СПГ-9, Копьё                        | 73 mm recoilless gun <a href="#">UA</a>                                   |      |
| SPN-4 SPN-4M, 1RL248-4                                  | HEART ACHE                         |                                     | radar jammer                                                              |      |
| Strizh-3                                                | Strizh-3                           |                                     | UAV jammer <a href="#">UA</a>                                             |      |
| Stupor                                                  | Stupor                             |                                     | UAV jammer <a href="#">UA</a>                                             |      |
| Supercam S350                                           |                                    |                                     | reconnaissance UAV <a href="#">UA</a>                                     |      |
| Syuzhet-MF                                              | Syuzhet-MF                         |                                     | meteorological station <i>trial</i>                                       |      |
| T-12 2A19                                               |                                    | T-12, 2A19                          | 100 mm anti-tank gun <i>stored</i>                                        |      |
| T-14 Object 148                                         | Armata                             | T-14, Объект 148                    | main battle tank · of Armata <i>trial</i>                                 |      |
| T-15                                                    | Armata                             | T-15                                | infantry fighting vehicle · of Armata <i>trial</i>                        |      |
| T-54 Object 137                                         |                                    | T-54, Объект 137                    | main battle tank <i>legacy</i> <a href="#">UA</a>                         |      |
| T-54-T                                                  |                                    |                                     | armoured recovery vehicle <i>legacy</i>                                   |      |
| T-55                                                    |                                    | T-55                                | main battle tank <i>stored</i> <a href="#">UA</a>                         |      |
| T-55A                                                   |                                    | T-55A                               | main battle tank · of T-55 <i>stored</i> <a href="#">UA</a>               |      |
| T-55AM T-55AMV                                          |                                    | T-55AM, T-55AMB                     | main battle tank · of T-55 <i>legacy</i>                                  |      |
| T-62 Object 166                                         |                                    | T-62, Объект 166                    | main battle tank <i>legacy</i> <a href="#">UA</a>                         |      |
| T-62M                                                   |                                    | T-62M                               | main battle tank · of T-62 <a href="#">UA</a>                             |      |
| T-62MV                                                  |                                    | T-62MB                              | main battle tank · of T-62 <a href="#">UA</a>                             |      |
| T-64A Object 434                                        |                                    | T-64A, Объект 434                   | main battle tank <a href="#">UA</a>                                       |      |
| T-64B Object 447, T-64BV                                |                                    | T-64S, Объект 447                   | main battle tank <a href="#">UA</a>                                       |      |
| T-64BV                                                  |                                    |                                     | main battle tank · of T-64B <a href="#">UA</a>                            |      |
| T-64BVK                                                 |                                    |                                     | armoured command vehicle · of T-64B <i>legacy</i> <a href="#">UA</a>      |      |
| T-72 T-72 Ural, Object 172M                             | Ural                               | T-72, T-72 «Урал»                   | main battle tank <a href="#">UA</a>                                       |      |

| DESIGNATION                   | NAME                 | CYRILLIC                    | WHAT IT IS                                                                   | PAGE |
|-------------------------------|----------------------|-----------------------------|------------------------------------------------------------------------------|------|
| T-72A                         |                      | T-72A                       | main battle tank · of T-72 Ural <a href="#">UA</a>                           |      |
| T-72AV                        |                      | T-72AB                      | main battle tank · of T-72A <a href="#">UA</a>                               |      |
| T-72B Object 184              |                      | T-72B, Объект 184           | main battle tank · of T-72 Ural <a href="#">UA</a>                           |      |
| T-72B1 T-72B1K                |                      | T-72B1, T-72B1K             | main battle tank · of T-72B <a href="#">UA</a>                               |      |
| T-72B3                        |                      | T-72B3                      | main battle tank · of T-72B <a href="#">UA</a>                               |      |
| T-72B3A                       |                      |                             | main battle tank · of T-72B3 <i>legacy</i> <a href="#">UA</a>                |      |
| T-72B3M                       |                      | T-72B3M                     | main battle tank · of T-72B3 <a href="#">UA</a>                              |      |
| T-72BA                        |                      | T-72BA                      | main battle tank · of T-72B <a href="#">UA</a>                               |      |
| T-72BM                        |                      | T-72BM                      | main battle tank · of T-72B <i>legacy</i> <a href="#">UA</a>                 |      |
| T-72M1 T-72M                  |                      | T-72M1, T-72M               | main battle tank · of T-72 Ural <i>legacy</i>                                |      |
| T-80B Object 219R             |                      | T-80B, Объект 219P          | main battle tank <i>stored</i> <a href="#">UA</a>                            |      |
| T-80BV                        |                      | T-80BВ                      | main battle tank · of T-80B <a href="#">UA</a>                               |      |
| T-80BVK                       |                      | T-80BВК                     | armoured command vehicle · of T-80BV <i>legacy</i> <a href="#">UA</a>        |      |
| T-80BVM                       |                      | T-80BВМ                     | main battle tank · of T-80BV <a href="#">UA</a>                              |      |
| T-80U Object 219AS            |                      | T-80У, Объект 219AC         | main battle tank <a href="#">UA</a>                                          |      |
| T-80UE-1                      |                      | T-80УЕ-1                    | main battle tank · of T-80U <i>legacy</i> <a href="#">UA</a>                 |      |
| T-80UK                        |                      | T-80УК                      | armoured command vehicle · of T-80U <i>legacy</i> <a href="#">UA</a>         |      |
| T-80UM2                       |                      | T-80УМ2                     | main battle tank · of T-80U <i>legacy</i> <a href="#">UA</a>                 |      |
| T-90 Object 188               |                      | T-90, Объект 188            | main battle tank <a href="#">UA</a>                                          |      |
| T-90A                         |                      | T-90A                       | main battle tank · of T-90 <a href="#">UA</a>                                |      |
| T-90AK                        |                      | T-90AK                      | armoured command vehicle · of T-90A <a href="#">UA</a>                       |      |
| T-90M Proryv-3                | Proryv-3             | T-90M, Прорыв-3             | main battle tank · of T-90A <a href="#">UA</a>                               |      |
| T-90S T-90SK                  |                      | T-90C, T-90CK               | main battle tank · of T-90 <a href="#">UA</a>                                |      |
| Tachyon                       | Tachyon, Takhion     | Тахион                      | reconnaissance UAV <a href="#">UA</a>                                        |      |
| TBS-M3                        |                      | ТБС-М3                      | 220 mm thermobaric rocket, long range · of TOS-2 Tosochka <a href="#">UA</a> | 7    |
| TDA-2K TDA-2M                 |                      | ТДА-2К, ТДА-2М              | smoke generator vehicle <a href="#">UA</a>                                   |      |
| TDA-3                         |                      |                             | smoke generator vehicle <a href="#">UA</a>                                   |      |
| TDM-2K                        |                      |                             | smoke generator vehicle <a href="#">UA</a>                                   |      |
| Tetrazdr                      | Tetrazdr             |                             | UAV jammer <i>trial</i>                                                      |      |
| Tirada-2S Tirada-2.3, RB-371A | Tirada-2             | «Турпага-2С», «Турпага 2.3» | satellite communications jammer <a href="#">UA</a>                           |      |
| TM-46                         |                      | TM-46                       | anti-tank blast mine <i>legacy</i>                                           |      |
| TM-57                         |                      | TM-57                       | anti-tank blast mine <i>legacy</i>                                           |      |
| TM-62 TM-62M                  |                      | TM-62                       | anti-tank blast mine <a href="#">UA</a>                                      |      |
| TM-83                         |                      | TM-83                       | side-attack anti-tank mine                                                   |      |
| TMM-3 TMM                     |                      | TMM-3                       | truck-mounted bridgelayer <a href="#">UA</a>                                 | 31   |
| TMM-6                         |                      | TMM-6                       | 8x8 heavy bridgelayer <a href="#">UA</a>                                     | 31   |
| TMS-65 TMS-65M, TMS-65U       |                      | TMC-65                      | decontamination vehicle                                                      |      |
| Tobol                         | Tobol                | «Тобол»                     | satellite communications jammer <a href="#">UA</a>                           |      |
| Torn                          | Torn, Torn-MDM       |                             | SIGINT system <a href="#">UA</a>                                             |      |
| Toros                         | Toros                |                             | armoured utility vehicle                                                     |      |
| TOS-1 Object 634              | Buratino             | ТОС-1 «Буратино»            | 220 mm heavy flamethrower system <i>legacy</i> <a href="#">UA</a>            | 7    |
| TOS-1A Object 634B            | Solntsepek           | ТОС-1А «Солнцецёк»          | 220 mm heavy flamethrower system <a href="#">UA</a>                          | 7    |
| TOS-2                         | Tosochka             | ТОС-2 «Тосочка»             | 220 mm heavy flamethrower system <a href="#">UA</a>                          | 7    |
| TOS-3                         | Drakon               | ТОС-3 «Дракон»              | heavy flamethrower system, tracked <i>trial</i> <a href="#">UA</a>           | 7    |
| TS-3132                       | Gusar                |                             | armoured utility vehicle <a href="#">UA</a>                                  |      |
| TZ-74                         |                      |                             | decontamination and smoke vehicle <i>legacy</i>                              |      |
| TZM-T                         |                      | ТЗМ-Т                       | rocket transloader · of TOS-1A Solntsepek <a href="#">UA</a>                 | 7    |
| UAZ-452                       | Bukhanka             | УАЗ-452, Буханка            | utility van <a href="#">UA</a>                                               |      |
| UAZ-469 UAZ-3151              | Kozlik               | УАЗ-469                     | utility vehicle <a href="#">UA</a>                                           |      |
| UAZ-469RKh UAZ-469RKhB        |                      |                             | CBRN reconnaissance and decontamination vehicle <a href="#">UA</a>           |      |
| UAZ-39094                     | Fermer               |                             | utility pickup <a href="#">UA</a>                                            |      |
| UMPB D-30SN                   |                      |                             | jet-powered glide bomb <a href="#">UA</a>                                    |      |
| UMPK                          |                      | УМПК                        | glide bomb guidance kit <a href="#">UA</a>                                   |      |
| UMZ                           |                      | УМЗ                         | truck-mounted minelayer <a href="#">UA</a>                                   |      |
| UMZ-G UMZ-K, UMZ-T            | Kleshch-G, Kleshch-G |                             | universal tracked remote minelayer <i>trial</i>                              |      |
| UR-67                         |                      | УР-67                       | mine-clearing line charge <i>legacy</i> <a href="#">UA</a>                   |      |

| DESIGNATION                     | NAME                              | CYRILLIC             | WHAT IT IS                                                                   | PAGE |
|---------------------------------|-----------------------------------|----------------------|------------------------------------------------------------------------------|------|
| UR-77 Object 256, <b>M1979</b>  | Meteorit                          | УР-77                | mine-clearing line charge vehicle <a href="#">(UA)</a>                       | 32   |
| UR-83P                          |                                   | УР-83П               | portable mine-clearing line charge                                           |      |
| Ural-375D Ural-375              |                                   | Урал-375Д            | cargo truck 6x6 <i>legacy</i> <a href="#">(UA)</a>                           |      |
| Ural-4320 Ural-43203            |                                   | Урал-4320            | cargo truck 6x6 <a href="#">(UA)</a>                                         |      |
| Ural-4320-7810                  | Motovoz-M                         |                      | cargo truck <a href="#">(UA)</a>                                             |      |
| Ural-6370 Ural-6370-0121-30     |                                   |                      | cargo truck 6x6 <a href="#">(UA)</a>                                         |      |
| Ural-43206                      |                                   | Урал-43206           | cargo truck 4x4 <a href="#">(UA)</a>                                         |      |
| Ural-63095                      | Typhoon-U                         | Урал-63095, Тайфун-У | mine-resistant ambush-protected vehicle                                      |      |
| Ural-63706-0122 Ural-63706-0011 | Tornado-U                         |                      | cargo truck 6x6 <a href="#">(UA)</a>                                         |      |
| Ural-432009                     | Ural-VV                           | Урал-432009, Урал-BB | armoured truck <a href="#">(UA)</a>                                          |      |
| Ural-532301                     |                                   |                      | cargo truck with armoured cab <a href="#">(UA)</a>                           |      |
| Ural-542301                     |                                   |                      | tank transporter <a href="#">(UA)</a>                                        |      |
| Uran-6                          |                                   | Уран-6               | remote-controlled mine-clearing robot <a href="#">(UA)</a>                   |      |
| Volna Kupol Garant              | Volna Kupol Garant                |                      | satellite communications jammer <a href="#">(UA)</a>                         |      |
| Volnorez                        | Volnorez                          | «Волнорез»           | UAV jammer <a href="#">(UA)</a>                                              |      |
| VPK-7829 K-16, K-17             | Bumerang, Boomerang               | Бумеранг             | armoured personnel carrier <i>trial</i>                                      |      |
| VPK-233136 SBM                  | Tigr SBM                          | ВПК-233136           | armoured car of GAZ-233014 Tigr <a href="#">(UA)</a>                         |      |
| VSM-1                           |                                   | BCM-1                | helicopter remote mining system                                              |      |
| VT-72A                          |                                   |                      | armoured recovery vehicle                                                    |      |
| Z-STS                           | Akhmat, Remdiesel Z-STS<br>Akhmat |                      | armoured utility vehicle <a href="#">(UA)</a>                                |      |
| ZALA 421-16 Z-16                |                                   |                      | reconnaissance UAV <a href="#">(UA)</a>                                      |      |
| Zaslon-REB                      | Zaslon-REB                        |                      | cellular jammer                                                              |      |
| Zastava                         | Zastava                           | Застава              | reconnaissance UAV <a href="#">(UA)</a>                                      |      |
| Zavarukha                       | Zavarukha                         |                      | portable acoustic reconnaissance system <a href="#">(UA)</a>                 |      |
| ZIL-131                         |                                   | ЗИЛ-131              | cargo truck 6x6 <i>legacy</i> <a href="#">(UA)</a>                           |      |
| ZIL-157                         |                                   | ЗИЛ-157              | cargo truck 6x6 <i>legacy</i>                                                |      |
| ZSU-23-4 2A6                    | Shilka                            | 3CY-23-4, 2A6        | 23 mm self-propelled anti-aircraft gun <i>stored</i><br><a href="#">(UA)</a> |      |
| ZU-23-2 ZU-23, 2A13             |                                   | 3Y-23, 3Y-23-2       | 23 mm anti-aircraft gun <a href="#">(UA)</a>                                 |      |

Table 1b. Other names and designations

Names, NATO names and codes (in **BLUE CAPITALS**), and other designations, each pointing (→) to its row in Table 1. Page numbers as in Table 1. English translations of Russian names are in *italics*.

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| 1B44 → RPKM-1        |      | 1V152 → 1V126       |      | 2S3M1 → 2S3M        |      |
| 1B44-1 → RPKM-1      |      | 1V152M → 1V126      |      | 2S3M2 → 2S3M        |      |
| 1B75E → 1B75         | 20   | 1V153 → 1V126       |      | 2S6M → 2S6          | 17   |
| 1L13 → 1L13-3        |      | 1V153M → 1V126      |      | 2S6M1 → 2S6         | 17   |
| 1RL110 → P-35        |      | 1V168 → ASUNO-S     |      | 2S9-1 → 2S9         |      |
| 1RL131 → P-18        |      | 1Zh3 → PRP-3        |      | 2S12A → 2S12        |      |
| 1RL133 → PSNR-5      | 36   | 2A6 → ZSU-23-4      |      | 2S33 → 2S19M2       | 8    |
| 1RL133-3 → Kreda-1E  |      | 2A13 → ZU-23-2      |      | 3F5 → 1K113         | 9    |
| 1RL134 → P-19        |      | 2A18 → D-30         |      | 3M14K → 3M14        |      |
| 1RL138 → 1RL144M     | 17   | 2A18M → D-30        |      | 3M14T → 3M14        |      |
| 1RL139 → P-35        |      | 2A19 → T-12         |      | 3M55 → P-800        |      |
| 1RL144 → 1RL144M     | 17   | 2A29 → MT-12        |      | 3OF45-1 → 3OF45     |      |
| 1RL144M1 → 1RL144M   | 17   | 2A29R → MT-12R      |      | 3OF843B → OF-843B   |      |
| 1RL232 → SNAR-10     | 20   | 2A45MR → 2A45M      |      | 3UBK10 → 9K116      |      |
| 1RL232-1M → SNAR-10M | 20   | 2B5 → BM-21         | 4    | 3VF2 → 3F2          | 9    |
| 1RL248-4 → SPN-4     |      | 2B9M → 2B9          |      | 3VF4 → 1K113        | 9    |
| 1V13M → 1V13         | 35   | 2B17 → BM-21-1      | 4    | 3VO11 → 3-O-8       | 9    |
| 1V15 → 1V14          | 35   | 2B17-1 → BM-21-1    | 4    | 3VO13 → 3-O-13      |      |
| 1V15M → 1V14         | 35   | 2B17M → 9K51M       | 4    | 3VO14 → 3-O-13      |      |
| 1V16 → 1V14          | 35   | 2F77M → 2F77        | 17   | 3VO28 → 3-O-23      |      |
| 1V19 → 1V18          |      | 2K22M → 2K22        | 17   | 3VO29 → 3-O-23      |      |
| 1V121 → PRP-4        | 35   | 2K22M1 → 2K22       | 17   | 3VO30 → 3-O-23      |      |
| 1V125 → PRP-4        | 35   | 2K25 → 3OF39        |      | 3VOF32 → 3OF25      |      |
| 1V126M → 1V126       |      | 2K25M → 3OF39M      |      | 3VOF33 → 3OF25      |      |
| 1V145 → PRP-4M       | 36   | 2S1M → 2S34         |      | 3VOF34 → 3OF43      |      |

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| 3VOF35 → 30F44      |      |
| 3VOF39 → 30F29      | 7    |
| 3VOF40 → 30F29      | 7    |
| 3VOF42 → 30F43      |      |
| 3VOF49 → 30F49      |      |
| 3VOF58 → 30F45      |      |
| 3VOF63 → 30F38      |      |
| 3VOF64 → 30F39      |      |
| 3VOF64M → 30F39M    |      |
| 3VOF66 → 30F38      |      |
| 3VOF72 → 30F45      |      |
| 3VOF73 → 30F45      |      |
| 3VOF86 → 30F59      | 7    |
| 3VOF87 → 30F59      | 7    |
| 3VOF91 → 30F61      |      |
| 3VOF93 → 30F39      |      |
| 3VOF96 → 30F64      |      |
| 3VOF97 → 30F64      |      |
| 3VOF98 → 30F64      |      |
| 3VRB36 → 3RB30      |      |
| 3VRB37 → 3RB30      |      |
| 3VRB38 → 3RB30      |      |
| 5P85D → 5P85SD      |      |
| 5P85S → 5P85SD      |      |
| 5P85SM → 5P85SD     |      |
| 5V55K → 5V55R       |      |
| 5V55RM → 5V55R      |      |
| 6G3 → RPG-7         |      |
| 6P49 → 6P50         |      |
| 6P57 → 6P50         |      |
| 6P58 → 6P50         |      |
| 6P59 → 6P50         |      |
| 8K14 → 9K72         |      |
| 9A33BM3 → 9A33B     | 17   |
| 9A38 → 9A310        | 14   |
| 9A39M1 → 9A39       | 14   |
| 9A53-G → 9K51M      | 4    |
| 9A53-S → 9A54       | 6    |
| 9A83-1 → 9A83       | 13   |
| 9A83M2 → 9A83       | 13   |
| 9A85-2 → 9A85       | 13   |
| 9A310M1 → 9A310     | 14   |
| 9A331-1 → 9A331     | 15   |
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| 9K32M → 9K32        |      |
| 9K33M2 → 9K33       | 17   |
| 9K33M3 → 9K33       | 17   |
| 9K35M3 → 9K35       | 17   |
| 9K37M1 → 9K37       | 14   |
| 9K79-1 → 9K79       |      |
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| 9K111-1M → 9K111-1  |      |
| 9K111M → 9K111      |      |
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| 9K116-3 → 9K116     |      |
| 9K119M → 9K119      |      |
| 9K120 → 9K119       |      |
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| 9K123-1 → 9K123     |      |
| 9K128-1 → 9P162     |      |

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| 9K331M → 9K331      | 15   |
| 9K510 → 9P132       | 4    |
| 9M14 → 9K11         |      |
| 9M22U-1 → 9M22U     | 4    |
| 9M33M → 9M33        | 17   |
| 9M33M2 → 9M33       | 17   |
| 9M33M3 → 9M33       | 17   |
| 9M37M → 9M37        | 17   |
| 9M37MD → 9M37       | 17   |
| 9M79-1 → 9M79       |      |
| 9M82M → 9M82        | 13   |
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| 9M83M → 9M83        | 13   |
| 9M96E → 9M96        | 12   |
| 9M96E2 → 9M96       | 12   |
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| 9M111-2 → 9M111     |      |
| 9M111M → 9M111      |      |
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| 9M117M → 9M117      |      |
| 9M117M1 → 9M117     |      |
| 9M119M → 9M119      |      |
| 9M119M1 → 9M119     |      |
| 9M123F → 9M123      |      |
| 9M131M → 9M131      |      |
| 9M133F-1 → 9M133    |      |
| 9M311-1M → 9M311    | 17   |
| 9M330-2 → 9M330     | 15   |
| 9M338K → 9M338      | 15   |
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| 9P135 → 9K111       |      |
| 9P135M → 9K111-1    |      |
| 9P149M → 9K132      |      |
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| 9P163-2 → 9P163M-2  |      |
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| 9S52M → 9S52        | 36   |
| 9S52M1 → 9S52       | 36   |
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| 30N6 → 5N63S                 |      |
| 39N6E → 39N6                 |      |
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| 48N6DM → 48N6                | 12   |
| 48N6E → 48N6                 | 12   |
| 48N6E2 → 48N6                | 12   |
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| 50K6E → 50K6A                | 13   |
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| 50R6A → S-350                | 13   |
| 52-M-853 → M-160             |      |
| 53-OF-843B → OF-843B         |      |
| 55K6A → 55K6                 | 12   |
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| Altayets → Altaets           |      |
| Antey-300 → S-300V           | 13   |
| Argus → PRP-4A               | 36   |
| ARK-1 → 1RL239               | 19   |
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| Arkan → 9M117                |      |
| Armata → T-14                |      |
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| ARS-14KM → ARS-14            |      |
| Artek → R-166                |      |
| AS-4 → Kh-22                 |      |
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| AT-13 → 9K115-1        |      |
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| AT-15 → 9K123          |      |
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| AT-16 → 9A4172         |      |
| Ataka → 9M120          |      |
| Ataka-1 → 9M120-1      |      |
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| AZP-57 → S-60          |      |
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| Bal → 3K60             |      |
| Bal-E → 3K60           |      |
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| Beetle (English) → IRM |      |
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| Fagot → 9M111                                    |      |
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| FLAT FACE B → P-19       |      |
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| Tornado-G → 2B17M                 | 4    |
| Tornado-G → 9K51M                 | 4    |
| Tornado-S → 9A54                  | 6    |
| Tornado-S → 9K515                 | 6    |
| Tornado-U → Ural-63706-0122       |      |
| Tosochka → TOS-2                  | 7    |
| Tray (English) → 2B14             |      |
| Triumf → S-400                    | 12   |
| Triumfator-M → S-500              |      |
| Tulip (English) → 2S4             | 9    |
| Tunguska → 2K22                   | 17   |
| Tunguska-M1 → 2K22                | 17   |
| Typhoon-K → KamAZ-53949           |      |
| Typhoon-K → KamAZ-63968           |      |
| Typhoon-K 4x4 → KamAZ-53949       |      |
| Typhoon-K 6x6 → KamAZ-63968       |      |
| Typhoon-U → Ural-63095            |      |
| Typhoon-VDV → KamAZ-4386          |      |
| Tyulpan → 2S4                     | 9    |
| UAZ-469RKhB → UAZ-469RKh          |      |
| UAZ-3151 → UAZ-469                |      |
| Ulibka-M → RPMK-1                 |      |
| Ulybka → RPMK-1                   |      |
| UMZ-K → UMZ-G                     |      |
| UMZ-T → UMZ-G                     |      |
| UNSh → K1Sh1                      |      |
| Uragan → 9K57                     | 5    |
| Uragan → 9P140                    | 5    |
| Uragan-1 → 9K57-1                 | 5    |
| Uragan-1M → 9A53                  | 5    |
| Uragan-1M → 9K512                 | 5    |
| Ural → T-72                       |      |
| Ural-375 → Ural-375D              |      |
| Ural-6370-0121-30 → Ural-6370     |      |
| Ural-43203 → Ural-4320            |      |
| Ural-63706-0011 → Ural-63706-0122 |      |
| Ural-VV → Ural-432009             |      |
| Utyos → NSV                       |      |
| V-500R → 5V55R                    |      |
| Val → PRP-3                       |      |
| Vampir → RPG-29                   |      |
| Vasilek → 2B9                     |      |
| Vedma → OZM-72                    |      |
| Vena → 2S31                       |      |
| Vepr → BMR-3M                     |      |
| Verba → 9K333                     |      |
| Vikhr → 9A4172                    |      |
| Vityaz → DT-10PM                  | 13   |
| Vityaz → DT-30                    | 13   |
| Vityaz → S-350                    | 13   |
| Vodnik → GAZ-39371                |      |
| Voronezh-DM → 77Ya6-DM            |      |
| VPK-233136 → GAZ-233014           |      |
| VPK Ural → AMN-590951             |      |
| VTS Ladoga → Ladoga               |      |
| Vystrel → KamAZ-43269             |      |

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| Witch (English) → OZM-72 |      |
| Yastreb-AV → 1K148       |      |
| YeSU TZ → ESU TZ         |      |
| Z-16 → ZALA 421-16       |      |
| ZAK-57 → 2S38            |      |
| Zemledeliye → ISDM       | 33   |
| Zhitel → R-330Zh         | 26   |
| Zhuk → IRM               |      |
| Zoopark-1 → 1L219        | 19   |
| Zoopark-1M → 1L260       | 19   |
| ZU-23 → ZU-23-2          |      |

Table 1c. Cyrillic designations and names

Sorted in Russian alphabetical order, each pointing (→) to its row in Table 1.

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| 1Б44 → 1В44                      | 6    | 2С3 «Акация» → 2S3           |      | 53-0Ф-В43В → 0F-В43В     |      |
| 1В119 → 1V119                    |      | 2С3М → 2S3М                  |      | 55Ж6МЕ → 55Zh6М          | 18   |
| 1В126 «Капустник-Б» → 1V126      |      | 2С40 → 2S40                  |      | 55Ж6М «Небо-М» → 55Zh6М  | 18   |
| 1В13 → 1V13                      | 35   | 2С43 «Мальва» → 2S43         | 8    | 55К6 → 55K6              | 12   |
| 1В152 → 1V152                    |      | 2С4 «Тюльпан» → 2S4          | 9    | 55К6Е → 55K6             | 12   |
| 1К113 «Смелычак» → 1K113         | 9    | 2С5 «Гуаццинт-С» → 2S5       | 7    | 55Р6М → S-500            |      |
| 1К148 «Ястреб-АВ» → 1K148        |      | 2С5М → 2S5М                  | 7    | 5736Е → 57Е6             | 16   |
| 1Л122 «Гармонь» → 1L122          |      | 2С6 → 2S6                    | 17   | 5В55К → 5V55R            |      |
| 1Л13 → 1L13-3                    |      | 2С6М → 2S6                   | 17   | 5В55Р → 5V55R            |      |
| 1Л219 «Зоонарк-1» → 1L219        | 19   | 2С6М1 → 2S6                  | 17   | 5Н63С → 5N63S            |      |
| 1Л219М → 1L219                   | 19   | 2С7М «Малка» → 2S7М          | 9    | 5Н84А «Оборона» → 5N84A  |      |
| 1Л259 → 1L259                    | 19   | 2С7 «Пион» → 2S7             |      | 5Н85А → 5P85SD           |      |
| 1Л260 → 1L260                    | 19   | 2С9 «Нона-С» → 2S9           |      | 5Н85С → 5P85SD           |      |
| 1Л2623 «Ртуть-БМ» → 1L262E       |      | 2Ф77 → 2F77                  | 17   | 5Н85СМ2-01 → 5P85SM2-01  | 12   |
| 1Л265 → 1L265                    | 22   | 2Ф77М → 2F77                 | 17   | 5Н85Т2 → 5P85T2          | 12   |
| 1Л266 → 1L266                    | 22   | 35Н6 «Каста» → 35N6          |      | 75Р6 → S-300PS           |      |
| 1Л269 «Красуха-2» → 1L269        | 21   | 39Н6Е «Каста-2Е2» → 39N6     |      | 85Я6 «Яер-2» → 85Ya6     | 24   |
| 1Л271 «Аистенок» → 1L271         | 20   | 39Н6 «Каста-2-2» → 39N6      |      | 91Н6Е → 91N6A            | 12   |
| 1РА131 → P-18                    |      | 35К13 → 3BK13                | 35   | 92Н6Е → 92N6A            | 12   |
| 1РА133-3 «Крего-1» → Kredo-1E    |      | 35К19 → 3BK19                |      | 96К6 → 96K6              | 16   |
| 1РА134 → P-19                    |      | 3ВФ4 → 1K113                 | 9    | 96А6Е → 96L6             | 18   |
| 1РА144М1 → 1RL144M               | 17   | 3К55 → 3K55                  |      | 9А310 → 9A310            | 14   |
| 1РА232-1М «Пантера» → SNAR-10M   | 20   | 3К60 → 3K60                  |      | 9А310М1-2 → 9A310M1-2    | 14   |
| 1РА232 «Леопард» → SNAR-10       | 20   | 3М-14 → 3M14                 |      | 9А316 → 9A316            | 14   |
| 1РА239 / АРК-1 / «Рысь» → 1RL239 | 19   | 3М16 → 3M16                  | 4    | 9А317 → 9A317            | 14   |
| 1РА257 «Красуха-4» → 1RL257      | 21   | 3-0-13 → 3-0-13              |      | 9А317М → 9A317M          | 14   |
| 1Т12-2М → 1T12-2M                | 6    | 3-0-23 → 3-0-23              |      | 9А330 → 9A330            | 15   |
| 2А13 → 2U-23-2                   |      | 3-0-8 → 3-0-8                | 9    | 9А331 → 9A331            | 15   |
| 2А18М → D-30                     |      | 30Ф24 → 30F24                | 35   | 9А331-1 → 9A331          | 15   |
| 2А19 → T-12                      |      | 30Ф25 → 30F25                |      | 9А331М2 → 9A331M         | 15   |
| 2А29 → МТ-12                     |      | 30Ф29 → 30F29                | 7    | 9А335 → 9A33B            | 17   |
| 2А36 «Гуаццинт-Б» → 2A36         | 10   | 30Ф38 → 30F38                |      | 9А35 → 9A35              | 17   |
| 2А6 → ZSU-23-4                   |      | 30Ф39 → 30F39                |      | 9А38 → 9A310             | 14   |
| 2А65 «Мста-Б» → 2A65             | 10   | 30Ф39М → 30F39M              |      | 9А39 → 9A39              | 14   |
| 2Б11 → 2B11                      |      | 30Ф43 → 30F43                |      | 9А39М1 → 9A39            | 14   |
| 2Б14 «Поднос» → 2B14             |      | 30Ф44 → 30F44                |      | 9А51 → 9A51              |      |
| 2Б16 «Нона-К» → 2B16             |      | 30Ф45 → 30F45                |      | 9А52 → 9A52              | 6    |
| 2Б17 → ВМ-21-1                   | 4    | 30Ф49 → 30F49                |      | 9А52-2 → 9A52-2          | 6    |
| 2Б17М → 2B17M                    | 4    | 30Ф50 → 30F50                |      | 9А52-2Т → 9A52-2T        | 6    |
| 2Б26 → 2B26                      | 4    | 30Ф56 → 30F56                | 35   | 9А52-4 → 9A52-4          | 6    |
| 2Б5 → ВМ-21                      | 4    | 30Ф59 → 30F59                | 7    | 9А53 → 9A53              | 5    |
| 2Б9 «Василёк» → 2B9              |      | 30Ф61 → 30F61                |      | 9А54 → 9A54              | 6    |
| 2К22 → 2K22                      | 17   | 30Ф64 → 30F64                |      | 9А82 → 9A82              | 13   |
| 2К22М1 «Тунгуска-М1» → 2K22      | 17   | 30Ф75 → 30F75                |      | 9А83 → 9A83              | 13   |
| 2К22М «Тунгуска-М» → 2K22        | 17   | 3РБ30 → 3RB30                |      | 9А84 → 9A84              | 13   |
| 2С12А → 2S12                     |      | 3Ф2 → 3F2                    | 9    | 9А85 → 9A85              | 13   |
| 2С12 «Сани» → 2S12               |      | 3Ф5 → 1K113                  | 9    | 9К11 → 9K11              |      |
| 2С19М1 → 2S19M1                  | 8    | 40Н6Е → 40N6                 | 12   | 9К111 → 9K111            |      |
| 2С19М2 → 2S19M2                  | 8    | 40Р6 → S-400                 | 12   | 9К111-1 → 9K111-1        |      |
| 2С19 «Мста-С» → 2S19             |      | 48Н6АМ → 48N6                | 12   | 9К114 → 9K114            |      |
| 2С1 «Гвоздика» → 2S1             | 35   | 48Н6Е → 48N6                 | 12   | 9К115 → 9K115            |      |
| 2С23 «Нона-СВК» → 2S23           |      | 48Н6Е2 → 48N6                | 12   | 9К115-1 → 9K115-1        |      |
| 2С25 → 2S25                      |      | 48Я6-К1 «Погонет» → 48Ya6-K1 | 18   | 9К115-2 → 9K115-2        |      |
| 2С31 «Вена» → 2S31               |      | 50К6А → 50K6A                | 13   | 9К116 → 9K116            |      |
| 2С33 «Мста-СМ» → 2S19M2          | 8    | 50Н6А → 50N6A                | 13   | 9К119М → 9K119           |      |
| 2С34 «Хоста» → 2S34              |      | 50П6А → 50P6A                | 13   | 9К123 → 9K123            |      |
| 2С35 «Колалиция-СВ» → 2S35       | 8    | 50Р6А → S-350                | 13   | 9К123-1 → 9K123          |      |
| 2С38 → 2S38                      |      | 51П6 → 51P6                  | 12   | 9К128-1 → 9P162          |      |

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| 9K132 → 9K132                  |      | 9M27K1 → 9M27K1          | 5    | 9П132 → 9П132                | 4    |
| 9K135 → 9K135                  |      | 9M27K2 → 9M27K2          | 5    | 9П133 → 9П133                |      |
| 9K310 → 9K310                  |      | 9M27K3 → 9M27K3          | 5    | 9П138 → 9П138                |      |
| 9K317 «Бук-М2» → 9K317         | 14   | 9M27C → 9M27S            |      | 9П139 → 9П139                | 4    |
| 9K317M «Бук-М3» → 9K317M       | 14   | 9M27D → 9M27F            | 5    | 9П140 → 9П140                | 5    |
| 9K32 → 9K32                    |      | 9M28A → 9M28D            | 4    | 9П148 → 9П148                |      |
| 9K33 → 9K33                    | 17   | 9M28K → 9M28K            | 4    | 9П149 → 9П149                |      |
| 9K330 → 9K330                  | 15   | 9M28C → 9M28S            | 4    | 9П149M → 9K132               |      |
| 9K331 «Тор-М1» → 9K331         | 15   | 9M28D → 9M28F            | 4    | 9П151 → 9K115                |      |
| 9K332 3PK «Тор-М2» → 9K332     | 15   | 9M311 → 9M311            | 17   | 9П157 → 9П157-2              |      |
| 9K333 → 9K333                  |      | 9M313 → 9M313            |      | 9П157-2 → 9П157-2            |      |
| 9K338 → 9K338                  |      | 9M317 → 9M317            | 14   | 9П162 → 9П162                |      |
| 9K33M2 → 9K333                 | 17   | 9M33 → 9M33              | 17   | 9П163-3 → 9П163-3            |      |
| 9K33M3 → 9K333                 | 17   | 9M330 → 9M330            | 15   | 9П163M-1 → 9П163M-1          |      |
| 9K34 → 9K34                    |      | 9M331 → 9M331            | 15   | 9П163M-2 → 9П163M-2          |      |
| 9K35M3 «Стрела-10M3» → 9K35    | 17   | 9M333 → 9M333            | 17   | 9П78-1 → 9П78-1              | 11   |
| 9K35 «Стрела-10» → 9K35        | 17   | 9M335 → 9M335            | 16   | 9C15M → 9S15M                | 13   |
| 9K37 → 9K37                    | 14   | 9M336 → 9M336            |      | 9C18 → 9S18                  | 14   |
| 9K37M1-2 «Бук-М1-2» → 9K37M1-2 | 14   | 9M338 → 9M338            | 15   | 9C18M1 → 9S18                | 14   |
| 9K37M1 «Бук-М1» → 9K37         | 14   | 9M33M3 → 9M333           | 17   | 9C19M2 → 9S19M2              | 13   |
| 9K38 → 9K38                    |      | 9M340 → 9M340            |      | 9C32 → 9S32                  | 13   |
| 9K512 «Ураган-1M» → 9K512      | 5    | 9M342 → 9M342            |      | 9C36M → 9S36                 | 14   |
| 9K515 → 9K515                  | 6    | 9M37 → 9M37              | 17   | 9C457 → 9S457                | 13   |
| 9K51 «Град» → 9K51             | 4    | 9M38 → 9M38              | 14   | 9C470 → 9S470                | 14   |
| 9K51M → 9K51M                  | 4    | 9M39 → 9M39              |      | 9C510M → 9S510               | 14   |
| 9K51M «Торнадо-Г» → 9K51M      | 4    | 9M42 → 9M42              | 4    | 9C52 → 9S52                  | 36   |
| 9K52 → 9K52                    |      | 9M43 → 9M43              | 4    | 9C552 → 9S552                | 11   |
| 9K54 «Град-Б» → 9K54           |      | 9M51 → 9M51              | 5    | 9C729M1 «Сленок-1» → 9S729M1 | 6    |
| 9K55-1 «Град-1» → 9K55-1       | 4    | 9M519 → 9M519            | 4    | 9C920 → 9S920                | 11   |
| 9K55 «Град-1» → 9K55           |      | 9M521 → 9M521            | 4    | 9C931 → 9S931                | 37   |
| 9K57 «Ураган» → 9K57           | 5    | 9M522 → 9M522            |      | 9C932-1 → 9S932-1            | 37   |
| 9K58 «Смерч» → 9K58            | 6    | 9M525 → 9M525            | 6    | 9T2175 → 9T217               | 17   |
| 9K59 «Прима» → 9K59            |      | 9M528 → 9M528            | 6    | 9T218 → 9T218                |      |
| 9K72 → 9K72                    |      | 9M529 → 9M529            | 6    | 9T232M → 9T232M              |      |
| 9K720 → 9K720                  | 11   | 9M531 → 9M531            | 6    | 9T234 → 9T234                | 6    |
| 9K79 → 9K79                    |      | 9M533 → 9M533            |      | 9T234-2 → 9T234-2            | 6    |
| 9K79-1 → 9K79                  |      | 9M534 → 9M534            | 6    | 9T234-2T → 9T234-2T          | 6    |
| 9K81 → S-300V                  | 13   | 9M53K → 9M53K            |      | 9T234-4 → 9T234-4            | 6    |
| 9M100 → 9M100                  | 13   | 9M53C → 9M53S            |      | 9T244 → 9T244                | 15   |
| 9M111 → 9M111                  |      | 9M53D → 9M53F            |      | 9T249 → 9T249                | 5    |
| 9M113 → 9M113                  |      | 9M542 → 9M542            | 6    | 9T250 → 9T250                | 11   |
| 9M113M → 9M113M                |      | 9M544 → 9M544            | 6    | 9T255 → 9T255                | 6    |
| 9M114 → 9M114                  |      | 9M549 → 9M549            | 6    | 9T450 → 9T450                |      |
| 9M115 → 9M115                  |      | 9M55K → 9M55K            | 6    | 9T451 → 9T451                | 4    |
| 9M117 → 9M117                  |      | 9M55K1 → 9M55K1          | 6    | 9T452 → 9T452                | 5    |
| 9M119M → 9M119                 |      | 9M55K4 → 9M55K4          | 6    | «Автомобаз» → 1L222          |      |
| 9M119M1 → 9M119                |      | 9M55K5 → 9M55K5          | 6    | Азень → RPG-26               |      |
| 9M120 → 9M120                  |      | 9M55C → 9M55S            | 6    | AGC-17 → AGS-17              |      |
| 9M120-1 → 9M120-1              |      | 9M55D → 9M55F            | 6    | AGC-30 → AGS-30              |      |
| 9M123 → 9M123                  |      | 9M59 → 9M59              | 5    | AGC-40 → AGS-40              |      |
| 9M123D → 9M123                 |      | 9M723 → 9M723            | 11   | A3П-57 → S-60                |      |
| 9M131 → 9M131                  |      | 9M728 → 9M728            | 11   | Ak8egyk → R-168              |      |
| 9M133 → 9M133                  |      | 9M729 → 9M729            | 11   | «Алмаз» → Altaets            |      |
| 9M14 → 9K11                    |      | 9M79 → 9M79              |      | AMH-233114 → AMN-233114      | 24   |
| 9M217 → 9M217                  | 4    | 9M82 → 9M82              | 13   | AMH-590951 → AMN-590951      |      |
| 9M218 → 9M218                  | 4    | 9M82M → 9M82             | 13   | Армата → Armata              |      |
| 9M22 → 9M22                    | 4    | 9M83 → 9M83              | 13   | APC-14KM → ARS-14            |      |
| 9M22M → 9M22M                  |      | 9M96E → 9M96             | 12   | Армек → R-166                |      |
| 9M22C → 9M22S                  | 4    | 9M96E2 → 9M96            | 12   | ACH-233115 → ASN-233115      | 24   |
| 9M22Y → 9M22U                  | 4    | 9П122 → 9П122            |      | Астролог → MZKT-7930         | 16   |
| 9M27A → 9M27D                  | 5    | 9П125 → 9П125            |      | ATC-59Г → ATS-59G            |      |
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| БМК-460 → BMK-460        |      |
| БМО-Т → BMO-T            |      |
| БМП-1 → BMP-1            |      |
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| БМП-1П → BMP-1P          |      |
| БМП-1У → BMP-1U          |      |
| БМП-2 → BMP-2            |      |
| БМП-2Д → BMP-2D          |      |
| БМП-2М → BMP-2M          |      |
| БМП-3 → BMP-3            |      |
| БМП-3М → BMP-3M          |      |
| БМП-3Ф → BMP-3F          |      |
| БМПТ → BMPT              |      |
| БМПТ-72 → BMPT-72        |      |
| БМР-3М → BMR-3M          |      |
| БПМ-97 → KamAZ-43269     |      |
| БРДМ-2 → BRDM-2          |      |
| БРДМ-2А → BRDM-2A        |      |
| БРДМ-2М → BRDM-2M        |      |
| БРДМ-2МС → BRDM-2MS      |      |
| БРМ-1 → BRM-1K           |      |
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| БРЭМ-2 → BREM-2          |      |
| БРЭМ-Д → BREM-D          |      |
| БРЭМ-К → BREM-K          |      |
| БРЭМ-Л → BREM-L          |      |
| БС-3 → BS-3              |      |
| БТМ-3 → BTM-3            |      |
| БТР-50 → BTR-50          |      |
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| БТР-82А → BTR-82A        |      |
| БТР-82АМ → BTR-82AM      |      |
| БТР-82АТ → BTR-82AT      |      |
| БТР-90 → BTR-90          |      |
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| БТС-4В → BTS-4V             |      |
| «Булат» → SBA-60K2          |      |
| Бумеранг → VPK-7829         |      |
| Буханка → UAZ-452           |      |
| Вепрь → BMR-3M              |      |
| «Вербя» → 9K333             |      |
| «Волнорез» → Volnerez       |      |
| «Воронж-ДМ» → 77Ya6-DM      |      |
| ВПК-233136 → VPK-233136     |      |
| ВПК-Урал → AMN-590951       |      |
| ВСМ-1 → VSM-1               |      |
| Выстрел → KamAZ-43269       |      |
| ГАЗ-2330 → GAZ-233014       |      |
| ГАЗ-233014 → GAZ-233014     |      |
| ГАЗ-233036 → GAZ-233036     |      |
| ГАЗ-3307 → GAZ-3307         |      |
| ГАЗ-3308 «Саяга» → GAZ-3308 |      |
| ГАЗ-41-08 → BRDM-2          |      |
| ГАЗ-51 → GAZ-51             |      |
| ГАЗ-5903 → BTR-80           |      |
| ГАЗ-5923 → BTR-90           |      |
| ГАЗ-66 → GAZ-66             |      |
| Гамма-ДЕ → 67N6             |      |
| Гарпун-3 → Garpun-3         |      |
| Герань-1 → Geran-1          |      |
| Герань-2 → Geran-2          |      |
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| Д-20 → D-20                 |      |
| Д-30 → D-30                 |      |
| Д-30А → D-30                |      |
| Д-44 → D-44                 |      |
| Деймос-Н → R-142N           |      |
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| Инвар → 9M119               |      |
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| КЗООП → 3K55                |      |
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| КамАЗ-43269 → KamAZ-43269   |      |
| КАМАЗ-5350 → KamAZ-5350     |      |
| КАМАЗ-53501 → KamAZ-53501   |      |
| КамАЗ-53949 → KamAZ-53949   |      |
| КАМАЗ-6350 → KamAZ-6350     |      |
| КАМАЗ-63501 → KamAZ-63501   |      |
| КамАЗ-63968 → KamAZ-63968   |      |
| КАМАЗ-65225 → KamAZ-65225   |      |
| КАМАЗ-6560 → KamAZ-6560     |      |
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| Кинжал → Kh-47M2            |      |
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| «Китолов-2М» → 30F69M       |      |
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| Штурм → 9K114                    |      |
| Элерон-3 → Eleron-3              |      |
| Элерон-3СВ → Eleron-3            |      |
| Зльбрус → 9K72                   |      |

**Table 2. Unit type to equipment**

Tag gives basis, date and confidence. See page 3 for explanations.

| UNIT TYPE                             | EQUIPMENT                                | HOLDING SUBUNIT AND NUMBER                                                                                                                                                                                                                                                                                              | TAG         | PAGE |
|---------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------|
| aerospace forces air defence regiment | 96L6 all-altitude acquisition radar      | 3rd SAM battalion. INA (photographed deployed and stowed)                                                                                                                                                                                                                                                               | O 2010-02 M | 18   |
|                                       | 96L6 all-altitude acquisition radar      | S-400 'battery' in Armada's usage; APA export diagram lists it as a battery component. 1 per S-400 battery (Armada, term undefined; APA lists 96L6E as a battery component without a count); S-300PMU2 count INA                                                                                                        | R 2014 M    | 18   |
|                                       | Buk-M1 / Buk-M2 / Buk-M3                 | 4 Aerospace Forces regiments with Buk-M1-2, Buk-M2 and S-300V; about 80 Buk-M1-2/Buk-M2 in Aerospace Forces holdings                                                                                                                                                                                                    | R 2024 M    | 14   |
|                                       | Pantsir-S1                               | Pantsir vehicles deployed with each S-400 battalion (Armada); Lenta: batteries attached to S-400 regiments. 3 per S-400 battalion (Armada, 2024)                                                                                                                                                                        | D 2024 H    | 16   |
|                                       | Pantsir-S1                               | Pantsir-S battery. 6 Pantsir-S delivered to Eastern MD air defence units on Kamchatka, Dec 2015; the regiment was reported due one battery                                                                                                                                                                              | O 2015-12 M | 16   |
|                                       | Pantsir-S1                               | Pantsir-S1 gun-missile battalion; each battery covers one S-400 battalion                                                                                                                                                                                                                                               | O 2020-02 M | 16   |
| air defence missile battalion         | Pantsir-S1                               | INA per regiment; IISS lists 9 SAM regts with Pantsir-S1 in the 45th, 4 in the 14th, 3 in the 4th and 4 in the 11th AF&AD Army                                                                                                                                                                                          | R 2024 M    | 16   |
|                                       | S-400 Triumf                             | IISS: 13 Aerospace Forces regts with S-400 and Pantsir-S1, plus 2 mixed with S-300                                                                                                                                                                                                                                      | R 2024 M    | 12   |
|                                       | S-350 Vityaz                             | Battalion set: 1 command post 50K6A, 2-3 multifunction radars 50NGA, 12 launchers 50P6A. 12 launchers (12 missiles each) per battalion, per Russian MoD statement                                                                                                                                                       | R 2020 M    | 13   |
| air defence missile brigade           | S-400 Triumf                             | IISS: 2 SAM bn in Belarus; 1 SAM bty in Syria                                                                                                                                                                                                                                                                           | R 2024 M    | 12   |
|                                       | Buk-M1 / Buk-M2 / Buk-M3                 | 4 air defence missile battalions, each 3 batteries; brigade command post Polyana-D4. Buk-M1: 6 TELARs, 3 loader-launchers, 1 CP, 1 search radar per battalion; 24 TELARs per 4-battalion brigade (JIT: 53rd Bde had 3 operational battalions in 2014)                                                                   | D 2016 H    | 14   |
|                                       | Buk-M1 / Buk-M2 / Buk-M3                 | battalion TELAR. INA per battalion; brigade had 3 operational battalions in 2014 (JIT)                                                                                                                                                                                                                                  | O 2014-07 M | 14   |
|                                       | Buk-M1 / Buk-M2 / Buk-M3                 | air defence missile battalion (brigade set of Buk-M3). Buk-M3: up to 6 TELARs or 9S36M radars and up to 12 launchers per battalion                                                                                                                                                                                      | R 2019 M    | 14   |
|                                       | Buk-M1 / Buk-M2 / Buk-M3                 | air defence missile battalion                                                                                                                                                                                                                                                                                           | R 2018 L    | 14   |
|                                       | Polyana-D4                               | brigade command post (combat control post MP06 and command-staff vehicle MP02)                                                                                                                                                                                                                                          | D 2021-12 H | 36   |
|                                       | Polyana-D4                               | brigade command post; vehicle captioned as 9S52M Polyana-D4M                                                                                                                                                                                                                                                            | O 2010-12 M | 36   |
|                                       | S-300V4                                  | air defence missile battalion (up to 4 per brigade, under Polyana-D4M1 brigade CP); battalion CP 9S457M controls up to 4 batteries; WEG: an upper-level 9S52 Polyana-D4 command post may integrate S-300V with Buk into one brigade. S-300V battery: 1 9S32, 4 9A83, 2 9A82, 1 9A84, 2 9A85; S-300V4 battery counts INA | D 2017 M    | 13   |
|                                       | S-300V4                                  | INA. IISS lists S-300V and S-300V4 in Army holdings without a count and counts 16 SAM brigades, but does not say which hold them; Izvestia reports S-300V4 brigades under the OSK                                                                                                                                       | R 2024 M    | 13   |
|                                       | Pantsir-S1                               | 30 held by the Navy (IISS p. 187)                                                                                                                                                                                                                                                                                       | R 2024 M    | 16   |
| air defence missile regiment          | Pantsir-S1                               | INA (1 SAM regt with S-400 and Pantsir-S1)                                                                                                                                                                                                                                                                              | R 2024 M    | 16   |
|                                       | S-400 Triumf                             | about 12 launchers reported at the unit site; 4 launchers and 96L6 and 92N6 radars reported destroyed                                                                                                                                                                                                                   | O 2025-11 M | 12   |
|                                       | S-400 Triumf                             | air defence missile battalion (division). usually 2 battalions of 8 launchers each (16 per regiment); 2012 press report relating MoD press service                                                                                                                                                                      | R 2012 M    | 12   |
|                                       | S-400 Triumf                             | Navy: regiments under fleet air defence divisions. IISS: 64 S-400 in Navy; 3 SAM regts: 2 Baltic Fleet, 1 Pacific Fleet (naval aviation)                                                                                                                                                                                | R 2024 M    | 12   |
|                                       | Strela-10 and Osa short-range SAM        | division air defence regiment (Osa-AKM, being deactivated in 2019)                                                                                                                                                                                                                                                      | D 2019 L    | 17   |
| airborne division                     | Barnaul-T                                | air defence missile regiment. one full divisional set                                                                                                                                                                                                                                                                   | O 2018-06 M | 37   |
|                                       | Barnaul-T                                | air defence missile regiment                                                                                                                                                                                                                                                                                            | O 2018 L    | 37   |
|                                       | Leer-2                                   | EW subunit (FOI: VDV EW companies formed from 2016). INA; MoD announced 10 complexes for VDV EW units, Dec 2012 (title only, state claim)                                                                                                                                                                               | R 2018 M    | 24   |
| anti-tank battalion                   | R-145BM Chayka command and staff vehicle | ACV listed as 'BTR-60PU/BRDM-2'; battery placement unclear in OCR. 5 per battalion (BTR-60PU or BRDM-2)                                                                                                                                                                                                                 | D 1991 L    | 34   |
| army corps                            | Iskander-M                               | corps missile brigade. 1 missile brigade (12 launchers)                                                                                                                                                                                                                                                                 | O 2021 M    | 11   |

| UNIT TYPE                 | EQUIPMENT                                                                       | HOLDING SUBUNIT AND NUMBER                                                                                                                                                                                           | TAG         | PAGE |
|---------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------|
| artillery brigade         | 2A36 Giatsint-B 152 mm towed gun                                                | gun battalion (152 mm) of the army artillery brigade; 2A36 or 2S5 or M-46. 36 per brigade (two 18-tube gun battalions) or 48 (four 24-tube battalions)                                                               | D 1991 M    | 10   |
|                           | 2A36 Giatsint-B 152 mm towed gun                                                | gun brigade of the front artillery division; 2A36 or 2S7. 72 per brigade                                                                                                                                             | D 1991 M    | 10   |
|                           | 2A65 Msta-B                                                                     | artillery battalions (several per brigade; 2S19 or 2A65)                                                                                                                                                             | D 2019 M    | 10   |
|                           | 2S35 Koalitsiya-SV                                                              | self-propelled artillery battalions. INA: planned replacement of the brigade's 2S19 by 2S35, reported 2017, not carried out by late 2024                                                                             | R 2019 L    | 8    |
|                           | 2S5 Giatsint-S                                                                  | gun battalion (152 mm). 18 per battalion (24 in forward armies); 2S5 or 2A36 or M-46                                                                                                                                 | D 1991 M    | 7    |
|                           | 2S7 Pion / 2S7M Malka                                                           | 2S7 battalion (1Z17, 2017); 2S7M Malka (mil.ru via RDP18). 1 battalion; 12 x 2S7 (RUWIKI, 2019)                                                                                                                      | O 2018-04 M | 9    |
|                           | 2S7 Pion / 2S7M Malka                                                           | 12 x 2S7 each (RUWIKI, 2019)                                                                                                                                                                                         | O 2019 L    | 9    |
| artillery regiment        | R-145BM Chayka command and staff vehicle                                        | 1 per brigade                                                                                                                                                                                                        | O 1994 L    | 34   |
|                           | Uragan-1M                                                                       | MRL battalion (Uragan 'being succeeded' by Uragan-M1; no holding confirmed)                                                                                                                                          | R 2019-10 L | 5    |
|                           | BM-21 Grad 122 mm MRL                                                           | one Grad MRL battalion in division field artillery regiment, being replaced by Tornado-G                                                                                                                             | D 2019 M    | 4    |
|                           |                                                                                 |                                                                                                                                                                                                                      |             |      |
| combined arms army        | 1B75 Penicillin                                                                 | 1L271 in rocket, missile and tube artillery units                                                                                                                                                                    | R 2017 L    | 20   |
|                           | Iskander-M                                                                      | army missile brigade. 1 missile brigade (12 launchers) per army                                                                                                                                                      | D 2024 H    | 11   |
|                           | Iskander-M                                                                      | army missile brigade. 12 launchers per brigade                                                                                                                                                                       | O 2023 M    | 11   |
|                           | Palantin                                                                        | EW units of a Western MD combined arms army in Voronezh Oblast (army not named)                                                                                                                                      | R 2019-04 M | 25   |
|                           | Palantin                                                                        | EW troops of a Guards combined arms army, Western MD, Luga range, Leningrad Oblast (Palantin-K, Oct 2019); Feb 2022 exercise in Leningrad Oblast names only the Western MD                                           | R 2022-02 M | 25   |
|                           | Pole-21                                                                         | Army EW specialists; first used in an exercise with battalion tactical groups, with R-934BMV and Silok-01 (MoD via AIF)                                                                                              | O 2021-04 M | 28   |
|                           | Polyana-D4                                                                      | army air defence and aviation command post (SKP PVO i aviatsii)                                                                                                                                                      | R 2021-12 M | 36   |
| command brigade           | R-149 command and staff vehicles                                                | R-149AKSh-1                                                                                                                                                                                                          | O 2022-03 L | 34   |
| engineer-sapper battalion | IMR-2 / IMR-2M / IMR-3M combat engineer vehicle                                 | mine clearing platoon (brigade engineer battalion). 2 IMR-3M per platoon                                                                                                                                             | D 2016 M    | 32   |
|                           | ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer             | obstacle platoon (brigade engineer battalion per RWW; division engineer battalion per WEG). 3 GMZ-3 per obstacle platoon                                                                                             | D 2016 M    | 33   |
|                           | TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge | brigade engineer battalion; sets almost always attached to the movement support detachment (OOD). 4 TMM-3 truck-launched bridge sets per brigade engineer battalion, each laying a 10 m span; together a 42 m bridge | D 2024-02 M | 31   |
|                           | UR-77 Meteorit                                                                  | mine clearing platoon (organic engineer battalion of a maneuver brigade). 2 per platoon                                                                                                                              | D 2016 M    | 32   |
| engineer-sapper brigade   | IMR-2 / IMR-2M / IMR-3M combat engineer vehicle                                 | 2 delivered Nov 2018, 1 more due first half 2019 (Central MD press service, state claim)                                                                                                                             | R 2018-11 L | 32   |
|                           | IMR-2 / IMR-2M / IMR-3M combat engineer vehicle                                 | INA; listed with UR-77, BMR-3 and GMZ-3 (English Wikipedia, citing a 2023 Russian regional news item)                                                                                                                | R 2023-06 L | 32   |
|                           | ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer             |                                                                                                                                                                                                                      | D 2024 L    | 33   |
|                           | PMP heavy folding pontoon bridge park                                           | pontoon bridge battalion of the army or front engineer brigade (FM 100-2-3 'Engineer Brigade, Army or Front'). one full set: 32 river and 4 shore links on KrAZ-255, 15 BMK-150 boats                                | D 1991 M    | 30   |
| engineer-sapper regiment  | UR-77 Meteorit                                                                  |                                                                                                                                                                                                                      | R 2023 L    | 32   |
|                           | ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer             |                                                                                                                                                                                                                      | D 2024 L    | 33   |
|                           | TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge | engineer and bridge subunit (press-release wording)                                                                                                                                                                  | O 2020-11 M | 31   |
| EW battalion              | Infaua                                                                          | EW battalions of the Airborne Forces (VDV) (WEG 2015); RIA 2012 and FOI say VDV EW units                                                                                                                             | R 2015 M    | 29   |
|                           | Krasukha-2 / Krasukha-4                                                         | independent REB-S battalion of an Air and Air Defence Army (VKS); FOI infers, does not name Krasukha there                                                                                                           | R 2018 L    | 21   |
|                           | Leer-3                                                                          | Leer-3 drone/EW company                                                                                                                                                                                              | R 2019 M    | 27   |

| UNIT TYPE                            | EQUIPMENT                                                                      | HOLDING SUBUNIT AND NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                       | TAG         | PAGE |
|--------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------|
| EW brigade                           | Krasukha-2 / Krasukha-4                                                        | REB-S (counter-air) EW battalion. INA; usually one Krasukha-2 and one Krasukha-4 per brigade [reported 2022]                                                                                                                                                                                                                                                                                                                                                                     | D 2018 H    | 21   |
|                                      | Krasukha-2 / Krasukha-4                                                        | part of 1 Krasukha-4 complex (command-post container found near Makariv); Armada estimates 4 complete Krasukha systems in theatre                                                                                                                                                                                                                                                                                                                                                | R 2022-03 M | 21   |
|                                      | Leer-3                                                                         | Leer-3 platoon. platoon operates the complex; number per brigade INA                                                                                                                                                                                                                                                                                                                                                                                                             | D 2024 H    | 27   |
|                                      | Moskva-1                                                                       | REB-S (counter-air EW) battalion, reported                                                                                                                                                                                                                                                                                                                                                                                                                                       | D 2018 H    | 22   |
|                                      | Moskva-1                                                                       | INA (ARM22 lists one Moskva-1 among five brigade systems)                                                                                                                                                                                                                                                                                                                                                                                                                        | R 2023-09 L | 22   |
|                                      | Murmansk-BN                                                                    | Probably a company of the REB-N (anti-ground) battalion (FOI assessment)                                                                                                                                                                                                                                                                                                                                                                                                         | D 2018 H    | 22   |
|                                      | Murmansk-BN                                                                    | At least 1 complex per brigade (Armada); 2 antenna groups observed at 16th EW bde, vehicle count at 19th suggests 2 (PAS imagery, 2019)                                                                                                                                                                                                                                                                                                                                          | O 2019 M    | 22   |
| EW company                           | R-330Zh Zhitel                                                                 | REB-S (anti-aircraft communications and navigation) battalion                                                                                                                                                                                                                                                                                                                                                                                                                    | D 2018 M    | 26   |
|                                      | R-330B Mandat-B and R-934BMV                                                   | EW company of a motor rifle or tank brigade: R-330B squad in the VHF-band jamming platoon; R-934B squad (VHF aircraft) in the radio-communications jamming platoon                                                                                                                                                                                                                                                                                                               | D 2016 H    | 24   |
|                                      | R-330Zh Zhitel                                                                 | EW company, jamming platoon (radio, GPS, cellular, radio-controlled fuses), jamming squad. 1 (single station on RWW chart; no count stated)                                                                                                                                                                                                                                                                                                                                      | D 2016 M    | 26   |
|                                      | R-330Zh Zhitel                                                                 | EW company of motor rifle or tank division [sources confirm the company exists; Zhitel holding not stated]                                                                                                                                                                                                                                                                                                                                                                       | D 2018 L    | 26   |
|                                      | R-330Zh Zhitel                                                                 | EW company deployed to Ukraine. 2 in the deployed company                                                                                                                                                                                                                                                                                                                                                                                                                        | R 2014 L    | 26   |
| high-power artillery brigade         | 2S4 Tyulpan                                                                    | 2S4 battalion (ATP figure 5-18 shows two 2S4 groups; subunit name not given). ATP fig. 5-18: groups of 6 and 4 2S4 (OCR uncertain); 8 reported in 45th brigade (2017)                                                                                                                                                                                                                                                                                                            | D 2024 H    | 9    |
|                                      | 2S4 Tyulpan                                                                    | heavy mortar battalion (240-mm); brigade allocated to a front (Soviet RVGK)                                                                                                                                                                                                                                                                                                                                                                                                      | D 1991 M    | 9    |
|                                      | 2S7 Pion / 2S7M Malka                                                          | 2S7 battalions (45th bde: two battalions of eight, 2017). ATP fig. 5-18: 24 x 2S7 per brigade, 8 x 2S7 battalion block (OCR uncertain); 16 reported in 45th bde, 2017                                                                                                                                                                                                                                                                                                            | D 2024-02 H | 9    |
|                                      | 2S7 Pion / 2S7M Malka                                                          | 16 x 2S7 Pion (with 8 x 2S4 Tyulpan)                                                                                                                                                                                                                                                                                                                                                                                                                                             | O 2017-12 M | 9    |
| military district                    | S-400 Triumf                                                                   | SAM regiments of the district's Air Force and Air Defence Army. IISS late 2024, SAM regiments listed with S-400 in each district's Air Force and Air Defence Army: 45th (Leningrad MD) 5, plus 4 listed with S-300PS, S-300PM1 and S-400; 14th (Central MD) 4; 4th (Southern MD) 3; 11th (Eastern MD) 4 listed with S-300PS and S-400, plus 1 with S-300V and S-400; 6th (Moscow MD) none (Moscow area: 5 regiments under Aerospace Defence Command). Launchers per district INA | D 2024 H    | 12   |
|                                      | SNAR-10 / SNAR-10M1 Leopard                                                    | artillery units of the tank formation stationed in the South Urals (MoD wording; formation not named)                                                                                                                                                                                                                                                                                                                                                                            | R 2019-01 M | 20   |
|                                      | SNAR-10 / SNAR-10M1 Leopard                                                    | artillery reconnaissance subunits (SNAR-10M Pantera, deliveries by end March 2016)                                                                                                                                                                                                                                                                                                                                                                                               | R 2016-03 L | 20   |
| missile battalion<br>missile brigade | Iskander-M                                                                     | 4 launchers                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | O 2008 M    | 11   |
|                                      | Iskander-M                                                                     | 3 missile battalions of 2 batteries each; 2 launchers per battery. 12 launchers and 12 transloaders per brigade; 4 of each per battalion                                                                                                                                                                                                                                                                                                                                         | D 2021 H    | 11   |
| motor rifle brigade                  | 2S19 Msta-S                                                                    | two self-propelled artillery battalions, each of three batteries of 6 howitzers; ATP notes 2S3 may be held instead. 18 per battalion (6 per battery), 36 per brigade; ATP: some battalions up to 24 systems                                                                                                                                                                                                                                                                      | D 2024 M    | 8    |
|                                      | Barnaul-T                                                                      | brigade AD subunit; 80 quotes a 'missile division' (battalion) commander                                                                                                                                                                                                                                                                                                                                                                                                         | O 2021-12 M | 37   |
|                                      | BM-21 Grad 122 mm MRL                                                          | MRL battalion; 6 launchers per battery. 18 per battalion                                                                                                                                                                                                                                                                                                                                                                                                                         | D 2016 M    | 4    |
|                                      | BM-21 Grad 122 mm MRL                                                          | BTG, attached BM-21 battery. 1 battery                                                                                                                                                                                                                                                                                                                                                                                                                                           | O 2022-04 M | 4    |
|                                      | Borisoglebsk-2                                                                 | EW company                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | D 2024-02 H | 23   |
|                                      | Borisoglebsk-2                                                                 | EW subunit                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | O 2018 M    | 23   |
|                                      | PMP heavy folding pontoon bridge park                                          | engineer battalion, pontoon bridge company, one pontoon park; RWW calls it 'PP-61' but its figures (268 m at 60 t; 90- and 120-t bridges) match PP-91, not PMP. 6 BMK boats, 7 PTS-2                                                                                                                                                                                                                                                                                             | D 2016 L    | 30   |
|                                      | PRP-4A Argus / PRP-4M Deytery                                                  | fire control and artillery reconnaissance battery, artillery spotting platoon                                                                                                                                                                                                                                                                                                                                                                                                    | D 2016 M    | 36   |
|                                      | PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry | engineer battalion. 7 PTS-2 per engineer battalion (RWW names PTS-2, not PTS-4)                                                                                                                                                                                                                                                                                                                                                                                                  | D 2016 M    | 31   |
|                                      | R-330Zh Zhitel                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | R 2014 L    | 26   |
|                                      | SNAR-10 / SNAR-10M1 Leopard                                                    | fire control and artillery reconnaissance battery, radar platoon                                                                                                                                                                                                                                                                                                                                                                                                                 | D 2016 M    | 20   |

| UNIT TYPE                         | EQUIPMENT                                                    | HOLDING SUBUNIT AND NUMBER                                                                                                                                                                         | TAG         | PAGE |
|-----------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------|
| motor rifle division              | Strela-10 and Osa short-range SAM                            | air defence missile-artillery battalion, missile battery (SA-13). 6 launchers per battery                                                                                                          | D 2024 M    | 17   |
|                                   | Tor                                                          | air defence missile battalion, three batteries. 12 launcher vehicles per battalion (4 per battery)                                                                                                 | D 2024 M    | 15   |
|                                   | Tor                                                          | air defence missile battalion (Tor-M2DT); brigade link by location and retraining report only                                                                                                      | R 2019-11 L | 15   |
|                                   | Tornado-G                                                    | brigade artillery, MRL battalion. INA for Tornado-G; it replaces the BM-21, 18 per brigade MRL battalion (RWW p.100); 18 delivered to one brigade 2012 (RBASE)                                     | D 2019 M    | 4    |
|                                   | Tornado-G                                                    | 18                                                                                                                                                                                                 | O 2012-04 M | 4    |
|                                   | Tunguska                                                     | air defence missile-artillery battalion, Tunguska battery. 6 launcher vehicles per battery                                                                                                         | D 2016 M    | 17   |
|                                   | Zoopark-1 / Zoopark-1M                                       | fire control and artillery reconnaissance battery, radar platoon (with SNAR-10M1)                                                                                                                  | D 2016 M    | 19   |
|                                   | Borisoglebsk-2                                               | separate EW battalion                                                                                                                                                                              | O 2022-01 M | 23   |
|                                   | Leer-3                                                       | separate EW battalion                                                                                                                                                                              | R 2022-01 M | 27   |
|                                   | PMP heavy folding pontoon bridge park                        | engineer battalion, pontoon bridge company (also in tank divisions). half set: 16 river and 2 shore links and 1 service vehicle on KrAZ-214; 6 BMK-90/150 boats per company                        | D 1991 M    | 30   |
|                                   | R-149 command and staff vehicles                             | R-149AKSh-1                                                                                                                                                                                        | O 2022-03 L | 34   |
|                                   | Tor                                                          | air defence missile regiment. 16 per regiment (4 batteries of 4, bmpd); ATP: two battalions; ru.wikipedia: Tor-M2 'divisional' set of 3 batteries, 12 vehicles                                     | D 2024 H    | 15   |
|                                   | Tor                                                          | air defence missile regiment. one regimental set of Tor-M2 (16 combat vehicles)                                                                                                                    | O 2017-11 M | 15   |
|                                   | Tornado-G                                                    | artillery regiment, MRL battalion (replacing Grad battalion)                                                                                                                                       | D 2019 H    | 4    |
|                                   | Zoopark-1 / Zoopark-1M                                       |                                                                                                                                                                                                    | D 2024 L    | 19   |
| motor rifle regiment              | BM-21 Grad 122 mm MRL                                        | BTG, attached MRL element. 3 in one BTG                                                                                                                                                            | O 2022-04 M | 4    |
|                                   | BMP-1KSh command and staff vehicle                           | regimental command post. 1 in regimental HQ of a BMP-equipped motor rifle regiment (FM 100-2-3 table, OCR scrambled)                                                                               | D 1991 M    | 35   |
|                                   | BMP-1KSh command and staff vehicle                           | 501st signal company                                                                                                                                                                               | O 1986 L    | 35   |
|                                   | PU-12M7                                                      | air defence missile-artillery battery; also regimental air defence chief's post (ARMF: PU-12 family 1977-95; ORUZH: undated PU-12M7 sheet). 1 per battery (PU-12 family)                           | D 1995 M    | 37   |
|                                   | PU-12M7                                                      | regiment on BTR, Soviet 1980s establishment. 3 PU-12 per regiment                                                                                                                                  | D 1989 L    | 37   |
| pontoon-bridge regiment           | R-149 command and staff vehicles                             | R-149MA3                                                                                                                                                                                           | O 2022-09 M | 34   |
|                                   | PMP heavy folding pontoon bridge park                        | two pontoon bridge battalions (army or front/district level). three full sets: 96 river and 12 shore links on KrAZ-255                                                                             | D 1991 M    | 30   |
|                                   | RKhBZ brigade                                                | flamethrower battalion of 3 batteries, each 6 BM-1 and 6 T2M-T; RWW places these battalions in NBC defence brigades and regiments and does not state the echelon. 18 per battalion (6 per battery) | D 2016 M    | 7    |
| rocket artillery brigade          | TOS-1A Solntsepek                                            | TOS battalion of 3 companies of 6 launchers (ATP para 5-146). 18 per battalion                                                                                                                     | D 2024 M    | 7    |
|                                   | BM-27 Uragan                                                 | rocket launcher battalions. 54 per brigade (3 battalions of 18)                                                                                                                                    | D 1991 M    | 5    |
|                                   | BM-30 Smerch                                                 | three groups of 4 x 9A52-2 in ATP fig. 5-16; whether each group is a battalion or a battery is not readable from the text. 12 x 9A52-2 shown in ATP fig. 5-16 (3 groups of 4)                      | D 2024 M    | 6    |
| rocket artillery regiment         | BM-30 Smerch                                                 | 7 delivered April 2017, 5 more announced                                                                                                                                                           | O 2017-04 M | 6    |
|                                   | BM-30 Smerch                                                 |                                                                                                                                                                                                    | O 2026-09 M | 6    |
|                                   | BM-27 Uragan                                                 | three rocket launcher battalions. 54 per regiment (3 battalions of 18)                                                                                                                             | D 1991 M    | 5    |
|                                   | 1B75 Penicillin                                              | battalion fire control network, conceptual ATP fig. 5-3 (1L271)                                                                                                                                    | D 2024 M    | 20   |
|                                   | 1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) | each battery: one 1V13 (battery FDC) and one 1V14 (battery COP); battalion HQ: one 1V15 (battalion COP) and one 1V16 (battalion FDC). 8 per battalion: 3 1V13, 3 1V14, 1 1V15, 1 1V16              | D 2016 H    | 35   |
| self-propelled artillery regiment | 2S35 Koalitsiya-SV                                           | self-propelled howitzer battery. 6 per battery, 18 per battalion                                                                                                                                   | D 2016 M    | 8    |
|                                   | PRP-4A Argus / PRP-4M Deytery                                | battalion fire control platoon, artillery spotter squad                                                                                                                                            | D 2016 L    | 36   |
|                                   | 1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) | howitzer battalion with 2S19M1. at least 3 1V13 and 3 1V14 seen with one battalion                                                                                                                 | R 2021-09 L | 35   |
|                                   | 2S35 Koalitsiya-SV                                           | 8 2S35 on troop trials                                                                                                                                                                             | R 2020-05 M | 8    |
|                                   | BM-27 Uragan                                                 | Uragan rocket artillery battalion. 8 per battalion                                                                                                                                                 | R 2017-12 M | 5    |

| UNIT TYPE           | EQUIPMENT                                                                      | HOLDING SUBUNIT AND NUMBER                                                                                                                                                                                                                 | TAG         | PAGE |
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| tank army           | Uragan-1M                                                                      | rocket artillery battalion (planned replacement for 9K57 Uragan, MoD statement; no holding confirmed)                                                                                                                                      | R 2017-12 L | 5    |
|                     | Iskander-M                                                                     | army missile brigade. 1 missile brigade (12 launchers)                                                                                                                                                                                     | O 2021 M    | 11   |
| tank brigade        | 2S19 Msta-S                                                                    | two howitzer (self-propelled artillery) battalions, three batteries of 6. 18 per battalion, 36 per brigade                                                                                                                                 | D 2016 M    | 8    |
|                     | Borisoglebsk-2                                                                 | EW company                                                                                                                                                                                                                                 | D 2024-02 H | 23   |
|                     | PRP-4A Argus / PRP-4M Deytery                                                  | fire control and artillery reconnaissance battery, artillery spotting platoon                                                                                                                                                              | D 2016 L    | 36   |
|                     | PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry | engineer battalion. 7 PTS-2 per engineer battalion (RWW names PTS-2, not PTS-4)                                                                                                                                                            | D 2016 M    | 31   |
|                     | SNAR-10 / SNAR-10M1 Leopard                                                    | fire control and artillery reconnaissance battery, radar platoon                                                                                                                                                                           | D 2016 M    | 20   |
|                     | Strela-10 and Osa short-range SAM                                              | air defence missile-artillery battalion, missile battery (SA-13). 6 launchers per battery                                                                                                                                                  | D 2024 L    | 17   |
|                     | Tor                                                                            | air defence missile battalion, three batteries. 12 launcher vehicles per battalion (4 per battery)                                                                                                                                         | D 2024 M    | 15   |
|                     | Tunguska                                                                       | anti-aircraft missile-artillery battalion, SA-19 battery; ATP 4-161 gives this battalion to tank brigades, but the battery figure is for brigades generally. 6 per battery (generic brigade figure: RWW 'maneuver brigade', ATP fig. 4-19) | D 2024 L    | 17   |
| tank division       | Zoopark-1 / Zoopark-1M                                                         | fire control and artillery reconnaissance battery, radar platoon                                                                                                                                                                           | D 2016 M    | 19   |
|                     | 1B75 Penicillin                                                                | 1L271; holding subunit not stated                                                                                                                                                                                                          | O 2018-01 M | 20   |
| tank regiment       | Tor                                                                            | air defence missile regiment. one regimental set of Tor-M2 (16 combat vehicles)                                                                                                                                                            | O 2017-03 M | 15   |
|                     | Tornado-G                                                                      | artillery regiment, MRL battalion                                                                                                                                                                                                          | D 2019 H    | 4    |
|                     | BMP-1KSh command and staff vehicle                                             | regimental command post. INA (FM 100-2-3 table reads 0 for the tank regiment HQ)                                                                                                                                                           | D 1990 M    | 35   |
|                     | PU-12M7                                                                        | air defence missile-artillery battery; also regimental air defence chief's post (ARMF: PU-12 family 1977-95; ORUZH: undated PU-12M7 sheet). 1 per battery (PU-12 family)                                                                   | D 1995 M    | 37   |
| unit not identified | 2A36 Giatshint-B 152 mm towed gun                                              | IIS estimate: 150 in service, 350 in store (Ground Forces total)                                                                                                                                                                           | R 2024-12 M | 10   |
|                     | 2A36 Giatshint-B 152 mm towed gun                                              | IIS estimate: 50 across Naval Infantry; holding unit not stated                                                                                                                                                                            | R 2024-12 L | 10   |
|                     | 2A65 Msta-B                                                                    | Naval Infantry artillery; brigade not named. 50 (IIS 2024 figures)                                                                                                                                                                         | R 2024 M    | 10   |
|                     | 2A65 Msta-B                                                                    | Russian deployment in Syria; unit not named. 12 (IIS 2024 figures)                                                                                                                                                                         | R 2024 M    | 10   |
|                     | 2S5 Giatshint-S                                                                | Airborne Forces artillery; unit not named. 18 (IIS 2024 figures)                                                                                                                                                                           | R 2024 M    | 7    |
|                     | ARK-1M Rys                                                                     | division's reconnaissance battery (WEG wording, identical to Wikipedia; Soviet-era)                                                                                                                                                        | R 2024 L    | 19   |
|                     | Borisoglebsk-2                                                                 | EW units                                                                                                                                                                                                                                   | O 2018 M    | 23   |
|                     | Borisoglebsk-2                                                                 |                                                                                                                                                                                                                                            | O 2022-03 L | 23   |
| Infaua              | Infaua                                                                         | dedicated EW units and manoeuvre reconnaissance units                                                                                                                                                                                      | D 2024 L    | 29   |
|                     | Infaua                                                                         | Russian military base in Armenia: MoD release title, 27 May 2015 (via FOI); filmed at Alagyaz range, 2019 (state media)                                                                                                                    | R 2019-04 M | 29   |
|                     | Krasukha-2 / Krasukha-4                                                        | 1 Krasukha-4 reported hit                                                                                                                                                                                                                  | R 2023-11 L | 21   |
|                     | Krasukha-2 / Krasukha-4                                                        | 1 Krasukha-2 reported                                                                                                                                                                                                                      | R 2018-08 L | 21   |
| Leer-2              | Leer-2                                                                         | Ground Forces EW units ('possibly' in FOI; state media: Southern MD column Mar 2018)                                                                                                                                                       | R 2018 M    | 24   |
| Leer-2              | Leer-2                                                                         | Position 17 km behind line of contact, Novoluhanske, Donetsk region. 1 vehicle                                                                                                                                                             | R 2023-11 M | 24   |
| Leer-2              | Leer-2                                                                         | Navy EW centres                                                                                                                                                                                                                            | R 2018 L    | 24   |
| Leer-3              | Leer-3                                                                         |                                                                                                                                                                                                                                            | R 2015 M    | 27   |
| Leer-3              | Leer-3                                                                         | Leer-3 crews                                                                                                                                                                                                                               | R 2022-11 M | 27   |
| Leer-3              | Leer-3                                                                         |                                                                                                                                                                                                                                            | R 2020-11 M | 27   |
| Moskva-1            | Moskva-1                                                                       | 1 claimed destroyed                                                                                                                                                                                                                        | R 2022-09 L | 22   |
| Murmansk-BN         | Murmansk-BN                                                                    | Radio-jamming node (yзел пагуономех); its 1st section named at the 471st (KZ). 2 complexes at the Kamchatka centre (KZ, reported Jan 2016; FOI: delivered 2014)                                                                            | O 2019 H    | 22   |
| Nebo-M              | Nebo-M                                                                         | 4 Nebo-M and Nebo-UM combined                                                                                                                                                                                                              | R 2015 L    | 18   |
| Nebo-M              | Nebo-M                                                                         | air defence and radio-technical crews with a division-level command post (MoD release; unit not named)                                                                                                                                     | R 2017-09 L | 18   |
| Palantin            | Palantin                                                                       | EW subunits of the Eastern MD in Zabaykalsky Krai (planned deliveries by end 2020)                                                                                                                                                         | R 2020-07 L | 25   |

| UNIT TYPE | EQUIPMENT                                                                      | HOLDING SUBUNIT AND NUMBER                                                                                                                                    | TAG         | PAGE |
|-----------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------|
|           | PMP heavy folding pontoon bridge park                                          | INA; Oryx lists 47 PMP(-M) river links and 3 shore links lost                                                                                                 | R 2026-09 M | 30   |
|           | PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry | INA (PMM-2 held by Russian Army; no count given)                                                                                                              | R 2024 L    | 31   |
|           | R-145BM Chayka command and staff vehicle                                       | RKhBZ (NBC protection) troops; Wikipedia names the vehicle BTR-60PU/PUM and equates BTR-60PU with R-145BM                                                     | R 2022 L    | 34   |
|           | R-330B Mandat-B and R-934BMV                                                   | R-934B (2024), Borisoglebsk-2 R-934 station (2023-11) and R-934BMV (2023-10, 2024-11, 2025-04) reported hit by Ukrainian units; no Russian holding unit named | R 2025 M    | 24   |
|           | R-330Zh Zhitel                                                                 | 25 lost in Ukraine Feb 2022 to Sep 2026 (visually confirmed)                                                                                                  | R 2026-09 H | 26   |
|           | S-350 Vityaz                                                                   | training centre of the Anti-aircraft Missile Forces, Zhukov Military Academy of Aerospace Defence. first S-350 set (crews and instructors training)           | R 2020-02 M | 13   |



**Table 3a. Reach of indirect fire**

| Guns, howitzers, mortars and rocket launchers, including mine-laying rockets. Practical rows give observed employment ranges.                                                                      |              |          |                                         |      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------|-----------------------------------------|------|
| SYSTEM, MUNITION AND CONDITION                                                                                                                                                                     | RANGE        | MINIMUM  | TERM                                    | PAGE |
| <b>BM-30 Smerch</b> 9M544 and 9M549 guided<br>Tornado-S guided cluster rockets, inertial and GLONASS; Tornado-S guided rockets; crew statement                                                     | 120 km       | 20–40 km | maximum (claim),<br>minimum (claim)     | 6    |
| <b>CEP 5–10 m (inertial and GLONASS; official claim)</b>                                                                                                                                           |              |          |                                         |      |
| <b>Uragan-1M</b> 9M544 guided 300 mm<br>9M544 guided rocket; claim in Russian press (ZR gives 120 km as the system maximum without naming the rocket)                                              | 120 km       |          | claimed                                 | 5    |
| <b>BM-30 Smerch</b> Employment range, Tornado-S<br>guided rockets against a command post (2026–08) and a fortified strongpoint (2026–09); MoD figures via state media                              | about 100 km |          | practical (claim)                       | 6    |
| <b>BM-30 Smerch</b> 9M528 HE-frag<br>unguided extended-range rocket, separable HE-frag warhead                                                                                                     | 90 km        |          | maximum (claim)                         | 6    |
| <b>2S35 Koalitsiya-SV</b> 2S35 guided or extended-range shell<br>precision-guided or rocket-assisted projectile; munition not named                                                                | 80 km        |          | maximum (claim)                         | 8    |
| <b>BM-30 Smerch</b> 9M55K cluster<br>unguided, range-corrected rocket with fragmentation submunitions                                                                                              | 70 km        | 20 km    | maximum, minimum                        | 6    |
| <b>CEP 150 m (inertial range and attitude correction, no terminal guidance; also given for 9M528)</b>                                                                                              |              |          |                                         |      |
| <b>BM-30 Smerch</b> 9M55F HE-frag<br>unguided, separable HE-frag warhead                                                                                                                           | 70 km        | 20 km    | maximum, minimum                        | 6    |
| <b>BM-30 Smerch</b> Employment range, Smerch<br>unguided rockets, Ukraine 2022, used at or near maximum range                                                                                      | 35–70 km     |          | practical                               | 6    |
| <b>Uragan-1M</b> 9M55F HE-frag 300 mm<br>9M55F HE-frag rocket with trajectory correction                                                                                                           | 70 km        | 20 km    | maximum, minimum                        | 5    |
| <b>BM-27 Uragan</b> Employment range, observed<br>9K57 and 9K58 Smerch together, Donbas, May–June 2022; RUSI does not separate the two                                                             | 35–70 km     |          | practical                               | 5    |
| <b>2S7 Pion / 2S7M Malka</b> HE rocket-assisted<br>RAP projectile, 103 kg                                                                                                                          | 47.5 km      |          | maximum                                 | 9    |
| <b>2S35 Koalitsiya-SV</b> 2S35 standard HE-frag<br>standard (unassisted) rounds                                                                                                                    | 40 km        |          | maximum                                 | 8    |
| <b>BM-21 Grad 122 mm MRL</b> 9M521 HE-frag<br>9M521 HE-frag, export round for 9K51M Tornado-G; in 9K51 only with a manual electronic-fuze setter (RUWPI); not adopted by the Russian Army (MISSIL) | 40 km        | 15 km    | maximum (claim),<br>minimum (claim)     | 4    |
| <b>Tornado-G</b> 9M521 HE-frag<br>9M521 extended-range HE-frag rocket; attribution to 9M521 from RUWIKI (Karpenko), WEG, ZR and WWW1 give 40 km as system maximum                                  | 40 km        |          | maximum                                 | 4    |
| <b>2S7 Pion / 2S7M Malka</b> HE-frag<br>Conventional HE-frag, 110 kg projectile                                                                                                                    | 37.5 km      |          | maximum                                 | 9    |
| <b>Tornado-G</b> 9M522 HE-frag<br>9M522 rocket, separable HE-frag warhead                                                                                                                          | 37.5 km      |          | maximum                                 | 4    |
| <b>2S19 Msta-S</b> rocket-assisted<br>HE rocket-assisted projectile (HERA/RAP), type not named                                                                                                     | 36 km        |          | range                                   | 8    |
| <b>Uragan-1M</b> 9M27F HE-frag 220 mm<br>9M27F unguided HE-frag rocket                                                                                                                             | 35 km        | 10 km    | maximum effective,<br>minimum effective | 5    |
| <b>Uragan-1M</b> 9M27K cluster 220 mm<br>9M27K rocket, 30 9N210 fragmentation submunitions                                                                                                         | 35 km        |          | maximum effective                       | 5    |
| <b>BM-27 Uragan</b> 9M27F HE-frag<br>9M27F HE-frag rocket                                                                                                                                          | 35 km        | 10 km    | maximum, minimum                        | 5    |
| <b>BM-27 Uragan</b> 9M27K cluster<br>9M27K rocket, 30 9N210 HE-frag submunitions                                                                                                                   | 35 km        | 10 km    | maximum, minimum                        | 5    |
| <b>BM-27 Uragan</b> 9M27K2 AT mines<br>9M27K2 rocket, 24 PTM-1 anti-tank mines                                                                                                                     | 35 km        | 10 km    | maximum, minimum                        | 5    |
| <b>BM-27 Uragan</b> 9M27K3 AP mines<br>9M27K3 rocket, PFM-1 anti-personnel mines                                                                                                                   | 35 km        | 10 km    | maximum, minimum                        | 5    |
| <b>2A36 Giatsint-B 152 mm towed gun</b> rocket-assisted 3V0F41<br>active-rocket projectile 30F30                                                                                                   | 33.5 km      |          | maximum                                 | 10   |
| <b>2S5 Giatsint-S</b> rocket-assisted HE-frag 30F30<br>rocket-assisted HE-frag                                                                                                                     | 33 km        |          | maximum                                 | 7    |
| <b>2S5 Giatsint-S</b> HE-frag 30F59<br>HE-frag 30F59, improved aerodynamic shape, full charge                                                                                                      | 30.5 km      |          | maximum                                 | 7    |
| <b>2A65 Msta-B</b> OF61 HE-frag base bleed<br>OF61 (30F61) HE-frag with base-bleed gas generator, round 3V0F91                                                                                     | 29 km        |          | maximum (claim)                         | 10   |
| <b>2S19 Msta-S</b> HE-frag base bleed<br>30F61 HE-frag base bleed (round 3V0F91)                                                                                                                   | 29 km        |          | maximum                                 | 8    |

| SYSTEM, MUNITION AND CONDITION                                                                                                                                                                                           | RANGE   | MINIMUM | TERM                                    | PAGE |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------|-----------------------------------------|------|
| <b>2A36 Giatsint-B 152 mm towed gun</b> HE-frag 3VOF39<br>HE-frag 3OF29, full charge; ammunition not stated; listed with HE-frag maximum                                                                                 | 28.5 km | 8.6 km  | maximum, minimum                        | 10   |
| <b>2S5 Giatsint-S</b> HE-frag 3OF29<br>HE-frag 3OF29, full charge; HE-frag, indirect fire                                                                                                                                | 28.4 km | 8,600 m | maximum, minimum                        | 7    |
| <b>2A65 Msta-B</b> OF45 HE-frag<br>OF45 (3OF45) HE-frag projectile; OF45 HE-frag projectile                                                                                                                              | 24.7 km | 6.5 km  | maximum (claim),<br>minimum (claim)     | 10   |
| <b>2S19 Msta-S</b> HE-frag<br>3OF45 (OF-45) HE-frag, full charge                                                                                                                                                         | 24.7 km |         | maximum                                 | 8    |
| <b>2S35 Koalitsiya-SV</b> 2S43 HE-frag<br>2A64 gun; munition not stated by WEG or WPMALRU (Rostec: HE-frag more than 24 km)                                                                                              | 24.5 km |         | maximum                                 | 8    |
| <b>2S5 Giatsint-S</b> HE-frag 3OF29 reduced charge<br>HE-frag 3OF29 with reduced charge 4Zh48                                                                                                                            | 21.5 km |         | maximum                                 | 7    |
| <b>BM-21 Grad 122 mm MRL</b> 9M22U HE-frag<br>9M22U (M-21OF) HE-fragmentation rocket                                                                                                                                     | 20.4 km | 5 km    | maximum, minimum                        | 4    |
| <b>Tornado-G</b> 9M22U HE-frag<br>legacy Grad 9M22U HE-frag rocket                                                                                                                                                       | 20.4 km | 5 km    | maximum effective,<br>minimum effective | 4    |
| <b>2S5 Giatsint-S</b> Krasnopol guided 3OF39<br>laser-guided HE-frag Krasnopol                                                                                                                                           | 20 km   |         | maximum                                 | 7    |
| <b>2S19 Msta-S</b> guided<br>Krasnopol laser-guided projectile; laser designation by an observer required                                                                                                                | 20 km   |         | maximum                                 | 8    |
| <b>Tornado-G</b> 9M538 HE-frag<br>9M538 rocket, enhanced monobloc HE-frag warhead                                                                                                                                        | 20 km   |         | maximum (claim)                         | 4    |
| <b>Tornado-G</b> 9M539 HE-frag<br>9M539 rocket, separable HE-frag warhead; Tecmash statement at IDEX 2019; 9M539 rocket;<br>Tecmash statement at IDEX 2019                                                               | 20 km   | 5 km    | maximum (claim),<br>minimum (claim)     | 4    |
| <b>Tornado-G</b> 9M541 cluster<br>9M541 rocket, cluster warhead with HEAT-frag submunitions                                                                                                                              | 20 km   |         | maximum                                 | 4    |
| <b>2S4 Tyulpan</b> 308 rocket-assisted cluster mine<br>cluster mine, 14 submunitions                                                                                                                                     | 19.3 km | 7.1 km  | maximum, minimum                        | 9    |
| <b>2S4 Tyulpan</b> Sayda incendiary rocket-assisted mine<br>incendiary (napalm) rocket-assisted mine; incendiary rocket-assisted mine                                                                                    | 19 km   | 0.7 km  | maximum, minimum                        | 9    |
| <b>2S4 Tyulpan</b> 3F2 rocket-assisted HE mine<br>228 kg rocket-assisted HE mine; WEG and GS do not name the round, 3F2 per RUWIKI                                                                                       | 18 km   | 0.8 km  | maximum, minimum                        | 9    |
| <b>BM-21 Grad 122 mm MRL</b> 9M28F HE-frag<br>9M28F HE-fragmentation rocket                                                                                                                                              | 15 km   | 1.5 km  | maximum, minimum<br>effective           | 4    |
| <b>Tornado-G</b> 9M28F HE-frag<br>legacy Grad 9M28F HE-frag rocket                                                                                                                                                       | 15 km   | 1.5 km  | maximum effective,<br>minimum effective | 4    |
| <b>ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer</b> ISDM<br>mine rocket<br>cassette rocket with AP or AT mines                                                                                    | 15 km   | 5 km    | maximum (claim),<br>minimum (claim)     | 33   |
| <b>BM-27 Uragan</b> 9M51 thermobaric<br>9M51 thermobaric rocket                                                                                                                                                          | 13 km   | 5 km    | maximum, minimum                        | 5    |
| <b>2S4 Tyulpan</b> 53-F-864 HE mine<br>HE mine 130.7 kg, full charge; HE mine, smallest charge                                                                                                                           | 9.65 km | 0.8 km  | maximum, minimum                        | 9    |
| <b>2S4 Tyulpan</b> Smel'chak-M laser-guided mine<br>laser-guided mine, 134 kg                                                                                                                                            | 9.5 km  | 0.9 km  | maximum, minimum                        | 9    |
| <b>2S4 Tyulpan</b> 3FS Smel'chak laser-guided mine<br>laser-guided mine, target designated by 1D15 from 0.2-5 km<br>CEP 80-90% within a 2-3 m circle [semi-active laser homing, target lit in final 400-800 m of flight] | 9.2 km  | 3.6 km  | maximum, minimum                        | 9    |
| <b>TOS-1A Solntsepek</b> thermobaric rocket<br>MO.1.01.04M thermobaric rocket, 3.7 m, 217 kg                                                                                                                             | 6,000 m |         | maximum                                 | 7    |
| <b>TOS-1A Solntsepek</b> incendiary rocket<br>MO.1.01.04M.OP incendiary rocket, 3.7 m, 204 kg                                                                                                                            | 6,000 m |         | maximum effective                       | 7    |
| <b>TOS-1A Solntsepek</b> original short rocket<br>MO.1.01.04 thermobaric rocket, 3.3 m, 173 kg, fired from TOS-1A                                                                                                        | 2,700 m |         | maximum                                 | 7    |
| <b>TOS-1A Solntsepek</b> minimum range<br>MO.1.01.04M rocket                                                                                                                                                             |         | 600 m   | minimum                                 | 7    |

Table 3b. Reach of long-range missiles and one-way attack drones

| Ballistic and cruise missiles and one-way attack drones. CEP, where known, is printed under the munition.                                                                                                                   |              |         |                          |      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------|--------------------------|------|
| SYSTEM, MUNITION AND CONDITION                                                                                                                                                                                              | RANGE        | MINIMUM | TERM                     | PAGE |
| Iskander-M 9M723 ballistic missile<br>9M723, domestic Iskander-M; From the MIS Iskander-E (export) data table, max 280 km;<br>domestic 9M723 minimum not confirmed<br>CEP 10 to 30 m (With electro-optical terminal seeker) | up to 500 km | 50 km   | maximum, minimum         | 11   |
| Iskander-M 9M728 (R-500) cruise missile<br>9M728 cruise missile, Iskander-K                                                                                                                                                 | 500 km       |         | official maximum (claim) | 11   |

**Table 3c. Reach of direct fire**

Guns, cannons, machine guns and grenade launchers, smallest calibre first, with the vehicles that carry them. The headline range follows the manufacturer, then WEG, then other sources; 'Sources' gives the spread where they disagree. Line charges are in the engineering table.

| WEAPON, TARGET AND AMMUNITION                                                                                                                                                                                                                                                                                                                                                                                                                                            | TERM      | RANGE              | SOURCES          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------|------------------|
| <b>PKT</b> 7.62 mm vehicle machine gun<br>Carried by, e.g.: T-62M, T-62MV, T-72A, T-72B, T-72B3, T-72B3M, T-80BV, T-80U, T-80BVM, T-90, T-90A, T-90M, T-14, BMPT, BMP-1, BMP-2, BMP-3, BMD-1, BMD-2, BMD-3, BMD-4, BMD-4M, BTR-D, BTR-60PB, BTR-70, BTR-80, BTR-80A, BTR-82, BTR-82A, BTR-82AM, BRDM-2, BRM-1K, BRM-3K, MT-LB, GAZ-3937 Vodnik, ZS3, ZS3M Akatsiya, ZS3 Giatint-S (7), ZS34 Khosta, ZS25 Sprut-SD, 9K35 Strela-10 (17), SNAR-10 (20), PRP-4M (36), GMZ-3 |           |                    |                  |
| ground. ball and API (57-N-323S, 7BZ-3)                                                                                                                                                                                                                                                                                                                                                                                                                                  | effective | 1,500 m (disputed) | 1,000 to 2,000 m |
| ground. ball and API (57-N-323S, 7BZ-3)                                                                                                                                                                                                                                                                                                                                                                                                                                  | maximum   | 4,000 m            | 3,800 to 4,000 m |
| <b>DShKM</b> 12.7 mm heavy machine gun; anti-aircraft gun on older tanks<br>Carried by, e.g.: T-62M, T-62MV, T-55A                                                                                                                                                                                                                                                                                                                                                       |           |                    |                  |
| ground. API (B-32), API-T (BZT-44)                                                                                                                                                                                                                                                                                                                                                                                                                                       | effective | 2,000 m            | 2,000 to 3,500 m |
| air. API (B-32), API-T (BZT-44)                                                                                                                                                                                                                                                                                                                                                                                                                                          | effective | 1,000 m            |                  |
| ground. API (B-32)                                                                                                                                                                                                                                                                                                                                                                                                                                                       | maximum   | 7,850 m            | 2,500 to 7,850 m |
| <b>Kord 6P49</b> 12.7 mm heavy machine gun, vehicle version of the Kord<br>Carried by, e.g.: T-72B3 (2016 standard), T-90 (late production), T-90M, T-14, ZS35 Koalitsiya-SV (8), MT-LBVMK, GAZ-3937 Vodnik, Tigr, Tigr-M                                                                                                                                                                                                                                                |           |                    |                  |
| ground, personnel, light armour. B-32 API, BZT-44 API-T                                                                                                                                                                                                                                                                                                                                                                                                                  | effective | 2,000 m            |                  |
| air. B-32 API, BZT-44 API-T                                                                                                                                                                                                                                                                                                                                                                                                                                              | effective | 1,000 m (disputed) | 1,000 to 1,500 m |
| ground. B-32 API                                                                                                                                                                                                                                                                                                                                                                                                                                                         | maximum   | 7,850 m            | 6,000 to 7,850 m |
| <b>NSVT</b> 12.7 mm heavy machine gun, vehicle version of the NSV<br>Carried by, e.g.: T-72A, T-72B, T-72B3 (early), T-80BV, T-80U, T-90 (early production), T-62M (some), ZS19 Msta-S (8), BREM-1, MT-LBVM                                                                                                                                                                                                                                                              |           |                    |                  |
| ground. API (B-32), API-T (BZT-44)                                                                                                                                                                                                                                                                                                                                                                                                                                       | effective | 2,000 m            |                  |
| air. API (B-32), API-T (BZT-44)                                                                                                                                                                                                                                                                                                                                                                                                                                          | effective | 1,000 m (disputed) | 1,000 to 1,500 m |
| ground. API (B-32)                                                                                                                                                                                                                                                                                                                                                                                                                                                       | maximum   | 7,850 m            | 2,000 to 7,850 m |
| <b>KPVT</b> 14.5 mm heavy machine gun, vehicle version of the KPV<br>Carried by, e.g.: BTR-60PB, BTR-70, BTR-80, BTR-82, BRDM-2, MT-LBM 6MA (Object 6MA), GAZ-3937 Vodnik                                                                                                                                                                                                                                                                                                |           |                    |                  |
| ground, light armour. API, API-T (B-32, BZT)                                                                                                                                                                                                                                                                                                                                                                                                                             | effective | 2,000 m            |                  |
| air. API, API-T (B-32, BZT)                                                                                                                                                                                                                                                                                                                                                                                                                                              | effective | 1,500 m            | 1,000 to 1,500 m |
| ground. 14.5 mm (not specified)                                                                                                                                                                                                                                                                                                                                                                                                                                          | maximum   | 9,000 m            |                  |
| <b>2A14</b> 23 mm automatic cannon, twin mount ZU-23-2<br>Carried by, e.g.: Ural-4320, Ural-43206 (truck-mounted ZU-23-2), KamAZ trucks (truck-mounted ZU-23-2), MT-LB, MT-LBM 6MA4, BTR-ZD Skrezhet                                                                                                                                                                                                                                                                     |           |                    |                  |
| ground. BZT, OFZ                                                                                                                                                                                                                                                                                                                                                                                                                                                         | effective | 2,500 m            | 2,000 to 2,500 m |
| air. BZT, OFZ                                                                                                                                                                                                                                                                                                                                                                                                                                                            | effective | 2,500 m            |                  |
| <b>2A38M</b> 30 mm twin-barrel anti-aircraft automatic cannon<br>Carried by: ZS6M, ZS6M1 Tunguska-M, -M1 (17), Pantsir-S1, Pantsir-SM (96K6) (16)                                                                                                                                                                                                                                                                                                                        |           |                    |                  |
| air. HE-I, AP-T                                                                                                                                                                                                                                                                                                                                                                                                                                                          | effective | 4,000 m            | 2,000 to 4,000 m |
| ground                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | maximum   | 2,000 m            |                  |
| <b>2A42</b> 30 mm automatic cannon<br>Carried by, e.g.: BMP-2, BMP-2D, BMP-2M Berezhok, BMD-2, BMD-3, BTR-90, BTR-82AT, BMPT, BMPT-72, Ka-52, Mi-28N, Mi-28NM, Ka-29, T-15, Kurganets-25, VPK-7829 Bumerang                                                                                                                                                                                                                                                              |           |                    |                  |
| light armour. armour-piercing (BT, APDS)                                                                                                                                                                                                                                                                                                                                                                                                                                 | effective | 1,500 m            | 1,500 to 2,000 m |
| personnel. HE (OFZ, OT)                                                                                                                                                                                                                                                                                                                                                                                                                                                  | effective | 4,000 m            |                  |
| air. HE                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | effective | 2,000 m            | 2,000 to 4,000 m |
| <b>2A72</b> 30 mm automatic cannon<br>Carried by, e.g.: BMP-3, BMP-3M, BMD-4, BMD-4M, BTR-80A, BTR-82A, BTR-82AM, BRM-3K Rys, BTR-90M, MT-LBM 6MB, BMP-1AM Basurmanin, GAZ Vodnik with BPPU, Uran-9                                                                                                                                                                                                                                                                      |           |                    |                  |
| light armour                                                                                                                                                                                                                                                                                                                                                                                                                                                             | effective | 1,500 m            | 1,500 to 2,000 m |
| personnel. HE                                                                                                                                                                                                                                                                                                                                                                                                                                                            | effective | 4,000 m            | 2,000 to 4,000 m |
| air. HE                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | effective | 4,000 m (disputed) | 1,500 to 4,000 m |
| <b>AGS-17</b> 30 mm automatic grenade launcher<br>Carried by, e.g.: BMPT (Terminator), BMD-3, Mi-24 (Hind), Mi-8T, LuAZ-967M                                                                                                                                                                                                                                                                                                                                             |           |                    |                  |
| personnel, ground. Frag-HE VOG-17M                                                                                                                                                                                                                                                                                                                                                                                                                                       | effective | 1,200 m            | 800 to 1,730 m   |
| personnel, ground. Frag-HE VOG-17M                                                                                                                                                                                                                                                                                                                                                                                                                                       | maximum   | 1,730 m            |                  |
| personnel, ground. Frag-HE GPD-30 (extended range)                                                                                                                                                                                                                                                                                                                                                                                                                       | maximum   | 2,100 m            |                  |

| WEAPON, TARGET AND AMMUNITION                                                                                                                                                                                       | TERM        | RANGE                 | SOURCES            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------|--------------------|
| <b>AGS-30</b> 30 mm automatic grenade launcher<br>Carried by, e.g.: BMP-2M Berezhok                                                                                                                                 |             |                       |                    |
| personnel, ground. Frag-HE VOG-17M, VOG-30                                                                                                                                                                          | maximum     | 1,700 m               |                    |
| personnel, ground. Frag-HE GPD-30 [extended range]                                                                                                                                                                  | maximum     | 2,100 m               | 2,100 to 2,200 m   |
| <b>57 mm gun of AU-220M Baikal (2A91 or 2A90, disputed)</b> 57 mm automatic gun in AU-220M remote weapon station<br>Carried by, e.g.: BMP-3 'Derivatsiya' (with AU-220M), BRM-3K with AU-220M, 2S38 Derivatsiya-PVO |             |                       |                    |
| ground                                                                                                                                                                                                              | effective   | 3,000 m               |                    |
| air                                                                                                                                                                                                                 | effective   | 7,500 m               | 6,000 to 7,500 m   |
| ground                                                                                                                                                                                                              | maximum     | more than 14,000 m    | 12,000 to 14,500 m |
| <b>2A28</b> 73 mm low-pressure smoothbore gun<br>Carried by: BMP-1, BMP-1P, BRM-1K, BMD-1, BMD-1P, Improvised towed mount                                                                                           |             |                       |                    |
| armour. HEAT-FS PG-15V                                                                                                                                                                                              | effective   | 1,300 m               | 500 to 1,300 m     |
| armour. HEAT-FS PG-15V                                                                                                                                                                                              | maximum     | 1,300 m               |                    |
| ground. Frag-HE OG-15V                                                                                                                                                                                              | maximum     | 4,400 m               | 4,400 to 4,500 m   |
| <b>SPG-9</b> 73 mm recoilless gun<br>Carried by, e.g.: UAZ-469                                                                                                                                                      |             |                       |                    |
| armour. HEAT PG-9VS (PG-9V)                                                                                                                                                                                         | effective   | 1,300 m               | 800 to 1,300 m     |
| armour. tandem HEAT PG-9VNT                                                                                                                                                                                         | effective   | 1,000 m               | 700 to 1,000 m     |
| personnel, ground. Frag-HE OG-9VM1                                                                                                                                                                                  | maximum     | 4,500 m               |                    |
| <b>2A70</b> 100 mm rifled gun-launcher<br>Carried by, e.g.: BMP-3, BMP-3M, BMP-3K, BMP-3F, BMD-4M, BMD-4                                                                                                            |             |                       |                    |
| ground. HE-Frag 3UOF17 (3OF32)                                                                                                                                                                                      | effective   | 4,000 m               |                    |
| air. HE-Frag                                                                                                                                                                                                        | effective   | 4,000 m               |                    |
| ground. HE-Frag 3UOF19 (3OF70)                                                                                                                                                                                      | maximum     | 6,500 m               | 6,500 to 7,000 m   |
| <b>D-10T2S</b> 100 mm rifled tank gun of the T-54B/T-55, manually loaded<br>Carried by, e.g.: T-55, T-55A, T-55AM, T-55AMV                                                                                          |             |                       |                    |
| armour. APFSDS (BM-412M, BM-25 type)                                                                                                                                                                                | effective   | 2,000-2,500 m         |                    |
| armour. HEAT (BK-17)                                                                                                                                                                                                | effective   | 1,000 m               |                    |
| ground. HE-Frag (OF-32)                                                                                                                                                                                             | effective   | 2,500-4,000 m         | 2,500 to 6,000 m   |
| ground. HE-Frag                                                                                                                                                                                                     | maximum     | 15,000 m              | 14,600 to 16,000 m |
| <b>MT-12</b> 100 mm smoothbore towed anti-tank gun<br>Carried by, e.g.: MT-LB, Ural-4320, Ural-375D                                                                                                                 |             |                       |                    |
| armour. APFSDS (BM-412M)                                                                                                                                                                                            | effective   | 2,500 m               | 1,880 to 2,500 m   |
| armour. direct fire sight OP4M-40U                                                                                                                                                                                  | sighting    | 3,000 m               |                    |
| ground. HE-Frag (OF-15)                                                                                                                                                                                             | maximum     | 8,200 m               |                    |
| <b>2A20</b> 115 mm smoothbore tank gun of the T-62, manually loaded<br>Carried by: T-62M, T-62MV                                                                                                                    |             |                       |                    |
| armour. APFSDS (BM-6, BD/36-2)                                                                                                                                                                                      | effective   | 1,800 m               | 1,800 to 2,100 m   |
| armour. HEAT (3BK4)                                                                                                                                                                                                 | effective   | 1,500 m               | 1,200 to 1,500 m   |
| ground. HE-Frag (OF-27)                                                                                                                                                                                             | effective   | 4,000 m               | 4,000 to 6,000 m   |
| ground. HE-Frag                                                                                                                                                                                                     | maximum     | 9,000 m               |                    |
| <b>2A45M</b> 125 mm smoothbore towed anti-tank gun with auxiliary power unit<br>Carried by, e.g.: MT-LB                                                                                                             |             |                       |                    |
| armour. APFSDS (BM-42M)                                                                                                                                                                                             | effective   | 3,000 m<br>(disputed) | 2,000 to 3,000 m   |
| ground. HE-Frag OF-26                                                                                                                                                                                               | direct fire | 5,000 m               |                    |
| ground. HE-Frag                                                                                                                                                                                                     | maximum     | 12,200 m              | 12,000 to 12,200 m |
| <b>2A46M</b> 125 mm smoothbore tank gun and ATGM launcher, autoloader<br>Carried by, e.g.: T-72B, T-72BM, T-90, T-80BV, T-80U, T-80BVM, T-64BV                                                                      |             |                       |                    |
| armour. APFSDS (3BM42M type)                                                                                                                                                                                        | effective   | 3,000 m               | 2,500 to 4,000 m   |
| armour. HEAT (3BK29M, 3BK27 type)                                                                                                                                                                                   | effective   | 3,000 m               | 2,000 to 4,000 m   |
| ground. HE-Frag (3OF26)                                                                                                                                                                                             | effective   | 5,000 m               | 4,000 to 5,000 m   |
| ground. HE-Frag                                                                                                                                                                                                     | maximum     | 10,000 m              |                    |
| <b>2A46M-5</b> 125 mm smoothbore tank gun and ATGM launcher, improved-accuracy 2A46M<br>Carried by, e.g.: T-72B3, T-72B3M, T-90A, T-90M                                                                             |             |                       |                    |
| armour. APFSDS (3BM42M type)                                                                                                                                                                                        | effective   | 3,000-4,000 m         | 3,000 to 4,000 m   |
| armour. HEAT-MP (3BK29M)                                                                                                                                                                                            | effective   | 4,000 m               |                    |
| ground. HE-Frag (3OF26)                                                                                                                                                                                             | effective   | 5,000 m               |                    |

| WEAPON, TARGET AND AMMUNITION                                                                                                     | TERM      | RANGE                  | SOURCES           |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------|-------------------|
| air. HE-shrapnel, time-fuzed (Ainet)                                                                                              | effective | 4,000-5,000 m          |                   |
| ground. HE-Frag                                                                                                                   | maximum   | 10,000 m               |                   |
| <b>2A75</b> 125 mm smoothbore gun with autoloader, a version of the 2A46M tank gun<br>Carried by: 2S25 Sprut-SD, 2S25M Sprut-SDM1 |           |                        |                   |
| armour. APFSDS (BM-42M / 3BM42)                                                                                                   | effective | 3,000 m                |                   |
| armour. APFSDS, HEAT (BK-29M) and guided rounds with 1A40M-1 sight                                                                | sighting  | 4,000 m                |                   |
| ground. HE-Frag OF-26 (3OF26)                                                                                                     | effective | 5,000 m                |                   |
| <b>2A82-1M</b> 125 mm smoothbore tank gun of the T-14, in an unmanned turret<br>Carried by: T-14 Armata                           |           |                        |                   |
| armour. APFSDS                                                                                                                    | effective | 3,500+ m               | 3,500 to 5,000 m  |
| ground. HEAT, HE-Frag                                                                                                             | effective | 4,700 m                |                   |
| ground                                                                                                                            | maximum   | 12,000 m<br>(disputed) | 7,000 to 12,000 m |

Other direct-fire figures in the entries

| SYSTEM, MUNITION AND CONDITION                                                                                                               | RANGE                          | MINIMUM | TERM        | PAGE |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|---------|-------------|------|
| <b>2A65 Msta-B</b> Direct fire range<br>ATP Table E-2 and RWW table; projectile not stated; the same table's maximum-range row is misaligned | 2,130 m                        |         | direct fire | 10   |
| <b>2S19 Msta-S</b> secondary MG<br>12.7x108 mm, ground and air targets                                                                       | 2,000 m<br>ground; 1,500 m air |         | effective   | 8    |
| <b>2A36 Giatsint-B 152 mm towed gun</b> direct fire<br>ammunition and target not stated in source                                            | 1,530 m                        |         | direct fire | 10   |
| <b>2S5 Giatsint-S</b> HEAT BP-540<br>HEAT BP-540, direct fire                                                                                | 1,000 m                        |         | direct fire | 7    |

**Table 3d. Reach of air defence weapons**

| Missiles and guns. Range is horizontal; altitude is the vertical reach. Figures are against the target named in the condition.                                                                         |                         |         |               |                                               |      |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------|---------------|-----------------------------------------------|------|--|
| SYSTEM, MUNITION AND CONDITION                                                                                                                                                                         | RANGE                   | MINIMUM | ALTITUDE      | TERM                                          | PAGE |  |
| <b>S-300V4</b> 9M82MD<br>air targets incl. AEW aircraft; S-300V4 new long-range missile                                                                                                                | 400 km                  |         |               | claimed maximum (claim)                       | 13   |  |
| <b>S-400 Triumf</b> 40N6<br>aerodynamic targets (STRASAM); APA brochure figure is for export 40N6E, 215 nmi; 40N6; the 27 km often quoted is reported for the 48N6DM missile specifically              | 400 km                  |         | 30-35 km      | maximum (claim)                               | 12   |  |
| <b>S-400 Triumf</b> 48N6DM (48N6E3)<br>aerodynamic targets; RUSIAD names export 48N6E3 (Army Technology lineage); TASS 2018 state_claim: 48N6DM up to 250 km                                           | 250 km                  |         |               | maximum (claim)                               | 12   |  |
| <b>S-300V4</b> 9M82M<br>aircraft targets; S-300VM                                                                                                                                                      | 200 km                  |         | 25 m-30 km    | maximum (claim), maximum and minimum (claim)  | 13   |  |
| <b>S-350 Vityaz</b> 9M96D<br>extended-range domestic 9M96D; target type not stated by TWZ                                                                                                              | about 121 km (75 miles) |         |               | range                                         | 13   |  |
| <b>S-350 Vityaz</b> 9M96E2<br>aerodynamic target; export missile                                                                                                                                       | 120 km                  | 1 km    | 5 m to 30 km  | maximum (claim), minimum (claim)              | 13   |  |
| <b>S-400 Triumf</b> 9M96E2<br>export 9M96E2; aerodynamic targets; APA gives 64.8 nmi; export 9M96E2; APA gives 15 ft to 100 kft                                                                        | 120 km                  |         | 0.005-30 km   | range (claim)                                 | 12   |  |
| <b>S-300V4</b> 9M82<br>aircraft and jammer targets; S-300V; aircraft targets; S-300V                                                                                                                   | 100 km                  | 13 km   | 1-30 km       | maximum, minimum (claim), maximum and minimum | 13   |  |
| <b>S-300V4</b> 9M83<br>aircraft targets; S-300V                                                                                                                                                        | 75 km                   | 6 km    | 250 m-25 km   | maximum, minimum, maximum and minimum         | 13   |  |
| <b>Buk-M1 / Buk-M2 / Buk-M3</b> 9M317M (Buk-M3)<br>air target                                                                                                                                          | 2.5-70 km               |         | 35 km         | maximum effective, maximum                    | 14   |  |
| <b>S-350 Vityaz</b> S-350E system<br>aerodynamic targets; export exhibit figure, also given by Russian MoD for the domestic S-350 (state claim); aerodynamic targets; export system brochure           | 1.5 to 60 km            |         | 10 m to 30 km | maximum (claim)                               | 13   |  |
| <b>S-400 Triumf</b> 48N6 (ballistic targets)<br>ballistic missile targets, 48N6 series                                                                                                                 | 60 km                   |         |               | range                                         | 12   |  |
| <b>Buk-M1 / Buk-M2 / Buk-M3</b> 9M317 (Buk-M2)<br>aircraft; air target                                                                                                                                 | 3-50 km                 |         | 15 m-25 km    | engagement zone, intercept altitude           | 14   |  |
| <b>Pantsir-S1</b> Pantsir-SM missile<br>Pantsir-SM with new missile; KBP chief designer, 2016                                                                                                          | 40 km                   |         |               | claimed (claim)                               | 16   |  |
| <b>S-300V4</b> 9M83/9M82 vs ballistic missile<br>ballistic targets; S-300V and S-300VM; ballistic targets; S-300V (S-300VM: 1-30 km)                                                                   | 40 km                   |         | 1-25 km       | maximum, maximum and minimum (claim)          | 13   |  |
| <b>S-350 Vityaz</b> 9M96E<br>aerodynamic target; export missile                                                                                                                                        | 40 km                   | 1 km    | 5 m to 20 km  | maximum (claim), minimum (claim)              | 13   |  |
| <b>S-400 Triumf</b> 9M96E<br>export 9M96E; aerodynamic targets; APA gives 21.6 nmi; export 9M96E; APA gives 15 ft to 66 kft                                                                            | 40 km                   |         | 0.005-20 km   | range (claim)                                 | 12   |  |
| <b>Buk-M1 / Buk-M2 / Buk-M3</b> 9M38M1 (Buk-M1)<br>air target                                                                                                                                          | 3.25-34 km              |         | 30 m-22 km    | maximum, intercept altitude                   | 14   |  |
| <b>Pantsir-S1</b> Pantsir-S1M missile<br>export Pantsir-S1M; new missile, designation not given by source; High Precision Systems, 2019                                                                | 30 km                   |         |               | claimed (claim)                               | 16   |  |
| <b>S-350 Vityaz</b> S-350E system vs ballistic<br>ballistic targets                                                                                                                                    | 1.5 to 30 km            |         | 2 to 25 km    | maximum (claim)                               | 13   |  |
| <b>S-350 Vityaz</b> 9M96<br>basic domestic 9M96; target type not stated by TWZ                                                                                                                         | about 29 km (18 miles)  |         |               | range                                         | 13   |  |
| <b>S-400 Triumf</b> S-400 (missile not stated)<br>reported engagement altitude band                                                                                                                    |                         |         | 0.1-27 km     | reported                                      | 12   |  |
| <b>Buk-M1 / Buk-M2 / Buk-M3</b> 9M317 vs ballistic and cruise missiles<br>ALCM-type cruise missile at 30 m (MISM2; ROE gives 100 m, unchecked); Lance-type tactical ballistic missile 15-20 km (MISM2) | 20 km                   |         |               | maximum (claim)                               | 14   |  |
| <b>Pantsir-S1</b> 57E6 missile<br>aircraft-type targets; KBP figures for export 57E6-E; KBP figures for export 57E6-E; 5 m only in optical mode over water                                             | 20 km                   | 1.2 km  | 5 m to 15 km  | maximum (claim), minimum (claim)              | 16   |  |
| <b>Tor</b> 9M338K<br>Tor-M2 domestic, air targets                                                                                                                                                      | 16 km                   |         | 10 m to 10 km | maximum (claim)                               | 15   |  |

| SYSTEM, MUNITION AND CONDITION                                                                                                                                                            | RANGE       | MINIMUM | ALTITUDE          | TERM                                             | PAGE |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|-------------------|--------------------------------------------------|------|
| <b>Strela-10 and Osa short-range SAM</b> 9M33M3<br>Osa-AKM missile; Osa-AKM missile; WEG figure                                                                                           | 15 km       |         | 4 km              | operational range,<br>maximum flight<br>altitude | 17   |
| <b>Tor</b> 9M330/9M331<br>aircraft and PGM targets, Tor/Tor-M1                                                                                                                            | 12 km       | 1.5 km  | 10 m to 6 km      | maximum,<br>minimum                              | 15   |
| <b>Tor</b> 9M331 (Tor-M2E)<br>Tor-M2E export; target 300 m/s; Tor-M2E export; target 300 m/s (10 m to 6 km<br>at 600 m/s)                                                                 | 1 to 12 km  |         | 10 m to 10 km     | maximum (claim)                                  | 15   |
| <b>S-350 Vityaz</b> 9M100<br>IR-homing short-range missile; fielding unconfirmed                                                                                                          | 10 km       |         |                   | maximum (claim)                                  | 13   |
| <b>Tunguska</b> 9M311-M1 missile<br>Tunguska-M1, air target; Tunguska-M1, air target; APA gives 2.5 km for 2K22 and<br>2K22M too; air target; ceiling for 9M311 family                    | 10 km       | 2.5 km  | 3.5 km            | maximum,<br>minimum                              | 17   |
| <b>Pantsir-S1</b> TKB-1055 small missile<br>anti-drone round, up to 48 per vehicle in cannon-less versions                                                                                | 7 km        |         |                   | claimed (claim)                                  | 16   |
| <b>Strela-10 and Osa short-range SAM</b> 9M37M/9M333<br>Strela-10 engagement zone, all variants; target closing at up to 415 m/s or<br>receding at 310 m/s; Strela-10; 9M333 minimum 10 m | 0.8 to 5 km |         | 25 m to 3.5<br>km | engagement<br>envelope                           | 17   |
| <b>Pantsir-S1</b> 30 mm 2A38M gun<br>air and ground targets; air targets                                                                                                                  | 4 km        | 200 m   | 0 to 3 km         | maximum (claim),<br>minimum (claim)              | 16   |
| <b>Tunguska</b> 30 mm 2A38M guns<br>air target                                                                                                                                            | 4 km        | 200 m   | 3 km              | maximum,<br>minimum                              | 17   |

Table 3e. Reach of sensors

Detection, tracking and location ranges, grouped by what the sensor is for. Target type and size are part of the figure: a radar that sees a bomber at 600 km sees a small drone much closer.

| REACH                                                               | SYSTEM, DATUM AND CONDITION                                                                                                                                                                                       | TARGET                     | TERM                         | PAGE |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------|------|
| <b>Ground surveillance</b>                                          |                                                                                                                                                                                                                   |                            |                              |      |
| up to 40 km                                                         | <b>SNAR-10 / SNAR-10M1 Leopard</b><br>Detection range, SNAR-10M1 radar. tanks and vehicles; Russian MoD figure; instrumented range 0.2–40 km                                                                      | tank, vehicle              | maximum (claim)              | 20   |
| 30 km                                                               | <b>SNAR-10 / SNAR-10M1 Leopard</b><br>Detection range, SNAR-10 radar. moving ships, without MTI                                                                                                                   | ship                       | maximum (claim)              | 20   |
| 20 km                                                               | <b>1B75 Penicillin</b><br>Detection range, 1L271. tank-type moving ground target                                                                                                                                  | tank                       | maximum (claim)              | 20   |
| 16 km                                                               | <b>SNAR-10 / SNAR-10M1 Leopard</b><br>Detection range, SNAR-10 radar. moving vehicles, without MTI; 10 km with MTI                                                                                                | vehicle                    | maximum (claim)              | 20   |
| 15 km                                                               | <b>SNAR-10 / SNAR-10M1 Leopard</b><br>Detection range, SNAR-10M1 radar. moving person; developer figure                                                                                                           | person                     | claimed (claim)              | 20   |
| 12 km                                                               | <b>PRP-4A Argus / PRP-4M Deytery</b><br>Detection range, radar (moving tank-type target). moving tank-type target                                                                                                 | tank                       | up to                        | 36   |
| up to 10,000 m                                                      | <b>1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)</b><br>Detection range, 1V14 optical reconnaissance. day                                                                                          | not stated                 | maximum                      | 35   |
| 10 km                                                               | <b>PRP-4A Argus / PRP-4M Deytery</b><br>Recognition range, laser rangefinder. up to three tank-type targets; large targets incl. buildings 25 km; WEG names the 1D14–1 periscopic rangefinder                     | tank                       | up to (claim)                | 36   |
| 10 km                                                               | <b>PRP-4A Argus / PRP-4M Deytery</b><br>Detection range, PSNR-5M radar on PRP-4M. moving tank, PRP-4M application; Jane's 2001 export data, confidence reported                                                   | tank                       | range (claim)                | 36   |
| 8 km                                                                | <b>PRP-4A Argus / PRP-4M Deytery</b><br>Detection range, optical-electronic (tank-type target, day). tank-type target, daytime                                                                                    | tank                       | up to                        | 36   |
| 7 km                                                                | <b>PRP-4A Argus / PRP-4M Deytery</b><br>Detection range, radar (person). single soldier                                                                                                                           | person                     | up to                        | 36   |
| 3 km                                                                | <b>PRP-4A Argus / PRP-4M Deytery</b><br>Detection range, optical-electronic (tank-type target, night). tank-type target, night                                                                                    | tank                       | up to                        | 36   |
| up to 2,500 m                                                       | <b>1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)</b><br>Detection range, 1V14 optical reconnaissance (night). night                                                                                | not stated                 | maximum                      | 35   |
| 2 km                                                                | <b>PRP-4A Argus / PRP-4M Deytery</b><br>Detection range, stationary tank, engine off (sensor not stated). stationary tank-type target, engine not running; ARMYR and RWW name no sensor, WEG lists it under radar | tank                       | no less than (claim)         | 36   |
| 750 m                                                               | <b>1B75 Penicillin</b><br>Minimum range, 1L271. mortar firing positions; 200 m for moving ground targets                                                                                                          | mortar                     | minimum (claim)              | 20   |
| <b>Counter-battery</b>                                              |                                                                                                                                                                                                                   |                            |                              |      |
| 65 km                                                               | <b>Zoopark-1 / Zoopark-1M</b><br>Detection range, tactical missile, 1L260. Launch positions of Lance / ATACMS type missiles                                                                                       | tactical missile           | up to                        | 19   |
| 45 km                                                               | <b>Zoopark-1 / Zoopark-1M</b><br>Detection range, rocket artillery, 1L260. MLRS firing position (DOGS, unverified, specifies M270 type)                                                                           | rocket launcher, artillery | up to                        | 19   |
| 40 km                                                               | <b>Zoopark-1 / Zoopark-1M</b><br>Detection range, tactical missile, 1L219M. Tactical missile launch position                                                                                                      | tactical missile           | detection of launch position | 19   |
| mortar 22, artillery 20, MLRS 35, tactical missile 40 km            | <b>Zoopark-1 / Zoopark-1M</b><br>Fire-correction range, 1L219M. Adjusting friendly fire, by weapon type                                                                                                           | not stated                 | fire correction range        | 19   |
| 30 km                                                               | <b>ARK-1M Rys</b><br>Detection range, ARK-1M vs battlefield missiles. Missile (WEG)                                                                                                                               | not stated                 | unspecified                  | 19   |
| 30 km                                                               | <b>Zoopark-1 / Zoopark-1M</b><br>Detection range, rocket artillery, 1L219M. MLRS firing position                                                                                                                  | rocket launcher, artillery | detection of launch position | 19   |
| 23–27 km                                                            | <b>Zoopark-1 / Zoopark-1M</b><br>Detection range, tube artillery, 1L260. 155–203 mm gun firing positions (23 km for 155 mm, 27 km for 203 mm)                                                                     | artillery                  | up to                        | 19   |
| Iskander 80, Smerch 70, Uragan 50, Grad 40, 203 mm 32, 152 mm 27 km | <b>Zoopark-1 / Zoopark-1M</b><br>Own-fire tracking range, 1L260. Tracking friendly fire, by weapon                                                                                                                | not stated                 | up to                        | 19   |

| REACH                                | SYSTEM, DATUM AND CONDITION                                                                                                                                                                          | TARGET                             | TERM                                  | PAGE |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------------------------------------|------|
| 25 km                                | <b>ARK-1M Rys</b><br>Detection range, ARK-1M vs MRL rockets. Rocket (WEG); mode not stated                                                                                                           | rocket launcher                    | unspecified                           | 19   |
| up to 25 km                          | <b>1B75 Penicillin</b><br>Detection range, 1B75. radius; artillery, mortar and MLRS firing positions, sound and thermal                                                                              | rocket launcher, mortar, artillery | maximum (claim)                       | 20   |
| 20 km                                | <b>Zoopark-1 / Zoopark-1M</b><br>Detection range, mortar, 1L219M. Mortar firing position; calibre not stated                                                                                         | mortar                             | detection of launch position          | 19   |
| 20 km                                | <b>Zoopark-1 / Zoopark-1M</b><br>Detection range, mortar, 1L260. Mortar firing position; calibre not stated. ATP lists this figure under '1L219M Zoopark-1M'                                         | mortar                             | range as listed by ATP (no qualifier) | 19   |
| 15 km                                | <b>1B75 Penicillin</b><br>Detection range, 1L271 (fire adjustment). by shell bursts; by trajectory 5 km                                                                                              | shell bursts                       | maximum (claim)                       | 20   |
| 15 km                                | <b>Zoopark-1 / Zoopark-1M</b><br>Detection range, tube artillery, 1L219M. Field artillery firing position                                                                                            | artillery                          | detection of launch position          | 19   |
| 13 km                                | <b>ARK-1M Rys</b><br>Detection range, ARK-1M vs mortars. Mortar (WEG); mode not stated                                                                                                               | mortar                             | unspecified                           | 19   |
| 10 km                                | <b>SNAR-10 / SNAR-10M1 Leopard</b><br>Detection range, SNAR-10 radar. shell impact (100 mm), without MTI                                                                                             | shell bursts                       | maximum (claim)                       | 20   |
| 10 km                                | <b>SNAR-10 / SNAR-10M1 Leopard</b><br>Detection range, SNAR-10M1 radar. shell burst on land; 20 km on water; developer figure                                                                        | shell bursts                       | claimed (claim)                       | 20   |
| 8 km                                 | <b>ARK-1M Rys</b><br>Detection range, ARK-1M vs artillery (howitzers). Artillery (WEG), howitzers (JREW02); target and mode not stated by WEG                                                        | artillery                          | unspecified                           | 19   |
| 5 km                                 | <b>1B75 Penicillin</b><br>Detection range, 1L271 (mortar positions). mortar firing positions, 81-120 mm                                                                                              | mortar                             | maximum (claim)                       | 20   |
| <b>SIGINT and DF</b>                 |                                                                                                                                                                                                      |                                    |                                       |      |
| 400 km                               | <b>Moskva-1</b><br>Detection range, airborne emitters. passive; emitting aircraft and cruise missiles; 360 deg sector                                                                                | cruise missile, emitters           | up to (claim)                         | 22   |
| up to 30 km                          | <b>Repellent-1</b><br>Detection range, UAV control-link signals. ground-station command signals; UAV telemetry up to 10 km; target size and altitude not given                                       | UAV, emitters                      | up to (claim)                         | 28   |
| 6 km                                 | <b>Leer-3</b><br>Detection range, GSM base stations. cellular base stations, GSM and possibly 3G/4G, measured from the UAV                                                                           | UAV, cell base station             | within                                | 27   |
| <b>Laser designation</b>             |                                                                                                                                                                                                      |                                    |                                       |      |
| up to 7,000 m                        | <b>1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)</b><br>Range, laser target illumination (1V14). for laser-guided (corrected) munitions                                               | not stated                         | maximum                               | 35   |
| <b>Air tracking and fire control</b> |                                                                                                                                                                                                      |                                    |                                       |      |
| 120 km                               | <b>Buk-M1 / Buk-M2 / Buk-M3</b><br>Detection range, 9A317 / 9A317M TELAR radar (M2/M3). RCS 1-2 m2 at 3 km altitude and 18-20 km vs same target at 10-15 m (M2); M3: RCS 1-2 m2, altitude not stated | RCS 1-2 m <sup>2</sup>             | detection range                       | 14   |
| 80 km                                | <b>Buk-M1 / Buk-M2 / Buk-M3</b><br>Detection range, 9S35 TELAR radar (M1). RCS 2 m2; 100 km vs RCS 3 m2                                                                                              | RCS 2 m <sup>2</sup>               | detection range                       | 14   |
| 80 km                                | <b>Strela-10 and Osa short-range SAM</b><br>Detection range, Dog Ear (PPRU-1M) battery radar. Strela-10 battery acquisition radar, F/G-band; target type not stated; tracking 35 km (JREW02)         | not stated                         | acquisition range                     | 17   |
| 36 km                                | <b>Pantsir-S1</b><br>Detection range, IRS1 search radar. 2 m2 target; S-band; up to 20 targets tracked                                                                                               | RCS 2 m <sup>2</sup>               | maximum (claim)                       | 16   |
| 28 km                                | <b>Pantsir-S1</b><br>Detection range, IRS2-1 tracking radar. target size not stated; WEG and Armada                                                                                                  | not stated                         | maximum (claim)                       | 16   |
| 17-26 km                             | <b>Pantsir-S1</b><br>Detection range, optical post. F-16 autotracking, 10 km visibility; 13-15 km for HARM missile                                                                                   | fighter, anti-radiation missile    | auto-tracking (claim)                 | 16   |
| 20 km                                | <b>Strela-10 and Osa short-range SAM</b><br>Detection range, Land Roll tracking radar. Osa; target type not stated in source                                                                         | not stated                         | tracking range                        | 17   |
| 20 km                                | <b>Tor</b><br>Detection range, TV sight. target not stated                                                                                                                                           | not stated                         | maximum                               | 15   |
| 15 km                                | <b>Tor</b><br>Detection range, tracking radar (Tor/Tor-M1). 60-degree sector; target and variant not stated                                                                                          | not stated                         | maximum                               | 15   |
| 13 km                                | <b>Tunguska</b><br>Detection range, 1RL144M tracking radar. J-band tracker                                                                                                                           | not stated                         | maximum                               | 17   |
| <b>Air surveillance</b>              |                                                                                                                                                                                                      |                                    |                                       |      |
| 600 km                               | <b>Nebo-M</b><br>Detection range, Nebo-M. complex; target type, size and altitude not stated                                                                                                         | not stated                         | maximum (claim)                       | 18   |
| 600 km                               | <b>Nebo-M</b><br>Altitude coverage, Nebo-M. complex, ballistic targets implied; not stated                                                                                                           | ballistic missile                  | maximum (claim)                       | 18   |

| REACH         | SYSTEM, DATUM AND CONDITION                                                                                                                                             | TARGET                             | TERM                         | PAGE |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------------------------------|------|
| 580-600 km    | <b>S-400 Triumf</b><br>Detection range, 91N6E. target RCS >10 m <sup>2</sup> (strategic bomber)                                                                         | RCS 10 m <sup>2</sup> , bomber     | detection/<br>tracking       | 12   |
| 330 km        | <b>S-300V4</b><br>Detection range, 9S15 Bill Board. target size not stated by WEG                                                                                       | not stated                         | maximum                      | 13   |
| 300 km        | <b>96L6 all-altitude acquisition radar</b><br>Detection range. Range band 5 to 300 km for detected targets; target size and altitude not stated                         | not stated                         | instrumented<br>(claim)      | 18   |
| 300 km        | <b>S-400 Triumf</b><br>Detection range, 96L6E. export 96L6E in S-300PMU family, target not stated; JREW-p1877, PDF p. 1877                                              | not stated                         | max (claim)                  | 12   |
| 200 km        | <b>Nebo-M</b><br>Detection range, Podlet-K1. main mode, 10-200 km; target size not stated                                                                               | not stated                         | maximum (claim)              | 18   |
| 175 km        | <b>S-300V4</b><br>Detection range, 9S19 High Screen. ballistic missile search; target size not stated                                                                   | ballistic missile                  | maximum                      | 13   |
| 160 km        | <b>Buk-M1 / Buk-M2 / Buk-M3</b><br>Detection range, 9S18M1-3 / 9S18M3 search radar (M2/M3). RCS 1-2 m <sup>2</sup> ; Buk-M2 9S18M1-3                                    | RCS 1-2 m <sup>2</sup>             | detection range              | 14   |
| 100-150 km    | <b>Buk-M1 / Buk-M2 / Buk-M3</b><br>Detection range, 9S18M1 search radar (M1). target size and altitude not stated                                                       | not stated                         | detection range              | 14   |
| 150 km        | <b>S-300V4</b><br>Detection range, 9S32 Grill Pan. 2 m <sup>2</sup> RCS target, manual mode (140 km automatic)                                                          | RCS 2 m <sup>2</sup>               | maximum<br>acquisition       | 13   |
| 120 km        | <b>Buk-M1 / Buk-M2 / Buk-M3</b><br>Detection range, 9S36 mast radar (M2). RCS 1-2 m <sup>2</sup> at 3 km altitude; 30-35 km vs same target at 10-15 m                   | RCS 1-2 m <sup>2</sup>             | detection range              | 14   |
| 100 km        | <b>96L6 all-altitude acquisition radar</b><br>Maximum target altitude. All-altitude mode, aerodynamic targets                                                           | aircraft                           | maximum (claim)              | 18   |
| 75 km         | <b>Pantsir-S1</b><br>Detection range, Pantsir-SM radar. target size not stated                                                                                          | not stated                         | claimed (claim)              | 16   |
| 70 km         | <b>Buk-M1 / Buk-M2 / Buk-M3</b><br>Detection range, 9S36M mast radar (M3). cruise missile flying at 5 m                                                                 | cruise missile                     | detection range              | 14   |
| 40 km         | <b>Barnaul-T</b><br>Detection range, 1L122-1E. aircraft, helicopters, cruise missiles, UAVs at up to 700 m/s; export 1L122-1E brochure figure                           | cruise missile,<br>helicopter, UAV | maximum (1-40<br>km) (claim) | 37   |
| 32 km         | <b>Tor</b><br>Detection range, surveillance radar (Tor-M2). air targets, size not stated                                                                                | aircraft                           | detection (claim)            | 15   |
| 30 km         | <b>Strela-10 and Osa short-range SAM</b><br>Detection range, Land Roll surveillance radar. Osa; target type not stated in source                                        | not stated                         | acquisition range            | 17   |
| 25-27 km      | <b>Tor</b><br>Detection range, surveillance radar (Tor/Tor-M1). aircraft; WEG does not state the variant                                                                | aircraft                           | maximum                      | 15   |
| 18 km         | <b>Tunguska</b><br>Detection range, 1RL144 surveillance radar. E-band; low-flying search mode for targets at about 15 m (JREW02)                                        | not stated                         | maximum                      | 17   |
| 9-15 km       | <b>Tor</b><br>Detection range, surveillance radar (Tor/Tor-M1), UAV. UAVs                                                                                               | UAV                                | maximum                      | 15   |
| 12 km         | <b>Tor</b><br>Detection range, surveillance radar (Tor/Tor-M1), helicopter. helicopters                                                                                 | helicopter                         | maximum                      | 15   |
| 10 km         | <b>Barnaul-T</b><br>Detection altitude, 1L122-1E. export 1L122-1E; 'at least 10 km'                                                                                     | not stated                         | maximum (claim)              | 37   |
| 10 km         | <b>Nebo-M</b><br>Minimum range, Podlet-K1. air targets, lower limit of stated 10-200 km detection zone                                                                  | aircraft                           | minimum (claim)              | 18   |
| 10 km         | <b>Nebo-M</b><br>Altitude coverage, Podlet-K1. maximum detection altitude                                                                                               | not stated                         | maximum (claim)              | 18   |
| 0.45 to 10 km | <b>Strela-10 and Osa short-range SAM</b><br>Detection range, 9S86 Snap Shot. Strela-10 range-only radar; target type not stated; minimum 0.45 km                        | not stated                         | maximum                      | 17   |
| 1 km          | <b>Barnaul-T</b><br>Minimum detection range, 1L122-1E. export 1L122-1E                                                                                                  | not stated                         | minimum (claim)              | 37   |
| <b>Other</b>  |                                                                                                                                                                         |                                    |                              |      |
| up to 100 km  | <b>PU-12M7</b><br>Coordinates measurement envelope, PU-12M7 (connected radars). coordinates measurement envelope from connected radars; PU-12M7 has no radar of its own | not stated                         | maximum (claim)              | 37   |

**Table 3f. Reach of communications**

Radio and data-link ranges. Ranges depend on antenna, terrain and whether the vehicle is halted.

| REACH       | SYSTEM, DATUM AND CONDITION                                                                                                                 | TERM                | PAGE |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------|
| 450 km      | <b>R-149 command and staff vehicles</b><br>Range, HF air-ground Prima, R-149MA1. HF to aircraft                                             | maximum (claim)     | 34   |
| 350 km      | <b>R-145BM Chayka command and staff vehicle</b><br>Range, R-130M dipole. HF, halted only, symmetrical dipole                                | range               | 34   |
| 350 km      | <b>R-149 command and staff vehicles</b><br>Range, HF R-168-100KBE, R-149MA1. halted, mast antenna; 50 km (20 km moving) on ASH-4 whip       | maximum (claim)     | 34   |
| 350 km      | <b>R-149 command and staff vehicles</b><br>Range, HF, R-149BMR. HF band 1.5-30 MHz                                                          | maximum (claim)     | 34   |
| 15-300 km   | <b>R-149 command and staff vehicles</b><br>Range, HF R-168-SKV, R-149AKSh-1. 1.5-30 MHz, 2 sets                                             | maximum             | 34   |
| 220-250 km  | <b>1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)</b><br>Radio range, R-111. 1V15 and 1V16, vehicle radio                     | stated              | 35   |
| 130 km      | <b>1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)</b><br>Radio range, R-130MT. 1V15 and 1V16, HF vehicle radio                | stated              | 35   |
| 120 km      | <b>R-149 command and staff vehicles</b><br>Range, VHF air-ground Prima, R-149MA1. VHF to aircraft                                           | maximum (claim)     | 34   |
| 75 km       | <b>R-145BM Chayka command and staff vehicle</b><br>Range, R-130M zenith antenna. HF, halted or moving, symmetrical zenith-radiation antenna | range               | 34   |
| 60 km       | <b>R-149 command and staff vehicles</b><br>Range, VHF, R-149BMR. VHF band 30-108 MHz                                                        | maximum (claim)     | 34   |
| 25-60 km    | <b>R-149 command and staff vehicles</b><br>Range, VHF R-168-100U-2, R-149AKSh-1. 30-108 MHz, 2 sets                                         | maximum             | 34   |
| 55 km       | <b>R-149 command and staff vehicles</b><br>Range, VHF R-168-100UE-2 on mast, R-149MA1. halted, mast antenna                                 | maximum (claim)     | 34   |
| 40 km       | <b>1B75 Penicillin</b><br>Radio range, 1B75. to external subscribers                                                                        | maximum (claim)     | 20   |
| up to 40 km | <b>PU-12M7</b><br>Data link range, PU-12M7 by radio. radio link to higher and subordinate posts                                             | maximum (claim)     | 37   |
| 25-40 km    | <b>R-149 command and staff vehicles</b><br>Range, VHF R-168-25U-2, R-149MA3. 30-108 MHz, 2 sets                                             | maximum             | 34   |
| 35 km       | <b>R-145BM Chayka command and staff vehicle</b><br>Range, R-111 3.4 m whip. VHF, halted or moving                                           | range               | 34   |
| up to 30 km | <b>PU-12M7</b><br>Data link range, 9S737 to higher post. radio                                                                              | maximum (claim)     | 37   |
| 20-25 km    | <b>1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)</b><br>Radio range. 1V13 and 1V14, vehicle radios                           | stated              | 35   |
| 25 km       | <b>R-149 command and staff vehicles</b><br>Range, VHF R-168-100UE-2, R-149MA1. on the move, vehicle antenna                                 | maximum (claim)     | 34   |
| 20 km       | <b>Barnaul-T</b><br>Data link range, 9S932-1. stationary; 10 km on the move (9S933: 3 km / 1 km)                                            | communication range | 37   |
| 20 km       | <b>R-145BM Chayka command and staff vehicle</b><br>Range, R-123MT 4 m whip. VHF, halted or moving                                           | range               | 34   |
| 6-20 km     | <b>R-149 command and staff vehicles</b><br>Range, UHF R-168-MRA, R-149MA3. 1.5-1.75 GHz data radio                                          | maximum             | 34   |
| up to 15 km | <b>PU-12M7</b><br>Data link range, PU-12M7 by field cable. field telephone cable                                                            | maximum (claim)     | 37   |
| 6 km        | <b>PRP-4A Argus / PRP-4M Deytery</b><br>Dismounted observation post, distance from vehicle. dismounted observation post                     | maximum             | 36   |
| up to 5 km  | <b>PU-12M7</b><br>Data link range, 9S737 to fire units. radio                                                                               | maximum (claim)     | 37   |

**Table 3g. Reach of electronic attack**

| Jamming ranges. Most are manufacturer claims. |                                                                                                                                                                                    |                                   |      |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------|
| REACH                                         | SYSTEM, DATUM AND CONDITION                                                                                                                                                        | TERM                              | PAGE |
| 5,000 km                                      | <b>Murmansk-BN</b><br>Jamming range, HF communications. HF (shortwave) radio links, skywave; ship, aircraft and ground emitters                                                    | effective (Armada's term) (claim) | 22   |
| about 1,000 km wide strip                     | <b>Palantín</b><br>Jamming frontage, radio communications. ground target: radio communications and radio reconnaissance; battalion tactical group exercise near Voronezh, Sep 2019 | claimed (claim)                   | 25   |
| 150–300 km                                    | <b>Krasukha-2 / Krasukha-4</b><br>Krasukha-4: jamming range. air and ground-based radar targets; target size and altitude not stated                                               | maximum (claim)                   | 21   |
| up to 250 km                                  | <b>Krasukha-2 / Krasukha-4</b><br>Krasukha-2: jamming range. air target: AEW aircraft radar; target altitude not stated                                                            | up to                             | 21   |
| 250 km                                        | <b>Moskva-1</b><br>Jamming range, airborne S-band radar. air targets; ARM23 says the 1L267 'can engage targets' per unnamed sources; Moskva-1 itself has no jammer (FOI)           | up to                             | 22   |
| 50 km or more                                 | <b>R-330Zh Zhitel</b><br>Jamming range, airborne users. User equipment installed on aircraft                                                                                       | claimed minimum (claim)           | 26   |
| up to 30 km                                   | <b>Repellent-1</b><br>Jamming range, small UAV control and telemetry links. air target: small UAVs; telemetry and satellite-navigation links; control link up to 10 km             | up to (claim)                     | 28   |
| up to 25 km                                   | <b>Pole-21</b><br>Jamming range, GNSS receivers (ground or air not stated). WEG (Pole-21E); Russian press gives 25 km per jamming module for the domestic Pole-21                  | effective                         | 28   |
| 25 km                                         | <b>R-330Zh Zhitel</b><br>Jamming range. Ground user equipment (WEG gives no target type; matches manufacturer's ground-user figure)                                                | maximum effective                 | 26   |
| 22 km                                         | <b>2S19 Msta-S</b><br>Range, jammer round. 3RB30 radio jamming projectile; jamming radius 700 m                                                                                    | range (claim)                     | 8    |
| up to 20 km                                   | <b>Palantín</b><br>Jamming range, UAV control links. air target: UAV radio control links                                                                                           | maximum (claim)                   | 25   |
| up to 6 km                                    | <b>Leer-3</b><br>Jamming range, cellular terminals (ground). radius of the zone of blocked GSM terminals around the UAV                                                            | up to (claim)                     | 27   |
| 3 km                                          | <b>Infaua</b><br>Jamming range, target not stated. WEG 'Disruption Range'; target and conditions not given                                                                         | disruption range                  | 29   |

Table 3h. Reach of UAVs

How far from its control point a UAV operates, and at what altitude.

| REACH         | SYSTEM, DATUM AND CONDITION                                                                               | TERM          | PAGE |
|---------------|-----------------------------------------------------------------------------------------------------------|---------------|------|
| 120 km        | Leer-3<br>Practical range, Orlan-10 from control vehicle. radius of UAV employment from the control point | up to (claim) | 27   |
| up to 5,000 m | Leer-3<br>Altitude, Orlan-10. operating altitude                                                          | up to (claim) | 27   |

Table 3i. Deployment distances

Where the system has been seen deployed or employed relative to the line of contact, and how far its elements spread. These are observations and reports, not rules; they indicate what may lie at a given depth.

| REACH                            | SYSTEM, DATUM AND CONDITION                                                                                     | TERM              | PAGE |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------|------|
| up to 100 km                     | <b>S-400 Triumf</b><br>Fire unit distance from command post. 98Zh6E fire unit from 55K6E, export composition    | maximum (claim)   | 12   |
| 15 to 30 km                      | <b>Borisoglebsk-2</b><br>Employment distance from line of contact. Donbas 2014-15, with Leer-3, Zhitel, R-934UM | practical         | 23   |
| 25 km                            | <b>BM-27 Uragan</b><br>Distance from front line, observed. one launcher hit by FPV drone, reported c. 2024-07   | practical         | 5    |
| about 15 km from enemy positions | <b>2S4 Tyulpan</b><br>Employment distance. reported MoD video, Ukraine, Nov 2022                                | practical (claim) | 9    |
| 5-10 km behind forward edge      | <b>Tor</b><br>Deployment distance, Tor-M1 in defence. first defensive position; 5-10 km between vehicles        | practical         | 15   |

**Table 3j. Engineering equipment**

What engineer vehicles can do, grouped by task: gap crossing, breaching, mine laying, earthmoving and transport across water. Mine-laying rockets also appear under indirect fire.

| SYSTEM, DATUM AND CONDITION                                                                                              | CAPABILITY                                    | PAGE |
|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------|
| <b>Gap crossing</b>                                                                                                      |                                               |      |
| <b>PMP heavy folding pontoon bridge park</b>                                                                             | 227 m                                         | 30   |
| Bridge, full set, 60-t. 32 river + 4 shore links, 6.5 m roadway                                                          |                                               |      |
| Bridge, full set, 20-t. split half-links, 3.29 m roadway                                                                 | 382 m                                         |      |
| Bridge, half set, 60-t. 16 river + 2 shore links                                                                         | 119 m                                         |      |
| Bridge, half set, 20-t. 16 river + 2 shore links, split                                                                  | 191 m                                         |      |
| Load capacity, bridge, full-width bridge; 20 t on split-link bridge                                                      | 60 t                                          |      |
| Roadway width, 60-t bridge                                                                                               | 6.5 m                                         |      |
| Rafts, 2, 3, 4, 5+1 shore, 8+1 shore links                                                                               | 40, 60, 80, 110, 170 t                        |      |
| Raft assembly, 40-t. 2 river links, working party 6                                                                      | 8 min                                         |      |
| Maximum current, PMP                                                                                                     | 2 m/s                                         |      |
| Tank speed on bridge, tanks on 60-t bridge                                                                               | up to 30 km/h                                 |      |
| Working party, full bridge, full bridge set                                                                              | approx. 65                                    |      |
| <b>TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge</b>                                   | up to 40 m wide, 3 m deep                     | 31   |
| Obstacle, TMM-3 bridge set. Set of four spans; ATP gives 42 m total length                                               |                                               |      |
| Load class, TMM-3. Wheeled and tracked loads; tracked vehicles at 15 km/h on bridge                                      | 60 t                                          |      |
| Bridge geometry, TMM-3. Spans can be connected to 10.5, 21, 31.5 or 42 m (RWW); ATP gives 10 m per set and 42 m together | 4 spans of 10.5 m; roadway 3.8 m              |      |
| Bridge, MTU-72. Single-span bridge; ENT72 and RUWM support only the 18 m span and 50 t                                   | 20 m long, 3.3 m wide, spans 18 m; 6,400 kg   |      |
| Load class, MTU-72                                                                                                       | 50 t                                          |      |
| <b>Breaching and mine clearing</b>                                                                                       |                                               |      |
| <b>IMR-2 / IMR-2M / IMR-3M combat engineer vehicle</b>                                                                   | 6 km/h                                        | 32   |
| Mine plough lane clearing, IMR-2M. AT mines, lane clearing rate (WEG term)                                               |                                               |      |
| <b>UR-77 Meteorit</b>                                                                                                    | 200 to 500 m                                  | 32   |
| Range, UZP-77 line charge. charge launch distance, UZP-77; UZ-67 charge 200 to 350 m (WEG)                               |                                               |      |
| Cleared lane length, UZP-77. anti-tank minefield; UZ-67 charge 75 to 80 m                                                | 80 to 90 m                                    |      |
| Cleared lane width, one line charge                                                                                      | up to 6 m                                     |      |
| Line charge length, UZP-77; UZ-67 is 83 m                                                                                | 93 m                                          |      |
| Cleared lane length, ZRShch. lane 6 m wide; charge in 6 m sections of 41 kg                                              | 100 m                                         |      |
| Time to clear a path, one full demining cycle                                                                            | 3 to 5 min                                    |      |
| <b>Mine laying</b>                                                                                                       |                                               |      |
| <b>ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer</b>                                               | 15 km                                         | 33   |
| Range, ISDM mine rocket. cassette rocket with AP or AT mines                                                             |                                               |      |
| Minimum range, ISDM mine rocket. cassette rocket with AP or AT mines                                                     | 5 km                                          |      |
| Reload, ISDM. container swap by transloader at a technical position                                                      | several minutes                               |      |
| Mine load, GMZ-3                                                                                                         | 208 AT mines                                  |      |
| Minefield length per load, GMZ-3. one vehicle, one load                                                                  | 1 km (pressure mines); 2 km (influence mines) |      |
| Platoon minefield, GMZ-3. 3 GMZ-3 of obstacle platoon                                                                    | 1,200 m, three rows, 624 mines in 26 min      |      |
| Laying speed, GMZ-3                                                                                                      | 16 km/h surface; 6 km/h buried                |      |
| Laying rate, GMZ-3                                                                                                       | 8 mines/min surface; 4 mines/min buried       |      |
| Mine spacing, GMZ-3                                                                                                      | 5 or 10 m                                     |      |
| <b>Earthmoving</b>                                                                                                       |                                               |      |
| <b>IMR-2 / IMR-2M / IMR-3M combat engineer vehicle</b>                                                                   | 2 t                                           | 32   |
| Boom lifting capacity                                                                                                    |                                               |      |
| Boom reach, IMR-2M                                                                                                       | 8.4 m                                         |      |
| Dozer capacity, IMR-2 earthmoving                                                                                        | 200 to 250 m <sup>3</sup> /h                  |      |
| <b>Transport across water</b>                                                                                            |                                               |      |
| <b>PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry</b>                                    | 12 t                                          | 31   |
| Payload on land, PTS-4                                                                                                   |                                               |      |
| Payload on water, PTS-4                                                                                                  | 18 t                                          |      |
| Payload, PMM-2 single ferry. WEG; RUWM gives 42.5 t for the PMM-2M                                                       | 42.5 t                                        |      |
| Payload, PMM-2 two joined. WEG gives 85 t for two PMM-2; RUWM gives 85 t for two PMM-2M, assembly 8 min                  | 85 t                                          |      |

Table 4. Mobility and time into action

| SYSTEM                                                                         | CODE    | WEIGHT           | FORDING                                | INTO ACTION                                                         | OUT OF ACTION    | PAGE |
|--------------------------------------------------------------------------------|---------|------------------|----------------------------------------|---------------------------------------------------------------------|------------------|------|
| 1B75 Penicillin                                                                | WH-AT   | INA              | INA                                    | 5 min                                                               | INA              | 20   |
| 1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)                   | TR-A    | 14.8 t           | Amphibious                             | INA                                                                 | INA              | 35   |
| 2A36 Giatsint-B 152 mm towed gun                                               | TW      | 9,760 kg         | INA                                    | 4 min                                                               | 4 min            | 10   |
| 2A65 Msta-B                                                                    | TW      | 7,000 kg         | INA                                    | 2 to 2.5 min                                                        | 2 min            | 10   |
| 2S19 Msta-S                                                                    | TR      | 42 t             | 1.2 m unprepared; 5 m prepared         | 1.5 min                                                             | 1.5 min          | 8    |
| 2S35 Koalitsiya-SV                                                             | TR      | 48-55 t          | 2.1 m                                  | 1.5 min                                                             | under 1 min      | 8    |
| 2S4 Tyulpan                                                                    | TR      | 27.5 t           | 1.0 m                                  | 25 min                                                              | 25 min           | 9    |
| 2S5 Giatsint-S                                                                 | TR      | 28.2 t           | 1.05 m                                 | less than 2 min                                                     | less than 2 min  | 7    |
| 2S7 Pion / 2S7M Malka                                                          | TR      | 46,500 kg        | 1.2 m                                  | 5-6 min                                                             | 3-5 min          | 9    |
| 96L6 all-altitude acquisition radar                                            | WH-AT   | INA              | 1.3 m                                  | 5 min                                                               | 5 min            | 18   |
| ARK-1M Rys                                                                     | TR      | 15.7 t           | 6 km/h                                 | 5 min                                                               | 5 min            | 19   |
| Barnaul-T                                                                      | TR-A    | 11,000 kg        | 5 km/h                                 | 5 min                                                               | 7 min            | 37   |
| BM-21 Grad 122 mm MRL                                                          | WH-AT   | 13.7 t           | 1.5 m                                  | 3 min                                                               | 2 min            | 4    |
| BM-27 Uragan                                                                   | WH-AT   | 20 t             | 1.2 m                                  | 3 min                                                               | 3 min            | 5    |
| BM-30 Smerch                                                                   | WH-AT   | 43.7 t           | 1.1 m                                  | 3 min                                                               | 3 min            | 6    |
| BMP-1KSh command and staff vehicle                                             | TR      | 13 t             | 7 km/h                                 | INA                                                                 | INA              | 35   |
| Borisoglebsk-2                                                                 | TR-A    | 11,000 kg        | 5 km/h                                 | 15 min or less                                                      | 15 min or less   | 23   |
| Buk-M1 / Buk-M2 / Buk-M3                                                       | TR      | 34 t             | 1 m                                    | 5 min                                                               | 5 min            | 14   |
| IMR-2 / IMR-2M / IMR-3M combat engineer vehicle                                | TR      | 44.3 t           | 1.2 m unprepared; 5.0 m with equipment | 6 min                                                               | INA              | 32   |
| Infaua                                                                         | WH-AT-A | 13,614 kg        | 9 km/h                                 | INA                                                                 | INA              | 29   |
| ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer            | WH-AT   | INA              | 1.8 m                                  | INA                                                                 | INA              | 33   |
| Iskander-M                                                                     | WH-AT   | 42.85 t          | 1.4 m                                  | 16 min to first launch from the march; 4 min from highest readiness | INA              | 11   |
| Krasukha-2 / Krasukha-4                                                        | WH-AT   | 39,000 kg        | 1.75 m                                 | less than 20 min                                                    | less than 20 min | 21   |
| Leer-2                                                                         | WH-AT   | 7,800 kg         | 1.3 m                                  | INA                                                                 | INA              | 24   |
| Leer-3                                                                         | WH-AT   | INA              | INA                                    | 30 min                                                              | INA              | 27   |
| Moskva-1                                                                       | WH-AT   | 12,450 kg        | 1.75 m                                 | 45 min                                                              | INA              | 22   |
| Murmansk-BN                                                                    | WH-AT   | up to 35 t       | INA                                    | 72 h                                                                | INA              | 22   |
| Nebo-M                                                                         | WH-AT   | INA              | INA                                    | 12 m                                                                | 15 min           | 18   |
| Palantin                                                                       | WH-AT   | INA              | INA                                    | about 30 min                                                        | INA              | 25   |
| Pantsir-S1                                                                     | WH-AT   | 14.85 t          | 1.8 m                                  | 3 min                                                               | INA              | 16   |
| PMP heavy folding pontoon bridge park                                          | WH-AT   | INA              | INA                                    | 7 m of bridge per minute                                            | a few minutes    | 30   |
| Pole-21                                                                        | NA      | INA              | INA                                    | INA                                                                 | INA              | 28   |
| Polyana-D4                                                                     | WH-AT   | 13.2 t           | 1.2 m                                  | 35 min                                                              | 35 min           | 36   |
| PRP-4A Argus / PRP-4M Deytery                                                  | TR-A    | 13.8 t           | 7 km/h                                 | 2 min                                                               | INA              | 36   |
| PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry | TR-A    | 33.1 t           | INA                                    | 5 min                                                               | INA              | 31   |
| PU-12M7                                                                        | WH-AT-A | 13.6 t           | 10 km/h                                | 3 min                                                               | INA              | 37   |
| R-145BM Chayka command and staff vehicle                                       | WH-AT-A | 10.3 t           | 10 km/h                                | INA                                                                 | INA              | 34   |
| R-149 command and staff vehicles                                               | WH-AT-A | 15,000-15,050 kg | 10 km/h                                | 10 min on the move / 30 min halted                                  | INA              | 34   |
| R-330B Mandat-B and R-934BMV                                                   | WH-AT   | 11,000 kg        | INA                                    | 30 min                                                              | INA              | 24   |
| R-330Zh Zhitel                                                                 | WH-AT   | INA              | INA                                    | 40 min or less                                                      | 40 min or less   | 26   |
| Repellent-1                                                                    | WH-AT   | 23,650 kg        | 1.0 m                                  | 15 min                                                              | INA              | 28   |
| S-300V4                                                                        | TR      | 50 t             | 1.2 m                                  | 5 min                                                               | 5 min            | 13   |
| S-350 Vityaz                                                                   | WH-AT   | 22 t             | 1.2 m                                  | 5 min                                                               | 5 min            | 13   |

| SYSTEM                                                                          | CODE  | WEIGHT    | FORDING | INTO ACTION            | OUT OF ACTION       | PAGE |
|---------------------------------------------------------------------------------|-------|-----------|---------|------------------------|---------------------|------|
| S-400 Triumf                                                                    | WH-AT | INA       | INA     | 5-10 min               | INA                 | 12   |
| SNAR-10 / SNAR-10M1 Leopard                                                     | TR    | 12.7 t    | INA     | 5 min                  | INA                 | 20   |
| Strela-10 and Osa short-range SAM                                               | TR    | 12.1 t    | 1.2 m   | 3 to 5 min             | 3 to 5 min          | 17   |
| TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge | WH-AT | 19,500 kg | 1.0 m   | 72 min                 | 8 min               | 31   |
| Tor                                                                             | TR    | 32 t      | 1.0 m   | less than 3 min        | less than 5 min     | 15   |
| Tornado-G                                                                       | WH-AT | 14 t      | 1.75 m  | 2 min                  | 3 min               | 4    |
| TOS-1A Sointsepek                                                               | TR    | 44.3 t    | 1.2 m   | 90 s from halt to fire | INA                 | 7    |
| Tunguska                                                                        | TR    | 34 t      | 1.0 m   | not more than 5 min    | not more than 5 min | 17   |
| UR-77 Meteorit                                                                  | TR-A  | 15.5 t    | 5 km/h  | INA                    | 30 to 40 min        | 32   |
| Uragan-1M                                                                       | WH-AT | 40 t      | 1.4 m   | 3 min                  | 3 min               | 5    |
| Zoopark-1 / Zoopark-1M                                                          | TR    | 38.1 t    | INA     | 5 min                  | 5 min               | 19   |

Table 5. Source register

Codes are printed at the foot of each entry. Web sources are coded by publisher and numbered: APA Air Power Australia, AR Army Recognition, ARMADA Armada International, BCAT Bellingcat, BMPD bmpd blog, CSIS CSIS Missile Threat, FAS Federation of American Scientists, GS GlobalSecurity, MIL Militaryny, MISS missilery.info, ORYX Oryx, RDP Russian Defense Policy, RUSI RUSI web, TASS TASS, TOPWAR Topwar, TWZ The War Zone, VPK VPK, VRF Vooruzhenie Rossii, WEG WEG online (ODIN), WMC Wikimedia Commons, WP Wikipedia (language after the hyphen), WSPOT WarSpotting. Other codes name a single document.

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42–44, 3GIBMALS, 2023–06–07.  
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Argumenty i Fakty, 2022–07–06.

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Ukraine? Argumenty i Fakty, 2022.

[https://aif.ru/society/army/chto\\_zh\\_aistyonok\\_vedyot\\_kontrabatareynuyu\\_borbu\\_na\\_ukraine](https://aif.ru/society/army/chto_zh_aistyonok_vedyot_kontrabatareynuyu_borbu_na_ukraine)

**AIF3** Voda — ne pregrada. Boiyci «Centra» otrabotali navedenie pontonnoy pererapy. Argumenty i fakty, 2025–02–26.

<https://aif.ru/politics/russia/voda-ne-pregrada-boiyci-centra-otrabotali-navedenie-pontonnoy-pererapy>

**AIF4** Chto izvestno o neproniisatseom kupole zashchity ot dronov

Pole–21? Argumenty i Fakty, 2021–04–17.

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**AIF5** Chto predstavlyayet soboy zennitnyy raketnyy kompleks C–350 «Vyutėz»?. Argumenty i Fakty.

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Argumenty i Fakty, 2022–06–07.

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**APA5** Russian EW systems: R–330B/T, R–934B/BM. Air Power Australia,

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[https://odin.tradoc.army.mil/WEG/Asset/2S35\\_Koalitsiya-SV\\_Russian\\_152mm\\_Self-Propelled\\_Howitzer\\_SPH](https://odin.tradoc.army.mil/WEG/Asset/2S35_Koalitsiya-SV_Russian_152mm_Self-Propelled_Howitzer_SPH)

**WEG2** 2S4 Tyulpan [M-1975] Russian 240mm Self-Propelled Heavy Mortar (local extract). US Army TRADOC, ODIN WEG, 2025-05-06.

[https://odin.tradoc.army.mil/WEG/Asset/2S4\\_Tyulpan\\_\[M-1975\]\\_Russian\\_240mm\\_Self-Propelled\\_Heavy\\_Mortar](https://odin.tradoc.army.mil/WEG/Asset/2S4_Tyulpan_[M-1975]_Russian_240mm_Self-Propelled_Heavy_Mortar)

**WEG3** 2S5 Giatshint-S Russian 152mm Self-Propelled Gun (SPG), WEG extract. US Army TRADOC, 2025-03-11.

[https://odin.tradoc.army.mil/WEG/Asset/2S5\\_Giatshint-S\\_Russian\\_152mm\\_Self-Propelled\\_Gun\\_SPG](https://odin.tradoc.army.mil/WEG/Asset/2S5_Giatshint-S_Russian_152mm_Self-Propelled_Gun_SPG)

**WEG4** 2S7 Pion [M1975] Russian 203mm SPG, WEG extract. US Army TRADOC, 2025-04-22.

[https://odin.tradoc.army.mil/WEG/Asset/2S7\\_Pion\\_\[M1975\]\\_Russian\\_203mm\\_Self-Propelled\\_Gun\\_SPG](https://odin.tradoc.army.mil/WEG/Asset/2S7_Pion_[M1975]_Russian_203mm_Self-Propelled_Gun_SPG)

**WEG5** 96L6E [Cheese Board] Russian Early-Warning and Acquisition Radar (WEG extract). US Army TRADOC ODIN, 2024-10-23.

[https://odin.tradoc.army.mil/WEG/Asset/96L6E\\_\[Cheese\\_Board\]\\_Russian\\_Early-Warning\\_and\\_Acquisition\\_Radar](https://odin.tradoc.army.mil/WEG/Asset/96L6E_[Cheese_Board]_Russian_Early-Warning_and_Acquisition_Radar)

**WEG6** WEG: ARK-1M RYS Russian Artillery Locating Radar (local extract, last modified 2024-10-22). US Army TRADOC, 2024-10-22.

[https://odin.tradoc.army.mil/WEG/Asset/ARK-1M\\_RYS\\_Russian\\_Artillery\\_Locating\\_Radar](https://odin.tradoc.army.mil/WEG/Asset/ARK-1M_RYS_Russian_Artillery_Locating_Radar)

**WEG7** Barnaul-T Russian Air Defense Command and Control System. US Army TRADOC, ODIN WEG, 2025-06-10.

[https://odin.tradoc.army.mil/WEG/Asset/Barnaul-T\\_Russian\\_Air\\_Defense\\_Command\\_and\\_Control\\_System](https://odin.tradoc.army.mil/WEG/Asset/Barnaul-T_Russian_Air_Defense_Command_and_Control_System)

**WEG8** WEG extract: BMP-1KSh [BMP M1978], last modified 2024-08-07. US Army TRADOC, ODIN Worldwide Equipment Guide, 2024-08-07.

[https://odin.tradoc.army.mil/WEG/Asset/BMP-1KSh\\_\[BMP\\_M1978\]\\_Russian\\_Amphibious\\_Command\\_Post\\_and\\_Staff\\_Infantry\\_Fighting\\_Vehicle\\_\[IFV\]](https://odin.tradoc.army.mil/WEG/Asset/BMP-1KSh_[BMP_M1978]_Russian_Amphibious_Command_Post_and_Staff_Infantry_Fighting_Vehicle_[IFV])

**WEG9** Borisoglebsk-2 [RB-301B] Russian Amphibious Multipurpose Jamming Complex. US Army TRADOC, ODIN WEG, 2025-06-05.

[https://odin.tradoc.army.mil/WEG/Asset/Borisoglebsk-2\\_\[RB-301B\]\\_Russian\\_Amphibious\\_Multipurpose\\_Jamming\\_Complex](https://odin.tradoc.army.mil/WEG/Asset/Borisoglebsk-2_[RB-301B]_Russian_Amphibious_Multipurpose_Jamming_Complex)

**WEG10** 9K37 Buk-M1 [SA-11 Gadfly] Russian Medium-Range Surface-to-Air Missile System. US Army TRADOC, ODIN Worldwide Equipment Guide, 2025-06-10.

[https://odin.tradoc.army.mil/WEG/Asset/9K37\\_Buk-M1\\_\[SA-11\\_Gadfly\]\\_Russian\\_Medium-Range\\_Surface-to-Air\\_Missile\\_System](https://odin.tradoc.army.mil/WEG/Asset/9K37_Buk-M1_[SA-11_Gadfly]_Russian_Medium-Range_Surface-to-Air_Missile_System)

**WEG11** Buk-M2 [SA-17 Grizzly] Russian Medium-Range Air Defense Missile System. US Army TRADOC, ODIN Worldwide Equipment Guide, 2025-06-10.

[https://odin.tradoc.army.mil/WEG/Asset/Buk-M2\\_\[SA-17\\_Grizzly\]\\_Russian\\_Medium-Range\\_Air\\_Defense\\_Missile\\_System](https://odin.tradoc.army.mil/WEG/Asset/Buk-M2_[SA-17_Grizzly]_Russian_Medium-Range_Air_Defense_Missile_System)

**WEG12** 9K317M Buk-M3 [SA-27] Russian Medium-Range Air Defense Missile System. US Army TRADOC, ODIN Worldwide Equipment Guide, 2025-06-10.

[https://odin.tradoc.army.mil/WEG/Asset/9K317M\\_Buk-M3\\_\[SA-27\]\\_Russian\\_Medium-Range\\_Air\\_Defense\\_Missile\\_System](https://odin.tradoc.army.mil/WEG/Asset/9K317M_Buk-M3_[SA-27]_Russian_Medium-Range_Air_Defense_Missile_System)

**WEG13** WEG extract: BM-21 Grad Russian 122mm MRL (last modified 2024-12-19). US Army TRADOC, 2024-12.

[https://odin.tradoc.army.mil/WEG/Asset/BM-21\\_Grad\\_Russian\\_122mm\\_Multiple\\_Rocket\\_Launcher\\_\[MRL\]](https://odin.tradoc.army.mil/WEG/Asset/BM-21_Grad_Russian_122mm_Multiple_Rocket_Launcher_[MRL])

**WEG14** WEG extract groza.json: R-934UM2, R-330B, R-934B pages. US Army TRADOC, 2025-06-05.

[https://odin.tradoc.army.mil/WEG/Asset/R-934B\\_Russian\\_VHF\\_Air\\_Communication\\_Automated\\_Jamming\\_Station](https://odin.tradoc.army.mil/WEG/Asset/R-934B_Russian_VHF_Air_Communication_Automated_Jamming_Station)

**WEG15** WEG extract: IMR-2, IMR-2M, IMR-3M pages (pilot/weg\_extract/imr.json). US Army TRADOC, 2024-10.

[https://odin.tradoc.army.mil/WEG/Asset/IMR-2\\_Russian\\_Combat\\_Engineering\\_Vehicle](https://odin.tradoc.army.mil/WEG/Asset/IMR-2_Russian_Combat_Engineering_Vehicle)

**WEG16** Infaua [RB-531B] Russian 8x8 Amphibious Electronic Warfare System. TRADOC WEG, 2025-06-05.

[https://odin.tradoc.army.mil/WEG/Asset/Infaua\\_\[RB-531B\]\\_Russian\\_8x8\\_Amphibious\\_Electronic\\_Warfare\\_System](https://odin.tradoc.army.mil/WEG/Asset/Infaua_[RB-531B]_Russian_8x8_Amphibious_Electronic_Warfare_System)

**WEG17** Krasuha Russian Mobile Electronic Warfare Jamming System (WEG extract). US Army TRADOC, ODIN, 2025-06-05.

[https://odin.tradoc.army.mil/WEG/Asset/Krasuha\\_Russian\\_Mobile\\_Electronic\\_Warfare\\_Jamming\\_System](https://odin.tradoc.army.mil/WEG/Asset/Krasuha_Russian_Mobile_Electronic_Warfare_Jamming_System)

**WEG18** WEG: Leer-2 Russian 4x4 Mobile EW Vehicle (local extract). US Army TRADOC, 2025-06-05.

[https://odin.tradoc.army.mil/WEG/Asset/Leer-2\\_Russian\\_4x4\\_Mobile\\_Electronic\\_Warfare\\_\[EW\]\\_Vehicle](https://odin.tradoc.army.mil/WEG/Asset/Leer-2_Russian_4x4_Mobile_Electronic_Warfare_[EW]_Vehicle)

**WEG19** Moskva-1 [1L265] Russian Electronic Warfare Vehicle. US Army TRADOC ODIN WEG, 2025-06-06.

[https://odin.tradoc.army.mil/WEG/Asset/Moskva-1\\_\[1L265\]\\_Russian\\_Electronic\\_Warfare\\_Vehicle](https://odin.tradoc.army.mil/WEG/Asset/Moskva-1_[1L265]_Russian_Electronic_Warfare_Vehicle)

**WEG20** Worldwide Equipment Guide: 2S19 Msta-S, 2S19M1 (Msta-SM1) and 2S19M2 (Msta-SM2) 152mm SPH records. US Army TRADOC (ODIN), 2025-04-03 [2S19]; 2025-04-07 [2S19M1, 2S19M2].

[https://odin.tradoc.army.mil/WEG/Asset/2S19\\_Msta-S\\_Russian\\_152mm\\_Self-Propelled\\_Howitzer\\_\[SPH\]](https://odin.tradoc.army.mil/WEG/Asset/2S19_Msta-S_Russian_152mm_Self-Propelled_Howitzer_[SPH])

**WEG21** Murmansk-BN Russian Long-Range Communications Jamming System. US Army TRADOC, ODIN WEG, 2025-06-05.

[https://odin.tradoc.army.mil/WEG/Asset/Murmansk-BN\\_Russian\\_Long-Range\\_Communications\\_Jamming\\_System](https://odin.tradoc.army.mil/WEG/Asset/Murmansk-BN_Russian_Long-Range_Communications_Jamming_System)

**WEG22** 48Ya6-K1 Podlet K1 Russian Mobile Long-Range High-Altitude Surveillance Radar (local extract). US Army TRADOC, ODIN WEG, 2025-04-08.

[https://odin.tradoc.army.mil/WEG/Asset/48Ya6-K1\\_Podlet\\_K1\\_Russian\\_Mobile\\_Long-Range\\_High-Altitude\\_Surveillance\\_Radar](https://odin.tradoc.army.mil/WEG/Asset/48Ya6-K1_Podlet_K1_Russian_Mobile_Long-Range_High-Altitude_Surveillance_Radar)

**WEG23** Pantsir-S1 [SA-22 Greyhound] Russian Short-Range Air Defense Gun/Missile System. US Army TRADOC ODIN WEG, 2025-06-10.

[https://odin.tradoc.army.mil/WEG/Asset/Pantsir-S1\\_\[SA-22\\_Greyhound\]\\_Russian\\_Short-Range\\_Air\\_Defense\\_Gun:Missile\\_System](https://odin.tradoc.army.mil/WEG/Asset/Pantsir-S1_[SA-22_Greyhound]_Russian_Short-Range_Air_Defense_Gun:Missile_System)

**WEG24** PMP Russian Heavy Folding Pontoon Bridge (WEG online). US Army TRADOC G-2, 2024-10-16.

[https://odin.tradoc.army.mil/WEG/Asset/PMP\\_Russian\\_Heavy\\_Folding\\_Pontoon\\_Bridge](https://odin.tradoc.army.mil/WEG/Asset/PMP_Russian_Heavy_Folding_Pontoon_Bridge)

**WEG25** Pole-21E Russian RF Jammer. US Army TRADOC WEG, 2025-06-05.

[https://odin.tradoc.army.mil/WEG/Asset/Pole-21E\\_Russian\\_RF\\_Jammer](https://odin.tradoc.army.mil/WEG/Asset/Pole-21E_Russian_RF_Jammer)

**WEG26** 9S52 Polyana-D4 ASU Russian 6x6 Air Defense Command and Control Vehicle (local extract). US Army TRADOC, ODIN WEG, 2025-06-10. [https://odin.tradoc.army.mil/WEG/Asset/9S52\\_Polyana-D4\\_ASU\\_Russian\\_6x6\\_Air\\_Defense\\_Command\\_and\\_Control\\_Vehicle](https://odin.tradoc.army.mil/WEG/Asset/9S52_Polyana-D4_ASU_Russian_6x6_Air_Defense_Command_and_Control_Vehicle)

**WEG27** WEG: PRP-4A (Argus) Russian Amphibious Artillery Reconnaissance Vehicle. TRADOC, 2024-10-11. [https://odin.tradoc.army.mil/WEG/Asset/PRP-4A\\_\[Argus\]\\_Russian\\_Amphibious\\_Artillery\\_Reconnaissance\\_Vehicle](https://odin.tradoc.army.mil/WEG/Asset/PRP-4A_[Argus]_Russian_Amphibious_Artillery_Reconnaissance_Vehicle)

**WEG28** WEG extract: PTS-4 and PMM-2 pages. US Army TRADOC, WEG (local extract), 2024. [https://odin.tradoc.army.mil/WEG/Asset/PTS-4\\_Russian\\_Amphibious\\_Cargo\\_Vehicle](https://odin.tradoc.army.mil/WEG/Asset/PTS-4_Russian_Amphibious_Cargo_Vehicle)

**WEG29** PU-12M7 Russian 8x8 Mobile Command and Control Vehicle; 9S737 Ranzhir Russian 6x6 Mobile Command Center. US Army TRADOC, ODIN WEG, 2024-08-26. [https://odin.tradoc.army.mil/WEG/Asset/PU-12M7\\_Russian\\_8x8\\_Mobile\\_Command\\_and\\_Control\\_Vehicle](https://odin.tradoc.army.mil/WEG/Asset/PU-12M7_Russian_8x8_Mobile_Command_and_Control_Vehicle)

**WEG30** WEG BTR-60-R-145BM Command Post and Communication Center Vehicle. US Army TRADOC, 2024-08-07. [https://odin.tradoc.army.mil/WEG/Asset/BTR-60-R-145BM\\_Russian\\_8x8\\_Command\\_Post\\_and\\_Communication\\_Center\\_Vehicle](https://odin.tradoc.army.mil/WEG/Asset/BTR-60-R-145BM_Russian_8x8_Command_Post_and_Communication_Center_Vehicle)

**WEG31** Worldwide Equipment Guide sheets: R-149AKSh-1 (mod. 2025-04-23), R-149BMR (2025-01-08), R-149MA3 (2025-04-10). US Army TRADOC (ODIN), 2025. [https://odin.tradoc.army.mil/WEG/Asset/R-149AKSh-1\\_Russian\\_6x6\\_Command\\_and\\_Signal\\_Vehicle](https://odin.tradoc.army.mil/WEG/Asset/R-149AKSh-1_Russian_6x6_Command_and_Signal_Vehicle)

**WEG32** Repellent Russian 8x8 Anti-Drone Warfare Complex. TRADOC WEG (local extract), 2025-06-05. [https://odin.tradoc.army.mil/WEG/Asset/Repellent\\_Russian\\_8x8\\_Anti-Drone\\_Warfare\\_Complex](https://odin.tradoc.army.mil/WEG/Asset/Repellent_Russian_8x8_Anti-Drone_Warfare_Complex)

**WEG33** Worldwide Equipment Guide: S-300V (SA-12a Gladiator) and S-300V (SA-12b Giant) Russian Long-Range Air Defense Missile System (local extract pilot/weg\_extract/s300v4.json). US Army TRADOC, 2025-06-10. [https://odin.tradoc.army.mil/WEG/Asset/S-300V\\_\[SA-12a\\_Gladiator\]\\_Russian\\_Long-Range\\_Air\\_Defense\\_Missile\\_System](https://odin.tradoc.army.mil/WEG/Asset/S-300V_[SA-12a_Gladiator]_Russian_Long-Range_Air_Defense_Missile_System)

**WEG34** Worldwide Equipment Guide: S-400 Triumph (SA-21 Growler) (local extract pilot/weg\_extract/s400.json). US Army TRADOC, 2025-11-17. [https://odin.tradoc.army.mil/WEG/Asset/S-400\\_Triumph\\_\[SA-21\\_Growler\]\\_Russian\\_Long-Range\\_Surface-to-Air\\_Missile\\_\[SAM\]\\_System](https://odin.tradoc.army.mil/WEG/Asset/S-400_Triumph_[SA-21_Growler]_Russian_Long-Range_Surface-to-Air_Missile_[SAM]_System)

**WEG35** 50R6 Vityaz (S-350E) Russian Medium-Range Air Defense Missile System (local extract). US Army TRADOC ODIN WEG, 2025-06-10. [https://odin.tradoc.army.mil/WEG/Asset/50R6\\_Vityaz\\_\[S-350E\]\\_Russian\\_Medium-Range\\_Air\\_Defense\\_Missile\\_System](https://odin.tradoc.army.mil/WEG/Asset/50R6_Vityaz_[S-350E]_Russian_Medium-Range_Air_Defense_Missile_System)

**WEG36** Worldwide Equipment Guide: 9A52-2 (BM-30 Smerch-M); 9A52-4 Tornado; 9A53-S Tornado-S (local extract). US Army TRADOC, 2024-10 to 2026-02. [https://odin.tradoc.army.mil/WEG/Asset/9A52-2\\_\[BM-30\\_Smerch-M\]\\_Russian\\_300mm\\_Multiple\\_Launch\\_Rocket\\_System](https://odin.tradoc.army.mil/WEG/Asset/9A52-2_[BM-30_Smerch-M]_Russian_300mm_Multiple_Launch_Rocket_System)

**WEG37** WEG: SNAR-10 and SNAR-10M1 (Big Fred) entries, last modified 2024-10-22. TRADOC, 2024-10. [https://odin.tradoc.army.mil/WEG/Asset/SNAR-10M1\\_\[Big\\_Fred\]\\_Russian\\_Battlefield\\_Surveillance\\_Radar](https://odin.tradoc.army.mil/WEG/Asset/SNAR-10M1_[Big_Fred]_Russian_Battlefield_Surveillance_Radar)

**WEG38** WEG extract: 9K35 Strela-10 and 9K33 Osa pages. US Army TRADOC, 2025-06-10. [https://odin.tradoc.army.mil/WEG/Asset/9K35\\_Strela-10\\_\[SA-13\\_Gopher\]\\_Russian\\_Short-Range\\_Surface-to-Air\\_Missile\\_System](https://odin.tradoc.army.mil/WEG/Asset/9K35_Strela-10_[SA-13_Gopher]_Russian_Short-Range_Surface-to-Air_Missile_System)

**WEG39** Worldwide Equipment Guide online: MTU-72 AVLБ and TMM-3 mobile bridging system (local extract tmm.json). TRADOC, 2024-10 to 2025-04. [https://odin.tradoc.army.mil/WEG/Asset/TMM-3\\_Russian\\_6x6\\_Mobile\\_Bridging\\_System](https://odin.tradoc.army.mil/WEG/Asset/TMM-3_Russian_6x6_Mobile_Bridging_System)

**WEG40** 9K330 Tor (SA-15 Gauntlet) Russian Short-Range Surface-to-Air Missile System. US Army TRADOC, Worldwide Equipment Guide (ODIN), 2025-06-10. [https://odin.tradoc.army.mil/WEG/Asset/9K330\\_Tor\\_\[SA-15\\_Gauntlet\]\\_Russian\\_Short-Range\\_Surface-to-Air\\_Missile\\_System](https://odin.tradoc.army.mil/WEG/Asset/9K330_Tor_[SA-15_Gauntlet]_Russian_Short-Range_Surface-to-Air_Missile_System)

**WEG41** 9A53-G Tornado Russian 122mm Multiple Rocket Launcher (local extract). US Army TRADOC, ODIN WEG, 2024-08-21. [https://odin.tradoc.army.mil/WEG/Asset/9A53-G\\_Tornado\\_Russian\\_122mm\\_Multiple\\_Rocket\\_Launcher](https://odin.tradoc.army.mil/WEG/Asset/9A53-G_Tornado_Russian_122mm_Multiple_Rocket_Launcher)

**WEG42** Worldwide Equipment Guide: TOS-1A Solntsepek, TOS-1 Buratino and TOS-2 220mm Heavy Flamethrower System records. US Army TRADOC (ODIN), 2024-09-05 (TOS-1A); 2024-10-10 (TOS-1); 2024-10-17 (TOS-2). [https://odin.tradoc.army.mil/WEG/Asset/TOS-1A\\_Solntsepek\\_Russian\\_220mm\\_Heavy\\_Flamethrower\\_System](https://odin.tradoc.army.mil/WEG/Asset/TOS-1A_Solntsepek_Russian_220mm_Heavy_Flamethrower_System)

**WEG43** WEG extracts: 2S6M1 Tunguska-M1 and 2S6M Tunguska-M (SA-19 Grison). US Army TRADOC, 2025-06-10. [https://odin.tradoc.army.mil/WEG/Asset/2S6M1\\_Tunguska-M1\\_Russian\\_30mm\\_Self-Propelled\\_Anti-Aircraft\\_System](https://odin.tradoc.army.mil/WEG/Asset/2S6M1_Tunguska-M1_Russian_30mm_Self-Propelled_Anti-Aircraft_System)

**WEG44** WEG entry: UR-77 Meteorit Russian Mine Clearing Vehicle (local extract ur77.json). US Army TRADOC, 2024-10-24. [https://odin.tradoc.army.mil/WEG/Asset/UR-77\\_Meteorit\\_Russian\\_Mine\\_Clearing\\_Vehicle](https://odin.tradoc.army.mil/WEG/Asset/UR-77_Meteorit_Russian_Mine_Clearing_Vehicle)

**WEG45** 9K512 (Uragan-1M) Russian 220mm/300mm Multiple Launch Rocket System (MRL), Worldwide Equipment Guide (local extract). US Army TRADOC G-2, 2025-03-27. [https://odin.tradoc.army.mil/WEG/Asset/9K512\\_\[Uragan-1M\]\\_Russian\\_220mm\\_300mm\\_Multiple\\_Launch\\_Rocket\\_System\\_\[MRL\]](https://odin.tradoc.army.mil/WEG/Asset/9K512_[Uragan-1M]_Russian_220mm_300mm_Multiple_Launch_Rocket_System_[MRL])

**WEG46** 9P140 (BM-27 Uragan) Russian 220mm Multiple Rocket Launch (MRL). US Army TRADOC, Worldwide Equipment Guide (ODIN), local extract, 2024-10-28. [https://odin.tradoc.army.mil/WEG/Asset/9P140\\_\[BM-27\\_Uragan\]\\_Russian\\_220mm\\_Multiple\\_Rocket\\_Launch\\_\[MRL\]](https://odin.tradoc.army.mil/WEG/Asset/9P140_[BM-27_Uragan]_Russian_220mm_Multiple_Rocket_Launch_[MRL])

**WEG47** ODIN WEG: GMZ-3 Russian Tracked Minelaying Vehicle; ISDM Zemledeliye Russian 8x8 Mine-Laying System (local extract). US Army TRADOC, 2024-10-16. [https://odin.tradoc.army.mil/WEG/Asset/ISDM\\_Zemledeliye\\_Russian\\_8x8\\_Mine-Laying\\_System](https://odin.tradoc.army.mil/WEG/Asset/ISDM_Zemledeliye_Russian_8x8_Mine-Laying_System)

**WEG48** R-330Zh Zhitel Russian Cellular Jamming and Direction Finding System. US Army TRADOC, ODIN Worldwide Equipment Guide, 2025-09-03. [https://odin.tradoc.army.mil/WEG/Asset/R-330Zh\\_Zhitel\\_Russian\\_Cellular\\_Jamming\\_and\\_Direction\\_Finding\\_System](https://odin.tradoc.army.mil/WEG/Asset/R-330Zh_Zhitel_Russian_Cellular_Jamming_and_Direction_Finding_System)

**WEG2015** Worldwide Equipment Guide 2015, Volume 1 Ground Systems, LV18/LV19 page and tier tables. TRADOC, 2015. <https://archive.org/details/USArmyWorldwideEquipmentGuide2015>

**WMC1** Wikimedia Commons photographs of 9616E (MAKS-2011, MAKS-2009, VVS 100th anniversary 2012, Patriot Park 2016). Wikimedia Commons, 2009-2016. <https://commons.wikimedia.org/wiki/Category:9616E>

**WMC2** MasteringTheArmament2018-13.jpg (Russian MoD photo, caption). Russian MoD via Wikimedia Commons, 2018. <https://commons.wikimedia.org/wiki/File:MasteringTheArmament2018-13.jpg>

**WMC3** File:ElectronicWarfareExercise2018-01.jpg (MoD caption: 74th Separate Guards MR Brigade exercise with Borisoglebsk-2). Wikimedia Commons / Russian MoD, 2018. <https://commons.wikimedia.org/wiki/File:ElectronicWarfareExercise2018-01.jpg>

**WMC4** File:Krasukha-2 (Kpacuxa-2) Unloaded.jpg. Wikimedia Commons (Vitaly V. Kuzmin), 2014-12-11. [https://commons.wikimedia.org/wiki/File:Krasukha-2\\_\[Kpacuxa-2\]\\_Unloaded.jpg](https://commons.wikimedia.org/wiki/File:Krasukha-2_[Kpacuxa-2]_Unloaded.jpg)

**WMC5** File:1L269E Krasukha-2 Army-2018.jpg. Wikimedia Commons (Boevaya mashina), 2018-08-22. [https://commons.wikimedia.org/wiki/File:1L269E\\_Krasukha-2\\_Army-2018.jpg](https://commons.wikimedia.org/wiki/File:1L269E_Krasukha-2_Army-2018.jpg)

**WMC6** File:1RL257E Krasukha-4 Army-2018.jpg. Wikimedia Commons (Boevaya mashina), 2018-08-22. [https://commons.wikimedia.org/wiki/File:1RL257E\\_Krasukha-4\\_Army-2018.jpg](https://commons.wikimedia.org/wiki/File:1RL257E_Krasukha-4_Army-2018.jpg)

**WMC7** File:1RL257E Krasukha-4 Army-2022 2022-08-20 2459.jpg. Wikimedia Commons (Mike1979 Russia), 2022-08-20. [https://commons.wikimedia.org/wiki/File:1RL257E\\_Krasukha-4\\_Army-2022\\_2022-08-20\\_2459.jpg](https://commons.wikimedia.org/wiki/File:1RL257E_Krasukha-4_Army-2022_2022-08-20_2459.jpg)

**WMC8** File:TB2015ExhibitionP2-47.jpg (R8-341V Leer-3 ECM system with UAV). Wikimedia Commons, 2015. <https://commons.wikimedia.org/wiki/File:TB2015ExhibitionP2-47.jpg>

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**WMC10** File:Пазверьивание ПЗБ Палантин на учениях.jpg. Wikimedia Commons (Ministry of Defence of Russia), 2019-03. [https://commons.wikimedia.org/wiki/File:Пазверьивание\\_ПЗБ\\_Палантин\\_на\\_учениях.jpg](https://commons.wikimedia.org/wiki/File:Пазверьивание_ПЗБ_Палантин_на_учениях.jpg)

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